

Matrix Spencer: Ten Standard Deviations Suffice and An Almost-Linear Time Algorithm for Dense Input*

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Abstract

The Matrix Spencer conjecture asserts that for all symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$ there are signs $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ with $\|\sum_{i=1}^n \varepsilon_i A_i\| = O(\sqrt{n})$. Random signs give only the matrix-concentration bound $O(\sqrt{n \log n})$, and the conjecture was known only under rank, block-diagonal, or Frobenius-norm restrictions. We prove it: a signing of discrepancy below $10\sqrt{n}$ always exists. We also give a randomized algorithm that finds a signing of discrepancy below $13\sqrt{n}$ with failure probability at most p using $n^{3+o(1)}$ $\text{polylog}(1/p)$ arithmetic operations in the real-arithmetic model, which matches the size n^3 of the dense input up to subpolynomial factors.

The existence proof is a partial-coloring argument with one new estimate: a hereditary Gaussian small-ball bound for the spectral body $\{x \in \mathbb{R}^n : \|\sum_i x_i A_i\| \leq R\}$, proved by interpolating a log-partition function from a diagonal model to the noncommutative one under a matrix-weighted Poincaré inequality. Proving the conjecture and bringing the constant below 10 are different problems. The partial-coloring argument loses a large factor twice, when it turns Gaussian measure into signs through a union bound and when it proves the small-ball estimate at a radius far larger than necessary. We remove the first loss by a lossless coding of Gaussian measure into signs and the second by smooth spectral barriers with certified coefficients. The algorithms project Gaussian points onto a smoothed version of the spectral body, whose derivatives are traces against one Gibbs matrix. Four successively cheaper ways of maintaining that matrix bring the running time down to $n^{3+o(1)}$.

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1 Introduction

Spencer’s “six standard deviations suffice” theorem [Spe85] states that every family of n sets on an n -element ground set admits a $\{-1, 1\}$ -coloring of discrepancy at most $6\sqrt{n}$, removing the $\sqrt{\log n}$ loss of independent random signs. The proof is a partial-coloring argument with an entropy count and gives no algorithm. Bansal [Ban10] gave the first randomized polynomial-time algorithm attaining the same asymptotic bound, through a random walk guided by semidefinite programs; Lovett and Meka [LM15] gave a simpler constructive proof by their edge walk; and Rothvoss [Rot17] showed that projecting a Gaussian point onto the intersection of the cube with any symmetric convex body of Gaussian measure $e^{-\Omega(n)}$ yields a partial coloring, so that the only question left about a body is its Gaussian measure.

The Matrix Spencer conjecture is the noncommutative form of Spencer’s theorem [Spe85]. For symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$ it asks for signs $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ with $\|\sum_{i=1}^n \varepsilon_i A_i\| = O(\sqrt{n})$. Diagonal matrices recover the set-system case, and random signs give only the matrix-concentration bound $O(\sqrt{n \log n})$ (Tropp [Tro15, Chapter 4]), so the conjecture asks whether the $\sqrt{\log n}$ can be removed for matrices as Spencer removed it for sets. Zouzias [Zou12] extended the hyperbolic-cosine method to matrices; Hopkins, Raghavendra, and Shetty [HRS22] proved the conjectured bound under $\|A_i\|_F \leq n^{1/4}$; Dadush, Jiang, and Reis [DJR22] proved the low-rank and block-diagonal cases through a partial-coloring framework and reduced the general case to a geometric covering question; and Bansal, Jiang, and Meka [BJM23] handled rank at most $n/\log^3 n$. The general case has remained open.

This paper answers two questions. Does the conjecture hold? And if it does, can a signing be found in time almost linear in the size of the input, which for n dense $n \times n$ matrices is n^3 ? We answer both affirmatively. A signing of discrepancy below $10\sqrt{n}$ always exists, and a randomized algorithm finds a signing of discrepancy below $12.40\sqrt{n}$, with failure probability at most p , in $n^{3+o(1)}$ $\text{polylog}(1/p)$ arithmetic operations, which matches the input size up to subpolynomial factors. The two answers rest on two results proved first: an existence bound $156000\sqrt{n}$ whose per-round structure each of the four algorithms implements. The second is a chain of four algorithms with running times $n^{2+\omega/2}$, $n^{1+\omega(1,1/2,1)}$, $n^{3+\gamma_*}$, and $n^{3+o(1)}$, each removing one obstacle left by the previous one. Proving the conjecture, reaching a constant near 10, and reaching the running time $n^{3+o(1)}$ are three separate problems, and Section 2 says how each is solved.

1.1 Our results

Theorem 1.1 (Matrix Spencer; informal version of Theorems 5.1 and A.1). *There is a universal constant $C < 10$ such that, for all symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$, there is a vector $\varepsilon \in \{-1, 1\}^n$ satisfying $\|\sum_{i=1}^n \varepsilon_i A_i\| \leq C\sqrt{n}$. Theorem 5.1 proves the bound with $C = 156000$, and Theorem A.1 sharpens it to $C = 9.9996 < 10$.*

Theorem 1.2 (Algorithm; informal version of Theorem B.1). *For dense input matrices and every $p \in (0, 1/2)$, a signing of discrepancy below $12.40\sqrt{n}$ can be found with failure probability at most p using*

$$n^{3+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model (Theorem B.1).

Remark 1.3 (Reading Theorem 1.2). The algorithm of Theorem B.1 either outputs a signing or reports failure, with failure probability at most p . It is designed for large n ; for $n \leq 10^{24}$ it draws independent uniform random signs and checks the norm, repeating until a trial is accepted (Lemma B.11). It is built on the algorithm of Theorem 9.1, which finds a signing of

Section	Fixed-multiplier solver	Exponent	Numerical value
6	full Gibbs matrix, accelerated blocks	$2 + \omega/2$	< 3.185670
7	selected-coordinate Gibbs oracle	$1 + \omega(1, 1/2, 1)$	< 3.042776
8	accelerated block oracle	$3 + \gamma_\star$	≤ 3.018320
9	recursive Gibbs tracking	$3 + o(1)$	3, unconditionally

Table 1: The four algorithms. Each finds a signing of discrepancy at most $156000\sqrt{n}$ with failure probability at most p ; an exponent e means $n^{e+o(1)}$ $\text{polylog}(1/p)$ arithmetic operations in the real-arithmetic model of Section 3, so every entry hides $n^{o(1)}$ and $\text{polylog}(1/p)$ factors. Appendix B lowers the constant of the last algorithm to 12.40 at the same exponent. The numerical values use $\omega < 2.371339$, $\omega(1, 0.4, 1) < 2.009280$, and $\omega(1, 1/2, 1) < 2.042776$ from Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1]; the last row uses only $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$.

discrepancy at most $156000\sqrt{n}$ within the same running time. Three algorithms lead up to that one (Theorems 6.1, 7.1, and 8.1); they find a signing of discrepancy at most $156000\sqrt{n}$ in $n^{2+\omega/2+o(1)}$, $n^{1+\omega(1,1/2,1)+o(1)} < n^{3.042776+o(1)}$, and $n^{3+\gamma_\star+o(1)} \leq n^{3.018320+o(1)}$ operations, up to the same $\text{polylog}(1/p)$ factor, where

$$\gamma_\star := \min_{0 \leq b \leq 1/2} \max\{\omega(1, b, 1) + b - 5/2, 1/2 - b\} \leq 0.018320;$$

in particular $\gamma_\star = 0$ whenever $\omega(1, 1/2, 1) = 2$.

The exponents can be read as follows. The dense input has n^3 entries, so $n^{3+o(1)}$ is the time to read it, up to subpolynomial factors. Every algorithm runs the partial-coloring loop of the existence proof: $O(\log n)$ rounds whose active dimensions m_t satisfy $\sum_t m_t = O(n)$; in each round a bounded expected number of Gaussian projections onto a smoothed spectral body; in each projection a bisection over one Lagrange multiplier; and for each fixed multiplier a strongly convex minimization over a box whose derivatives are traces against a Gibbs matrix $\exp(aB(x)/R)$, $B(x) := \sum_i x_i A_i$, with R the radius of the spectral body and a a smoothing parameter. The four algorithms differ only in that last step, and Section 2.2 says what each of them changes there. The last of them uses only two facts from fast matrix multiplication, $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$, and they are theorems of Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] and of Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1], not hypotheses. Table 1 collects the four bounds.

Theorem 1.4 (Bit complexity; informal version of Theorem 10.1). *The recursive Gibbs-tracking algorithm of Section 9, which for every $p \in (0, 1/2)$ finds a signing of discrepancy at most $156000\sqrt{n}$ with failure probability at most p in $n^{3+o(1)} \text{polylog}(1/p)$ arithmetic operations, can be run on fixed-point words of $O(\log n)$ bits. Under the known bounds $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$, where $\alpha_{\text{dual}} := \sup\{b : \omega(1, b, 1) = 2\}$ is the dual matrix-multiplication exponent (in the asymptotic-rank sense of Definition 10.2), it then uses $n^{3+o(1)} \text{polylog}(1/p) + \tilde{O}(n^3 L_{\text{in}})$ bit operations on inputs whose entries have bit length L_{in} .*

Dense input means that each A_i is given as an $n \times n$ array of real numbers, and the real-arithmetic model charges unit cost for every arithmetic operation and comparison on real numbers; both are fixed in Section 3. Theorem 1.2, the theorem behind it, and the four theorems of Remark 1.3 count arithmetic operations, not bit operations; Theorem 1.4 and Section 10 supply the bit complexity for the cubic-time algorithm at the constant 156000 only, and no fixed-point word bound is claimed for the algorithm of Appendix B.

Proving the conjecture, bringing the constant near 10, and bringing the running time to $n^{3+o(1)}$ are three separate problems. Section 2 treats them in turn: the proof is one new estimate, a hereditary Gaussian small-ball bound for the spectral body, inside the partial-coloring frame of Rothvoss [Rot17] and of Dadush, Jiang, and Reis [DJR22]; the constant is lowered by removing two losses of that frame, the union bound that converts Gaussian measure into signs and the radius at which the small-ball estimate is proved (Section 2.1); and the running time comes down in four stages, each removing one obstacle left by the previous one (Section 2.2).

1.2 Previous work

Constructive discrepancy. Spencer’s bound [Spe85] is proved by partial coloring with an entropy argument, which gives no algorithm. Bansal [Ban10] obtained the bound algorithmically by a random walk whose steps are guided by semidefinite programs, and Lovett and Meka [LM15] by the edge walk, a Brownian motion inside the cube that sticks to the constraint hyperplanes it meets. Rothvoss [Rot17] showed that projecting a Gaussian point onto the intersection of the cube with a symmetric convex body of Gaussian measure $e^{-\Omega(n)}$ yields a partial coloring with a constant fraction of the coordinates frozen; this is the step that Lemma 5.4 quantifies.

Matrix Spencer. Zouzias [Zou12] extended the hyperbolic-cosine method to matrices, an early step toward the noncommutative setting. Two works from 2022 went below the matrix-concentration bound under hypotheses. Hopkins, Raghavendra, and Shetty [HRS22] proved the conjectured bound when $\|A_i\|_F \leq n^{1/4}$, which includes matrices of rank at most $\sqrt{n}$, using ideas from quantum communication. Dadush, Jiang, and Reis [DJR22] developed a mirror-descent partial-coloring framework, proved the low-rank and block-diagonal cases, and reduced the general problem to a geometric covering question. Bansal, Jiang, and Meka [BJM23] then handled rank at most $n/\log^3 n$ by combining partial coloring with the refined noncommutative Khintchine inequality of Bandeira, Boedihardjo, and van Handel [BBH23] for correlated Gaussian matrices.

Organization. Section 2 is an informal overview, first of the proof and of the constant, then of the four algorithms. Section 3 fixes notation and the computational model and collects the matrix-analytic tools used in Section 4. Section 4 proves the hereditary Gaussian small-ball estimate with explicit constants, and Section 5 combines it with a refreshed Gaussian projection step to prove the discrepancy bound $156000\sqrt{n}$. Section 6 gives the $n^{2+\omega/2+o(1)}$ algorithm, Section 7 the $n^{1+\omega(1,1/2,1)+o(1)}$ algorithm based on a selected-coordinate Gibbs oracle, Section 8 the accelerated block algorithm with exponent $3+\gamma_*$, and Section 9 the $n^{3+o(1)}$ algorithm, which tracks the Gibbs matrix recursively. Section 10 proves that the cubic-time algorithm keeps its exponent on fixed-point words of $O(\log n)$ bits. Appendix A proves the existence bound $9.9996\sqrt{n}$, and Appendix B the cubic-time algorithm with constant 12.40.

2 Technical overview

This section treats the discrepancy bound in two steps, proving the conjecture and sharpening the constant, and then the running time in four steps, from $n^{2+\omega/2}$ to $n^{3+o(1)}$.

2.1 Proving the conjecture and sharpening the constant

Proving the conjecture and reaching a constant near 10 are separate challenges. The first needs one new estimate inside a known frame, and the frame does not care about the constant: the partial-

coloring argument of Section 5 tolerates a radius coefficient in the hundreds and a guaranteed freezing fraction of 2.8 percent per round, and gives $156000\sqrt{n}$. The second needs every loss the frame tolerates to be located and paid for exactly, which is what Appendix A does, with two ingredients, a lossless Gaussian-to-sign coding lemma, Kleitman’s cube-diameter theorem [Kle66] applied to a cube-translate average, and smooth spectral barriers with certified scalar data, and one that is bookkeeping, a seven-stage schedule with hereditary radii. The next three paragraphs prove the conjecture; the four paragraphs from ‘Where the constant comes from’ to ‘A seven-stage schedule’ sharpen the constant; the last carries a small constant over to the algorithm.

The hereditary small-ball estimate. The partial-coloring step needs $\gamma_m(K_S) \geq e^{-cm}$, with a small explicit c , for the spectral body

$$K_S := \{x \in \mathbb{R}^S : \|\sum_{i \in S} x_i A_i\| \leq c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}}\}, \quad m := |S|, \quad \ell_{\text{asp}} := \max\{1, \log(n/m)\},$$

uniformly over every active set S , since the active set is random and changes after every round; matrix concentration bounds the typical norm of $\sum_{i \in S} \xi_i A_i$ by $O(\sqrt{m \log n})$ and says nothing about the probability of the atypical event that the norm is $O(\sqrt{m} \ell_{\text{asp}})$. After rescaling by the target radius, define $\mathcal{A}(\xi) := (c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}})^{-1} \sum_{i \in S} \xi_i A_i$. The probability $\Pr[\|\mathcal{A}(\xi)\| < 1]$ is the endpoint of a log-partition function interpolating from an explicit diagonal model to the full noncommutative one, and its second derivative is the difficulty. A spectral 0–1 split separates the eigenvalues near the barriers ± 1 from the regular part of the spectrum; the edge component is absorbed by the curvature of the barrier, the regular component is small because the variance proxy of the rescaled matrices is small, and together they verify the positive-curvature hypothesis of a matrix-weighted Poincaré inequality, which controls the weighted resolvent forms bounding the interpolation Hessian; resolvent moments of order $\tau(\ell_{\text{asp}}) := \lceil 256\ell_{\text{asp}}/17 \rceil$ bound the remaining noncommutative products. The result is $\gamma_m(K_S) \geq \exp(-0.003712m)$ for all $1 \leq m \leq n$ (Lemma 4.1). The coefficient $c_{\text{rad}}(\ell_{\text{asp}})$ takes six values, from 279 at $\tau(\ell_{\text{asp}}) = 16$ to 185 for $\tau(\ell_{\text{asp}}) \geq 44$ (Table 2): as the moment order grows with the aspect ratio the most efficient barrier parameters change, and the six choices produce the single exponent 0.003712 that the projection step consumes; the division is quantitative, not conceptual.

From Gaussian measure to signs. A measure bound for K_S is not yet a partial coloring. Lemma 5.4 projects a Gaussian point onto K_S intersected with the coordinate box Q_y of the current fractional coloring y and compares its distance to $K_S \cap Q_y$ with its distances to all low-codimension coordinate sections of K_S : on an event of constant probability, valid for every active size $m \geq N_0 := 3 \cdot 10^8$, fewer than δm tight facets would define a coordinate section of K_S inside a relaxation of the constraint set, which on this event is strictly closer to the Gaussian point than the box, a contradiction. Hence, if $\gamma_m(K_S) \geq \exp(-0.00372m)$, one trial produces $x \in K_S$ with $y + 2x/\varepsilon_{\text{box}} \in [-1, 1]^m$ and at least δm coordinates of the new coloring equal to ± 1 , with probability greater than 0.12, for $\delta = 0.0281672$ and $\varepsilon_{\text{box}} = 0.4235$; the gap between 0.003712 and 0.00372 is the room later spent on smoothing the body. A successful round freezes at least δm variables and adds at most $(2c_{\text{rad}}(\ell_{\text{asp}})/\varepsilon_{\text{box}})e^{-u/2}\ell_{\text{asp}}\sqrt{n}$ to the operator norm, $u := \log(n/m)$. The lemma is a statement about one round, not about the trajectory, and it needs $m \geq N_0$.

Summing the increments. Each round freezes at least δm coordinates, and possibly far more, so the successive values of u are separated by at least $\lambda := -\log(1 - \delta)$ but need not lie on a grid. While $u < 1$, one has $c_{\text{rad}} = 279$, the increment $(2 \cdot 279/\varepsilon_{\text{box}})e^{-u/2}\sqrt{n}$ is monotone in u , and at most 35 such rounds occur, so an exact geometric series sums them. For $u \geq 1$ the

summand $c_{\text{rad}}(u)ue^{-u/2}$ is neither monotone nor continuous, since c_{rad} jumps at the five regime endpoints; the separated-sum estimate (Lemma 5.5) bounds every λ -separated sum by the sum of the suprema of the summand over consecutive bins of length λ , which rational Taylor enclosures evaluate. All projection rounds contribute less than $154709.3\sqrt{n}$; below N_0 , independent rounding with the fractional coloring as mean and the matrix Laplace-transform bound of Tropp [Tro15, Chapter 3] contribute at most $6.5\sqrt{n}$ with probability at least $1/2$; failed trials are discarded and repeated. Hence a signing of discrepancy at most $156000\sqrt{n}$ exists (Theorem 5.1). The algorithms of Sections 6–9 retain this bound; the smaller constants are the subject of the rest of this subsection.

Figure 1 shows how the small-ball estimate, projection, and summation fit together.

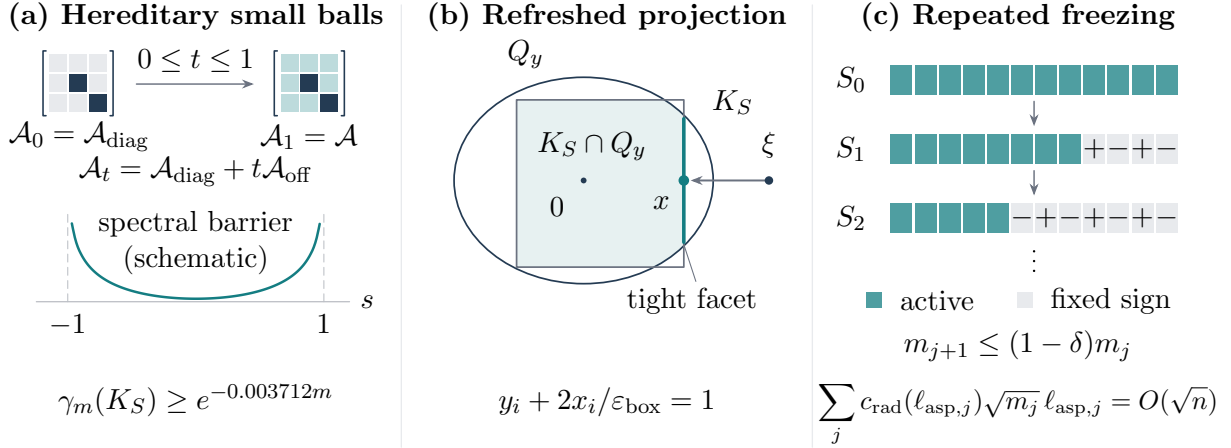


Figure 1: The partial-coloring mechanism (Lemmas 4.1 and 5.4). (a) The normalized interpolation $\mathcal{A}_t$ (Definition 4.4) gives Gaussian mass for every active set S . (b) On a successful projection trial, tight facets of Q_y become fixed signs after affine rescaling. (c) The active sets shrink geometrically until terminal rounding applies; $m_j := |S_j|$ and $\ell_{\text{asp},j} := \max\{1, \log(n/m_j)\}$. The barrier, geometry, and coordinate counts are schematic.

Where the constant comes from. The bound 156000 dominates the sum over the rounds of the increments

$$\frac{2c_{\text{rad}}(\ell_{\text{asp}})}{\varepsilon_{\text{box}}} e^{-u/2} \ell_{\text{asp}} \sqrt{n},$$

plus $6.5\sqrt{n}$ for the terminal rounding, and slack sits in two factors of the increment and in the number of rounds. The radius coefficient c_{rad} , between 185 and 279, is the radius at which Lemma 4.1 proves its small-ball exponent, and the exponent 0.003712 it proves there is used only through the threshold 0.00372 of the projection lemma; a proof that reached the same exponent at a smaller radius would lower every increment in proportion. The factor $2/\varepsilon_{\text{box}}$, with $\varepsilon_{\text{box}} = 0.4235$, is the box scale at which the Gaussian point's distance to $K_S \cap Q_y$ is compared with its distances to coordinate sections of K_S . The number of rounds comes from the same comparison: it is a union bound over all coordinate sets of size at most δm , $e^{h(\delta)m}$ of them with h the binary entropy, and its entropy reserve $\sqrt{2h(\delta)}$ together with the $\sqrt{\delta + 2(0.00372)}$ term of the section distance forces $\delta = 0.0281672$. The small δ makes the rounds only $\lambda = -\log(1 - \delta)$ apart in u , so at most 35 rounds start before u reaches 1 and contribute up to $36550.1\sqrt{n}$, and the rounds at $u \geq 1$ add a further $118159.2\sqrt{n}$. The coding lemma of the next paragraph removes the union bound, and with it both the factor $2/\varepsilon_{\text{box}} = 4.72$, which becomes $1/s_\star = 1.48$ with $s_\star = 0.67448$ (Lemma A.3), and

the small δ ; the smooth barriers of the paragraph after it lower the radius; the seven-stage schedule removes no factor and only organizes the rounds.

Lossless Gaussian-to-sign coding. The projection step converts Gaussian measure into signs at a cost: the measure bound must survive a union bound over which coordinates freeze, and only a fraction δ of them is guaranteed to. Lemma A.3 converts measure into signs without the union bound. It compares the Gaussian measure of the body with a translated density that is peaked (Lemma A.2) and reads a sign pattern off the translate by Kleitman’s cube-diameter theorem [Kle66]; the entropy the argument needs is a reserve certified separately (Lemma A.32), and it is spent once, so a stage signs most of its active coordinates instead of the 2.8 percent that the projection lemma guarantees. The lemma is a statement about Gaussian measure and signs only; it does not compute the signs, and this is why the algorithms of Sections 6–9 cannot use it.

Smooth barriers and certified scalar data. With the coding lemma in place, the constant is decided by the radius at which the small-ball exponent can be proved. In Appendix A the estimate of stage k is proved at radius $\sqrt{w_k n / \eta_k}$, where $w_k n$ bounds the active count at that stage, and the stage increments $\sqrt{w_k / \eta_k} / s_\star$ sum to less than 10 (Table 5), against $c_{\text{rad}} \geq 185$ before. The spectral 0–1 split of Lemma 4.1 is replaced by smooth barriers, even power series with certified coefficients (Definition A.30). Three quantities are then bounded separately: the curvature of the barrier along the spectrum, by resolvent means and parity-sensitive polynomial energies; the noncommutative products, by covariance-power moments; and the Gaussian integral, by a radial integration from the origin whose finite terms are exact Gaussian moments, after which a determinant factor returns the estimate to the standard Gaussian. At the last four stages the barrier is the two-component family of Definition A.15, with a base coefficient and a tail coefficient; the first three stages carry explicit coefficient tables. Every scalar inequality that a stage needs is an all-domain certificate with exact rational data: the coefficient tables are part of the proof, and the text is the argument that they suffice.

A seven-stage schedule. As the active set shrinks, the aspect ratio n/m grows and the best barrier changes with it, exactly as the six regimes of Table 2 change with $\tau(\ell_{\text{asp}})$; the sharpened proof uses seven stages with fixed spectral parameters and radial budgets (Definition A.30). Each stage’s estimate must be hereditary, valid for every smaller active set met during the signing (Lemma A.33), and beyond the seventh stage a uniform small-ball estimate at large aspect ratio continues the signing to the end. Lemma A.34 sums the seven increments and the entire remaining tail: the first stage, on all n coordinates, costs $7.0527\sqrt{n}$, the second $2.4207\sqrt{n}$, the remaining five together $0.5255\sqrt{n}$, and the tail $0.0006\sqrt{n}$ (Table 5), so there is a signing of discrepancy at most $9.9996\sqrt{n}$ (Theorem A.1). The first stage decides the constant. Spencer’s constant 6 in the diagonal case [Spe85] is below this, and what the route can give is an open question, stated at the end of the section.

The algorithmic constant. Appendix B recovers a small constant for the cubic-time algorithm without the coding lemma, with two kinds of projection. The near-integral step projects a scaled Gaussian point onto the smoothed body inside a shrunken box, and a Gaussian-width bound in terms of the small-ball exponent (Lemma B.14) shows that the output is nearly integral in a Gaussian-weighted sense (Lemma B.30), so this step needs no count of tight facets. The remaining fractional mass is rounded one binary digit at a time by an unbiased repair at four fixed levels: each digit is fixed by a helper signing of the coordinates that carry it, drawn with a fresh random

orientation so that the repair is unbiased (Lemma B.31). A helper is a repair call one level down, and at depth zero it is a certified suffix signing, which projects toward a point sampled from a Euclidean ball, does count tight facets, more than $m/4$ per call (Lemma B.9), and reuses the seven original small-ball estimates of Appendix A (Definition A.15), not the refined stages of Table 5, on a geometric schedule of hereditary radii; Lemma B.12 bounds its discrepancy. All projections run through the certified multiplier wrapper and recursive solver of Section 9, so the running time stays $n^{3+o(1)}$ polylog($1/p$) and the discrepancy is below $12.40\sqrt{n}$ (Lemma B.35, Theorem B.1). The algorithm computes Euclidean projections only; the polar bodies and coordinate sections of the analysis never appear in it.

2.2 From polynomial time to $n^{3+o(1)}$

Four algorithms take the round's projection from a convex program with one spectral constraint, solvable in polynomial time but far above n^3 , down to $n^{3+o(1)}$, and each removes one obstacle left by the previous one. The first replaces the spectral constraint by a smooth one and batches all derivatives through one Gibbs matrix, and its cost is a dense diagonalization at every block query. The second forms the Gibbs matrix densely only at snapshots and sketches its increment between them, and its exponent is $1 + \omega(1, 1/2, 1)$, which is 3 only under an open conjecture. The third shares one sketch across a block of coordinates and narrows the gap to $\gamma_\star \leq 0.018320$, and its loss is that the increment is sketched afresh at every query. The fourth tracks the Gibbs matrix across coordinate steps, so that the variance of an update is paid for by the descent energy of the step, and reaches $n^{3+o(1)}$ using only $\omega < 5/2$ and $\alpha_{\text{dual}} > 1/4$. A fifth step, not about the exponent, runs the fourth algorithm on words of $O(\log n)$ bits.

The $n^{2+\omega/2}$ algorithm. The trajectory needs a point of $K_S \cap Q_y$ within a small distance ρ of the exact projection, not the projection itself. Two ideas survive in all four algorithms: smooth the spectral constraint, and batch all requested derivatives through one matrix function. With $R := c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ and $B(x) := \sum_{i \in S} x_i A_i$, the smooth body is

$$D := \{x \in \mathbb{R}^S : h(x) \leq 1 - s\}, \quad h(x) := \frac{1}{a} \log(\text{tr}[\exp(aB(x)/R)] + \text{tr}[\exp(-aB(x)/R)]),$$

with $a := 2 \cdot 10^6 \log(2n)$ and $s := 5 \cdot 10^{-7}$; it satisfies $(1 - 2s)K_S \subseteq D \subseteq (1 - s)K_S$ and $\gamma_m(D) \geq (1 - 10^{-6})^m \gamma_m(K_S) \geq \exp(-0.00372m)$, so the projection lemma applies to D unchanged. One Lagrange multiplier $\lambda \geq 0$ reduces the projection of ξ onto $D \cap Q_y$ to minimizing the strongly convex $\frac{1}{2}\|x - \xi\|_2^2 + \lambda h(x)$ over the box; λ is found by bisection, which inexact minimization does not derail; a radial rescaling toward the origin repairs slight infeasibility, and coordinates within 2ρ of a facet are snapped to it, the slack s keeping the snapped point in K_S (Theorem 6.1). For a block of ℓ coordinates the block gradient of h is $\tilde{O}(\ell)$ -Lipschitz, so accelerated random block-coordinate descent needs $\tilde{O}(m/\sqrt{\ell})$ block queries; one query diagonalizes $B(x)$ in $n^{\omega+o(1)}$ operations and forms the signed Gibbs matrix W_x , the matrix with $\partial_i h(x) = R^{-1} \text{tr}[W_x A_i]$, after which the ℓ requested derivatives are Frobenius products costing $O(\ell n^2)$ in total. A projection costs $\tilde{O}((m/\sqrt{\ell})(n^{\omega+o(1)} + \ell n^2) + mn^2)$, the balance $\ell \asymp n^{\omega-2}$ gives $mn^{1+\omega/2+o(1)}$ per round, and $\sum_t m_t = O(n)$ over the rounds gives $n^{2+\omega/2+o(1)}$. Nothing cheaper than the full Gibbs matrix is used.

Figure 2 shows how one Gibbs matrix supplies a block of derivatives.

The selected-coordinate Gibbs oracle. Every query above constructs a full Gibbs matrix, although a coordinate method consumes one derivative per step. Put $C(x) := \text{diag}(B(x)/R, -B(x)/R)$,

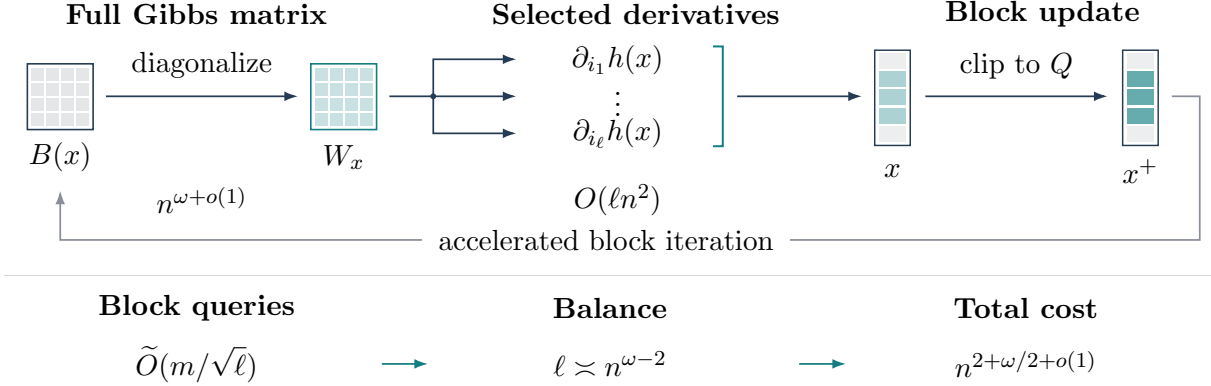


Figure 2: Full-Gibbs block-coordinate method (Theorem 6.1). One approximate diagonalization of $B(x) := \sum_{i \in S} x_i A_i$ supplies the signed Gibbs matrix W_x for all ℓ spectral derivatives in the selected block I . The columns schematize selected-block clipping in the coordinate box $Q = Q_y$ within the accelerated recurrence. The choice $\ell \asymp n^{\omega-2}$ balances diagonalization and Frobenius products. Block-query counts are per fixed-multiplier solve in the dominant rounds; the total cost covers all partial-coloring rounds. Logarithmic accuracy and confidence factors are suppressed.

which stacks the two sides of the spectral constraint, so that $\Phi(x) := \text{tr}[\exp(aC(x) - abI)]$ with $b := 1 - s$ satisfies $h(x) \leq b$ exactly when $\Phi(x) \leq 1$, and minimize $F_\nu(x) := \frac{1}{2}\|x - \xi\|_2^2 + \nu\Phi(x)$ over the box, with minimizer x_ν . The Gibbs matrix $P_x := \exp(aC(x) - abI)$ is formed densely, to inverse-polynomial accuracy, at a snapshot w , and the oracle estimates only the increment $P_x - P_w$: a Gaussian range finder of rank $\tilde{\Theta}(\sqrt{m})$ captures its leading part, and $\tilde{\Theta}(\sqrt{m})$ fresh Gaussian probes estimate the deflated residual. The generalized matrix Pinsker inequality [BGG⁺25, Theorem 1.3] and the Bregman divergence of F_ν convert the trace norm of the residual into the average-coordinate variance bound

$$\mathbb{E}[|\hat{g}_i - \partial_i F_\nu(x)|^2 \mid x, w] \leq \frac{\eta}{m} (\Delta_x + \Delta_w), \quad \Delta_x := F_\nu(x) - F_\nu(x_\nu),$$

for a small universal constant η . The factor $1/m$ is the analytic gain: a loopless variance-reduced coordinate method reduces the objective gap by a constant factor in $\tilde{O}(m)$ selected-coordinate queries with a bounded expected number of dense snapshot refreshes. The oracle is finite and certified: squared polynomial actions replace the exponential by positive-semidefinite surrogates on a guarded spectral interval, fresh one-sided trace guards keep the iterate on the level set where the variance bound holds, candidates and multiplier brackets are accepted only from outward-rounded intervals, and a trial that exhausts its cap returns FAIL, its probability charged to the budget p . One query applies the surrogates to $\tilde{O}(\sqrt{m})$ vectors and reconstructs two dense matrices from rank- $\tilde{O}(\sqrt{m})$ factors, so it costs $\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + n^2$ up to subpolynomial factors, and $\sum_t m_t = O(n)$ with the monotonicity of $\mathcal{T}_{\text{mat}}$ gives $n^{1+\omega(1, 1/2, 1)+o(1)} < n^{3.042776+o(1)}$ (Theorem 7.1). The speedup comes from a cheaper certified Gibbs oracle, not from a new acceleration theorem.

Figure 3 shows the snapshot and increment calculations used for a selected derivative.

The accelerated block algorithm. The exponent $1 + \omega(1, 1/2, 1)$ equals 3 only if the dual exponent $\alpha_{\text{dual}} := \sup\{b : \omega(1, b, 1) = 2\}$ is at least $1/2$, which is open. Section 8 narrows the gap: one sketch of width $\tilde{O}(\sqrt{m/\ell})$ serves a block of $\ell = n^{1-2b}$ coordinates, Catalyst acceleration [LMH18] with regularization $\kappa = \ell$ brings the block queries to $\tilde{O}(m/\sqrt{\ell})$ at the price of $\tilde{O}(\sqrt{\ell})$ dense prox-

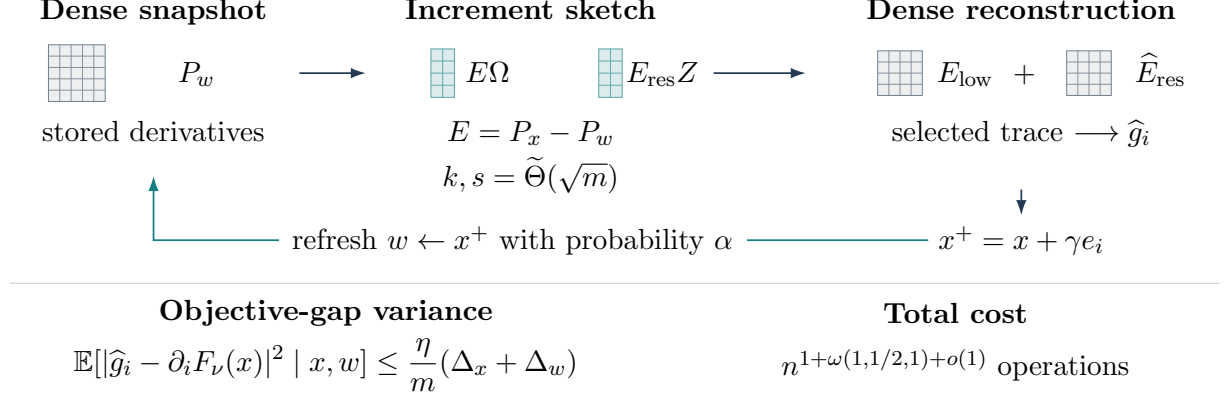


Figure 3: Selected-coordinate Gibbs oracle in the ideal exact-action analysis (Theorem 7.1). Gaussian range and residual probes Ω, Z sketch $E := P_x - P_w$ with widths $2k, s$, where $Z := [z_1, \dots, z_s]$ collects the residual probes. The dense residual reconstruction is $\hat{E}_{\text{res}} := E_{\text{res}} Z Z^\top / s$; its selected trace combines with E_{low} and the stored snapshot derivative. The refresh rate is $\alpha := 1/(m\bar{L})$, with $\bar{L}$ the coordinate curvature including the trust-step correction, and $\Delta_z := F_\nu(z) - F_\nu(x_\nu)$. The snapshot stays fixed between refreshes. The variance bound is conditional on x, w and averages over the uniformly selected coordinate and oracle randomness on the guarded level. The total cost covers all partial-coloring rounds and suppresses logarithmic accuracy and confidence factors.

imal subproblems, a safe-level invariant keeps the extrapolated centers where the sketch variance bound holds, and balancing $(n/\sqrt{\ell})\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n/\ell} \rceil)$ against $n^3\sqrt{\ell}$ gives $3 + \gamma_\star$, $\gamma_\star \leq 0.018320$ at $b_\star = 0.48168\dots$ (Theorem 8.1). The loss $\gamma_\star$ is the price of sketching $P_x - P_w$ afresh at every query.

Figure 4 shows the tradeoff between shared block sketches and dense proximal solves.

Recursive Gibbs tracking. Section 9 instead tracks the Gibbs matrix across coordinate steps: it maintains a dense $\hat{P}_t$ and, after the step $x_{t+1} = x_t + d_t e_{i_t}$, adds an unbiased range-finder-and-probe estimate $\hat{\Delta}_t$ of $P_{x_{t+1}} - P_{x_t}$ whose variance is proportional to d_t^2 (Lemma 9.5); the squared step length is exactly what the negative step-energy term of coordinate descent pays for, and an independent refresh $\hat{P}_{t+1} := P_{x_{t+1}}$ with probability π keeps the error from accumulating. With $\mathcal{E}_t$ the tracking error in the gradient seminorm and L the coordinate curvature, the coupled potential $V_t := \Delta(x_t) + \mathcal{E}_t/(\pi L)$ contracts,

$$\mathbb{E}_t[V_{t+1}] \leq (1 - \alpha)V_t - \frac{L}{8}\mathbb{E}_t[d_t^2], \quad \alpha := \frac{1}{mL}$$

(Lemma 9.6). The guard is built around the level $H := 64U$, $U := F_\nu(0)$: a phase starts from a certified point z with $F_\nu(z) \leq H/16$; the ideal process is stopped at the first proposal above $5H/16$, which happens with probability at most $1/20$; every proposal gets a fresh certificate accurate to $H/48$ and is rejected above $H/3$; and the refreshes are capped at $M_{\text{ref}} = \lceil C(T\pi + \log(1/\delta_{\text{ph}})) \rceil$. A phase of $T = \lceil 16/\alpha \rceil$ queries then halves the objective gap with a universal positive probability (Lemma 9.4), and J halving phases with repetition and certified comparison return an ε_{opt} -accurate solution with the interval certificates the multiplier wrapper requires (Lemma 9.3). With π of order $n^{-1/2}$ and sketch width $n^{1/4+o(1)}$, a nonrefresh query costs $n^{2+o(1)}$ because $\alpha_{\text{dual}} > 1/4$, a refresh costs $n^{\omega+o(1)}$ at rate $n^{-1/2}$, and $\sum_t m_t = O(n)$ gives $(n^3 + n^{\omega+1/2} + n^\omega)n^{o(1)} = n^{3+o(1)}$ because $\omega < 5/2$ (Lemma 9.9, Theorem 9.1). The two bounds are theorems of Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] and of Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1], not hypotheses.

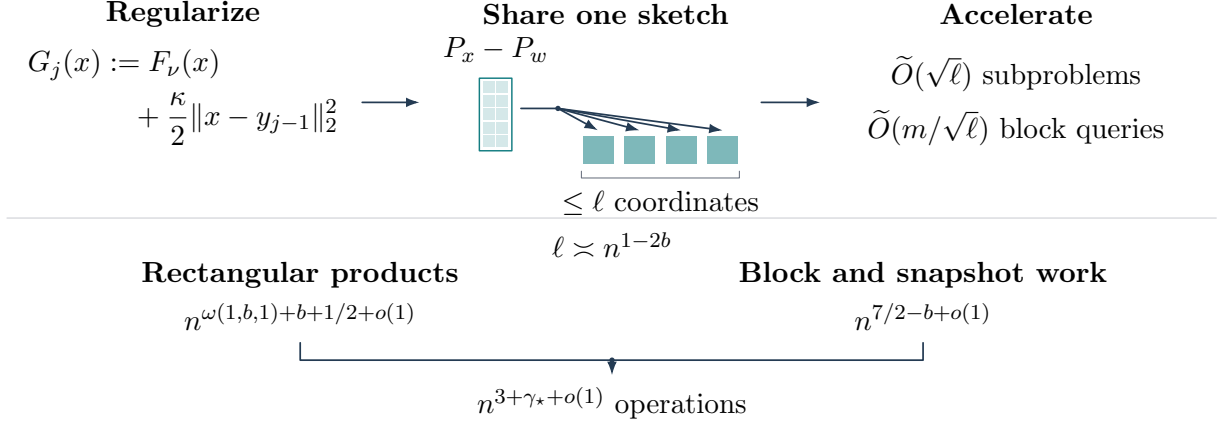
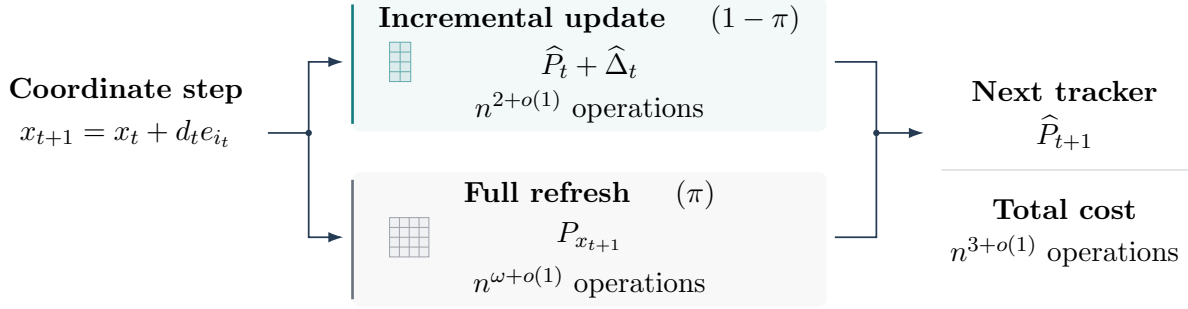


Figure 4: Accelerated block method (Theorem 8.1). Here $\kappa := \ell$, y_{j-1} is the Catalyst [LMH18] center, and w is the snapshot. A fresh range sketch and residual probes of the ideal difference $P_x - P_w$, with widths $\tilde{O}(\sqrt{m/\ell})$, serve the whole block. The upper counts are per fixed-multiplier solve in the accelerated regime. The safe-level invariant keeps accepted iterates on the guarded Gibbs trace level. The block size is $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$. Choosing $1/3 \leq b \leq 1/2$ balances the lower costs over all partial-coloring rounds; γ_* minimizes the larger exponent minus three. Logarithmic accuracy and confidence factors are suppressed.

Figure 5 shows the two tracker updates and the variance estimate that links them to coordinate descent.

Fixed-point words. The real-arithmetic analysis never controls three things that a finite word forces one to control: decisions taken on rounded quantities, random objects that cannot be drawn exactly, and the size of the tracked matrix, which the gradient seminorm does not see. Section 10 specifies a finite version of the cubic-time algorithm as a list of replacements of its primitives, proves one lemma per primitive, and assembles them (Theorem 10.1). Every number is a fixed-point word of $W := I + f$ bits, with f fractional and I integer bits given by explicit formulas in $\log_2(1/\varepsilon_{\text{opt}})$, $\log_2(H_{\max}/\underline{\nu})$, $\log_2(2n)$, the top spectral scale λ_{top} , and the spectral bound M_{sp} , so that $W = O(\log n)$ independent of p ; inputs are rounded to $b_{\text{in}} := 2\lceil \log_2 n \rceil + 10$ bits and parameters in the direction that preserves the inequality they enter, Gaussians are exact samples of rounded Gaussians, the exponential is a scaled Taylor power, products are exact bilinear algorithms rounded once, a refresh is forced every $\lceil 1/\pi' \rceil$ steps, the trace guard is additive, and comparisons act on outward-rounded enclosures. The bridge to the real-arithmetic proof is a few explicit constants: the additive guard accepts a proposal y when $q(y) + \nu U(y) \leq H$ for an upper estimate $U(y) \leq \frac{5}{3}\Phi(y) + H/(9\nu)$, and one phase-independent δ_g with $32m\delta_g^2 \leq \varepsilon_{\text{opt}}/(6250J)$ replaces the per-phase requirement $C_3m\delta_{g,j}^2 \leq \varepsilon_{\text{ph},j}/(64J)$. The total is $n^{3+o(1)} \text{polylog}(1/p) + \tilde{O}(n^3 L_{\text{in}})$ bit operations on inputs of bit length L_{in} . Here $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$ enter as hypotheses in the asymptotic-rank sense (Definition 10.2), because the fast products need exact bilinear algorithms with rational coefficients, which the laser-method proofs of both bounds deliver. Theorem 6.1 is not covered: its diagonalization step is analyzed by Sobczyk [Sob25] only in the real-arithmetic model.

Composition. The four running times compose in the same way: $O(\log n)$ rounds with active dimensions m_t , $\sum_t m_t = O(n)$; per round a bounded expected number of projections, each trial succeeding with probability greater than 0.1; per projection polylogarithmically many multiplier



$$\begin{array}{ccc}
 \text{Step-sensitive variance} & & \text{Coupled contraction} \\
 \mathbb{E}_t \|\hat{\Delta}_t - (P_{x_{t+1}} - P_{x_t})\|_{\mathcal{G}}^2 \leq \frac{CL^2}{ks} \mathbb{E}_t[d_t^2] & \longrightarrow & \mathbb{E}_t[V_{t+1}] \leq (1 - \alpha)V_t - \frac{L}{8} \mathbb{E}_t[d_t^2]
 \end{array}$$

Figure 5: Recursive Gibbs tracking in the ideal exact-action analysis (Lemmas 9.5 and 9.6). The step-sensitive variance is absorbed by the descent energy. Here $\Delta(x) := \Delta_x$, $\mathcal{E}_t := \|\hat{P}_t - P_{x_t}\|_{\mathcal{G}}^2$, $V_t := \Delta(x_t) + \mathcal{E}_t/(\pi L)$, $\mathcal{G}$ is the coordinate-gradient seminorm, L is the coordinate curvature, $\alpha := 1/(mL)$, and $\pi = \pi_m := \min\{1, \max\{n^{-1/2}, 4\alpha\}\}$. The range-finder rank and residual-probe count are $k = s = \lceil C\pi_m^{-1/2} \rceil = O(n^{1/4})$, with C large enough that $ks \geq K_0/\pi_m$. Conditional expectations are given the history before choosing i_t . The total cost covers all partial-coloring rounds. The displayed costs suppress $\text{polylog}(1/p)$ confidence factors and use $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$ (Lemma 9.9).

tests; and per fixed-multiplier solve the cost computed in each of Sections 6–9. Summing over the rounds with $\sum_t m_t = O(n)$ and the monotonicity of $\mathcal{T}_{\text{mat}}$ gives the exponents $2+\omega/2$, $1+\omega(1, 1/2, 1)$, $3 + \gamma_*$, and 3; the repetitions that amplify the constant success probability to $1 - p$ and the union bound over all randomized calls give the factor $\text{polylog}(1/p)$.

What is shared and what is not. The four algorithms follow the same pipeline. The outer partial-coloring trajectory, the smoothed body D with its constants a and s , the radial repair and snapping rule, the terminal rounding with its $6.5\sqrt{n}$, and the discrepancy calculation are proved once, in Sections 5 and 6, and the algorithms of Sections 7–9 further share the certified multiplier wrapper, the finite positive-semidefinite surrogates, and the guard-and-certificate discipline of Section 7. They differ in one place, the solver of the smooth projection: Section 6 runs accelerated block-coordinate descent on a dense Gibbs matrix, and Sections 7–9 change only the fixed-multiplier part of the solver, each with its own variance bound, level-set invariant, and cost count. The bit-complexity analysis of Section 10 keeps the wrapper and the guards and replaces the polynomial surrogate by a scaled-Taylor power (Lemma 10.8). There is no general theorem behind the four; each section proves its own.

Open questions. What is the smallest constant this route can give? The existence bound 9.9996 of Appendix A is still larger than Spencer’s 6 in the diagonal case [Spe85]. For the projection-based algorithms, a smaller radius coefficient $c_{\text{rad}}(\ell_{\text{asp}})$ at the same exponent, or a projection lemma with a larger δ , would lower the discrepancy bound. Can the running time be made proportional to the number of nonzero input entries, given that a dense Gibbs matrix is what all four solvers maintain in some form? Is there a deterministic polynomial-time algorithm with an explicit constant, all four algorithms here being randomized in the Gaussian draw and in their solvers? And what is the bit

complexity of the $n^{2+\omega/2}$ algorithm, whose diagonalization step is analyzed only in real arithmetic?

There is a gap of roughly 2.4 between the existence constant of Appendix A (9.9996) and the algorithmic constant of Appendix B (12.40). We do not know how to close it. Is there a fine-grained lower bound, constants $c, c' > 0$ such that every algorithm finding a signing of discrepancy at most $c\sqrt{n}$ needs $\Omega(n^{3+c'})$ time in the worst case?

Independent and concurrent work. After completing a draft of this paper, the authors became aware of the independent and concurrent work of [AS26], which proves a similar result.

AI Disclosure. The AI tools used in the preparation of this paper are ChatGPT Pro 5.6, Codex Sol 5.6, Claude Fable 5 and Claude Fable 5.1. The proof of the existence of such a signing comes mainly from AI. The majority of the ideas for improving the constants down to below 10 mainly come from the authors.

Authors develop the main ideas for the efficient algorithm. In particular, the $n^{2+\omega/2}$ time algorithm is developed by the authors. The algorithms with runtime $n^{1+\omega(1,1/2,1)}$ and $n^{3.018}$ are designed by authors prompting the LLMs with ideas from [CLS19]. The final $n^{3+o(1)}$ time algorithm comes mainly from authors as well.

The bit complexity of the algorithm is analyzed by LLMs. The proofs have been rewritten by the authors for presentation, and have been carefully verified by the authors, who take full responsibility for their correctness.

3 Preliminaries

3.1 Notation and conventions

We use n for the number of input matrices, N for their matrix dimension, and $m = |S|$ for the number of active matrices in a partial-coloring step. In proofs that pad the matrices to order n , the effective matrix dimension is $N = n$.

For an integer $r \geq 1$, $[r] := \{1, \dots, r\}$. If S is a finite set, $\mathbb{R}^S$ is the Euclidean space of real vectors indexed by S . We write $\mathbb{R}^{N \times N}$ for the space of real $N \times N$ matrices, I_N and 0_N for the identity and zero matrices, and suppress their subscripts when the dimension is clear.

For $A \in \mathbb{R}^{N \times N}$, $A^\top$ is its transpose, $\text{tr}[A]$ its trace, and $|A| := (A^\top A)^{1/2}$. The unadorned norm $\|A\|$ is the operator norm; for $1 \leq q < \infty$ the Schatten norms are

$$\|A\|_{S_q} := (\text{tr}[|A|^q])^{1/q}, \quad \|A\|_{S_\infty} := \|A\|,$$

and we write $\|A\|_F := \|A\|_{S_2}$ for the Frobenius norm and $\|A\|_1 := \|A\|_{S_1} = \text{tr}[|A|]$ for the trace norm. The Frobenius inner product is $\langle A, B \rangle_F := \text{tr}[A^\top B]$. For symmetric matrices A and B , $A \preceq B$ means that $B - A$ is positive semidefinite.

If $X = U \text{diag}(\lambda_1, \dots, \lambda_N) U^\top$ is symmetric and f is scalar-valued on its spectrum, then $f(X) := U \text{diag}(f(\lambda_1), \dots, f(\lambda_N)) U^\top$. We write $\text{diag}(A_1, \dots, A_k)$ for a block-diagonal matrix, $A \oplus B := \text{diag}(A, B)$, $\text{vec}(A)$ for column-wise vectorization, and $A \otimes B$ for the Kronecker product; thus $\text{vec}(AXB) = (B^\top \otimes A) \text{vec}(X)$ whenever the product is defined.

For vectors, $\langle x, y \rangle := x^\top y$, $\|x\|_2 := \sqrt{\langle x, x \rangle}$, and $\|x\|_1 := \sum_i |x_i|$. For a nonempty closed set $K \subseteq \mathbb{R}^m$,

$$d(x, K) := \inf_{z \in K} \|x - z\|_2.$$

For $c \geq 0$, $cK := \{cx : x \in K\}$, and K is symmetric when $K = -K$. For open sets U and V , $U \subseteq V$ means that the closure of U is compact and contained in V . We write $\mathbb{1}[\mathcal{E}]$ for the indicator of an event $\mathcal{E}$ and $a_+ := \max\{a, 0\}$.

Unless stated otherwise, $\xi := (\xi_1, \dots, \xi_m) \sim \mathcal{N}(0, I_m)$ is a standard Gaussian vector and

$$\gamma_m(K) := \Pr[\xi \in K].$$

For a probability law μ , $\mathbb{E}_\mu[X]$, $\Pr_\mu[\mathcal{E}]$, $\mathbb{V}_\mu[X]$, and $\text{Cov}_\mu[X]$ are the expectation, probability, variance, and covariance under μ , so that

$$\mathbb{V}_\mu[X] := \mathbb{E}_\mu[(X - \mathbb{E}_\mu[X])^2], \quad \text{Cov}_\mu[X] := \mathbb{E}_\mu[(X - \mathbb{E}_\mu[X])(X - \mathbb{E}_\mu[X])^\top].$$

All logarithms are natural.

For a differentiable map F , $D^j F(X)[H_1, \dots, H_j]$ is its j th Fréchet derivative, while ∂_i , ∂_{i_1, i_2} , and ∇^j denote coordinate derivatives. For $F : \mathbb{R}^m \rightarrow \mathbb{R}^{N \times N}$,

$$\|\nabla^j F(x)\|_{\text{HS}}^2 := \sum_{i_1=1}^m \cdots \sum_{i_j=1}^m \|\partial_{i_1} \cdots \partial_{i_j} F(x)\|_F^2,$$

with the convention $\|\nabla^0 F(x)\|_{\text{HS}} := \|F(x)\|_F$.

Constants. A symbol $C > 0$ or $c > 0$ without a subscript denotes a universal constant, C large and c small, whose value may change from line to line. The letter $p \in (0, 1/2)$ is reserved for the failure probability of a randomized algorithm, and a subscripted p_0 , p_{clean} , or p_s is the failure budget of the subroutine it names. The letter ε is reserved for the sign vector $\varepsilon \in \{-1, 1\}^n$ and its coordinates; the box scale of the partial-coloring step is $\varepsilon_{\text{box}} := 0.4235$, and every tolerance or accuracy carries a subscript naming its role or level (ε_{opt} , ε_{op} , $\varepsilon_{\text{ph}, j}$, $\dots$); ε_i with a bare index is always a sign coordinate; the local gap variables of the divided-difference proof in Section 4 (Lemma 4.6, re-run in the explicit specialization) are ε_{gap} and δ_{gap} . A subscripted constant keeps, within the section that introduces it, the value fixed there, and every explicit numerical constant (such as 156000, 0.003712, or $N_0 = 3 \cdot 10^8$) denotes the exact number written.

3.2 Computational model

The input to every algorithm in this paper is dense: the n symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ are given entry by entry, as n^3 real numbers, and n^3 is the input size against which the running times of Sections 6–9 are measured. The algorithms use no structure of the matrices beyond the standing hypotheses (symmetric, $\|A_i\| \leq 1$); the phrase ‘for dense input matrices’ in a theorem statement refers to this comparison with the input size, not to a hypothesis on the A_i . Running times are counted in the real-arithmetic model: each addition, subtraction, multiplication, division, or comparison of two real numbers costs one arithmetic operation, as does each draw of a random real number that is uniform on $[0, 1]$ or standard Gaussian; the numbers themselves are exact, so the count does not depend on their bit length. Every running-time bound in Sections 6–9 is a count of arithmetic operations in this model, not a bit-complexity bound; bit complexity is the subject of Section 10, which runs the cubic-time algorithm on fixed-point words of $O(\log n)$ bits.

Matrix products are charged at their asymptotic cost. For positive integers r, s, t , define

$$\mathcal{T}_{\text{mat}}(r, s, t) := \min\{T : \text{every } r \times s \text{ matrix can be multiplied by every } s \times t \text{ matrix using at most } T \text{ arithmetic operations}\}.$$

For $a, b, c \geq 0$, define the rectangular matrix-multiplication exponent by

$$\omega(a, b, c) := \inf\{\tau \geq 0 : \mathcal{T}_{\text{mat}}(\lceil N_{\text{mult}}^a \rceil, \lceil N_{\text{mult}}^b \rceil, \lceil N_{\text{mult}}^c \rceil) \leq N_{\text{mult}}^{\tau+o(1)}\}.$$

Thus $\omega := \omega(1, 1, 1)$ is the matrix-multiplication exponent. Transposition and permutation of the matrix-multiplication tensor imply that $\omega(a, b, c)$ is invariant under permutations of (a, b, c) . The dual matrix-multiplication exponent is

$$\alpha_{\text{dual}} := \sup\{b \in [0, 1] : \omega(1, b, 1) = 2\},$$

the largest aspect ratio at which an $N_{\text{mult}} \times N_{\text{mult}}^b$ by $N_{\text{mult}}^b \times N_{\text{mult}}$ product still costs $N_{\text{mult}}^{2+o(1)}$. Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] prove $\alpha_{\text{dual}} > 0.321334$, and Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25, Table 1] prove $\omega < 2.371339$; the unconditional cubic-time bound of Section 9 and the bit-complexity analysis of Section 10 use only $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$.

Finally, $\tilde{O}(\cdot)$ suppresses factors polylogarithmic in n and in the accuracy and failure parameters. Because $\omega(a, b, c)$ is an infimum, a running time stated as $n^{\tau+o(1)}$ with τ an expression in $\omega(\cdot, \cdot, \cdot)$ carries an unavoidable $n^{o(1)}$ factor; where such a bound is written $\tilde{O}(n^\tau)$, this factor is included in the notation.

3.3 Tensor Gram inequalities

The small-ball argument of Section 4 bounds its noncommutative products through the following Gram forms.

Fact 3.1 (Tensor Gram inequalities). *Let $A_1, \dots, A_m, B_1, \dots, B_m \in \mathbb{R}^{N \times N}$ be symmetric, let $\mathcal{H}$ be a finite-dimensional Hilbert space, and let $z_1, \dots, z_m \in \mathbb{R}^N \otimes \mathcal{H}$. Then*

$$\begin{aligned} \left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (A_{i_2} A_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| &\leq \sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[A_{i_1} A_{i_2}] \langle z_{i_1}, z_{i_2} \rangle, \\ \left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (A_{i_2} B_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| &\leq \left(\sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[A_{i_1} A_{i_2}] \langle z_{i_1}, z_{i_2} \rangle \right)^{1/2} \left(\sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[B_{i_1} B_{i_2}] \langle z_{i_1}, z_{i_2} \rangle \right)^{1/2}. \end{aligned}$$

The two Gram forms on the right-hand sides are nonnegative. Moreover, if $U_1, \dots, U_m$ are symmetric and $\sum_{i=1}^m U_i^2 \preceq sI$, then

$$\left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (U_{i_2} U_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| \leq s \sum_{i=1}^m \|z_i\|_2^2.$$

Proof. Choose an orthonormal basis of $\mathcal{H}$ and write the coordinates of $z_i \in \mathbb{R}^N \otimes \mathcal{H}$ as $(z_i)_{a'h}$ with $a' \in [N]$ and $h \in [\dim \mathcal{H}]$. Define the four-index arrays

$$T_{aba'h}^A := \sum_{i=1}^m (A_i)_{ab} (z_i)_{a'h}, \quad T_{aba'h}^B := \sum_{i=1}^m (B_i)_{ab} (z_i)_{a'h},$$

and equip such arrays with the inner product $\langle S, T \rangle := \sum_{a,b,a',h} S_{aba'h} T_{aba'h}$ and the norm $\|T\|_2 := \langle T, T \rangle^{1/2}$, the square root of the sum of the squared entries. Since each A_i is symmetric, $\text{tr}[A_{i_1} A_{i_2}] = \sum_{a,b} (A_{i_1})_{ab} (A_{i_2})_{ab}$, and expanding $\langle z_{i_1}, z_{i_2} \rangle = \sum_{a',h} (z_{i_1})_{a'h} (z_{i_2})_{a'h}$ gives

$$\sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[A_{i_1} A_{i_2}] \langle z_{i_1}, z_{i_2} \rangle = \|T^A\|_2^2,$$

and likewise the second Gram form equals $\|T^B\|_2^2$; both are nonnegative. Let σ be the cyclic shift of the three $[N]$ -indices, $(\sigma T)_{pqrh} := T_{rpqh}$, which permutes the entries of an array and is therefore an isometry for $\|\cdot\|_2$. Writing out the first left-hand side coordinatewise,

$$\begin{aligned} \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (A_{i_2} A_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle &= \sum_{a,e,b,h} \left(\sum_{i_1} (A_{i_1})_{eb} (z_{i_1})_{ah} \right) \left(\sum_{i_2} (A_{i_2})_{ae} (z_{i_2})_{bh} \right) \\ &= \sum_{a,e,b,h} T_{ebah}^A T_{aebh}^A = \langle T^A, \sigma T^A \rangle, \end{aligned}$$

and the same computation with B_{i_1} in place of A_{i_1} identifies the second left-hand side as $\langle T^B, \sigma T^A \rangle$. Cauchy–Schwarz and the isometry of σ give $|\langle T^A, \sigma T^A \rangle| \leq \|T^A\|_2^2$ and $|\langle T^B, \sigma T^A \rangle| \leq \|T^B\|_2 \|T^A\|_2$, which are the first two inequalities.

For the last inequality, $2|\langle x, y \rangle| \leq \|x\|_2^2 + \|y\|_2^2$ applied to $x = (U_{i_2} \otimes I_{\mathcal{H}}) z_{i_1}$ and $y = (U_{i_1} \otimes I_{\mathcal{H}}) z_{i_2}$ gives

$$\begin{aligned} \left| \sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (U_{i_2} U_{i_1} \otimes I_{\mathcal{H}}) z_{i_2} \rangle \right| &\leq \frac{1}{2} \sum_{i_1=1}^m \sum_{i_2=1}^m (\|(U_{i_2} \otimes I_{\mathcal{H}}) z_{i_1}\|_2^2 + \|(U_{i_1} \otimes I_{\mathcal{H}}) z_{i_2}\|_2^2) \\ &= \sum_{i_1=1}^m \langle z_{i_1}, \left(\sum_{i_2=1}^m U_{i_2}^2 \otimes I_{\mathcal{H}} \right) z_{i_1} \rangle \leq s \sum_{i_1=1}^m \|z_{i_1}\|_2^2. \end{aligned}$$

Here the equality interchanges i_1 and i_2 in the second of the two sums, and the last inequality is $\sum_{i_2=1}^m U_{i_2}^2 \preceq sI$. $\square$

3.4 Divided differences

The spectral Hessian of a trace function depends on a divided difference of its derivative, including when eigenvalues coincide.

Definition 3.2 (First divided difference). Let $J \subseteq \mathbb{R}$ be an interval and let $f \in C^2(J)$. For $x, y \in J$, define

$$f'[x, y] := \begin{cases} \frac{f'(x) - f'(y)}{x - y}, & x \neq y, \\ f''(x), & x = y. \end{cases}$$

We call $f'[x, y]$ the first divided difference of f' .

3.5 Positive-semidefinite inequalities

Fact 3.3 (Basic PSD inequalities). *Let all matrices below have compatible dimensions.*

(a) *If X and Y are symmetric with $X \preceq Y$, then $R^\top X R \preceq R^\top Y R$ for every matrix R . Consequently, if $0 \preceq P \preceq I$ and M is symmetric, then $MPM \preceq M^2$. Loewner inequalities may also be added termwise.*

(b) *For symmetric matrices A and B ,*

$$(A + B)^2 \preceq 2A^2 + 2B^2, \quad (A - B)^2 \preceq 2A^2 + 2B^2.$$

Proof. For part (a), $R^\top(Y - X)R \succeq 0$ because $Y - X \succeq 0$; the bound $MPM \preceq M^2$ is the case $R = M$, $X = P$, $Y = I$, and termwise addition is the case of sums of positive semidefinite differences. For part (b), expand the positive semidefinite matrices $(A - B)^2$ and $(A + B)^2$ to obtain, respectively, $AB + BA \preceq A^2 + B^2$ and $-AB - BA \preceq A^2 + B^2$; substituting these into $(A + B)^2 = A^2 + B^2 + AB + BA$ and $(A - B)^2 = A^2 + B^2 - AB - BA$ proves the two claims. $\square$

Table 2: The six small-ball regimes. A row applies when its range of moment orders contains $\tau(\ell_{\text{asp}}) = \lceil 256\ell_{\text{asp}}/17 \rceil$; every terminating decimal denotes the corresponding exact rational number. The last two columns are an upper bound for $10^9 E$ and a lower bound for $10^9 \Delta_{\text{curv}}$, the normalized small-ball exponent and the curvature margin of the row, both defined in the proof of Lemma 4.1.

τ	c_{rad}	b	D	A	$10^9 E$	$10^9 \Delta_{\text{curv}}$
16	279	0.400840	323	$4.7 \cdot 10^6$	3711824	118.0
17–19	266	0.412210	365	$5.6 \cdot 10^6$	3517582	1156.8
20–26	240	0.435510	497	$11.3 \cdot 10^6$	2376271	2086.7
27–30	200	0.481310	1713	$31 \cdot 10^6$	2980959	629.4
31–43	190	0.494840	6210	$122.5 \cdot 10^6$	2730038	834.3
$\tau \geq 44$	185	0.499855	221000	$228 \cdot 10^9$	51714	168.4

4 Hereditary Gaussian small balls

This section proves the hereditary small-ball estimate consumed by Section 5, Lemma 4.1 below.

The partial-coloring argument needs a Gaussian measure bound for every active subfamily. The radius may depend on the aspect ratio, while the exponential loss must stay uniform across subfamilies.

Lemma 4.1 (Six-regime hereditary small ball; Table 2). *Let $1 \leq m \leq N$, let $A_1, \dots, A_m \in \mathbb{R}^{N \times N}$ be symmetric contractions, and put*

$$\ell_{\text{asp}} = \max\{1, \log(N/m)\}, \quad c_{\text{cut}} := \frac{256}{17}, \quad \tau(\ell_{\text{asp}}) := \lceil c_{\text{cut}} \ell_{\text{asp}} \rceil.$$

Let $c_{\text{rad}}(\ell_{\text{asp}})$ be the entry c_{rad} of the row of Table 2 whose range of moment orders contains $\tau(\ell_{\text{asp}})$. Then

$$\Pr\left[\left\|\sum_{i=1}^m \xi_i A_i\right\| \leq c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}}\right] \geq \exp(-0.003712m).$$

The rest of the section proves Lemma 4.1. Sections 4.1–4.7 build the small-ball estimate with universal constants, Section 4.8 tracks every constant for one parameter set and proves the radius 279, and Section 4.9 repeats that computation for each row of Table 2.

4.1 Normalization and the interpolating partition function

The matrix series is first rescaled so that its quadratic variance is small. A diagonal-to-full interpolation converts the small-ball probability into estimates for a log-partition function.

Definition 4.2 (Barrier potential). For an integer $\tau \geq 2$ and parameters $\kappa_1 \geq \kappa_0 > 0$, define on $(-1, 1)$

$$\phi_0(s) := -\log(1 - s^2), \quad \phi_{1,\tau}(s) := \sum_{k=\tau}^{\infty} \frac{s^{2k}}{k}, \quad \phi(s) := \kappa_0 \phi_0(s) + (\kappa_1 - \kappa_0) \phi_{1,\tau}(s).$$

Extend ϕ by $+\infty$ outside $(-1, 1)$, and, for every symmetric matrix X , define

$$\Phi(X) := \text{tr}[\phi(X)].$$

We normalize the target radius to one so that the small-ball event matches the domain of the barrier potential.

Definition 4.3 (Standing normalization). For integers $1 \leq m \leq N$, symmetric contractions $A_1, \dots, A_m \in \mathbb{R}^{N \times N}$, and a parameter $c_{\text{rad}} > 0$, define

$$\ell_{\text{asp}} := \max\{1, \log(N/m)\}, \quad r_{\text{sb}} := c_{\text{rad}} \sqrt{m} \ell_{\text{asp}}, \quad \hat{A}_i := r_{\text{sb}}^{-1} A_i, \quad \rho := (c_{\text{rad}}^2 \ell_{\text{asp}}^2)^{-1}.$$

Here ℓ_{asp} denotes the logarithmic aspect ratio floored at one, $c_{\text{rad}} > 0$ is the radius multiplier, and r_{sb} is the target small-ball radius. The matrices $\hat{A}_i$ are the rescaled coefficient matrices, and ρ is their scalar variance bound. Unless explicitly redefined, these symbols retain this meaning throughout Section 4.

The diagonal matrix series has a scalar Gaussian description. Turning on the off-diagonal entries continuously connects this endpoint to the full series.

Definition 4.4 (Diagonal-to-full interpolation). Let $A_1, \dots, A_m, \hat{A}_1, \dots, \hat{A}_m, \ell_{\text{asp}}, r_{\text{sb}}$, and ρ be defined as in Definition 4.3. Let $\xi_1, \dots, \xi_m$ be independent standard Gaussian random variables. Fix an integer $\tau \geq 2$ and parameters $\kappa_1 \geq \kappa_0 > 0$, and let Φ, ϕ be as in Definition 4.2. Define $\mathcal{A} := \mathcal{A}(\xi) := \sum_{i=1}^m \xi_i \hat{A}_i$. Fix an orthonormal basis and write $\mathcal{A} = \mathcal{A}_{\text{diag}} + \mathcal{A}_{\text{off}}$, where $\mathcal{A}_{\text{diag}}$ and $\mathcal{A}_{\text{off}}$ are respectively the diagonal and off-diagonal parts of $\mathcal{A}$ in this basis. Likewise, $\hat{A}_{i,\text{diag}}$ and $\hat{A}_{i,\text{off}}$ denote the diagonal and off-diagonal parts of $\hat{A}_i$ in the same basis. For $t \in [0, 1]$, define

$$\begin{aligned} \mathcal{A}_t &:= \mathcal{A}_{\text{diag}} + t \mathcal{A}_{\text{off}}, \quad \hat{A}_i(t) := \hat{A}_{i,\text{diag}} + t \hat{A}_{i,\text{off}}, \\ a_t &:= \mathbb{E}[e^{-\Phi(\mathcal{A}_t)}], \quad a(t) := \log a_t. \end{aligned}$$

The interpolation argument reduces the probability estimate to the diagonal endpoint and an integrated lower bound on the curvature of the log-partition function.

Lemma 4.5 (Normalized interpolation). *Let $A_i, \xi_i, \hat{A}_i, \mathcal{A}, \mathcal{A}_{\text{diag}}, \mathcal{A}_{\text{off}}, \mathcal{A}_t, \hat{A}_i(t), a_t$, and $a(t)$ be defined as in Definition 4.4. Assume moreover that $\kappa_1 \geq 2$. Then the following statements hold.*

- (i) *For every $t \in [0, 1]$, we have $\mathcal{A}_t = \sum_{i=1}^m \xi_i \hat{A}_i(t)$ and $a_t > 0$.*
- (ii) *For $t \in [0, 1]$, define $d\mu_t := a_t^{-1} e^{-\Phi(\mathcal{A}_t)} d\gamma_m$. Then $a'(t) = -\mathbb{E}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]]$.*
- (iii) *For every $t \in [0, 1]$, $a''(t) = \mathbb{V}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]] - \mathbb{E}_{\mu_t}[D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]]$.*
- (iv) *The normalized matrices satisfy $\sum_{i=1}^m \hat{A}_i^2 \preceq \rho I_N$, and the initial derivative satisfies $a'(0) = 0$.*
- (v) *The Gaussian small-ball probability satisfies $\Pr[\|\sum_{i=1}^m \xi_i A_i\| < r_{\text{sb}}] \geq a_1$.*

Proof. Proof of (i). The identity for $\mathcal{A}_t$ follows directly from the definitions of $\mathcal{A}_t$ and $\hat{A}_i(t)$. For every $t \in [0, 1]$, continuity and $\mathcal{A}_t(0) = 0$ show that a neighborhood of the origin lies in $\{\xi : \|\mathcal{A}_t(\xi)\| < 1\}$. Thus $a_t > 0$.

Proof of (ii). By (i), the measure μ_t is well defined. Put $G_t := D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]$ and $\Delta_t := 1 - \|\mathcal{A}_t\|$. Here and in the rest of the section, $c_{\kappa_1, \kappa_0, \tau}$ denotes a constant depending only on κ_1, κ_0, τ , whose value may change from line to line. The identity $\phi = \kappa_1 \phi_0 - (\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} s^{2k}/k$ shows that, on $\{\|\mathcal{A}_t\| < 1\}$, $e^{-\Phi(\mathcal{A}_t)} \leq c_{\kappa_1, \kappa_0, \tau} \Delta_t^{\kappa_1}$. After extension by zero, the integrand of a_t against γ_m and its first t -derivative are bounded, uniformly for $t \in [0, 1]$, by $c(1 + \|\xi\|_2)^2$ with c depending

on $\kappa_1, \kappa_0, \tau, N, \rho$; this bound is γ_m -integrable, so dominated convergence applies without a moving cutoff. Since $\mathcal{A}'_t = \mathcal{A}_{\text{off}}$, the chain rule gives

$$a'_t = - \int G_t e^{-\Phi(\mathcal{A}_t)} d\gamma_m = -a_t \mathbb{E}_{\mu_t}[G_t].$$

Consequently,

$$a'(t) = \frac{a'_t}{a_t} = -\mathbb{E}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]],$$

which is (ii).

Proof of (iii). Retain $G_t := D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]$ from (ii). Since $\mathcal{A}'_t = \mathcal{A}_{\text{off}}$ and $\mathcal{A}''_t = 0$, the chain rule gives

$$G'_t = D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}].$$

The second t -derivative of the extended integrand is bounded by

$$c_{\kappa_1, \kappa_0, \tau} N \|\mathcal{A}_{\text{off}}\|_F^2 \Delta_t^{\kappa_1 - 2}$$

on $\{\|\mathcal{A}_t\| < 1\}$. Since $\kappa_1 \geq 2$ and $\mathcal{A}_{\text{off}}$ is linear in ξ , this is dominated by a Gaussian-integrable polynomial, uniformly for $t \in [0, 1]$. Differentiating once more therefore gives

$$\begin{aligned} a''_t &= \int (G_t^2 - G'_t) e^{-\Phi(\mathcal{A}_t)} d\gamma_m \\ &= a_t \mathbb{E}_{\mu_t}[G_t^2 - D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]]. \end{aligned}$$

Using $a(t) = \log a_t$ and the formula for a'_t/a_t from (ii),

$$\begin{aligned} a''(t) &= \frac{a''_t}{a_t} - \left(\frac{a'_t}{a_t}\right)^2 \\ &= \mathbb{E}_{\mu_t}[G_t^2] - \mathbb{E}_{\mu_t}[G_t]^2 - \mathbb{E}_{\mu_t}[D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]] \\ &= \mathbb{V}_{\mu_t}[D\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}]] - \mathbb{E}_{\mu_t}[D^2\Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]], \end{aligned}$$

which is (iii).

Proof of (iv). Since $A_i^2 \preceq I_N$, the normalized matrices satisfy $\sum_{i=1}^m \widehat{A}_i^2 \preceq \rho I_N$. Moreover, $\mathcal{A}_0$ and $\phi'(\mathcal{A}_0)$ are diagonal whereas $\mathcal{A}_{\text{off}}$ has zero diagonal, so (ii) gives $a'(0) = 0$.

Proof of (v). The definition of ϕ gives $0 \leq e^{-\Phi(X)} \leq \mathbb{1}[\|X\| < 1]$, so $a_1 \leq \Pr[\|\mathcal{A}(\xi)\| < 1]$. Since $\mathcal{A}(\xi) = r_{\text{sb}}^{-1} \sum_{i=1}^m \xi_i A_i$, this is exactly the stated probability bound. $\square$

4.2 Divided-difference estimates

Divided-difference estimates quantify the scalar barrier near the spectral edge and in the interior, in the notation of Definition 3.2.

Lemma 4.6 (Divided differences). *Let $0 < w < 1/20$, let $\tau \geq 2$ be an integer, and let ϕ_0 and $\phi_{1,\tau}$ be as in Definition 4.2. For every $u, v \in (-1, 1)$, the following statements hold.*

(i)

$$\phi'_{1,\tau}[u, v] \geq \frac{1}{4} \sum_{\sigma \in \{-1, 1\}} \frac{\mathbb{1}[\min\{1 + \sigma u, 1 + \sigma v\} \leq w/\tau]}{(1 + \sigma u)(1 + \sigma v)}.$$

(ii)

$$\phi'_{1,\tau}[u, v] \leq 2\tau \max\{|u|, |v|\}^{2\tau-2} \phi'_0[u, v].$$

Proof. By Definition 3.2,

$$\phi'_0[u, v] = \frac{1}{(1-u)(1-v)} + \frac{1}{(1+u)(1+v)}$$

and

$$\phi'_{1,\tau}(s) = \frac{2s^{2\tau-1}}{1-s^2}, \quad \phi''_{1,\tau}(s) = 2s^{2\tau-2} \frac{(2\tau-1) - (2\tau-3)s^2}{(1-s^2)^2}.$$

Proof of (i). For $\sigma \in \{-1, 1\}$, define

$$A_\sigma := \frac{\mathbb{1}[\min\{1 + \sigma u, 1 + \sigma v\} \leq w/\tau]}{(1 + \sigma u)(1 + \sigma v)}.$$

We prove the stronger one-sided estimate $\phi'_{1,\tau}[u, v] \geq A_\sigma/2$ for each σ . First consider $\sigma = -1$. Since $\phi''_{1,\tau} \geq 0$, the estimate is trivial when $A_{-1} = 0$. Otherwise, symmetry in u, v allows us to assume $u \geq v$. Put $\delta_u := 1 - u$ and $\delta_v := 1 - v$, so $0 < \delta_u \leq \delta_v$ and $\delta_u \leq w/\tau$. We first treat $u > v$; the case $u = v$ follows by continuity. If $\delta_v \leq 1/(4\tau)$, then every $s \in [v, u]$ satisfies

$$s^{2\tau-2} \geq (1 - 1/(4\tau))^{2\tau-2} \geq 1/2.$$

The displayed formula for $\phi''_{1,\tau}$ and $(1+s)^2 \leq 4$ therefore give $\phi''_{1,\tau}(s) \geq 1/(2(1-s)^2)$. Hence

$$\phi'_{1,\tau}[u, v] = \frac{1}{u-v} \int_v^u \phi''_{1,\tau}(s) ds \geq \frac{1}{2(u-v)} \left(\frac{1}{1-u} - \frac{1}{1-v} \right) = \frac{1}{2\delta_u \delta_v}.$$

If $\delta_v > 1/(4\tau)$, Bernoulli's inequality yields

$$u^{2\tau-1} \geq (1 - w/\tau)^{2\tau-1} \geq 1 - 2w.$$

Thus $\phi'_{1,\tau}(u) \geq (1 - 2w)/\delta_u$. Also $\phi'_{1,\tau}(v) \leq 2/\delta_v$: this is immediate when $v \leq 0$, and when $v > 0$ it follows from $v^{2\tau-1}/(1+v) \leq 1$. Since $\delta_u/\delta_v < 4w$ and $0 < w < 1/20$,

$$\begin{aligned} \phi'_{1,\tau}[u, v] &\geq \frac{(1 - 2w)/\delta_u - 2/\delta_v}{\delta_v - \delta_u} \\ &\geq \frac{1 - 10w}{\delta_u \delta_v} > \frac{1}{2\delta_u \delta_v}. \end{aligned}$$

The first inequality uses the endpoint derivative bounds; the second uses $\delta_u/\delta_v < 4w$ and $\delta_v - \delta_u < \delta_v$, with positive numerator; the last uses $w < 1/20$. This proves $\phi'_{1,\tau}[u, v] \geq A_{-1}/2$. Because $\phi_{1,\tau}$ is even, $\phi'_{1,\tau}[-u, -v] = \phi'_{1,\tau}[u, v]$; applying the one-sided estimate to $(-u, -v)$ gives $\phi'_{1,\tau}[u, v] \geq A_1/2$. Therefore

$$\phi'_{1,\tau}[u, v] \geq \frac{1}{2} \max\{A_{-1}, A_1\} \geq \frac{1}{4} (A_{-1} + A_1),$$

which is (i).

Proof of (ii). For every s between u and v ,

$$\phi''_{1,\tau}(s) \leq 2\tau |s|^{2\tau-2} \phi''_0(s) \leq 2\tau \max\{|u|, |v|\}^{2\tau-2} \phi''_0(s).$$

Integrating this inequality along the segment from u to v proves (ii). $\square$

4.3 Spectral Hessian and the 0–1 decomposition

The scalar estimates yield spectral Hessian and 1-component variance bounds.

Definition 4.7 (Spectral 0–1 decomposition). Fix an arbitrary $\kappa_3 \in (0, 1/20)$, retained throughout the general estimates below. Fix $t \in [0, 1]$, let $\widehat{A}_i(t)$ be as in Definition 4.4, and let X be symmetric with $\|X\| < 1$. For $\sigma \in \{-1, 1\}$, define

$$Y_\sigma := I + \sigma X, \quad C^\sigma(Y) := Y_\sigma^{-1/2}(\sigma Y)Y_\sigma^{-1/2}.$$

Let P_σ be the spectral projection of Y_σ onto $[0, \kappa_3/\tau]$, put $Q_\sigma := I - P_\sigma$, and define

$$C_1^\sigma(Y) := Q_\sigma C^\sigma(Y) Q_\sigma, \quad C_0^\sigma(Y) := C^\sigma(Y) - C_1^\sigma(Y).$$

Here index 0 denotes the edge-touching component, whereas index 1 denotes the regular component supported on Q_σ . For the coefficient matrices at time t , define

$$C_i^\sigma := C^\sigma(\widehat{A}_i(t)), \quad C_{i,1}^\sigma := Q_\sigma C_i^\sigma Q_\sigma, \quad C_{i,0}^\sigma := C_i^\sigma - C_{i,1}^\sigma.$$

The tensor Gram inequalities apply separately to the edge and regular components through the following quadratic forms.

Definition 4.8 (Spectral quadratic forms). Let $C_{i,0}^\sigma$ and $C_{i,1}^\sigma$ be defined in Definition 4.7. For $z_1, \dots, z_m$ in a finite-dimensional Hilbert space, define

$$\begin{aligned} f_{0,\sigma}(z) &:= \sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[C_{i_1,0}^\sigma C_{i_2,0}^\sigma] \langle z_{i_1}, z_{i_2} \rangle, \\ f_{1,\sigma}(z) &:= \sum_{i_1=1}^m \sum_{i_2=1}^m \text{tr}[C_{i_1,1}^\sigma C_{i_2,1}^\sigma] \langle z_{i_1}, z_{i_2} \rangle. \end{aligned}$$

The barrier supplies strong curvature on the edge component, while the regular component retains a bounded quadratic variance. These two controls enter the curvature calculation for the resolvent weights.

Lemma 4.9 (Spectral 0–1 estimates). *Let the notation be as in Definitions 4.7 and 4.8. Then the following statements hold.*

(i) *For every symmetric Y ,*

$$D^2\Phi(X)[Y, Y] \geq \sum_{\sigma \in \{-1, 1\}} \left(\frac{\kappa_1}{4} \|C_0^\sigma(Y)\|_F^2 + \kappa_0 \|C_1^\sigma(Y)\|_F^2 \right).$$

(ii) *The 1-components satisfy*

$$\sum_{i=1}^m (C_{i,1}^\sigma)^2 \preceq \kappa_3^{-2} \tau^2 \rho I.$$

(iii) *The forms $f_{0,\sigma}$ and $f_{1,\sigma}$ are nonnegative, and*

$$\sum_{i_1=1}^m \sum_{i_2=1}^m D^2\Phi(X)[\widehat{A}_{i_1}(t), \widehat{A}_{i_2}(t)] \langle z_{i_1}, z_{i_2} \rangle \geq \sum_{\sigma \in \{-1, 1\}} \left(\frac{\kappa_1}{4} f_{0,\sigma}(z) + \kappa_0 f_{1,\sigma}(z) \right).$$

Proof. Proof of (i). If $X = U \operatorname{diag}(\lambda_1, \dots, \lambda_N) U^\top$ and $f \in C^2(-1, 1)$, the Daleckii–Krein formula (Bhatia [Bha97, Theorem V.3.3]) gives

$$D^2 \operatorname{tr}[f(X)][Y, Y] = \sum_{a=1}^N \sum_{b=1}^N f'[\lambda_a, \lambda_b] |(U^\top Y U)_{ab}|^2,$$

where $f'[\lambda_a, \lambda_b]$ is the divided difference of Definition 3.2, equal to $f''(\lambda_a)$ on the diagonal. Apply Lemma 4.6(i) with $w = \kappa_3 \in (0, 1/20)$ to the tail term. The logarithmic part is exactly the sum of the full squared Frobenius norms, while the tail supplies one quarter of each 0-component norm. Thus the 0-component coefficient is $\kappa_0 + (\kappa_1 - \kappa_0)/4 \geq \kappa_1/4$, which is (i).

Proof of (ii). The Schur multiplier $\mathcal{S}_t$ that fixes diagonal entries and multiplies off-diagonal entries by t is unital and completely positive, because its symbol $(1-t)I + tJ$, with J the all-ones matrix, is positive semidefinite with unit diagonal for $t \in [0, 1]$. Kadison's inequality gives

$$\sum_{i=1}^m \hat{A}_i(t)^2 \preceq \sum_{i=1}^m \mathcal{S}_t(\hat{A}_i^2) = \mathcal{S}_t\left(\sum_{i=1}^m \hat{A}_i^2\right) \preceq \rho I.$$

Moreover, Q_σ commutes with Y_σ , and $Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq (\tau/\kappa_3) Q_\sigma$. Therefore

$$\begin{aligned} \sum_{i=1}^m (C_{i,1}^\sigma)^2 &\preceq \frac{\tau}{\kappa_3} Q_\sigma Y_\sigma^{-1/2} \left(\sum_{i=1}^m \hat{A}_i(t) Q_\sigma \hat{A}_i(t) \right) Y_\sigma^{-1/2} Q_\sigma \\ &\preceq \frac{\tau}{\kappa_3} Q_\sigma Y_\sigma^{-1/2} \left(\sum_{i=1}^m \hat{A}_i(t)^2 \right) Y_\sigma^{-1/2} Q_\sigma \\ &\preceq \frac{\tau \rho}{\kappa_3} Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq \frac{\tau^2 \rho}{\kappa_3^2} I. \end{aligned}$$

Here the first inequality is the definition of $C_{i,1}^\sigma$, the commutation of Q_σ with Y_σ , and Fact 3.3(a) applied to $Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq (\tau/\kappa_3) Q_\sigma$; the second is Fact 3.3(a) applied to $0 \preceq Q_\sigma \preceq I$; the third uses $\sum_{i=1}^m \hat{A}_i(t)^2 \preceq \rho I$ and again Fact 3.3(a), followed by $Q_\sigma Y_\sigma^{-1} Q_\sigma \preceq (\tau/\kappa_3) Q_\sigma \preceq (\tau/\kappa_3) I$.

Proof of (iii). Fact 3.1 shows that both forms are nonnegative. Expand the z_i in an orthonormal basis and apply (i) to each resulting linear combination of the matrices $\hat{A}_i(t)$. Summing over the basis proves (iii). $\square$

4.4 The matrix-weighted Poincaré estimate

Convex matrix weights require a Poincaré inequality that respects their noncommutative geometry. Positive curvature turns weighted variance into a Dirichlet-energy bound. For a positive definite weight A on an open set $\Gamma \subseteq \mathbb{R}^{q_1}$, a smooth field $f : \Gamma \rightarrow \mathbb{R}^{q_2}$ has finite energy if $\int f^\top A f$ and $\int \sum_{i=1}^{q_1} (\partial_i f)^\top A (\partial_i f)$ are finite.

Lemma 4.10 (Matrix-weighted Poincaré criterion). *Let $\Gamma \subseteq \mathbb{R}^{q_1}$ be open and convex, let $q_3 > 0$, and let $A : \Gamma \rightarrow \mathbb{R}^{q_2 \times q_2}$ be smooth, positive definite, and integrable. Define*

$$\mathcal{K}_{i_1, i_2}(A) := (\partial_{i_1} A) A^{-1} (\partial_{i_2} A) - \partial_{i_1, i_2} A.$$

If, for every $x_1, \dots, x_{q_1} \in \mathbb{R}^{q_2}$,

$$\sum_{i_1=1}^{q_1} \sum_{i_2=1}^{q_1} x_{i_1}^\top \mathcal{K}_{i_1, i_2}(A) x_{i_2} \geq q_3 \sum_{i=1}^{q_1} x_i^\top A x_i,$$

then every smooth $f : \Gamma \rightarrow \mathbb{R}^{q_2}$ of finite energy satisfies

$$\int f^\top A f - \left(\int A f \right)^\top \left(\int A \right)^{-1} \left(\int A f \right) \leq q_3^{-1} \int \sum_{i=1}^{q_1} (\partial_i f)^\top A (\partial_i f).$$

Proof. First let $\Gamma' \Subset \Gamma$ be bounded, convex, and have smooth boundary, with outward unit normal ν and an orthonormal tangential frame $e_1, \dots, e_{q_1-1}$ along $\partial\Gamma'$. On $L^2(\Gamma', A)$ define

$$\langle u, v \rangle_A := \int_{\Gamma'} u^\top A v, \quad \mathcal{L}u := - \sum_{i=1}^{q_1} (\partial_i^2 u + A^{-1} (\partial_i A) \partial_i u).$$

Since A is smooth and uniformly positive definite on $\overline{\Gamma'}$, this is a uniformly elliptic divergence-form system. For smooth vector fields satisfying $\partial_\nu u = 0$, integration by parts gives

$$\langle v, \mathcal{L}u \rangle_A = \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i v)^\top A (\partial_i u). \quad (1)$$

A second integration by parts, followed by the product rule, gives the following identity. To keep the noncommutative order explicit, expand $\mathcal{L}u = -A^{-1} \sum_i \partial_i (A \partial_i u)$ before integrating and never commute A with its derivatives. Pairing each term containing one derivative of A with its transpose leaves the ordered product $(\partial_{i_1} A) A^{-1} (\partial_{i_2} A)$; differentiating the remaining coefficient gives $-\partial_{i_1, i_2} A$. Thus

$$\begin{aligned} \|\mathcal{L}u\|_A^2 &= \int_{\Gamma'} \sum_{i_1=1}^{q_1} \sum_{i_2=1}^{q_1} (\partial_{i_1, i_2} u)^\top A (\partial_{i_1, i_2} u) \\ &\quad + \int_{\Gamma'} \sum_{i_1=1}^{q_1} \sum_{i_2=1}^{q_1} (\partial_{i_1} u)^\top \mathcal{K}_{i_1, i_2}(A) (\partial_{i_2} u) \\ &\quad + \int_{\partial\Gamma'} \sum_{a=1}^{q_1-1} \sum_{b=1}^{q_1-1} S_{ab} (\partial_{e_a} u)^\top A (\partial_{e_b} u). \end{aligned}$$

The interior zeroth-order term in this calculation is

$$(\partial_{i_1} A) A^{-1} (\partial_{i_2} A) - \partial_{i_1, i_2} A = \mathcal{K}_{i_1, i_2}(A);$$

all first-order terms cancel in pairs. The boundary term is the tangential Hessian of the Neumann condition, and S is the second fundamental form of $\partial\Gamma'$ in the frame $e_1, \dots, e_{q_1-1}$, with the outward-normal convention. Convexity makes this term nonnegative. The curvature hypothesis in the lemma therefore implies

$$\|\mathcal{L}u\|_A^2 \geq q_3 \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i u)^\top A (\partial_i u). \quad (2)$$

Let

$$\bar{f}_{\Gamma'} := \left(\int_{\Gamma'} A \right)^{-1} \int_{\Gamma'} A f, \quad h := f - \bar{f}_{\Gamma'}.$$

Then $\int_{\Gamma'} A h = 0$. The bilinear form on the right side of Eq. (1) is coercive modulo constant vectors. Lax–Milgram gives a weak solution, unique modulo constants, of

$$\mathcal{L}u = h, \quad \partial_\nu u = 0,$$

and elliptic regularity makes it smooth when f is smooth. Eq. (1) and Eq. (2) give

$$q_3 \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i u)^\top A(\partial_i u) \leq \|h\|_A^2$$

and

$$\|h\|_A^2 = \langle h, \mathcal{L}u \rangle_A = \int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i h)^\top A(\partial_i u) \leq \left(\int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i f)^\top A(\partial_i f) \right)^{1/2} \left(\int_{\Gamma'} \sum_{i=1}^{q_1} (\partial_i u)^\top A(\partial_i u) \right)^{1/2}.$$

If $h \neq 0$, division by $\|h\|_A$ proves the desired inequality on Γ' ; if $h = 0$, it is immediate.

Choose an increasing smooth convex exhaustion $\Gamma_k \Subset \Gamma$. The left side on Γ_k is

$$\inf_{c \in \mathbb{R}^{q_2}} \int_{\Gamma_k} (f - c)^\top A(f - c).$$

Integrability of A and $f^\top A f$ shows that the minimizing constants and these infima converge to their counterparts on Γ , while the Dirichlet integrals increase to the integral on Γ , which gives the inequality on Γ for every smooth finite-energy field. $\square$

4.5 One- and two-slot resolvent Poincaré inequalities

The matrix-weighted criterion yields Poincaré inequalities for the weights $e^{-\Phi(X) - \|\xi\|_2^2/2} Y_\sigma^{-1}$ and $e^{-\Phi(X) - \|\xi\|_2^2/2} Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}$ on the small-ball domain. A slot is one Kronecker factor of the weight, so these are the one- and two-slot inequalities.

Definition 4.11 (Weighted resolvent energy). In the setting of Definition 4.4, fix $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$ and $Y_\sigma := I + \sigma X$ for $\sigma \in \{-1, 1\}$, and let μ_t be defined as in Lemma 4.5(ii). For $\sigma_1, \sigma_2 \in \{-1, 1\}$ and every matrix field $U = U(\xi)$ for which the following expectation is finite, define

$$\mathcal{E}_{\sigma_1, \sigma_2}(U) := \mathbb{E}_{\mu_t}[\text{vec}(U)^\top (Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}) \text{vec}(U)].$$

Each resolvent factor contributes a negative term to the normalized curvature of the matrix weight. The spectral 0–1 estimates allow the barrier Hessian to absorb these terms.

Lemma 4.12 (One- and two-slot resolvent Poincaré inequalities). Fix $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$ and $Y_\sigma := I + \sigma X$, and let μ_t be defined as in Lemma 4.5(ii). Let κ_3 be as in Definition 4.7. Assume $\kappa_1 > 8$, $\kappa_0(\kappa_1 - 8) > 16$, and $\tau^2 \rho \leq \kappa_3^2/4$. Define

$$c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho) := \frac{1}{1 - 2\tau^2 \rho / \kappa_3^2}.$$

In the following statements, c_1 denotes this value.

(i) For every $\sigma \in \{-1, 1\}$ and every smooth vector field $v : \mathbb{R}^m \rightarrow \mathbb{R}^N$ of finite energy, define $S_\sigma := \mathbb{E}_{\mu_t}[Y_\sigma^{-1}]$ and $b_\sigma(v) := \mathbb{E}_{\mu_t}[Y_\sigma^{-1} v]$. Then

$$\mathbb{E}_{\mu_t}[v^\top Y_\sigma^{-1} v] - b_\sigma(v)^\top S_\sigma^{-1} b_\sigma(v) \leq c_1 \sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i v)^\top Y_\sigma^{-1} (\partial_i v)].$$

(ii) For every $\sigma_1, \sigma_2 \in \{-1, 1\}$ and every smooth vector field $f : \mathbb{R}^m \rightarrow \mathbb{R}^{N^2}$ of finite energy, define

$$\mathcal{Y}_{\sigma_1, \sigma_2} := Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}, \quad S_{\sigma_1, \sigma_2} := \mathbb{E}_{\mu_t}[\mathcal{Y}_{\sigma_1, \sigma_2}], \quad b_{\sigma_1, \sigma_2}(f) := \mathbb{E}_{\mu_t}[\mathcal{Y}_{\sigma_1, \sigma_2} f].$$

Then

$$\mathbb{E}_{\mu_t}[f^\top \mathcal{Y}_{\sigma_1, \sigma_2} f] - b_{\sigma_1, \sigma_2}(f)^\top S_{\sigma_1, \sigma_2}^{-1} b_{\sigma_1, \sigma_2}(f) \leq c_1 \sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i f)^\top \mathcal{Y}_{\sigma_1, \sigma_2} (\partial_i f)].$$

Proof. We apply Lemma 4.10 to the small-ball weights.

Step 1: curvature of the two-slot weight. Put $X := \mathcal{A}_t(\xi)$ and define the two-slot weights

$$A_{\sigma_1, \sigma_2}(\xi) := e^{-\|\xi\|_2^2/2 - \Phi(X)} (Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}), \quad \sigma_1, \sigma_2 \in \{-1, 1\}.$$

Define the normalized curvature blocks by

$$\tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2} := A_{\sigma_1, \sigma_2}^{-1/2} \mathcal{K}_{i_1, i_2}(A_{\sigma_1, \sigma_2}) A_{\sigma_1, \sigma_2}^{-1/2}.$$

Direct differentiation, using $\partial_i X = \hat{A}_i(t)$, gives

$$\tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2} = \delta_{i_1, i_2} I + D^2 \Phi(X) [\hat{A}_{i_1}(t), \hat{A}_{i_2}(t)] I - C_{i_2}^{\sigma_2} C_{i_1}^{\sigma_2} \otimes I - I \otimes C_{i_2}^{\sigma_1} C_{i_1}^{\sigma_1}. \quad (3)$$

Fix $z_1, \dots, z_m \in \mathbb{R}^N \otimes \mathbb{R}^N$. For either negative slot, expand $C_i^\sigma = C_{i,0}^\sigma + C_{i,1}^\sigma$, so that $C_{i_2}^\sigma C_{i_1}^\sigma$ splits into four 0/1 products. Fact 3.1 bounds the 00 product by $f_{0,\sigma}(z)$, each mixed product by $\sqrt{f_{0,\sigma}(z)f_{1,\sigma}(z)}$, and, together with Lemma 4.9(ii), the 11 product by $\kappa_3^{-2} \tau^2 \rho \sum_i \|z_i\|_2^2$:

$$|\sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, (C_{i_2}^\sigma C_{i_1}^\sigma \otimes I) z_{i_2} \rangle| \leq f_{0,\sigma}(z) + 2\sqrt{f_{0,\sigma}(z)f_{1,\sigma}(z)} + \kappa_3^{-2} \tau^2 \rho \sum_{i=1}^m \|z_i\|_2^2.$$

We have

$$f_{0,\sigma} + 2\sqrt{f_{0,\sigma}f_{1,\sigma}} \leq \frac{\kappa_1}{8} f_{0,\sigma} + \frac{8}{\kappa_1 - 8} f_{1,\sigma} \leq \frac{\kappa_1}{8} f_{0,\sigma} + \frac{\kappa_0}{2} f_{1,\sigma},$$

where the first step follows from Young's inequality with parameter $(\kappa_1 - 8)/8 > 0$, and the second step follows from $\kappa_0(\kappa_1 - 8) > 16$ and $f_{1,\sigma} \geq 0$. By Lemma 4.9(iii), this bounds the 00 and mixed products in each slot by one half of the barrier Hessian quadratic form. Applying the estimate to both tensor slots (to the second after the coordinate swap of $\mathbb{R}^N \otimes \mathbb{R}^N$, which preserves every $\langle z_{i_1}, z_{i_2} \rangle$) in Eq. (3) gives

$$\sum_{i_1=1}^m \sum_{i_2=1}^m \langle z_{i_1}, \tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2} z_{i_2} \rangle \geq (1 - 2\kappa_3^{-2} \tau^2 \rho) \sum_{i=1}^m \|z_i\|_2^2.$$

Thus the conditions

$$\kappa_1 > 8, \quad \kappa_0(\kappa_1 - 8) > 16, \quad \tau^2 \rho \leq \kappa_3^2/4 \quad (4)$$

imply that $\lambda := 1 - 2\kappa_3^{-2} \tau^2 \rho$ satisfies $\lambda \geq 1/2$ and

$$[\tilde{\mathcal{K}}_{i_1, i_2}^{\sigma_1, \sigma_2}]_{i_1, i_2} \succeq \lambda I.$$

Step 2: curvature of the one-slot weight. For the one-slot weight, define

$$A_\sigma(\xi) := e^{-\|\xi\|_2^2/2 - \Phi(X)} Y_\sigma^{-1}, \quad \tilde{\mathcal{K}}_{i_1, i_2}^\sigma := A_\sigma^{-1/2} \mathcal{K}_{i_1, i_2}(A_\sigma) A_\sigma^{-1/2}.$$

Direct differentiation gives

$$\tilde{\mathcal{K}}_{i_1, i_2}^\sigma = \delta_{i_1, i_2} I + D^2 \Phi(X) [\hat{A}_{i_1}(t), \hat{A}_{i_2}(t)] I - C_{i_2}^\sigma C_{i_1}^\sigma.$$

We have

$$[\tilde{\mathcal{K}}_{i_1, i_2}^\sigma]_{i_1, i_2} \succeq (1 - \kappa_3^{-2} \tau^2 \rho) I \succeq \lambda I,$$

where the first step follows from the same spectral 0–1 estimates with a single negative slot, and the second step follows from $\kappa_3^{-2} \tau^2 \rho \geq 0$. The three conditions in Eq. (4) are exactly the hypotheses of the lemma.

Step 3: the domain and integrability. We apply the curvature criterion on

$$\Omega_t := \{\xi \in \mathbb{R}^m : \|\mathcal{A}_t(\xi)\| < 1\}.$$

This set is open, convex, and contains the origin. The weights are smooth and positive definite on Ω_t . If $\Delta(\xi) := 1 - \|\mathcal{A}_t(\xi)\|$, then the boundary expansion of ϕ gives

$$e^{-\Phi(\mathcal{A}_t(\xi))} \leq c_{\kappa_1, \kappa_0, \tau} \Delta(\xi)^{\kappa_1}.$$

Since each resolvent has norm at most $\Delta(\xi)^{-1}$, the one-slot and two-slot weights are bounded by a Gaussian density times, respectively, $c_{\kappa_1, \kappa_0, \tau} \Delta^{\kappa_1-1}$ and $c_{\kappa_1, \kappa_0, \tau} \Delta^{\kappa_1-2}$. Thus they and their products with polynomial test fields are integrable because $\kappa_1 > 8$. Apply Lemma 4.10 with $q_1 = m$, $\Gamma = \Omega_t$, and $q_3 = \lambda = 1 - 2\kappa_3^{-2} \tau^2 \rho$. For (i), take $q_2 = N$ and $A = A_\sigma$; for (ii), take $q_2 = N^2$ and $A = A_{\sigma_1, \sigma_2}$. Since

$$d\mu_t(\xi) = \frac{e^{-\|\xi\|_2^2/2 - \Phi(X)}}{(2\pi)^{m/2} a_t} d\xi,$$

dividing each resulting inequality by $(2\pi)^{m/2} a_t$ gives the corresponding assertion with

$$c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho) = q_3^{-1} = \lambda^{-1}.$$

□

4.6 The weighted resolvent estimate

The one- and two-slot inequalities yield the resolvent estimate used in the moment argument.

Lemma 4.13 (Weighted resolvent estimate). *Fix $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$ and $Y_\sigma := I + \sigma X$, and let $\mathcal{E}_{\sigma_1, \sigma_2}$ be defined as in Definition 4.11. Let $0 < \kappa_3 < 1/20$ be as in Definition 4.7, and let $\tau \geq 2$ be an integer. Assume $\kappa_1 > 8$, $\kappa_0(\kappa_1 - 8) > 16$, and $\tau^2 \rho \leq \kappa_3^2/4$. Let $c_1 = c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho)$ be as defined in Lemma 4.12, and define*

$$\eta := (1 + 8c_1(1 + 2c_1\rho))\rho, \quad c_2(c_1, \rho) := \frac{2(1 + \rho)}{1 - 2(1 + \rho)(\rho + \eta/(1 - \eta))}.$$

The denominators are positive, and $c_2(c_1, \rho) < 3$ throughout the admissible parameter range. Hereafter c_2 denotes this value. For every $\sigma_1, \sigma_2 \in \{-1, 1\}$ and every smooth matrix field U of finite energy,

$$\mathcal{E}_{\sigma_1, \sigma_2}(U) \leq c_2 \mathbb{E}_{\mu_t} [\|U\|_F^2] + c_2 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U).$$

Proof. Retain the notation $X := \mathcal{A}_t(\xi)$. The assumptions $0 < \kappa_3 < 1/20$ and $\tau \geq 2$ give

$$\rho \leq \frac{\kappa_3^2}{4\tau^2} < \frac{1}{6400}, \quad c_1 = \frac{1}{1 - 2\tau^2\rho/\kappa_3^2}, \quad 1 \leq c_1 \leq 2.$$

The first bound is the hypothesis on $\tau^2\rho$; the second uses $\kappa_3^2 < 1/400$ and $\tau^2 \geq 4$. The formula for c_1 is its definition in Lemma 4.12, and $c_1 \leq 2$ follows from $\tau^2\rho \leq \kappa_3^2/4$. In particular,

$$\eta \leq 17\rho + 64\rho^2 < \frac{1}{200}.$$

The first step uses $c_1 \leq 2$, and the second uses $\rho < 1/6400$. It follows that

$$2(1 + \rho)(\rho + \eta/(1 - \eta)) < 2(1 + 1/6400)(1/6400 + 1/199) < 1/50.$$

The first step uses $\eta/(1 - \eta) < 1/199$ and $\rho < 1/6400$; the second is a rational comparison. Thus both denominators in the definition of c_2 are positive, and

$$c_2(c_1, \rho) < \frac{2(1 + 1/6400)}{1 - 1/50} < 3.$$

The first step uses the preceding denominator bound, and the second is a rational comparison. We retain the parameter-dependent coefficients below, using the one-slot and two-slot estimates of Lemma 4.12(i)–(ii) with coefficient c_1 .

Step 1: the projected mean of a one-slot weight. Put

$$S_\sigma := \mathbb{E}_{\mu_t}[Y_\sigma^{-1}].$$

Evenness gives $\mathbb{E}_{\mu_t}[X] = 0$, hence $\mathbb{E}_{\mu_t}[Y_\sigma] = I$. Operator convexity of inversion yields $S_\sigma \succeq I$. For $z \in \mathbb{R}^N$, apply the one-slot inequality to

$$v_{\sigma,z} := (Y_\sigma - I)z.$$

Since $Y_\sigma^{-1}v_{\sigma,z} = (I - Y_\sigma^{-1})z$, $\partial_i v_{\sigma,z} = \sigma \hat{A}_i(t)z$, and $\mathbb{E}_{\mu_t}[Y_\sigma] = I$, direct expansion of the weighted variance gives

$$I - S_\sigma^{-1} \preceq c_1 \sum_{i=1}^m \hat{A}_i(t) S_\sigma \hat{A}_i(t).$$

If $L_\sigma := \|S_\sigma\|$, then

$$1 - L_\sigma^{-1} \leq c_1 \rho L_\sigma.$$

To exclude the large root, for $u \in [0, 1]$ define

$$X_u := uX, \quad Y_{\sigma,u} := I + \sigma X_u, \quad Z_{t,u} := \mathbb{E}[e^{-\Phi(X_u)}], \quad d\mu_{t,u} := Z_{t,u}^{-1} e^{-\Phi(X_u)} d\gamma_m,$$

$$S_{\sigma,u} := \mathbb{E}_{\mu_{t,u}}[Y_{\sigma,u}^{-1}], \quad L_\sigma(u) := \|S_{\sigma,u}\|.$$

The function $c_1(\kappa_0, \kappa_1, \kappa_3, \tau, \rho)$ is nondecreasing in ρ on the admissible range. Since $u^2\rho \leq \rho$, Lemma 4.12 applies to the coefficients $u\hat{A}_i(t)$ with a coefficient no larger than the fixed value c_1 at ρ . The same argument therefore gives

$$1 - L_\sigma(u)^{-1} \leq c_1 u^2 \rho L_\sigma(u).$$

Put $\Delta_u := 1 - \|X_u\|$ on $\{\|X_u\| < 1\}$. For $|s| < 1$, Definition 4.2 gives

$$\phi(s) = -\kappa_1 \log(1 - s^2) - (\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} \frac{s^{2k}}{k}.$$

Also $\phi(s) \geq 0$, since its power-series coefficients are positive. Choose an eigenvalue λ of X_u with $|\lambda| = \|X_u\|$. Then

$$\begin{aligned} e^{-\Phi(X_u)} &\leq e^{-\phi(\lambda)} \\ &\leq (1 - \lambda^2)^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k] \\ &\leq 2^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k] \Delta_u^{\kappa_1}. \end{aligned}$$

The first step uses $\phi \geq 0$ for all the other eigenvalues. The second uses $\kappa_1 \geq \kappa_0$ and $\lambda^{2k} \leq 1$. The last uses $1 - \lambda^2 = (1 - |\lambda|)(1 + |\lambda|) \leq 2\Delta_u$. Consequently, the integrand in the numerator defining $S_{\sigma,u}$ satisfies

$$\|e^{-\Phi(X_u)} Y_{\sigma,u}^{-1}\| \leq 2^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k] \Delta_u^{\kappa_1-1} \leq 2^{\kappa_1} \exp[(\kappa_1 - \kappa_0) \sum_{k=1}^{\tau-1} 1/k].$$

The first step uses $\|Y_{\sigma,u}^{-1}\| \leq \Delta_u^{-1}$, and the second uses $0 < \Delta_u \leq 1$ and $\kappa_1 > 8$. Extend the numerator and denominator integrands by zero outside $\{\|X_u\| < 1\}$. These extensions are continuous at the boundary, and the denominator integrand is bounded by 1. Dominated convergence, together with $Z_{t,u} > 0$, shows that $L_\sigma(u)$ is continuous. Since $L_\sigma(0) = 1$, this path stays on the smaller branch of the quadratic. Consequently,

$$I \preceq S_\sigma \preceq \frac{2}{1 + \sqrt{1 - 4c_1\rho}} I \preceq (1 + 2c_1\rho) I. \quad (5)$$

The last step uses $2/(1 + \sqrt{1 - 4x}) \leq 1 + 2x$, valid for $0 \leq x \leq (\sqrt{2} - 1)/2$ (squaring out, it is $4x^2 + 4x - 1 \leq 0$), applied with $x = c_1\rho \leq 2\rho \leq \kappa_3^2/8 < 1/3200$, since $\tau \geq 2$ and $0 < \kappa_3 < 1/20$.

Step 2: the two-slot expectation. For the two-slot weight, put

$$\mathcal{Y} := Y_{\sigma_2}^{-1} \otimes Y_{\sigma_1}^{-1}, \quad S := \mathbb{E}_{\mu_t}[\mathcal{Y}], \quad \mathcal{R} := \mathcal{Y}^{-1} = Y_{\sigma_2} \otimes Y_{\sigma_1}, \quad T := \mathbb{E}_{\mu_t}[\mathcal{R}].$$

The measure μ_t is a 1-strongly log-concave Gaussian tilt. Approximate the extended-valued potential monotonically by finite smooth even convex potentials. Brascamp–Lieb [BL76] applies to each approximation, and passage to the limit gives $\text{Cov}_{\mu_t}[\xi] \preceq I$. Factor this covariance as $UU^\top$ and set $N_k := \sum_{i=1}^m U_{ik} \hat{A}_i(t)$. Then

$$\sum_{k=1}^m N_k^2 \preceq \sum_{i=1}^m \hat{A}_i(t)^2 \preceq \rho I,$$

and the operator Cauchy–Schwarz inequality gives

$$\|\mathbb{E}_{\mu_t}[X \otimes X]\| = \left\| \sum_{k=1}^m N_k \otimes N_k \right\| \leq \left\| \sum_{k=1}^m N_k^2 \right\| \leq \rho.$$

The operator Cauchy–Schwarz inequality used here has a one-line proof: for $z = \text{vec}(Z)$, symmetry of N_k gives $\langle z, (N_k \otimes N_k)z \rangle = \text{tr}[Z^\top N_k Z N_k] = \langle N_k Z, Z N_k \rangle_F$, so Cauchy–Schwarz applied twice and $\sum_k \|N_k Z\|_F^2 = \text{tr}[Z^\top (\sum_k N_k^2) Z] \leq \|\sum_k N_k^2\| \|Z\|_F^2$ (and the same for $Z N_k$) bound $|\langle z, (\sum_k N_k \otimes N_k)z \rangle|$ by $\|\sum_k N_k^2\| \|z\|_2^2$. Therefore

$$T = I + \sigma_1 \sigma_2 \mathbb{E}_{\mu_t}[X \otimes X], \quad \|T - I\| \leq \rho. \quad (6)$$

Step 3: closeness of S to the identity. For $z \in \mathbb{R}^{N^2}$, apply the two-slot inequality to $\Psi_z := (\mathcal{R} - I)z$. Its weighted variance is $z^\top (T - S^{-1})z$. Moreover,

$$\partial_i \mathcal{R} = \sigma_2 \hat{A}_i(t) \otimes Y_{\sigma_1} + Y_{\sigma_2} \otimes \sigma_1 \hat{A}_i(t).$$

By $(U + V)^\top \mathcal{Y}(U + V) \preceq 2U^\top \mathcal{Y}U + 2V^\top \mathcal{Y}V$, $Y_\sigma \preceq 2I$, $\sum_{i=1}^m \hat{A}_i(t)^2 \preceq \rho I$, and Eq. (5),

$$\begin{aligned} \sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i \mathcal{R}) \mathcal{Y}(\partial_i \mathcal{R})] &\preceq 4 \left(\sum_{i=1}^m \hat{A}_i(t) S_{\sigma_2} \hat{A}_i(t) \right) \otimes I \\ &\quad + 4I \otimes \left(\sum_{i=1}^m \hat{A}_i(t) S_{\sigma_1} \hat{A}_i(t) \right) \\ &\preceq 8(1 + 2c_1 \rho) \rho I. \end{aligned}$$

The first step uses a factor 2 from the quadratic-form bound and a factor 2 from $Y_\sigma \preceq 2I$ in each tensor term. The second uses $S_\sigma \preceq (1 + 2c_1 \rho)I$ and $\sum_i \hat{A}_i(t)^2 \preceq \rho I$. The two-slot estimate with coefficient c_1 therefore gives

$$0 \preceq T - S^{-1} \preceq 8c_1(1 + 2c_1 \rho) \rho I.$$

Together with Eq. (6), this implies

$$\|S^{-1} - I\| \leq \rho + 8c_1(1 + 2c_1 \rho) \rho = \eta.$$

The first step is the triangle inequality, and the second is the definition of η . Since $\eta < 1$, we have $S^{-1} \succeq (1 - \eta)I$, hence $\|S\| \leq (1 - \eta)^{-1}$. Consequently,

$$\|S - I\| = \|S(I - S^{-1})\| \leq \|S\| \|I - S^{-1}\| \leq \frac{\eta}{1 - \eta}.$$

The first step is the identity $S(I - S^{-1}) = S - I$, the second is submultiplicativity, and the last uses the two preceding bounds. Also $S^{-1} \preceq T \preceq (1 + \rho)I$. We have proved

$$\|S^{-1} - I\| \leq \eta, \quad \|S - I\| \leq \frac{\eta}{1 - \eta}, \quad S^{-1} \preceq (1 + \rho)I. \quad (7)$$

Step 4: the projected mean and the conclusion. Now let $f := \text{vec}(U)$ and put

$$u := \mathbb{E}_{\mu_t}[\mathcal{Y}f], \quad w := \mathbb{E}_{\mu_t}[(\mathcal{Y} - I)f], \quad u = \mathbb{E}_{\mu_t}[f] + w.$$

The identity

$$(\mathcal{Y} - I)\mathcal{Y}^{-1}(\mathcal{Y} - I) = \mathcal{Y} - 2I + \mathcal{Y}^{-1}$$

and Eq. (6) and Eq. (7) give

$$0 \preceq \mathbb{E}_{\mu_t}[(\mathcal{Y} - I)\mathcal{Y}^{-1}(\mathcal{Y} - I)] = S - 2I + T \preceq (\rho + \eta/(1 - \eta))I.$$

The first step uses positive semidefiniteness of the integrand. The second follows from the displayed identity. The last uses $S - I \preceq \eta(1 - \eta)^{-1}I$ and $T - I \preceq \rho I$. Weighted Cauchy–Schwarz and ordinary Cauchy–Schwarz give, respectively,

$$\|w\|_2^2 \leq (\rho + \eta/(1 - \eta))\mathcal{E}_{\sigma_1, \sigma_2}(U), \quad \|\mathbb{E}_{\mu_t}[f]\|_2^2 \leq \mathbb{E}_{\mu_t}[\|U\|_F^2].$$

Therefore

$$\begin{aligned} u^\top S^{-1}u &\leq (1 + \rho)\|u\|_2^2 \\ &\leq 2(1 + \rho)(\|\mathbb{E}_{\mu_t}[f]\|_2^2 + \|w\|_2^2) \\ &\leq 2(1 + \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + 2(1 + \rho)(\rho + \eta/(1 - \eta))\mathcal{E}_{\sigma_1, \sigma_2}(U). \end{aligned}$$

The first step uses $S^{-1} \preceq (1 + \rho)I$. The second uses $u = \mathbb{E}_{\mu_t}[f] + w$ and $\|a + b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$. The last uses the two preceding Cauchy–Schwarz bounds. Substituting this projected-mean bound into Lemma 4.12(ii) yields

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(U) &\leq 2(1 + \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + c_1 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U) \\ &\quad + 2(1 + \rho)(\rho + \eta/(1 - \eta))\mathcal{E}_{\sigma_1, \sigma_2}(U). \end{aligned}$$

The denominator $1 - 2(1 + \rho)(\rho + \eta/(1 - \eta))$ is positive, as checked at the start of the proof. Moving the last term to the left and dividing gives

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(U) &\leq \frac{2(1 + \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + c_1 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U)}{1 - 2(1 + \rho)(\rho + \eta/(1 - \eta))} \\ &\leq c_2(c_1, \rho)\mathbb{E}_{\mu_t}[\|U\|_F^2] + c_2(c_1, \rho) \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U). \end{aligned}$$

The first step is this rearrangement. The last uses the displayed definition of $c_2(c_1, \rho)$ and $c_1 \leq 2 \leq 2(1 + \rho)$. $\square$

4.7 Polynomial resolvent moments and the interpolation Hessian

Repeated differentiation produces noncommutative polynomials in Gaussian matrix series. Schatten interpolation and Gaussian moment bounds control these polynomials uniformly over the interpolation path.

Lemma 4.14 (Polynomial resolvent and interpolation bounds). *In the notation of Definitions 4.4, 4.7, and 4.11, and with c_2 as in Lemma 4.13, define*

$$c_3 := 4c_2(1 + c_2), \quad c_4 := 2c_3.$$

These constants are uniformly bounded over the admissible parameter range, and the following statements hold.

(i) *For symmetric X, V with $\|X\| < 1$,*

$$D^2 \operatorname{tr}[\phi_{1, \tau}(X)][V, V] \leq 4\tau \sum_{\sigma \in \{-1, 1\}} \operatorname{vec}(X^{\tau-1}V)^\top (Y_\sigma^{-1} \otimes Y_\sigma^{-1}) \operatorname{vec}(X^{\tau-1}V).$$

(ii) Assume $\kappa_1 > 8$, $\kappa_0(\kappa_1 - 8) > 16$, and $\tau^2 \rho \leq \kappa_3^2/4$. For $t \in [0, 1]$, put $X := \mathcal{A}_t(\xi)$. Then, for $\sigma_1, \sigma_2 \in \{-1, 1\}$ and uniformly for $t \in [0, 1]$,

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) \leq N(c_3 \tau^2 \rho)^\tau, \quad \mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) \leq c_3 N \rho,$$

and

$$\mathbb{E}_{\mu_t}[D^2 \Phi(\mathcal{A}_t)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]] \leq c_4 \kappa_0 N \rho + c_4 \kappa_1 \tau N(c_3 \tau^2 \rho)^\tau.$$

Proof. *Proof of (i).* Diagonalize $X = U \text{diag}(\eta_1, \dots, \eta_N) U^\top$ and put $\tilde{V} := U^\top V U$. Then

$$\begin{aligned} D^2 \text{tr}[\phi_{1, \tau}(X)][V, V] &\leq 2\tau \sum_{\sigma \in \{-1, 1\}} \sum_{i_1=1}^N \sum_{i_2=1}^N \frac{\max\{|\eta_{i_1}|, |\eta_{i_2}|\}^{2\tau-2} |\tilde{V}_{i_1, i_2}|^2}{(1 + \sigma \eta_{i_1})(1 + \sigma \eta_{i_2})} \\ &\leq 2\tau \sum_{\sigma \in \{-1, 1\}} \sum_{i_1=1}^N \sum_{i_2=1}^N \frac{(|\eta_{i_1}|^{2\tau-2} + |\eta_{i_2}|^{2\tau-2}) |\tilde{V}_{i_1, i_2}|^2}{(1 + \sigma \eta_{i_1})(1 + \sigma \eta_{i_2})} \\ &= 4\tau \sum_{\sigma \in \{-1, 1\}} \sum_{i_1=1}^N \sum_{i_2=1}^N \frac{|\eta_{i_1}|^{2\tau-2} |\tilde{V}_{i_1, i_2}|^2}{(1 + \sigma \eta_{i_1})(1 + \sigma \eta_{i_2})} \\ &= 4\tau \sum_{\sigma \in \{-1, 1\}} \text{vec}(X^{\tau-1} V)^\top (Y_\sigma^{-1} \otimes Y_\sigma^{-1}) \text{vec}(X^{\tau-1} V). \end{aligned}$$

Here the first inequality is the upper divided-difference estimate of Lemma 4.6(ii); the second uses $\max\{u, v\}^{2\tau-2} \leq u^{2\tau-2} + v^{2\tau-2}$ for $u, v \geq 0$; the first equality holds because $\tilde{V}$ is symmetric, so that interchanging i_1 and i_2 shows that the two numerator contributions are equal; for the last equality, the transformed product has entries $\eta_{i_1}^{\tau-1} \tilde{V}_{i_1, i_2}$ and Y_σ^{-1} has diagonal entries $(1 + \sigma \eta_{i_1})^{-1}$, so vectorization gives the displayed sum, and orthogonal conjugation preserves the quadratic form.

Proof of (ii). We use c_2 from Lemma 4.13 without changing its value. Its defining denominator lies in $(0, 1]$ and its numerator is $2(1 + \rho)$, so $c_2 \geq 2$. The calculations below verify the stated choices of c_3 and c_4 .

Step 1: variance proxies. By Kadison's inequality for the unital completely positive diagonal projection, and since that projection is positive and $\sum_{i=1}^m \hat{A}_i^2 \preceq \rho I$,

$$\sum_{i=1}^m \hat{A}_{i, \text{diag}}^2 \preceq \text{diag}(\sum_{i=1}^m \hat{A}_i^2) \preceq \rho I,$$

and Fact 3.3(b) then gives

$$\sum_{i=1}^m \hat{A}_{i, \text{off}}^2 \preceq 4\rho I.$$

Together with $\sum_{i=1}^m \hat{A}_i(t)^2 \preceq \rho I$ from the proof of Lemma 4.9(ii), every family of coefficient matrices that occurs below has variance proxy at most $4\rho I$.

Step 2: the polynomial moments. We use the ordered-multi-index convention from Section 3 for all derivative tensors. Apply the defective Poincaré estimate of Lemma 4.13 to every derivative field of order j and sum over all ordered indices. Since $\nabla^{\tau+1}(X^{\tau-1} \mathcal{A}_{\text{off}}) = 0$, induction on the degree gives

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) \leq \sum_{j=0}^{\tau} c_2^{j+1} \mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1} \mathcal{A}_{\text{off}})\|_{\text{HS}}^2].$$

For $0 \leq j \leq \tau$, put $s := \tau - j$. We claim

$$\mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})\|_{\text{HS}}^2] \leq N(4\rho)^\tau \tau^{2j} (2s)^s \leq N(4\tau^2\rho)^\tau. \quad (8)$$

Here $0^0 := 1$. The first step will follow from the word estimates below. The second uses $2s \leq 2\tau \leq \tau^2$, since $\tau \geq 2$, and $j + s = \tau$. Let $g_1, \dots, g_j$ be independent standard Gaussian directions, also independent of ξ . Gaussian contraction of the derivative tensor gives

$$\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})(\xi)\|_{\text{HS}}^2 = \mathbb{E}_g[\|D^j(X^{\tau-1}\mathcal{A}_{\text{off}})(\xi)[g_1, \dots, g_j]\|_F^2].$$

Repeated use of the product rule writes the derivative as a sum of at most $(\tau)_j \leq \tau^j$ words. Each word has τ factors, exactly j evaluated at the distinct directions $g_1, \dots, g_j$, and s evaluated at the common vector ξ . Every factor is a copy of either X or $\mathcal{A}_{\text{off}}$.

For a direction factor $G := \sum_{i=1}^m g_i R_i$, define the completely positive map

$$\Phi_G(U) := \mathbb{E}_g[GUG] = \sum_{i=1}^m R_i U R_i.$$

The coefficient matrices are symmetric and satisfy $\sum_{i=1}^m R_i^2 \leq 4\rho I_N$ by Step 1. The map is positive, so its norm on S_∞ is attained at the identity (the Russo–Dye theorem; here Cauchy–Schwarz gives it directly, since for unit vectors x, y one has $|x^\top \Phi_G(U)y| \leq \|U\| \sum_i \|R_i x\|_2 \|R_i y\|_2 \leq \|U\| \|\Phi_G(I_N)\|$), which gives

$$\|\Phi_G\|_{S_\infty \rightarrow S_\infty} = \|\Phi_G(I_N)\| \leq 4\rho.$$

It is self-adjoint for the trace pairing. Duality therefore gives the same bound on $S_1 \rightarrow S_1$, and complex interpolation of Schatten classes gives

$$\|\Phi_G(U)\|_{S_q} \leq 4\rho \|U\|_{S_q}, \quad 1 \leq q \leq \infty. \quad (9)$$

Fix one word $\mathcal{B} := \tilde{B}_1 \cdots \tilde{B}_\tau$. Since its factors are symmetric,

$$\|\mathcal{B}\|_F^2 = \text{tr}[\mathcal{B}_\tau], \quad B_0 := I_N, \quad B_k := \tilde{B}_k B_{k-1} \tilde{B}_k.$$

All B_k are positive semidefinite. Conditional on ξ , define $\bar{B}_k := \mathbb{E}_g[B_k]$ for $0 \leq k \leq \tau$, so that $\bar{B}_0 = I_N$. For $1 \leq k \leq \tau$, we have

$$\bar{B}_k = \begin{cases} \Phi_{\tilde{B}_k}(\bar{B}_{k-1}), & \text{if } \tilde{B}_k \text{ is an independent-direction factor,} \\ \tilde{B}_k \bar{B}_{k-1} \tilde{B}_k, & \text{if } \tilde{B}_k \text{ is a common-}\xi \text{ factor.} \end{cases}$$

In the first case, the direction vector in $\tilde{B}_k$ appears nowhere in B_{k-1} and is independent of the other directions and of ξ , so conditional independence and Fubini give the first formula. In the second case, $\tilde{B}_k$ is fixed conditional on ξ . Each independent-direction factor preserves the current Schatten exponent at cost 4ρ by Eq. (9).

Now assume $s \geq 1$, and let $G_1(\xi), \dots, G_s(\xi)$ denote the common- ξ factors in their order in the word. For the ℓ -th such factor, we have

$$\frac{1}{s/\ell} = \frac{1}{s/(\ell-1)} + \frac{2}{2s},$$

and Schatten Hölder gives

$$\|G_\ell(\xi) U G_\ell(\xi)\|_{S_{s/\ell}} \leq \|G_\ell(\xi)\|_{S_{2s}}^2 \|U\|_{S_{s/(\ell-1)}},$$

where $s/0 := \infty$. Starting with $\|\overline{B}_0\|_{S_\infty} = 1$ and applying these bounds recursively, we end in S_1 . Since $\mathbb{E}_g[\|\mathcal{B}\|_F^2] = \text{tr}[\overline{B}_\tau]$, we obtain, conditionally on ξ and for $s \geq 1$,

$$\mathbb{E}_g[\|\mathcal{B}\|_F^2] \leq (4\rho)^j \prod_{\ell=1}^s \|G_\ell(\xi)\|_{S_{2s}}^2. \quad (10)$$

The case $s = 0$ is treated separately in Step 3 below.

Step 3: Gaussian moments under the tilt. Suppose first that $s \geq 1$. Scalar Hölder under μ_t , Hargé's theorem applied separately to each even convex trace moment, and the Gaussian trace-moment inequality for a Gaussian matrix series with variance proxy νI , $\mathbb{E}_\gamma[\text{tr}[|G|^{2s}]] \leq N(2s - 1)!!\nu^s \leq N(2s\nu)^s$ (a matrix Khintchine bound, Tropp [Tro15]; the proof of Lemma 4.15 derives it by Gaussian integration by parts), give

$$\begin{aligned} \mathbb{E}_{\mu_t}[\prod_{\ell=1}^s \|G_\ell(\xi)\|_{S_{2s}}^2] &\leq \prod_{\ell=1}^s (\mathbb{E}_{\mu_t}[\text{tr}[|G_\ell(\xi)|^{2s}]]^{1/s}) \\ &\leq \prod_{\ell=1}^s (\mathbb{E}_\gamma[\text{tr}[|G_\ell(\xi)|^{2s}]]^{1/s}) \\ &\leq N(8s\rho)^s. \end{aligned}$$

The first step is scalar Hölder, the second is Hargé's theorem, and the last uses the variance bound $\nu \leq 4\rho$ from Step 1 and $(2s - 1)!! \leq (2s)^s$. Here $\xi \mapsto \Phi(\mathcal{A}_t(\xi))$ is an even proper convex extended-valued function, so μ_t is centered and Hargé's convex-domination theorem [Har04] applies (or approximate the potential monotonically by finite even convex potentials, as for Brascamp–Lieb in the proof of Lemma 4.13). For a linear symmetric matrix series G and $s \geq 1$, $\xi \mapsto \text{tr}[|G(\xi)|^{2s}] = \|G(\xi)\|_{S_{2s}}^{2s}$ is even and convex. Thus every hypothesis used in the preceding display is satisfied.

For $s \geq 1$, combining this bound with Eq. (10) gives

$$\mathbb{E}_{\mu_t}[\mathbb{E}_g[\|\mathcal{B}\|_F^2]] \leq N(4\rho)^j (8s\rho)^s = N(4\rho)^\tau (2s)^s.$$

The first step combines the two estimates, and the second uses $j + s = \tau$. If $s = 0$, all factors are independent-direction factors. Applying the τ maps in S_∞ and then using $\text{tr}[U] \leq N\|U\|$ gives

$$\mathbb{E}_g[\|\mathcal{B}\|_F^2] \leq N(4\rho)^\tau.$$

This uses the bound 4ρ for each map and agrees with the preceding word bound under the convention $0^0 = 1$. There are at most $(\tau)_j \leq \tau^j$ words in the derivative. Cauchy–Schwarz for their sum therefore gives

$$\begin{aligned} \mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})\|_{\text{HS}}^2] &\leq (\tau)_j^2 N(4\rho)^\tau (2s)^s \\ &\leq N(4\rho)^\tau \tau^{2j} (2s)^s \\ &\leq N(4\tau^2\rho)^\tau. \end{aligned}$$

The first step combines the word bound with Cauchy–Schwarz. The second uses $(\tau)_j \leq \tau^j$. The last uses $2s \leq 2\tau \leq \tau^2$ and $j + s = \tau$. This proves Eq. (8).

Step 4: the linear field and the resolvent moments. For the linear field $\mathcal{A}_{\text{off}}$, Hargé's theorem and Step 1 give

$$\mathbb{E}_{\mu_t}[\|\mathcal{A}_{\text{off}}\|_F^2] \leq \mathbb{E}_\gamma[\|\mathcal{A}_{\text{off}}\|_F^2] = \sum_{i=1}^m \text{tr}[\widehat{A}_{i,\text{off}}^2] \leq 4N\rho.$$

The first step is Hargé's theorem, the second is the Gaussian second-moment identity, and the last is the variance bound from Step 1. Since $\partial_i \mathcal{A}_{\text{off}} = \hat{A}_{i,\text{off}}$ is constant, Lemma 4.13 gives

$$\sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i \mathcal{A}_{\text{off}}) \leq c_2 \sum_{i=1}^m \|\hat{A}_{i,\text{off}}\|_F^2 \leq 4c_2 N \rho.$$

The first step uses that all derivatives of the constant fields vanish, and the second uses Step 1. Applying Lemma 4.13 once more gives

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) &\leq c_2 \mathbb{E}_{\mu_t}[\|\mathcal{A}_{\text{off}}\|_F^2] + c_2 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i \mathcal{A}_{\text{off}}) \\ &\leq 4c_2(1 + c_2)N\rho = c_3 N\rho. \end{aligned}$$

The first step is Lemma 4.13, the second uses the two preceding bounds, and the equality is the definition of c_3 . For the degree- τ field $X^{\tau-1} \mathcal{A}_{\text{off}}$, the coefficients in the iteration from Step 2 satisfy

$$\sum_{j=0}^{\tau} c_2^{j+1} \leq c_2(1 + c_2)^{\tau} \leq [c_2(1 + c_2)]^{\tau} = (c_3/4)^{\tau}.$$

The first step uses the binomial expansion of $(1 + c_2)^{\tau}$ and $\binom{\tau}{j} \geq 1$ for $0 \leq j \leq \tau$. The second uses $c_2 \geq 1$ and $\tau \geq 2$. The last step is the definition of c_3 . Consequently,

$$\begin{aligned} \mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) &\leq \sum_{j=0}^{\tau} c_2^{j+1} \mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1} \mathcal{A}_{\text{off}})\|_{\text{HS}}^2] \\ &\leq N(4\tau^2 \rho)^{\tau} \sum_{j=0}^{\tau} c_2^{j+1} \\ &\leq N(c_3 \tau^2 \rho)^{\tau}. \end{aligned}$$

The first step is the iteration from Step 2, the second uses Eq. (8), and the last uses the preceding bound on the sum. Thus

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1} \mathcal{A}_{\text{off}}) \leq N(c_3 \tau^2 \rho)^{\tau}, \quad \mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) \leq c_3 N\rho. \quad (11)$$

Step 5: the interpolation Hessian. The logarithmic identity is

$$D^2 \text{tr}[\phi_0(X)][\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}] = \sum_{\sigma \in \{-1, 1\}} \text{vec}(\mathcal{A}_{\text{off}})^{\top} (Y_{\sigma}^{-1} \otimes Y_{\sigma}^{-1}) \text{vec}(\mathcal{A}_{\text{off}}).$$

Therefore

$$\begin{aligned} \mathbb{E}_{\mu_t}[D^2 \Phi(X)[\mathcal{A}_{\text{off}}, \mathcal{A}_{\text{off}}]] &\leq \kappa_0 \sum_{\sigma \in \{-1, 1\}} \mathcal{E}_{\sigma, \sigma}(\mathcal{A}_{\text{off}}) + 4\tau(\kappa_1 - \kappa_0) \sum_{\sigma \in \{-1, 1\}} \mathcal{E}_{\sigma, \sigma}(X^{\tau-1} \mathcal{A}_{\text{off}}) \\ &\leq 2c_3 \kappa_0 N\rho + 8\kappa_1 \tau N(c_3 \tau^2 \rho)^{\tau} \\ &\leq c_4 \kappa_0 N\rho + c_4 \kappa_1 \tau N(c_3 \tau^2 \rho)^{\tau}. \end{aligned} \quad (12)$$

The first step uses the logarithmic identity, part (i), and $\phi = \kappa_0 \phi_0 + (\kappa_1 - \kappa_0) \phi_{1, \tau}$ with $\kappa_1 \geq \kappa_0 > 0$. The second uses Eq. (11), $\kappa_1 - \kappa_0 \leq \kappa_1$, and the two values of σ , giving the factors 2 and $4 \cdot 2 = 8$. The last uses $c_4 = 2c_3 = 8c_2(1 + c_2) \geq 48 \geq 8$, since $c_2 \geq 2$. Since $X = \mathcal{A}_t(\xi)$, this is the last assertion of (ii). $\square$

4.8 Explicit numerical specialization

This subsection fixes numbers, not new estimates: most inequalities below are instances of the preceding lemmas with their constants tracked (Step 1 sharpens the cutoff and Step 4 re-derives the Gaussian moment bound directly), and the outcome is the radius $279\sqrt{m}\ell_{\text{asp}}$ with the exponent $3711824/10^9$.

Lemma 4.15 (Explicit hereditary small ball). *Let $1 \leq m \leq N$, and let $A_1, \dots, A_m \in \mathbb{R}^{N \times N}$ be symmetric matrices satisfying $\|A_i\| \leq 1$, and let $\xi_1, \dots, \xi_m$ be independent standard Gaussian random variables. Then*

$$\Pr\left[\left\|\sum_{i=1}^m \xi_i A_i\right\| \leq 279\sqrt{m} \max\{1, \log(N/m)\}\right] \geq \exp(-3711824m/10^9).$$

Proof. Step 1: a sharper edge cutoff. The general cutoff satisfies $0 < \kappa_3 < 1/20$. We repeat the curvature computation with the sharper cutoff below, which may exceed $1/20$, and retain convenient sufficient parameter thresholds for the numerical specialization. For any $0 < b < 1/2$, put

$$\hat{\kappa}_3 := \frac{b^2}{1+b}, \quad \hat{c}_0 := \frac{1-2b}{2}.$$

The proof of Lemma 4.6(i) extends to cutoff $w = \hat{\kappa}_3$ with coefficient $\hat{c}_0$ in place of $1/4$. We repeat the argument below, allowing this cutoff to exceed $1/20$. Splitting at b/τ , the near case uses $u^{2\tau-2} \geq 1 - 2b$. In the far case, if $\delta_{\text{gap}} \leq \hat{\kappa}_3/\tau$, then

$$x^{2\tau-1} \geq 1 - 2\hat{\kappa}_3, \quad \frac{\delta_{\text{gap}}}{b/\tau} \leq \frac{\hat{\kappa}_3}{b},$$

and

$$1 - 2\hat{\kappa}_3 - \frac{2\hat{\kappa}_3}{b} = 1 - 2b.$$

Combining the two one-sided estimates ($\sigma = \pm 1$) costs a factor of two. Thus the lower divided-difference estimate is valid with $\hat{\kappa}_3, \hat{c}_0$, while its upper constant remains 2. With spectral coefficient $\hat{c}_0$, the curvature calculation of Lemma 4.12 applies whenever $\hat{c}_0\kappa_1 > 2$ and $\kappa_0(\hat{c}_0\kappa_1 - 2) > 4$: use Young's inequality with parameter $\hat{c}_0\kappa_1/2 - 1$ to bound the 00 and mixed products in each slot by one half of the barrier Hessian quadratic form. For the numerical specialization, we retain the sufficient thresholds $\kappa_1 \geq A_*$ and $\kappa_1\kappa_0 \geq D_*$, with

$$A_* := 8/\hat{c}_0, \quad D_* := 32/\hat{c}_0, \quad \tau^2\rho \leq \hat{\kappa}_3^2/4.$$

Indeed, these thresholds give $\hat{c}_0\kappa_1 \geq 8$ and

$$\kappa_0(\hat{c}_0\kappa_1 - 2) = (\hat{c}_0\kappa_1\kappa_0)(1 - 2/(\hat{c}_0\kappa_1)) \geq 24 > 4,$$

where the first step follows by factoring, the second step follows from $\hat{c}_0\kappa_1\kappa_0 \geq 32$ and $\hat{c}_0\kappa_1 \geq 8$, and the third step is numerical.

Step 2: the parameters and the curvature checks. Take

$$b := \frac{10021}{25000}, \quad D := 323, \quad A := 4700000, \quad c_{\text{cut}} = \frac{256}{17}, \quad c_{\text{rad}} := 279, \quad \tau := \lceil c_{\text{cut}}\ell_{\text{asp}} \rceil.$$

Then $\hat{c}_0 = 2479/25000$, $\hat{\kappa}_3 = 100420441/875525000$, and $\tau/\ell_{\text{asp}} \leq 16$. Indeed, $\tau = 16$ on the first cell $[1, 17/16]$, and on the cell of any $\tau \geq 17$ the ratio τ/ℓ_{asp} is at most $c_{\text{cut}}\tau/(\tau - 1)$, whose largest value is

$$\frac{256}{17} \frac{17}{16} = 16.$$

The numerical curvature checks are

$$\frac{16^2}{279^2} < \frac{\hat{\kappa}_3^2}{4}, \quad \rho \leq \frac{1}{279^2} < \frac{1}{6400}, \quad A > A_*, \quad D > D_*.$$

Step 3: the explicit Poincaré constants. We now use the normalized interpolation framework of Lemma 4.5 and track every analytic constant in the weighted Poincaré and resolvent estimates of Lemma 4.13. The curvature lower bound is $1/2$ (the computation of Lemma 4.12 with spectral coefficient $\hat{c}_0$, cutoff $\hat{\kappa}_3$, and the sufficient thresholds A_* , D_*), so the weighted Poincaré coefficient is at most 2. We use this uniform upper bound throughout the numerical calculation. Under $\rho \leq 1/6400$, the one-slot calculation gives

$$1 - L_\sigma^{-1} \leq 2\rho L_\sigma, \quad L_\sigma \leq \frac{2}{1 + \sqrt{1 - 8\rho}} < 1.001.$$

In the two-slot calculation, the one-slot estimate and the displayed derivative of $\mathcal{R}$ from the proof of Lemma 4.13 give

$$\sum_{i=1}^m \mathbb{E}_{\mu_t}[(\partial_i \mathcal{R}) \mathcal{Y}(\partial_i \mathcal{R})] \preceq 8.008\rho I \preceq 8.01\rho I;$$

here each Kronecker term carries the factor $2 \cdot 2$ from the parallelogram bound and $Y_\sigma \preceq 2I$, and $\sum_{i=1}^m \hat{A}_i(t) S_\sigma \hat{A}_i(t) \preceq L_\sigma \rho I$ with $L_\sigma < 1.001$. The Poincaré coefficient 2 therefore gives

$$0 \preceq T - S^{-1} \preceq 16.02\rho I.$$

Since $\|T - I\| \leq \rho$,

$$\|S^{-1} - I\| \leq 17.02\rho, \quad \|S - I\| \leq \frac{17.02\rho}{1 - 17.02\rho} < 17.07\rho.$$

Consequently,

$$0 \preceq \mathbb{E}_{\mu_t}[(\mathcal{Y} - I) \mathcal{Y}^{-1}(\mathcal{Y} - I)] \preceq 18.07\rho I, \quad S^{-1} \preceq 1.003I.$$

Using $\|a + b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$ in the projected mean estimate gives

$$u^\top S^{-1} u \leq 2.006 \mathbb{E}_{\mu_t}[\|U\|_F^2] + 36.25\rho \mathcal{E}_{\sigma_1, \sigma_2}(U).$$

Hence

$$(1 - 36.25\rho) \mathcal{E}_{\sigma_1, \sigma_2}(U) \leq 2.006 \mathbb{E}_{\mu_t}[\|U\|_F^2] + 2 \sum_{i=1}^m \mathcal{E}_{\sigma_1, \sigma_2}(\partial_i U).$$

At $\rho \leq 1/6400$, division by $1 - 36.25\rho$ makes both coefficients smaller than

$$c_{\Pi} := 2.02.$$

Step 4: the explicit moment constants. We next make the Gaussian moment constant explicit. If $Z := \sum_{i=1}^m h_i R_i$ and $\sum_{i=1}^m R_i^2 \preceq \nu I$, Gaussian integration by parts gives

$$\mathbb{E}[\text{tr}[Z^{2q}]] = \sum_{i=1}^m \sum_{k=0}^{2q-2} \mathbb{E}[\text{tr}[R_i Z^k R_i Z^{2q-2-k}]].$$

Diagonalizing Z and applying scalar weighted AM–GM to each pair of eigenvalues bounds the sum over k by $(2q-1)\operatorname{tr}[R_i^2|Z|^{2q-2}]$; summing over i and using $\sum_{i=1}^m R_i^2 \preceq \nu I$ gives $\mathbb{E}[\operatorname{tr}[Z^{2q}]] \leq (2q-1)\nu\mathbb{E}[\operatorname{tr}[|Z|^{2q-2}]]$. Induction from $\mathbb{E}[\operatorname{tr}[Z^0]] = N$ gives

$$\mathbb{E}[\operatorname{tr}[|Z|^{2q}]] \leq N(2q-1)!!\nu^q \leq N(2q\nu)^q.$$

A word in the j th derivative of $X^{\tau-1}\mathcal{A}_{\text{off}}$ contains exactly j independent-direction factors and $s := \tau - j$ common- ξ factors. There is only one $\mathcal{A}_{\text{off}}$ factor in the word. Since the variance proxy of X is at most ρI and that of $\mathcal{A}_{\text{off}}$ is at most $4\rho I$, the completely positive word scan (with Hargé’s theorem passing from μ_t to γ , as in the proof of Lemma 4.14) and the preceding moment bound give

$$\mathbb{E}_{\mu_t, h}[\|W\|_F^2] \leq 4N\rho^j(2s\rho)^s.$$

There are at most τ^j words, so Cauchy–Schwarz gives

$$\mathbb{E}_{\mu_t}[\|\nabla^j(X^{\tau-1}\mathcal{A}_{\text{off}})\|_{\text{HS}}^2] \leq 4N\rho^j(2s\rho)^s\tau^{2j} \leq N(2\tau^2\rho)^\tau.$$

For the last step, after cancelling $N\rho^\tau$ it is enough to check $4 \leq 2^j(\tau^2/s)^s$. For $j = 0$ this is $4 \leq \tau^\tau$; for $j = 1$ it follows from $\tau \geq 2$; and for $j \geq 2$ it follows from $2^j \geq 4$.

For $\tau \geq 2$,

$$\sum_{j=0}^{\tau} c_{\Pi}^{j+1} < 4^\tau.$$

Indeed, the inequality is direct at $\tau = 2$, and the induction step uses $c_{\Pi}^{\tau+2} < 3 \cdot 4^\tau$. Iterating the defective Poincaré estimate therefore yields

$$\mathcal{E}_{\sigma_1, \sigma_2}(X^{\tau-1}\mathcal{A}_{\text{off}}) \leq N(8\tau^2\rho)^\tau.$$

For the linear field,

$$\mathcal{E}_{\sigma_1, \sigma_2}(\mathcal{A}_{\text{off}}) \leq 4c_{\Pi}(1 + c_{\Pi})N\rho < 25N\rho.$$

The two signs in the logarithmic Hessian contribute less than $50\kappa_0 N\rho$. Lemma 4.14(i) and the two signs contribute at most

$$8\kappa_1\tau N(8\tau^2\rho)^\tau \leq 50\kappa_1\tau N(50\tau^2\rho)^\tau.$$

Consequently the interpolation estimate holds with the single explicit constant

$$C_I := 50.$$

Step 5: the endpoint reduction and the ledger. Let $r := N/m$, $\rho := (279^2\ell_{\text{asp}}^2)^{-1}$, and choose the potential parameters

$$\kappa_{1,\text{pot}} := A \max\{1, r\rho\}, \quad \kappa_{0,\text{pot}} := D/\kappa_{1,\text{pot}}.$$

The interpolation Hessian estimate (12), with the constants tracked above, gives uniformly for $t \in [0, 1]$

$$a''(t) \geq -50\kappa_{0,\text{pot}}N\rho - 50\kappa_{1,\text{pot}}\tau N(50\tau^2\rho)^\tau.$$

We next bound the diagonal endpoint. Write $\mathcal{A}_0 = \operatorname{diag}(\zeta_1, \dots, \zeta_N)$, where ζ is a centered Gaussian vector with covariance Σ . The normalization gives $\operatorname{tr}[\Sigma] \leq N\rho$ and $\Sigma_{jj} \leq \rho$ for every j . Since $\kappa_{1,\text{pot}} \geq \kappa_{0,\text{pot}}$, for $|x| \leq 1/2$ we have

$$\phi(x) \leq 2\kappa_{0,\text{pot}}x^2 + \frac{2\kappa_{1,\text{pot}}}{\tau}4^{-\tau}.$$

Tilting the law of ζ by $e^{-2\kappa_{0,\text{pot}}\|\zeta\|_2^2}$ produces a centered Gaussian vector with covariance

$$\tilde{\Sigma} = \Sigma^{1/2}(I + 4\kappa_{0,\text{pot}}\Sigma)^{-1}\Sigma^{1/2} \preceq \Sigma,$$

and

$$\mathbb{E}[e^{-2\kappa_{0,\text{pot}}\|\zeta\|_2^2}] = \det(I + 4\kappa_{0,\text{pot}}\Sigma)^{-1/2} \geq e^{-2\kappa_{0,\text{pot}}N\rho}.$$

The Gaussian correlation inequality for the symmetric strips $\{|\zeta_j| \leq 1/2\}$, in the form proved by Šidák [Š67, Theorem 1] (the general inequality is Royen's [Roy14]), and the scalar Gaussian tail bound give

$$\Pr_{\tilde{\Sigma}}[|\zeta_j| \leq 1/2 \text{ for every } j] \geq (1 - 2e^{-1/(8\rho)})^N \geq e^{-4Ne^{-1/(8\rho)}};$$

here $2e^{-1/(8\rho)} \leq 1/2$ follows from $\rho \leq 1/6400$. Consequently,

$$a(0) \geq -2\kappa_{0,\text{pot}}N\rho - \frac{2\kappa_{1,\text{pot}}N}{\tau}4^{-\tau} - 4Ne^{-1/(8\rho)}.$$

By Lemma 4.5(iv), $a'(0) = 0$. Taylor's formula therefore gives

$$a(1) = a(0) + \int_0^1 (1-t)a''(t) dt.$$

Using $r = N/m \leq e^{\ell_{\text{asp}}}$ and $\max\{1, r\rho\}r \leq e^{\ell_{\text{asp}}} + e^{2\ell_{\text{asp}}}\rho$, and tracking the factor $\int_0^1 (1-t) dt = 1/2$, we obtain $a(1) \geq -mE(\ell_{\text{asp}})$, where

$$\begin{aligned} E(\ell_{\text{asp}}) &\leq \frac{27D}{A} \\ &\quad + 25A\tau(e^{\ell_{\text{asp}}} + \frac{e^{2\ell_{\text{asp}}}}{279^2\ell_{\text{asp}}^2})(\frac{50\tau^2}{279^2\ell_{\text{asp}}^2})^\tau \\ &\quad + \frac{2A}{\tau}4^{-\tau}(e^{\ell_{\text{asp}}} + \frac{e^{2\ell_{\text{asp}}}}{279^2\ell_{\text{asp}}^2}) + 4\exp(\ell_{\text{asp}} - 279^2\ell_{\text{asp}}^2/8). \end{aligned} \quad (13)$$

The first line combines the low-order diagonal loss $2D/A$ with the integrated interpolation loss $25D/A$.

Step 6: maximization over ℓ_{asp} . Put

$$P(\ell_{\text{asp}}) := e^{\ell_{\text{asp}}} + \frac{e^{2\ell_{\text{asp}}}}{279^2\ell_{\text{asp}}^2},$$

write $E_{\text{low}} := 27D/A$ for the first line of Eq. (13), and denote its second and third terms by $I(\tau, \ell_{\text{asp}})$ and $J(\tau, \ell_{\text{asp}})$. Since $(\log P)' \leq 2$,

$$\partial_{\ell_{\text{asp}}} \log I(\tau, \ell_{\text{asp}}) \leq 2 - 2\tau/\ell_{\text{asp}} < 0,$$

whereas $J(\tau, \ell_{\text{asp}})$ increases on $\ell_{\text{asp}} \geq 1$. Hence the maximum of I on a cell of τ is at its left endpoint, and the maximum of J is at its right endpoint.

The first ratio of consecutive cell maxima of I is

$$\frac{I(17, 17/16)}{I(16, 1)} < 0.187.$$

For every later transition, with $\tau \geq 17$, the ratio is at most

$$\frac{\tau+1}{\tau} e^{2/c_{\text{cut}}} \frac{50c_{\text{cut}}^2}{279^2} \left(\left(1 + \frac{1}{\tau}\right)^{\tau+1} \left(1 - \frac{1}{\tau}\right)^{\tau} \right)^2 \leq \frac{18}{17} e^{2/c_{\text{cut}}} \frac{50c_{\text{cut}}^2}{279^2} < 0.177.$$

Here the parenthesized factor is at most one because $\log(1-u) \leq -u$ and $\log(1+u) \leq u$ imply $(1-\tau^{-2})^{\tau} \leq (1+\tau^{-1})^{-1}$. Thus the global interpolation maximum is $I(16,1)$. The ratio of consecutive cell maxima of J is at most

$$\frac{e^{2/c_{\text{cut}}}}{4} < 0.286,$$

so the global diagonal maximum is the right endpoint $J(16, 17/16)$.

Step 7: rational evaluation. No numerical program is needed for the remaining evaluation. For rational $x \in [0, 17/8]$ and $K_{\text{Taylor}} := 30$, Taylor's formula gives the rational enclosure

$$\sum_{k=0}^{K_{\text{Taylor}}} \frac{x^k}{k!} < e^x < \sum_{k=0}^{K_{\text{Taylor}}} \frac{x^k}{k!} + \frac{x^{K_{\text{Taylor}}+1}}{(K_{\text{Taylor}}+1)!} \left(1 - \frac{x}{K_{\text{Taylor}}+2}\right)^{-1}.$$

At $x = 17/128, 1, 2, 17/16, 17/8$, this gives

$$\begin{aligned} e^{17/128} &< 1.142035847, & e &< 2.718281829, & e^2 &< 7.389056099, \\ e^{17/16} &< 2.893595945, & e^{17/8} &< 8.372897489. \end{aligned}$$

Substitution using exact rational arithmetic yields

$$10^9 E_{\text{low}} < 1855532.0, \quad 10^9 I(16, 1) < 1460469.6, \quad 10^9 J(16, 17/16) < 395822.3.$$

Finally, $279^2/8 - 1 > 25$, while $e > 2.7$ and $2.7^{25} > 4 \cdot 10^{10}$, so

$$10^9 \cdot 4 \exp(1 - 279^2/8) < 0.1.$$

The four bounds are strict and add up to exactly 3711824, so

$$E(\ell_{\text{asp}}) < \frac{3711824}{10^9}$$

for every $\ell_{\text{asp}} \geq 1$. The Taylor estimate of Step 5 then gives $a(1) > -3711824m/10^9$, and Lemma 4.5(v) yields

$$\Pr[\|\mathcal{A}(\xi)\| < 1] \geq a_1 = e^{a(1)} > \exp(-3711824m/10^9).$$

Undoing the normalization gives the asserted radius. $\square$

Remark 4.16 (Reading the ledger). The bound $E(\ell_{\text{asp}}) < 3711824/10^9$ is a maximum over the cells of τ , not a sum over them: on each cell the interpolation term $I(\tau, \ell_{\text{asp}})$ is largest at the left endpoint and the diagonal term $J(\tau, \ell_{\text{asp}})$ at the right endpoint, and the cell-to-cell ratios 0.187 and 0.286 make the first cell $[1, 17/16]$ the binding one. On that cell the three visible terms $27D/A$, $I(16, 1)$, and $J(16, 17/16)$ are 0.50, 0.39, and 0.11 of the total, and the strip term is below 10^{-10} . The rows of Table 2 behave alike (Table 3 in the next subsection records the components): in every row the interpolation term is matched against $27D/A$, to within one percent for $20 \leq \tau \leq 43$, while the diagonal term falls from eleven percent of E_0 in the first row to below 0.2 percent from the third row on. Two margins are thin by construction rather than by accident. In every row c_{rad} was chosen as the least integer for which the curvature check $(\tau_0/(c_{\text{rad}}\ell_0))^2 < \hat{\kappa}_3^2/4$ holds, so the margins Δ_{curv} of the last column lie between $1.18 \cdot 10^{-7}$ and $2.1 \cdot 10^{-6}$, and D exceeds its threshold $32/\hat{c}_0$ by less than 0.2 percent. Both checks are exact rational comparisons, so the thin margins are not near-failures.

Table 3: The first-cell endpoint calculation, scaled by 10^9 .

τ	$10^9(27D/A)$	10^9I_0	10^9J_0	10^9E_0
16	1855532	1460470	395823	3711824
17–19	1759822	1639172	118589	3517582
20–26	1187523	1184870	3879	2376271
27–30	1491968	1488990	1	2980959
31–43	1368735	1361304	1	2730038
$\tau \geq 44$	26172	25543	1	51714

4.9 Proof of the six-regime lemma

One parameter set is not optimal across aspect ratios: the six rows of Table 2 suffice for the constant 156000 of Section 5.

Proof of Lemma 4.1. The ledger of Lemma 4.15 allows the parameters to change when the moment order changes. Assign $(c_{\text{rad}}, b, D, A)$ from Table 2 according to $\tau(\ell_{\text{asp}}) = \lceil c_{\text{cut}}\ell_{\text{asp}} \rceil$; the cell of τ is $\ell_{\text{asp}} \in ((\tau - 1)/c_{\text{cut}}, \tau/c_{\text{cut}}]$, so $c_{\text{rad}}(\ell_{\text{asp}})$ is the step function of the lemma. Every terminating decimal in the table denotes the corresponding exact rational number. The endpoint exponent is also recorded component by component. Every entry in Table 3 is an outward integer upper bound of its own quantity, so the first three columns of a row may add up to slightly more than the last (by at most 2).

Step 1: the first-cell checks. For one row of the table, let τ_0 be the first moment order in that row and put

$$\ell_0 = \max\{1, (\tau_0 - 1)/c_{\text{cut}}\}, \quad r_0 = \tau_0/c_{\text{cut}}.$$

For the row's fixed parameters define

$$P_{c_{\text{rad}}}(x) = e^x + \frac{e^{2x}}{c_{\text{rad}}^2 x^2},$$

$$I_0 = 25A\tau_0 P_{c_{\text{rad}}}(\ell_0) \left(\frac{50\tau_0^2}{c_{\text{rad}}^2 \ell_0^2} \right)^{\tau_0}, \quad J_0 = \frac{2A}{\tau_0} 4^{-\tau_0} P_{c_{\text{rad}}}(r_0),$$

$$E_0 = \frac{27D}{A} + I_0 + J_0 + 4 \exp(\ell_0 - c_{\text{rad}}^2 \ell_0^2/8).$$

With $\widehat{\kappa}_3 = b^2/(1+b)$ and $\widehat{c}_0 = (1-2b)/2$ as in the proof of Lemma 4.15, direct rational comparison gives

$$A > \frac{8}{\widehat{c}_0}, \quad D > \frac{32}{\widehat{c}_0}, \quad \left(\frac{\tau_0}{c_{\text{rad}} \ell_0} \right)^2 < \frac{\widehat{\kappa}_3^2}{4}, \quad \frac{1}{c_{\text{rad}}^2 \ell_0^2} < \frac{1}{6400}.$$

Here

$$\Delta_{\text{curv}} = \frac{\widehat{\kappa}_3^2}{4} - \left(\frac{\tau_0}{c_{\text{rad}} \ell_0} \right)^2.$$

Its lower bounds are the last column of Table 2. The smallest value of $10^6(1/6400 - 1/(c_{\text{rad}}^2 \ell_0^2))$ is greater than 143.4. On a fixed cell τ/ℓ_{asp} decreases with ℓ_{asp} , and at successive left limits its supremum is $c_{\text{cut}}\tau/(\tau - 1)$, which decreases with τ . Likewise, $1/(c_{\text{rad}}^2 \ell_{\text{asp}}^2)$ decreases with ℓ_{asp} . Thus these first-cell checks imply the two inequalities throughout the row.

Step 2: the first cell controls the row. On a fixed- τ cell,

$$\frac{d}{d\ell_{\text{asp}}} \log I(\tau, \ell_{\text{asp}}) \leq 2 - \frac{2\tau}{\ell_{\text{asp}}} < 0.$$

The diagonal term increases because $P_{c_{\text{rad}}}$ increases, while the strip term decreases because

$$\frac{d}{d\ell_{\text{asp}}} (\ell_{\text{asp}} - c_{\text{rad}}^2 \ell_{\text{asp}}^2 / 8) = 1 - c_{\text{rad}}^2 \ell_{\text{asp}} / 4 < 0.$$

From one full cell to the next, the interpolation maximum is multiplied by at most

$$\frac{\tau + 1}{\tau} e^{2/c_{\text{cut}}} \frac{50c_{\text{cut}}^2}{c_{\text{rad}}^2},$$

and the diagonal maximum is multiplied by at most $e^{2/c_{\text{cut}}}/4$. The first quantity is below 0.387 in every row and decreases with τ ; the second is below 0.286. This uses the elementary monotonicity of $(1 + 1/\tau)^{\tau+1}$. Thus the normalized negative exponent throughout the row is at most E_0 .

Step 3: rational evaluation. Every numerical entry in the last two columns can be checked using rational arithmetic. For $0 \leq x \leq 6$ and $M = 40$, put

$$S_M(x) = \sum_{j=0}^M \frac{x^j}{j!}.$$

The geometric bound on the positive Taylor tail gives

$$S_M(x) < e^x < S_M(x) + \frac{x^{M+1}}{(M+1)!} \frac{1}{1 - x/(M+2)}.$$

Substitution of the exact table rationals gives Table 3 and the displayed bounds for $10^9 E_0$. For the strip term, exact rational comparison gives $c_{\text{rad}}^2 \ell_0^2 / 8 - \ell_0 > 1000$ in every row. Since $e > 2$ and $2^{10} > 10^3$, the strip term is less than 10^{-100} . Hence

$$E_0 < \frac{3711824}{10^9} < 0.003712.$$

Repeating the endpoint and Taylor calculation from the proof of Lemma 4.15 with the parameters of the current row gives $a(1) \geq -mE_0$. Hence Lemma 4.5(v) gives $\Pr[\|\mathcal{A}(\xi)\| < 1] \geq e^{-mE_0}$. Undoing the normalization, with $r_{\text{sb}} = c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$, proves the lemma. $\square$

5 Proof of the discrepancy bound

This section proves Theorem 5.1 from the six-regime small-ball estimate, Lemma 4.1, by repeated partial coloring; the two tools the proof consumes, the refreshed projection step (Lemma 5.4) and the separated-sum estimate (Lemma 5.5), are proved in the two subsections that follow. The argument is an existence proof in exact arithmetic, not an algorithm; the implementations and their running times are the subject of Sections 6–9.

5.1 Matrix Spencer discrepancy

Theorem 5.1 (Matrix Spencer discrepancy bound; formal version of Theorem 1.1). *For all symmetric matrices $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ with $\|A_i\| \leq 1$, there is a vector $\varepsilon \in \{-1, 1\}^n$ satisfying*

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| \leq 156000\sqrt{n}.$$

Proof. Step 1: one phase. Start with the zero fractional coloring. At each phase, let S be the active set, $m := |S|$, and $y \in (-1, 1)^S$ the restriction of the current fractional coloring to S . Put

$$u := \log(n/m), \quad \ell_{\text{asp}} := \max\{1, u\},$$

and take $c_{\text{rad}}(\ell_{\text{asp}})$ from Table 2. The set

$$K := \{x \in \mathbb{R}^S : \left\| \sum_{i \in S} x_i A_i \right\| \leq c_{\text{rad}}(\ell_{\text{asp}}) \sqrt{m} \ell_{\text{asp}}\}$$

is closed, convex, and symmetric, and Lemma 4.1, applied to the m contractions A_i , $i \in S$, in dimension $N = n$, gives $\gamma_m(K) \geq \exp(-0.003712m)$. Since $0.003712 < 0.00372$, the set K satisfies the hypothesis of Lemma 5.4 as long as $m \geq N_0$. A trial of that lemma is successful when at least δm coordinates of $y + 2x/\varepsilon_{\text{box}}$ become signs; on a successful trial the lemma gives $x \in K$, and the phase updates y to $y + 2x/\varepsilon_{\text{box}}$ and removes from S every coordinate that has become a sign. If m' is the new active-set size, then $m' \leq (1 - \delta)m$, and hence the next value $u' := \log(n/m')$ satisfies

$$u' - u \geq \lambda, \quad \lambda := -\log(1 - \delta).$$

The operator norm of the signed sum grows in this phase by at most $(2/\varepsilon_{\text{box}})c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$, the bound on $\left\| \sum_{i \in S} (2x_i/\varepsilon_{\text{box}})A_i \right\|$, which, with $\sqrt{m} = e^{-u/2}\sqrt{n}$, is at most

$$\frac{2c_{\text{rad}}(\ell_{\text{asp}})}{\varepsilon_{\text{box}}} e^{-u/2} \ell_{\text{asp}} \sqrt{n}. \tag{14}$$

The phases stop when $m < N_0$.

Step 2: the phases starting at $u \geq 1$. Let $u_0 < u_1 < \dots$ be the values of u at the starts of the phases. Then $u_0 = 0$, and Step 1 gives $u_{j+1} - u_j \geq \lambda$, so it remains to sum the increments (14) over an arbitrary λ -separated sequence. Put

$$\Gamma(u) := c_{\text{rad}}(u) u e^{-u/2} \quad (u \geq 1),$$

the coefficient of $2\sqrt{n}/\varepsilon_{\text{box}}$ in (14) for $u \geq 1$, and for $a \geq 1$ and $k \geq 0$ put

$$M_k(a) := \sup_{a+k\lambda \leq u < a+(k+1)\lambda} \Gamma(u).$$

Lemma 5.5 gives

$$\sum_j \Gamma(u_j) \leq \sum_{k=0}^{\infty} M_k(a)$$

for every λ -separated sequence all of whose terms are at least a . The six values of c_{rad} apply on

$$[1, 17/16], \quad (17/16, 323/256], \quad (323/256, 221/128],$$

$$(221/128, 255/128], \quad (255/128, 731/256], \quad (731/256, \infty).$$

Since c_{rad} is constant on each of these intervals and $ue^{-u/2}$ increases on $[1, 2]$ and decreases on $[2, \infty)$, the supremum of Γ over a bin is attained or approached at a bin endpoint, at $u = 2$, or at one of the five regime endpoints contained in that bin. After the bin containing $731/256$, the function Γ is decreasing, and the remaining sum is evaluated in closed form from $\sum_{j \geq 0} \rho^j$ and $\sum_{j \geq 0} j\rho^j$, where $\rho := e^{-\lambda/2}$. The order-40 rational Taylor enclosures of the proof of Lemma 4.1, together with the logarithmic and square-root enclosures recorded in the proof of Lemma 5.4, which enclose $\lambda = -\log(1 - \delta)$ and $\rho = \sqrt{1 - \delta}$, give

$$\sum_{k=0}^{\infty} M_k(1) < 25020.2, \quad \sum_{k=0}^{\infty} M_k(35\lambda) < 25020.2.$$

Hence the phases starting at or above either threshold contribute less than

$$\frac{2}{\varepsilon_{\text{box}}} \cdot 25020.2 < 118159.2$$

times $\sqrt{n}$ to the operator norm.

Step 3: the phases starting at $u < 1$. The same enclosures give

$$34\lambda < 1 < 35\lambda < 1.0001.$$

Since $u_j \geq j\lambda$, a phase with $u_j < 1$ has $j \leq 34$, so at most 35 phases start below 1. On such a phase $\ell_{\text{asp}} = 1$ and $c_{\text{rad}}(\ell_{\text{asp}}) = 279$, and $e^{-u_j/2} \leq \rho^j$ with $\rho = e^{-\lambda/2} = \sqrt{1 - \delta}$, so these phases are dominated by a geometric series, and exact geometric summation gives

$$\frac{2 \cdot 279}{\varepsilon_{\text{box}}} \sum_{j=0}^{33} \rho^j < 35739.4, \quad \frac{2 \cdot 279}{\varepsilon_{\text{box}}} \sum_{j=0}^{34} \rho^j < 36550.1.$$

If at most 34 phases start below 1, they contribute less than 35739.4, and every later phase starts at or above 1, so Step 2 with $a = 1$ bounds their contribution by 118159.2; the total is less than 153898.6. If 35 phases start below 1, they contribute less than 36550.1, and the continuation starts at $u_{35} \geq 35\lambda$, so Step 2 with $a = 35\lambda$ bounds it by 118159.2; the total is less than

$$154709.3.$$

In either case, all projection phases together contribute less than $154709.3\sqrt{n}$.

Step 4: cleanup rounding. When fewer than N_0 coordinates remain, round independently with the current fractional coloring as the mean: for $i \in S$, choose $\varepsilon_i \in \{-1, 1\}$ independently with $\mathbb{E}[\varepsilon_i] = y_i$ and put $X_i := (\varepsilon_i - y_i)A_i$. The scalar Hoeffding lemma, applied to the eigenvalues of A_i by functional calculus, gives

$$\mathbb{E}[e^{\theta X_i}] \preceq e^{\theta^2 A_i^2/2}.$$

Since $\sum_{i \in S} A_i^2 \preceq mI$, the matrix Laplace-transform method of Tropp [Tro15, Chapter 3] yields

$$\Pr[\|\sum_{i \in S} X_i\| \geq t] \leq 2n \exp(-t^2/(2m)).$$

Taking $t := \sqrt{2m(\log(2n) + \log 2)}$ makes the right-hand side $1/2$, so with probability at least $1/2$ the rounding contributes at most $t = \sqrt{2m \log(4n)} \leq 6.5\sqrt{n}$. Indeed, if $n \geq N_0$, then $m < N_0$ and

$(N_0/n) \log(4n)$ decreases with n , so $t^2/n < 2(N_0/n) \log(4n) \leq 2 \log(4N_0) < 6.5^2$; if $n < N_0$, no projection phase runs, $m = n$, and the same endpoint bound applies. Each projection trial succeeds with probability greater than 0.12 and the rounding with probability at least 1/2, so a run in which every trial succeeds exists: failed projection or rounding trials are discarded and repeated. Hence

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| < (154709.3 + 6.5)\sqrt{n} < 156000\sqrt{n}. \quad \square$$

Remark 5.2 (Smaller ambient dimension). The same conclusion holds for symmetric $N \times N$ contractions with $N \leq n$ by embedding A_i as $A_i \oplus 0_{n-N}$.

5.2 Refreshed projection

Throughout this subsection, $G \sim \mathcal{N}(0, 1)$ is a standard normal variable, $\phi(t) := (2\pi)^{-1/2} \exp(-t^2/2)$ is its density, and $\bar{\Phi}(t) := \Pr[G \geq t]$ is its upper tail. The projection step rests on two facts about Gaussian measure: a central section of a symmetric convex set is at least as likely as the set, and a set of Gaussian measure p is within distance $\sqrt{2 \log(1/p)}$ of a Gaussian point on average. Part (i) is the section inequality of the Gaussian partial-coloring framework (Rothvoss [Rot17]; Dadush, Jiang, and Reis [DJR22]); both proofs are short and are included.

Fact 5.3 (Gaussian sections and Gaussian distances). *Let $H \subseteq \mathbb{R}^m$ be a linear subspace, and let γ_H and ξ_H be the standard Gaussian measure and a standard Gaussian vector on H .*

(i) *If $C \subseteq \mathbb{R}^m$ is symmetric and convex, then $\gamma_H(C \cap H) \geq \gamma_m(C)$.*

(ii) *If $C \subseteq H$ is measurable with $\gamma_H(C) = q > 0$, then $\mathbb{E}[d(\xi_H, C)] \leq \sqrt{2 \log(1/q)}$.*

Proof. For (i), write $\mathbb{R}^m = H \oplus H^\perp$, so that $\gamma_m = \gamma_H \otimes \gamma_{H^\perp}$, and for $w \in H^\perp$ let $s(w) := \gamma_H((C - w) \cap H)$ be the Gaussian measure of the section of C by the translate $w + H$. Fubini gives $\gamma_m(C) = \int_{H^\perp} s(w) d\gamma_{H^\perp}(w)$. The function s is even, because $C = -C$ and γ_H is symmetric, and it is log-concave by Prékopa's theorem, because it is the marginal over $v \in H$ of the log-concave function $(v, w) \mapsto \mathbb{1}[v + w \in C] e^{-\|v\|_2^2/2}$ on $H \times H^\perp$, up to the normalizing constant of γ_H . An even log-concave function is maximized at the origin, since $s(0) \geq s(w)^{1/2} s(-w)^{1/2} = s(w)$. Hence $\gamma_m(C) \leq s(0) = \gamma_H(C \cap H)$.

For (ii), let ν be γ_H conditioned on C , so that $d\nu = q^{-1} \mathbb{1}[C] d\gamma_H$ and the relative entropy of ν with respect to γ_H is $\int \log(q^{-1}) d\nu = \log(1/q)$. Talagrand's transportation inequality for the standard Gaussian measure bounds the quadratic Wasserstein distance by $W_2(\nu, \gamma_H) \leq \sqrt{2 \log(1/q)}$. In any coupling (ξ_H, Y) of γ_H and ν the point Y lies in C almost surely, so $d(\xi_H, C) \leq \|\xi_H - Y\|_2$; taking expectations, applying $\mathbb{E}[\|\xi_H - Y\|_2] \leq (\mathbb{E}[\|\xi_H - Y\|_2^2])^{1/2}$, and passing to the infimum over couplings gives $\mathbb{E}[d(\xi_H, C)] \leq W_2(\nu, \gamma_H) \leq \sqrt{2 \log(1/q)}$. $\square$

The projection box encodes the current fractional coloring. Comparing the distance from a Gaussian point to this box with its distances to coordinate sections forces many box constraints to be tight.

Lemma 5.4 (Refreshed projection). *Let $m \geq N_0 = 3 \cdot 10^8$, let $K \subseteq \mathbb{R}^m$ be a symmetric closed convex set with*

$$\gamma_m(K) \geq \exp(-0.00372m),$$

and put

$$\delta := 0.0281672, \quad \varepsilon_{\text{box}} := 0.4235.$$

Given $y \in (-1, 1)^m$, put

$$R_i := \frac{\varepsilon_{\text{box}}}{2}(1 - y_i), \quad L_i := \frac{\varepsilon_{\text{box}}}{2}(1 + y_i) \quad (i \in [m]), \quad Q_y := \prod_{i=1}^m [-L_i, R_i],$$

sample $\xi \sim \mathcal{N}(0, I_m)$, and let x be the Euclidean projection of ξ onto $K \cap Q_y$; this is one Gaussian projection trial. Then, with probability greater than 0.12, the point $x \in K$ satisfies

$$y + 2x/\varepsilon_{\text{box}} \in [-1, 1]^m$$

and at least δm coordinates of $y + 2x/\varepsilon_{\text{box}}$ are in $\{-1, 1\}$.

Proof. Step 1: reduction to tight box constraints. Since $y_i \in (-1, 1)$, the numbers L_i and R_i lie in $(0, \varepsilon_{\text{box}})$, so $Q_y \subseteq [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m$ and

$$d(\xi, Q_y) \geq d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m).$$

The projection x lies in $K \cap Q_y$, so $-L_i \leq x_i \leq R_i$ for every i , which is $y + 2x/\varepsilon_{\text{box}} \in [-1, 1]^m$; and the i th coordinate of $y + 2x/\varepsilon_{\text{box}}$ is a sign exactly when $x_i \in \{-L_i, R_i\}$, that is, when one of the two box constraints of coordinate i is tight at x . Since $L_i, R_i > 0$, at most one constraint per coordinate is tight, so the tight constraints are indexed by a set of coordinates. It remains to show that, with probability greater than 0.12, more than δm coordinates carry a tight constraint.

Step 2: the distance to the cube. Put

$$\psi(t) := \mathbb{E}[(|G| - t)_+^2] = 2((1 + t^2)\bar{\Phi}(t) - t\phi(t)) \quad (t > 0),$$

and let $h(t) := -t \log t - (1-t) \log(1-t)$ be the binary entropy. The function $z \mapsto d(z, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)$ is 1-Lipschitz, and since $d(z, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)^2 = \sum_i (|z_i| - \varepsilon_{\text{box}})_+^2$,

$$\mathbb{E}[d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)^2] = ma(\varepsilon_{\text{box}}).$$

The Gaussian Poincaré inequality bounds the variance of a 1-Lipschitz function of ξ by 1, so $(\mathbb{E}[d])^2 \geq \mathbb{E}[d^2] - 1$ and

$$\mathbb{E}[d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)] \geq \sqrt{ma(\varepsilon_{\text{box}}) - 1}.$$

Step 3: the distances to the coordinate sections. For $I \subseteq [m]$, put $H(I) := \{z \in \mathbb{R}^m : z_i = 0 \text{ for every } i \in I\}$, the coordinate subspace in which the coordinates of I are frozen, and write $\xi_{H(I)}$ for the orthogonal projection of ξ onto $H(I)$, a standard Gaussian vector on $H(I)$. Fact 5.3(i) with $C = K$ and $H = H(I)$ gives $\gamma_{H(I)}(K \cap H(I)) \geq \gamma_m(K) \geq \exp(-0.00372m)$, and Fact 5.3(ii) with $C = K \cap H(I) \subseteq H(I)$ then gives

$$\mathbb{E}[d(\xi_{H(I)}, K \cap H(I))] \leq \sqrt{2(0.00372)m}.$$

Because $K \cap H(I) \subseteq H(I)$, Pythagoras gives $d(\xi, K \cap H(I))^2 = d(\xi_{H(I)}, K \cap H(I))^2 + \sum_{i \in I} \xi_i^2$. The first term is the square of a 1-Lipschitz function of $\xi_{H(I)}$, so the Gaussian Poincaré inequality bounds its expectation by $(\mathbb{E}[d(\xi_{H(I)}, K \cap H(I))])^2 + 1 \leq 2(0.00372)m + 1$, and the second term has expectation $|I|$. Jensen's inequality $\mathbb{E}[d] \leq (\mathbb{E}[d^2])^{1/2}$ then gives, uniformly over $|I| \leq \delta m$,

$$\mathbb{E}[d(\xi, K \cap H(I))] \leq \sqrt{(\delta + 2(0.00372))m + 1}.$$

Step 4: the separation constant and its enclosures. Dividing the lower bound of Step 2 and the upper bound of Step 3 by $\sqrt{m}$, and reserving a further $\sqrt{2h(\delta)}$ for the union bound of Step 5, the normalized separation at $m = N_0$ is

$$r := \sqrt{\psi(\varepsilon_{\text{box}}) - 1/N_0} - \sqrt{\delta + 2(0.00372) + 1/N_0} - \sqrt{2h(\delta)}.$$

The first square-root term increases with m and the second decreases, so the same lower separation persists for every $m \geq N_0$. Direct rational enclosures give

$$\begin{aligned} 0.36472389700743 &< \phi(\varepsilon_{\text{box}}) < 0.36472389700744, \\ 0.33596524927926 &< \overline{\Phi}(\varepsilon_{\text{box}}) < 0.33596524927928, \\ \psi(0.4235) &> 0.4835216045, \quad 0.1283122818 < h(0.0281672) < 0.1283122819, \\ 10^6 r &> 77.2, \quad 10^6(r - 6 \cdot 10^{-5}) > 17.2. \end{aligned}$$

The displayed enclosures are self-contained. One may use

$$\overline{\Phi}(x) = \frac{1}{2} - \frac{1}{\sqrt{2\pi}} \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)2^k k!}$$

for $0 \leq x \leq 1/2$, truncating after an even and an odd term, and

$$\log z = 2 \sum_{k=0}^{\infty} \frac{v^{2k+1}}{2k+1}, \quad v = \frac{z-1}{z+1}.$$

For $\log \delta$, first write $\log \delta = \log(32\delta) - 5 \log 2$. The value of π is enclosed by

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239), \quad \arctan z = \sum_{k=0}^{\infty} \frac{(-1)^k z^{2k+1}}{2k+1}.$$

Alternating upper and lower truncations, together with the first omitted term, give the displayed Gaussian enclosures; the same alternating rule applied to e^{-x} encloses $\phi(\varepsilon_{\text{box}})$. All logarithmic series have squared ratio at most $1/9$. Twenty terms give errors below 10^{-20} , and every square-root enclosure is checked by exact rational squaring.

Step 5: the good event. Choose the cube and section deviations both equal to $3 \cdot 10^{-5}$; their sum is less than r . The cube event is $d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m) \leq \mathbb{E}[d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m)] - 3 \cdot 10^{-5} \sqrt{m}$, and the section event is that some I with $|I| \leq \delta m$ has $d(\xi, K \cap H(I)) \geq \mathbb{E}[d(\xi, K \cap H(I))] + (\sqrt{2h(\delta)} + 3 \cdot 10^{-5}) \sqrt{m}$. Gaussian concentration for 1-Lipschitz functions, applied once to the cube distance and once to each of the at most $\sum_{k=0}^{\lfloor \delta m \rfloor} \binom{m}{k} \leq \exp(h(\delta)m)$ section distances, shows that the two failure probabilities are at most

$$\exp(-m(3 \cdot 10^{-5})^2/2)$$

and

$$\exp(-m(3 \cdot 10^{-5} \sqrt{2h(\delta)} + (3 \cdot 10^{-5})^2/2)).$$

At $m = N_0$ their sum is at most $e^{-0.135} + e^{-4558} < 0.874$, and it decreases with m . Hence, with probability greater than 0.126 , neither event occurs, and then, by Steps 2 and 3 and the monotonicity of Step 4,

$$d(\xi, Q_y) \geq d(\xi, [-\varepsilon_{\text{box}}, \varepsilon_{\text{box}}]^m) > d(\xi, K \cap H(I)) \quad \text{for every } I \subseteq [m] \text{ with } |I| \leq \delta m,$$

the two sides being separated by at least $(r - 6 \cdot 10^{-5}) \sqrt{m} > 0$.

Step 6: on the good event more than δm constraints are tight. On the event of Step 5, Q_y is farther from ξ than every section $K \cap H(I)$ with $|I| \leq \delta m$. Suppose that the tight box constraints at the projection x have coordinate set I with $|I| \leq \delta m$. Relax Q_y by discarding every nontight box constraint. The point x remains the projection of ξ onto the relaxed set: otherwise a sufficiently

short segment from x toward a closer relaxed point would satisfy all discarded constraints, which hold with slack at x , and contradict the definition of x . Moreover, $K \cap H(I)$ lies in the relaxed set because the zero value in every coordinate of I satisfies its retained halfspace. Hence

$$d(\xi, K \cap Q_y) \leq d(\xi, K \cap H(I)),$$

contradicting $d(\xi, K \cap Q_y) \geq d(\xi, Q_y) > d(\xi, K \cap H(I))$. Thus more than δm constraints are tight, and by Step 1 the affine rescaling $y + 2x/\varepsilon_{\text{box}}$ turns their coordinates into signs. $\square$

5.3 Separated sums

The phase parameters have a minimum spacing, but their exact locations depend on the projection outcomes. Interval suprema give a deterministic bound on the accumulated discrepancy.

Lemma 5.5 (Separated sum against bin suprema). *Let $F : [a, \infty) \rightarrow [0, \infty)$, and let $a \leq x_0 < x_1 < \dots$ satisfy $x_{k+1} - x_k \geq \lambda > 0$. Then*

$$\sum_{k=0}^{\infty} F(x_k) \leq \sum_{j=0}^{\infty} \sup_{a+j\lambda \leq u < a+(j+1)\lambda} F(u).$$

Proof. Every x_k lies in exactly one of the half-open bins $[a + j\lambda, a + (j+1)\lambda)$, $j \geq 0$, and each bin contains at most one point x_k , because consecutive points are at least λ apart. Bounding each $F(x_k)$ by the supremum of F over its bin and summing over the bins proves the claim. $\square$

6 The $n^{2+\omega/2}$ algorithm

Theorem 6.1 counts arithmetic operations in the real-arithmetic model of Section 3, not bit operations: no bound on the word length is claimed for the algorithm of this section. Section 10 proves a bit-complexity bound for the cubic-time algorithm of Section 9, and Remark 10.22 there records why the diagonalization step used below is not covered.

6.1 Statement and algorithm

Theorem 6.1 (Dense arithmetic implementation; formal version of the $n^{2+\omega/2+o(1)}$ variant in Remark 1.3). *Let $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ be symmetric with $\|A_i\| \leq 1$, and let $p \in (0, 1/2)$. Algorithm 1 returns, with failure probability at most p , a signing satisfying the conclusion of Theorem 5.1, using*

$$n^{2+\omega/2+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model of Section 3; with the $n^{o(1)}$ carried by ω absorbed into the soft- O , this is $\tilde{O}(n^{2+\omega/2})$. The working accuracy is inverse polynomial, and its contribution to the running time is polylogarithmic.

Algorithm 1 is the implementation. The rest of the section proves Theorem 6.1: Section 6.2 fixes the smoothed body and its constants, Section 6.3 solves one projection by an inexact block-coordinate method inside a robust bisection on the multiplier, Section 6.4 implements the oracle that the solver queries, Section 6.5 snaps the computed point and sums the round costs, and Section 6.6 handles the cleanup and assembles the proof.

Remark 6.2 (Reading the running time). One projection at active dimension m costs $\tilde{O}(m/\sqrt{\ell})$ block queries, each a diagonalization at $n^{\omega+o(1)}$ plus ℓ Frobenius products at $O(\ell n^2)$; the block size $\ell = n^{\omega-2}$ balances the two terms at $m n^{1+\omega/2+o(1)}$, and the active dimensions sum to $O(n)$ over the rounds. The exponent $2 + \omega/2$ is the price of one diagonalization per query, which the algorithms of Sections 7–9 avoid.

Algorithm 1 $n^{2+\omega/2}$ time algorithm for Matrix Spencer signing

```

1: procedure MAINDENSE( $A_1, \dots, A_n, n, p$ ) ▷ Theorem 6.1
2:    $y \leftarrow 0$ ,  $S \leftarrow [n]$ , and  $N_0 \leftarrow 3 \cdot 10^8$ 
3:   Fix the per-phase repetition cap  $O(\log \log n + \log(1/p))$  and the cleanup cap  $O(\log(1/p))$  of
   Sections 6.5 and 6.6
4:   while  $|S| \geq N_0$  do
5:      $m \leftarrow |S|$ ,  $u \leftarrow \log(n/m)$ ,  $\ell_{\text{asp}} \leftarrow \max\{1, u\}$ , and  $R \leftarrow c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ 
6:      $\ell \leftarrow \min\{m, \max\{1, \lceil n^{\omega-2} \rceil\}\}$ 
7:     repeat
8:       Sample  $\xi \sim \mathcal{N}(0, I_m)$ , rejecting draws with  $\|\xi\|_2^2 > C_g m$ , and solve Eq. (15) to the
       prescribed inverse-polynomial accuracy using block size  $\ell$ 
9:       Rescale the computed point toward 0 to restore exact smooth feasibility, and snap
       every coordinate within  $2\rho$  of a box facet; call the result  $\bar{x}$ 
10:      until at least  $\delta m$  coordinates of  $\bar{x}$  are snapped, or the repetition cap is reached
11:      if the repetition cap is reached without success then
12:        return FAIL
13:      end if
14:       $y_i \leftarrow y_i + 2\bar{x}_i/\varepsilon_{\text{box}}$  for every  $i \in S$ 
15:       $S \leftarrow \{i \in [n] : |y_i| < 1\}$ 
16:    end while
17:    repeat
18:      Set  $\varepsilon_i \leftarrow y_i$  for  $i \notin S$ , and independently sample  $\varepsilon_i \in \{-1, 1\}$  with  $\Pr[\varepsilon_i = 1] = (1 + y_i)/2$ 
      for  $i \in S$ 
19:       $E \leftarrow \sum_{i \in S} (\varepsilon_i - y_i) A_i$ ; diagonalize  $E/(2m)$  to spectral residual  $\zeta \leq$ 
       $\min\{10^{-2}, \sqrt{n}/(1000m)\}$  and orthogonality defect at most  $\zeta$ , with eigenvalue estimates
       $\beta_1, \dots, \beta_n$ , and put  $U \leftarrow 2m \max_j |\beta_j| + 0.01\sqrt{n}$ 
20:      until  $U \leq 6.5\sqrt{n}$ , or the cleanup cap is reached
21:      if  $U > 6.5\sqrt{n}$  then
22:        return FAIL
23:      end if
24:    return  $\varepsilon$ 
25: end procedure

```

6.2 The smoothed body and its constants

Step 1: the smoothed body. We first fix the constants. At an active set S of size m , put

$$u := \log(n/m), \quad \ell_{\text{asp}} := \max\{1, u\}, \quad R := c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}},$$

with the coefficient $c_{\text{rad}}(\ell_{\text{asp}})$ of Lemma 4.1 (Table 2), and use the same projection parameters at every phase:

$$(\delta, \varepsilon_{\text{box}}) := (0.0281672, 0.4235).$$

Put $B(x) := \sum_{i \in S} x_i A_i$ and

$$\chi := s := 5 \cdot 10^{-7}, \quad a := 2 \cdot 10^6 \log(2n), \quad b := 1 - s,$$

$$h(x) := \frac{1}{a} \log(\text{tr}[\exp(aB(x)/R)] + \text{tr}[\exp(-aB(x)/R)]), \quad D := \{x \in \mathbb{R}^S : h(x) \leq b\}.$$

The function h is the constraint function of the whole section. Since $\text{tr}[\exp(aZ)] + \text{tr}[\exp(-aZ)]$ lies between $e^{a\|Z\|}$ and $2ne^{a\|Z\|}$ for symmetric Z , and $\log(2n)/a = \chi$,

$$\frac{\|B(x)\|}{R} \leq h(x) \leq \frac{\|B(x)\|}{R} + \chi;$$

in particular $h(0) = \chi$, and with $K := \{x : \|B(x)\| \leq R\}$,

$$(1 - s - \chi)K \subseteq D \subseteq (1 - s)K.$$

Radial monotonicity of the Gaussian density and Lemma 4.1 give

$$\gamma_m(D) \geq (1 - 10^{-6})^m \gamma_m(K) \geq \exp(-0.00372m), \quad \text{because } 0.003712 - \log(1 - 10^{-6}) < 0.00372.$$

At the current fractional coloring y , define

$$R_i := \frac{\varepsilon_{\text{box}}}{2}(1 - y_i), \quad L_i := \frac{\varepsilon_{\text{box}}}{2}(1 + y_i), \quad Q := \prod_{i \in S} [-L_i, R_i].$$

One trial is the projection

$$\min\{f_\xi(x) : x \in Q, h(x) \leq b\}, \quad f_\xi(x) := \frac{1}{2}\|x - \xi\|_2^2. \quad (15)$$

Lemma 5.4, applied to the symmetric closed convex body D , gives exact-projection success probability greater than 0.12. The Gaussian-norm rejection and the numerical failure budgets below are chosen with total below 0.02, which leaves success probability greater than 0.1 per attempt.

Step 2: smoothness of h . For a symmetric matrix Z , let

$$\Phi(Z) := \frac{1}{a} \log \text{tr}[\exp(aZ)], \quad \rho_Z := \frac{\exp(aZ)}{\text{tr}[\exp(aZ)]}.$$

Differentiation of the trace exponential (Duhamel's formula) gives

$$D\Phi(Z)[H] = \text{tr}[\rho_Z H], \quad D^2\Phi(Z)[H, H] = a \int_0^1 \text{tr}[\rho_Z^{1-\theta} H \rho_Z^\theta H] d\theta - a(\text{tr}[\rho_Z H])^2.$$

To bound the integral, diagonalize ρ_Z with eigenvalues $r_1, \dots, r_{2n}$; in that basis the integral equals

$$\sum_{j,k=1}^{2n} |H_{jk}|^2 \int_0^1 r_j^{1-\theta} r_k^\theta d\theta,$$

and the inner integral is the logarithmic mean of r_j and r_k , which is at most $(r_j + r_k)/2$. Hence the integral is at most $\text{tr}[\rho_Z H^2] \leq \|H\|^2$, and

$$0 \leq D^2\Phi(Z)[H, H] \leq a\|H\|^2.$$

Apply this to the two-block representation of h and put $G := \sqrt{m}/R$. Since $\|A_i\| \leq 1$,

$$|\partial_i h(x)| \leq \frac{1}{R}, \quad \|\nabla h(x)\|_2 \leq G, \quad \partial_{ii}^2 h(x) \leq \frac{a}{R^2}.$$

6.3 The block solver and robust bisection

This subsection solves Eq. (15), for every block size $1 \leq \ell \leq m$, to inverse-polynomial Euclidean accuracy in

$$\tilde{O}\left(\frac{m}{\sqrt{\ell}}(n^{\omega+o(1)} + \ell n^2) + mn^2\right)$$

arithmetic operations, given a block-gradient oracle whose error and cost Section 6.4 supplies.

Step 3: the multiplier bracket and the path bounds. Let x_* be the minimizer in Eq. (15). If the box projection of ξ is feasible, it equals x_* and there is nothing to prove. Otherwise 0 is a strict Slater point, because $h(0) = \chi < b$, so there is a multiplier $\lambda_* > 0$ for which x_* is the unique minimizer over Q of

$$F_\lambda(x) := f_\xi(x) + \lambda h(x)$$

at $\lambda = \lambda_*$, and $h(x_*) = b$. Write x_λ for the unique minimizer of F_λ over Q . Comparison with 0 gives $\lambda_* \leq f_\xi(0)/(b - \chi)$. Define

$$\Lambda := \frac{2f_\xi(0)}{b - \chi}.$$

Optimality of x_Λ and comparison with 0 imply

$$f_\xi(x_\Lambda) + \Lambda(h(x_\Lambda) - b) \leq f_\xi(0) + \Lambda(\chi - b) = -f_\xi(0),$$

so $h(x_\Lambda) \leq (b + \chi)/2 < b$, and $[0, \Lambda]$ brackets λ_* . Draws with $\|\xi\|_2^2 > C_g m$ are rejected and resampled, an event of probability $e^{-\Omega(m)}$ for a suitable universal C_g ; on the accepted event, $\Lambda = O(m)$.

The map $\lambda \mapsto h(x_\lambda)$ is nonincreasing. Moreover, for $\lambda, \mu \geq 0$,

$$\|x_\lambda - x_\mu\|_2 \leq G|\lambda - \mu|. \quad (16)$$

Indeed, if $\lambda \geq \mu$ and $d := x_\lambda - x_\mu$, subtracting the two variational inequalities and taking the inner product with d gives, by monotonicity of ∇h and of the normal cone of Q ,

$$\|d\|_2^2 \leq -(\lambda - \mu)\langle \nabla h(x_\mu), d \rangle \leq G(\lambda - \mu)\|d\|_2.$$

Strong convexity also gives the residual estimate

$$\|x_\lambda - x_*\|_2^2 \leq |\lambda - \lambda_*| |h(x_\lambda) - b| : \quad (17)$$

apply 1-strong convexity once to F_λ and once to F_{λ_*} , at their respective minimizers, and add the two inequalities.

Step 4: the fixed-multiplier block solver. It remains to minimize a fixed F_λ . Fix an integer $1 \leq \ell \leq m$, partition the active coordinates into $q := \lceil m/\ell \rceil$ blocks $I_1, \dots, I_q$ of size at most ℓ , and use the Euclidean norm on each block. The Hessian estimate of Step 2 implies that, for every vector v supported on one block,

$$D^2 h(x)[v, v] \leq \frac{a}{R^2} \left\| \sum_{i \in S} v_i A_i \right\|^2 \leq \frac{a\ell}{R^2} \|v\|_2^2.$$

Consequently every block gradient of F_λ has the common Lipschitz constant

$$L_\ell := 1 + \frac{\Lambda a \ell}{R^2} = \tilde{O}(\ell), \quad \text{by } \Lambda = O(m), \ R^2 = \Omega(m), \ a = O(\log n).$$

The function F_λ is 1-strongly convex in the Euclidean norm, hence $1/L_\ell$ -strongly convex in the weighted norm $\|v\|_E^2 := L_\ell\|v\|_2^2$. The feasible set Q is a product over the blocks, and its Euclidean block proximal map is coordinatewise clipping.

The solver is the restarted accelerated random block-coordinate method with inexact block derivatives of Dvurechensky, Gasnikov, Tyurin, and Zholobov [DGTZ23, Corollary 3 and Theorem 2]. Under the preceding normalization its block-count parameter is q , its strong-convexity parameter is $1/L_\ell$, and one restart epoch has

$$K_\ell = O(q\sqrt{L_\ell})$$

block queries. Suppose that every returned block gradient has Euclidean error at most Δ_{or} . After J epochs the cited theorem gives

$$\mathbb{E}[F_\lambda(y^{(J)}) - F_\lambda(x_\lambda)] \leq \varepsilon_{\text{init}} 2^{-J} + 8(K_\ell + 2q)^2 \frac{\Delta_{\text{or}}^2}{L_\ell},$$

where $\varepsilon_{\text{init}}$ is an upper bound on the initial objective gap; on the accepted Gaussian event one may take $\varepsilon_{\text{init}} = n^{O(1)}$. For a target objective error ε_{obj} and a failure probability p_0 , choose

$$J := \lceil \log_2 \frac{2\varepsilon_{\text{init}}}{p_0\varepsilon_{\text{obj}}} \rceil, \quad \Delta_{\text{or}} \leq \frac{\sqrt{L_\ell p_0 \varepsilon_{\text{obj}}}}{4(K_\ell + 2q)}.$$

The expected objective gap is then at most $p_0\varepsilon_{\text{obj}}$, and Markov's inequality shows that the gap exceeds ε_{obj} with probability at most p_0 . All target errors below are inverse polynomial, so the required block-gradient accuracy is inverse polynomial as well, and the number of block queries is

$$\tilde{O}(q\sqrt{L_\ell}) = \tilde{O}(m/\sqrt{\ell}).$$

The accelerated recurrence forms affine combinations of feasible vectors and changes only the selected block in each proximal update. Maintain the corresponding matrix images, such as $B(x)$ and the auxiliary image $B(u)$, together with the coordinate vectors: an affine combination of matrix images costs $O(n^2)$ operations and a selected-block change costs $O(\ell n^2)$ operations, so acceleration introduces no hidden $O(mn^2)$ matrix reconstruction per query.

Step 5: robust bisection and exact feasibility. The multiplier search is made robust to the inexact solves as follows. Let $\rho > 0$ be the desired Euclidean accuracy, set $d := \rho/4$, and set

$$\tau := \min\left\{\frac{d^2}{1 + \Lambda}, \frac{d}{16\sqrt{m}}\right\}, \quad \sigma := \min\left\{\frac{d}{8}, \frac{\tau}{16G}\right\}.$$

At each bisection point, run the fixed-multiplier solver with target objective gap $\sigma^2/2$, which it attains except with the failure probability allocated above; strong convexity then gives $\|\hat{x}_\lambda - x_\lambda\|_2 \leq \sigma$. Evaluate $h(\hat{x}_\lambda)$ with absolute numerical error at most $\tau/16$. Because h is G -Lipschitz, these two bounds give a certified interval $\mathcal{I}_\lambda$ of length at most $\tau/4$ containing $h(x_\lambda)$.

If $\mathcal{I}_\lambda$ lies strictly above or strictly below b , monotonicity determines the half interval containing λ_* . If $b \in \mathcal{I}_\lambda$, then $|h(x_\lambda) - b| \leq \tau/4$, and Eq. (17) gives $\|x_\lambda - x_*\|_2 \leq d/2$. Otherwise continue until the multiplier bracket has length at most $d/(4G)$ and solve at its feasible endpoint; Eq. (16) then gives $\|x_\lambda - x_*\|_2 \leq d/4$. The number of fixed-multiplier solves is $O(\log(\Lambda G/d)) = O(\log(n/\rho))$.

The computed point $\hat{x}$ can be made exactly feasible for the smooth constraint. Compute a certified upper bound $\bar{h} \geq h(\hat{x})$ with numerical error at most $\tau/16$. If $\bar{h} \leq b$, set $\alpha := 1$; otherwise set

$$\alpha := \frac{b - h(0)}{\bar{h} - h(0)},$$

and in either case replace $\hat{x}$ by $\alpha\hat{x}$. Since $0, \hat{x} \in Q$, the new point remains in Q , and convexity gives $h(\alpha\hat{x}) \leq b$. In both stopping cases, after the preceding constant choices, $\bar{h} - b \leq \tau/2$, hence

$$(1 - \alpha)\|\hat{x}\|_2 \leq \tau\|\hat{x}\|_2 \leq 2\tau\sqrt{m} \leq \frac{d}{8}.$$

Combining the multiplier, optimization, and rescaling errors shows that the final feasible point z satisfies $\|z - x_*\|_2 \leq \rho$.

6.4 The diagonalization oracle

The solver of Section 6.3 needs the value $h(x)$ and the ℓ derivatives $\partial_i h(x)$, $i \in I$, of a selected block, each to inverse-polynomial accuracy; this subsection supplies them at cost $n^{\omega+o(1)} + O(\ell n^2)$ per query.

Step 6: the value and derivative oracle. At the current query point, diagonalize

$$B(x) = U \operatorname{diag}(\beta_1, \dots, \beta_n) U^\top$$

to spectral residual $\zeta := \|B(x) - \tilde{B}\|$ at most an inverse-polynomial tolerance fixed below, where $\tilde{B}$ is the matrix represented by the computed factorization. The diagonalization is applied after rescaling $B(x)$ by an upper bound on its norm, and the near-orthogonality defect $\|U^\top U - I\|$ of the computed factor is inverse polynomial and is charged to the matrix-arithmetic budget below. Put

$$t_j := \frac{a\beta_j}{R}, \quad M := \max\{|t_1|, \dots, |t_n|\}, \quad w_j^+ := e^{t_j - M}, \quad w_j^- := e^{-t_j - M},$$

so that

$$h(x) = \frac{1}{a} (M + \log \sum_{j=1}^n (w_j^+ + w_j^-)).$$

Define the common signed Gibbs matrix

$$W_x := U \operatorname{diag} \left(\frac{w_j^+ - w_j^-}{\sum_{k=1}^n (w_k^+ + w_k^-)} \right)_{j=1}^n U^\top, \quad \text{so that} \quad \partial_i h(x) = \frac{1}{R} \operatorname{tr}[W_x A_i]$$

for every active coordinate. Thus one diagonalization and one matrix multiplication form W_x , after which all derivatives in a selected block I are the ℓ Frobenius products $\operatorname{tr}[W_x A_i]$, $i \in I$, at $O(\ell n^2)$ operations. This batching is what the exponent depends on: forming $U^\top A_i U$ separately for every selected coordinate would lose the improvement.

Sobczyk's Hermitian diagonalization algorithm [Sob25, Corollary 2.4] computes the required approximate orthogonal diagonalization in

$$O(n^\omega \log n + n^2 \operatorname{polylog}(n/\zeta)) = n^{\omega+o(1)}$$

arithmetic operations, and forming W_x has the same asymptotic cost. The quadratic part of the block gradient costs $O(\ell)$ operations, and maintaining a matrix image after a block update costs $O(\ell n^2)$. Consequently one block query costs

$$n^{\omega+o(1)} + O(\ell n^2).$$

The numerical diagonalization supplies the inexact oracle that Step 4 requires. Put

$$C_B := \operatorname{diag}(B/R, -B/R), \quad C_{\tilde{B}} := \operatorname{diag}(\tilde{B}/R, -\tilde{B}/R).$$

The symmetrized relative-entropy identity and the quantum Pinsker inequality for the corresponding Gibbs states give

$$\|\rho_{C_B} - \rho_{C_{\tilde{B}}}\|_1 \leq 2a\|C_B - C_{\tilde{B}}\| \leq \frac{2a}{R}\|B - \tilde{B}\|.$$

Therefore the spectral residual $\zeta = \|B - \tilde{B}\|$ causes error at most

$$\frac{2\Lambda a\sqrt{\ell}}{R^2}\zeta$$

in the selected block gradient of F_λ , and error at most ζ/R in h . It suffices to request

$$\zeta \leq \frac{\Delta_{\text{or}} R^2}{4\Lambda a\sqrt{\ell}}$$

and to allocate the remaining half of Δ_{or} to scalar and matrix-arithmetic errors. All requested tolerances are inverse polynomial, so the spectral cost remains $n^{\omega+o(1)}$, and no eigengap is required.

The shift by M prevents overflow. Let ε_{sc} be the requested scalar accuracy, which is inverse polynomial. Terms below e^{-T} , with $T = O(\log(n/\varepsilon_{\text{sc}}))$, can be discarded, and range reduction followed by Taylor approximation evaluates the remaining scalar exponentials and the logarithm in $\text{polylog}(n/\varepsilon_{\text{sc}})$ arithmetic operations. Including the one-time $O(mn^2)$ construction of a matrix image, a fixed-multiplier solve therefore costs

$$\tilde{O}\left(\frac{m}{\sqrt{\ell}}(n^{\omega+o(1)} + \ell n^2) + mn^2\right),$$

and multiplier bisection changes this bound only by a polylogarithmic factor.

6.5 Snapping and round costs

Step 7: snapping. Let x_* be the exact minimizer and let z be a feasible point with $\|z - x_*\|_2 \leq \rho$, where

$$\rho := \min\left\{\frac{1}{8}, \frac{sR}{4(2m + \sqrt{m})}\right\}.$$

Snap each coordinate within 2ρ of a box facet. Every truly tight coordinate is snapped, and the snapped point $\bar{x}$ satisfies

$$\|B(\bar{x})\| \leq (1-s)R + 2m\rho < R.$$

Thus every accepted numerical increment belongs to the original body K and preserves the discrepancy certificate. The requested Euclidean accuracy is inverse polynomial and does not change the asymptotic solver bound. The repetition cap for one phase is

$$O(\log \log n + \log(1/p)),$$

and there are $O(\log n)$ phases.

Step 8: round costs. One successful round at active dimension m uses a constant number of projections, up to the repetition factor just recorded. Set

$$\ell_0 := \max\{1, \lceil n^{\omega-2} \rceil\}, \quad \ell(m) := \min\{m, \ell_0\}.$$

If $m \geq \ell_0$, the projection bound gives

$$\tilde{O}\left(\frac{m}{\sqrt{\ell_0}}(n^\omega + \ell_0 n^2) + mn^2\right) = m n^{1+\omega/2+o(1)}$$

operations for the round. Each round freezes at least a fixed fraction of the active coordinates, so the active dimensions satisfy $m_{k+1} \leq (1 - \delta)m_k$ and $\sum_{k=0}^{\infty} m_k = O(n)$. The rounds with $m_k \geq \ell_0$ therefore cost

$$n^{2+\omega/2+o(1)}.$$

For the remaining rounds, take one block of size $\ell = m$. Their total cost is bounded by

$$\tilde{O}(n^\omega \sqrt{\ell_0} + n^2 \ell_0^{3/2} + n^2 \ell_0) = n^{3\omega/2-1+o(1)},$$

which is lower order than $n^{2+\omega/2+o(1)}$ because $\omega < 3$. The one-time matrix-image constructions sum to at most $\tilde{O}(n^3)$ and are of no larger order. Hence all partial-coloring rounds together use

$$n^{2+\omega/2+o(1)} \text{polylog}(1/p)$$

arithmetic operations.

6.6 Cleanup and the proof of Theorem 6.1

Step 9: cleanup. When the active dimension first falls below $N_0 := 3 \cdot 10^8$, use independent biased rounding, and certify the cleanup error $E := \sum_{i \in S} (\varepsilon_i - y_i) A_i$ as follows. The matrix $E/(2m)$ is symmetric with $\|E/(2m)\| \leq 1$, because $|\varepsilon_i - y_i| \leq 2$ and $\|A_i\| \leq 1$. Diagonalize it by Sobczyk's algorithm [Sob25, Corollary 2.4] to spectral residual $\zeta \leq \min\{10^{-2}, \sqrt{n}/(1000m)\}$ and orthogonality defect at most ζ : the output is $\tilde{U}$ and $\beta_1, \dots, \beta_n$ with $\|E/(2m) - U \text{diag}(\beta) \tilde{U}^\top\| \leq \zeta$ and $\|\tilde{U}^\top \tilde{U} - I\| \leq \zeta$, at cost $n^{\omega+o(1)}$. Put

$$U := 2m \max_j |\beta_j| + 0.01\sqrt{n}.$$

Then

$$\|E\| \leq U \leq \|E\| + 0.03\sqrt{n}.$$

Indeed, write $\tilde{U} = VP$ with V orthogonal and $P := (\tilde{U}^\top \tilde{U})^{1/2}$, so that $\|P - I\| \leq \zeta$ and $\|P \text{diag}(\beta) P - \text{diag}(\beta)\| \leq (2 + \zeta)\zeta \max_j |\beta_j| \leq 3\zeta \max_j |\beta_j|$. Hence $\|E/(2m) - V \text{diag}(\beta) V^\top\| \leq \zeta(1 + 3 \max_j |\beta_j|)$, and Weyl's inequality for the orthogonally diagonalizable $V \text{diag}(\beta) V^\top$ gives $|\|E/(2m)\| - \max_j |\beta_j|| \leq \zeta(1 + 3 \max_j |\beta_j|)$. Since $\|E/(2m)\| \leq 1$ and $\zeta \leq 10^{-2}$, this forces $\max_j |\beta_j| \leq 1.05$ and then $|\|E\| - 2m \max_j |\beta_j|| \leq 4.2 \cdot 2m\zeta \leq 0.0084\sqrt{n}$, which gives both inequalities. Accept when $U \leq 6.5\sqrt{n}$.

The scalars $\varepsilon_i - y_i$ are independent, centered, and of range 2, so Hoeffding's lemma gives $\mathbb{E}[e^{\eta(\varepsilon_i - y_i)}] \leq e^{\eta^2/2}$ for every real η , hence $\mathbb{E}[\exp(\eta(\varepsilon_i - y_i) A_i)] \preceq \exp(\eta^2 A_i^2/2)$ by the functional calculus, and Tropp's matrix Laplace-transform method [Tro15, Chapter 3] yields the matrix Hoeffding bound

$$\Pr[\|E\| \geq t] \leq 2n \exp\left(-\frac{t^2}{2\|\sum_{i \in S} A_i^2\|}\right), \quad \left\|\sum_{i \in S} A_i^2\right\| \leq m < N_0.$$

Setting the right side equal to $1/2$, and using that $(N_0/n)(\log(2n) + \log 2)$ decreases for $n \geq N_0$, gives, together with the $0.03\sqrt{n}$ certification allowance,

$$\sqrt{2(\log(2N_0) + \log 2)} + 0.03 < 6.5.$$

For $n < N_0$ the algorithm rounds immediately with $m = n$ and $\log(4n) \leq \log(4N_0)$, so the bound is uniform in n , and one cleanup trial succeeds with probability at least $1/2$. Repetition contributes only $O(\log(1/p))$.

Proof of Theorem 6.1. Every accepted numerical increment lies in the body K (Step 7) and freezes at least δm coordinates, exactly as the exact projection of Lemma 5.4 does, so the increment bound Eq. (14) and the constant calculation in the proof of Theorem 5.1 apply unchanged, and the cleanup of Step 9 adds at most $6.5\sqrt{n}$; the returned signing has discrepancy at most $156000\sqrt{n}$. There are $P = O(\log n)$ phases (end of Step 7, by the geometric decrease of Step 8). One attempt succeeds with probability at least 0.1 (Step 1), so the repetition cap $\lceil \log(2P/p)/\log(10/9) \rceil = O(\log \log n + \log(1/p))$ of Step 7 makes a phase fail with probability at most $p/(2P)$; one cleanup trial succeeds with probability at least $1/2$ (Step 9), so the cleanup cap $\lceil \log_2(2/p) \rceil = O(\log(1/p))$ makes the cleanup fail with probability at most $p/2$; the numerical failure budgets of Steps 4 and 5 are part of the 0.02 allowance of Step 1. The total failure probability is at most $P \cdot p/(2P) + p/2 = p$. The arithmetic count is the sum of the round costs of Step 8 and the $O(\log(1/p))$ cleanup trials of Step 9, each of cost $n^{\omega+o(1)}$, which is $n^{2+\omega/2+o(1)}$ polylog($1/p$). $\square$

7 The selected-coordinate algorithm

The change from Section 6 is confined to the solver of each smooth projection Eq. (15): the outer partial-coloring trajectory, the smoothed spectral body, the projection accuracy, the snapping rule, and the cleanup are retained, and the running time is again counted in arithmetic operations, not in bit operations.

7.1 Statement and pseudocode

Theorem 7.1 (Selected-coordinate dense implementation; formal version of the $n^{1+\omega(1,1/2,1)+o(1)}$ variant in Remark 1.3). *For dense input matrices and $p \in (0, 1/2)$, a signing satisfying the conclusion of Theorem 5.1 can be found with failure probability at most p using*

$$n(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n))n^{o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model. In particular, the running time is at most

$$n^{1+\omega(1,1/2,1)+o(1)} \text{polylog}(1/p).$$

Using the bound $\omega(1, 1/2, 1) < 2.042776$ from [ADVW⁺25, Table 1] gives the numerical exponent $3.042776 + o(1)$.

Algorithm 2 is the implementation. Its outer loop is that of Algorithm 1; each projection is solved by the certified multiplier wrapper of Section 7.5 around the fixed-multiplier procedure COORDINATEMULTIPLIER, candidates are compared by COMPAREVALUEEXACT, and the final rounding is the shared procedure CERTIFIEDCLEANUP of Algorithm 3, called with its own share p_{clean} of the failure budget. The rest of the section proves Theorem 7.1: Section 7.2 sets up the fixed-multiplier problem and its selected-coordinate oracle, Sections 7.3 and 7.4 analyze one guarded candidate phase with exact and then with finite matrix actions, Section 7.5 gives the selection rule and the wrapper, and Section 7.6 transfers correctness, allocates the failure budget, and counts operations.

Algorithm 2 $n^{1+\omega(1,1/2,1)+o(1)}$ polylog($1/p$) time algorithm for Matrix Spencer signing

```

1: procedure MAINSELECTED( $A_1, \dots, A_n, n, p$ ) ▷ Theorem 7.1
2:    $y \leftarrow 0$ ,  $S \leftarrow [n]$ , and  $N_0 \leftarrow 3 \cdot 10^8$ 
3:   Allocate from  $p$  the per-solve budget  $\delta_\nu$ , the trial caps, and the cleanup budget  $p_{\text{clean}}$  as in
   the proof of Theorem 7.1
4:   while  $|S| \geq N_0$  do
5:      $m \leftarrow |S|$ ,  $u \leftarrow \log(n/m)$ ,  $\ell_{\text{asp}} \leftarrow \max\{1, u\}$ , and  $R \leftarrow c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ 
6:     repeat
7:       Sample an accepted  $\xi \sim \mathcal{N}(0, I_m)$  and keep it fixed through all multiplier tests
8:       If the certified interval for  $\Phi(\xi_Q)$  lies below 1, retain the box projection  $\xi_Q$ ; otherwise
       solve Eq. (15) by certified bisection of the multiplier over  $[\underline{\nu}, \Lambda]$ 
9:       At each tested  $\nu$ :  $\hat{x}_\nu \leftarrow \text{COORDINATEMULTIPLIER}(F_\nu, Q, m, a, R, \varepsilon_{\text{opt}}, \delta_\nu)$ ; update the
       bracket only from the outward-rounded interval for  $\Phi(\hat{x}_\nu)$ , or retain  $\hat{x}_\nu$  if that interval meets 1
10:      Apply the downward-rounded radial correction to the retained point, snap within  $2\rho$ ,
       and call the result  $\bar{x}$ 
11:      until at least  $\delta m$  coordinates of  $\bar{x}$  are snapped, or the prescribed trial cap is reached
12:      if the trial cap is reached without success then
13:        return FAIL
14:      end if
15:       $y_i \leftarrow y_i + 2\bar{x}_i/\varepsilon_{\text{box}}$  for every  $i \in S$ 
16:       $S \leftarrow \{i \in [n] : |y_i| < 1\}$ 
17:    end while
18:    return CERTIFIEDCLEANUP( $A_1, \dots, A_n, y, S, p_{\text{clean}}$ ) ▷ Algorithm 3
19: end procedure
20: procedure COORDINATEMULTIPLIER( $F_\nu, Q, m, a, R, \varepsilon_{\text{opt}}, \delta_\nu$ ) ▷  $\delta_\nu$  is the failure budget of one
   fixed-multiplier solve, written  $p_0$  in Section 7.2
21:    $F_0 \leftarrow F_\nu(0)$ ,  $H \leftarrow 16F_0$ ,  $J \leftarrow \max\{1, \lceil \log_2(4F_0/\varepsilon_{\text{opt}}) \rceil\}$ ,  $R_{\text{rep}} \leftarrow \lceil C_{\text{rep}} \log(J/\delta_\nu) \rceil$ , and  $z \leftarrow 0$ 
22:   for  $j = 1, \dots, J$  do
23:     Run  $R_{\text{rep}}$  independent guarded candidate phases from  $z$ , each of  $T$  selected-coordinate
     queries with  $k, s = \tilde{\Theta}(\sqrt{m})$ , finite positive semidefinite exponential actions, and fresh trace
     guards; let  $\mathcal{C}$  be  $z$  together with the returned points
24:      $z \leftarrow \text{COMPAREVALUEEXACT}(\mathcal{C}, \varepsilon_{\text{opt}}/(128J))$ 
25:   end for
26:   return  $z$ , or FAIL if a prescribed cap is reached
27: end procedure
28: procedure COMPAREVALUEEXACT( $\mathcal{C}, w$ ) ▷  $\mathcal{C}$  contains the old point  $z$ ;  $w$  is the interval width
29:   Enclose  $F_\nu(x)$  for every  $x \in \mathcal{C}$  in an outward-rounded interval of width at most  $w$ , by one
   dense matrix-function evaluation per point
30:   return the  $x \in \mathcal{C} \setminus \{z\}$  with the least upper endpoint among those whose upper endpoint
   lies below the lower endpoint of  $z$ ; if there is none, return  $z$ 
31: end procedure

```

Algorithm 3 Certified bounded-dimension cleanup

```
1: procedure CERTIFIEDCLEANUP( $A_1, \dots, A_n, y, S, p_{\text{clean}}$ )
2:   repeat
3:     Set  $\varepsilon_i \leftarrow y_i$  for  $i \notin S$ , and independently sample  $\varepsilon_i \in \{-1, 1\}$  with  $\Pr[\varepsilon_i = 1] = (1 + y_i)/2$ 
     for  $i \in S$ 
4:     Compute the deterministic spectral certificate  $U_{\text{cl}}$  of the cleanup error  $E = \sum_{i \in S} (\varepsilon_i - y_i)A_i$  as in Step 9 of Section 6.6 (diagonalize  $E/(2m)$  to residual  $\zeta \leq \min\{10^{-2}, \sqrt{n}/(1000m)\}$ 
     and orthogonality defect at most  $\zeta$ , and put  $U_{\text{cl}} := 2m \max_j |\beta_j| + 0.01\sqrt{n}$ , so that  $\|E\| \leq U_{\text{cl}} \leq \|E\| + 0.03\sqrt{n}$ )
5:     until  $U_{\text{cl}} \leq 6.5\sqrt{n}$ , or the cleanup cap prescribed by  $p_{\text{clean}}$  is reached
6:     if  $U_{\text{cl}} > 6.5\sqrt{n}$  then
7:       return FAIL
8:     end if
9:   return  $\varepsilon$ 
10: end procedure
```

Remark 7.2 (Where $2 + \omega/2$ comes from). The proof of Theorem 6.1 solves each projection by an accelerated block method with block size ℓ : a projection at active dimension m takes $\tilde{O}(m/\sqrt{\ell})$ block queries, each of which forms one dense Gibbs matrix at cost $n^{\omega+o(1)}$ and then reads ℓ block derivatives from it at cost $O(\ell n^2)$. Balancing the two terms at $\ell = n^{\omega-2}$ gives $m n^{1+\omega/2+o(1)}$ per projection, and $\sum_t m_t = O(n)$ over the rounds gives the exponent $2 + \omega/2$. Theorem 7.1 instead makes $\tilde{O}(m)$ single-coordinate queries per projection and never forms a Gibbs matrix inside a query: a query applies polynomial functions of two dense matrices to $\tilde{O}(\sqrt{m})$ vectors, at cost $\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + n^2$ up to subpolynomial factors, which is $n^{\omega(1,1/2,1)+o(1)}$ at $m = n$. Convexity of $b \mapsto \omega(1, b, 1)$, recalled in Section 8, gives $\omega(1, 1/2, 1) \leq (\omega(1, 0, 1) + \omega(1, 1, 1))/2 = 1 + \omega/2$, so the new exponent is never larger than the old one; with the current bounds it is 3.042776 against $2 + \omega/2 < 3.185670$, the latter using $\omega < 2.371339$ from [ADVW⁺25].

7.2 Setup and the fixed-multiplier problem

Retain from the proof of Theorem 6.1, at an active set S of size m with fractional coloring y : the box Q of Eq. (15), the matrix $B(x) := \sum_{i \in S} x_i A_i$, the scale $R = c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$, the smoothing constants a, b , and $\chi = 5 \cdot 10^{-7}$ together with the bound $h(x) \leq \|B(x)\|/R + \chi$, the norm rejection that discards a Gaussian draw ξ unless $\|\xi\|_2^2 \leq C_g m$, the projection accuracy ρ of the snapping step, the snapping rule, the approximate Hermitian diagonalization primitive, and the cleanup. Every object called *retained* below is one of these.

Fix one instance of Eq. (15). Define

$$C(x) := \text{diag}(B(x)/R, -B(x)/R), \quad D_x := aC(x) - abI_{2n},$$

$$P_x := \exp(D_x), \quad \Phi(x) := \text{tr}[P_x], \quad q(x) := \frac{1}{2}\|x - \xi\|_2^2.$$

Since $h(x) = b + a^{-1} \log \Phi(x)$, the smooth constraint is equivalent to

$$h(x) \leq b \iff \Phi(x) \leq 1,$$

and at the origin

$$\phi_0 := \Phi(0) = 2n \exp(-ab) < \frac{1}{4}.$$

The eigenvalues of $C(x)$ come in pairs $\pm\lambda$ and $e^{a\lambda} + e^{-a\lambda} \geq 2$, so $\Phi(x) \geq \phi_0$ for every x .

The fixed-multiplier problem. For a multiplier $\nu > 0$ consider the strongly convex box problem

$$F_\nu(x) := q(x) + \nu\Phi(x), \quad x_\nu := \operatorname{argmin}_{x \in Q} F_\nu(x),$$

and write

$$\Delta_x := F_\nu(x) - F_\nu(x_\nu)$$

for the objective gap. The wrapper of Section 7.5 reduces one projection to $O(\log(n/\rho))$ such problems; this subsection and the next two solve one of them.

Constants. We first fix the constants. Put

$$\eta := 10^{-3}, \quad d_0 := 2n, \quad d := \frac{\rho}{4},$$

where η is the relative accuracy of the coordinate oracle (only $\eta \leq 1/4$ and the value 10^{-3} in the crossing bounds of Sections 7.3 and 7.4 are used), d_0 is the order of D_x , and d is the radius the wrapper works to. For a fixed multiplier ν put

$$F_0 := F_\nu(0), \quad H := 16F_0;$$

H is the level below which every guarded iterate is kept. For a requested objective accuracy $\varepsilon_{\text{opt}} > 0$ and a failure budget $p_0 \in (0, 1/2)$ put

$$J := \max\{1, \lceil \log_2(4F_0/\varepsilon_{\text{opt}}) \rceil\}, \quad R_{\text{rep}} := \lceil C_{\text{rep}} \log(J/p_0) \rceil$$

for a sufficiently large universal constant C_{rep} : J is the number of halving phases and R_{rep} the number of independent candidates per phase. Unsubscripted constants C are universal and may change from line to line.

The failure budget and the accepted draw. Let $p_0 \in (0, 1/2)$ denote the failure budget allocated to one fixed-multiplier solve; the $O(\log(n/\rho))$ solves of one projection call share the numerical allowance of Section 6, and Section 7.6 does the accounting. Before the multiplier search, draw ξ once and apply the retained norm rejection. The acceptance probability is at least a universal constant, so the sampler is capped after $O(\log(1/p_0))$ draws; the cap fails with probability at most p_0 and costs a logarithmic factor. Fix the accepted ξ through all multiplier tests; every statement below is conditional on it. On the accepted draw $q(0) \leq C_g m/2$, and for every multiplier the wrapper tests $F_0 \leq q(0) + \Lambda\phi_0 = O(m)$ with the bracket endpoint $\Lambda = O(m)$ of Section 7.5.

The selected-coordinate Gibbs oracle. Put

$$E_i := \partial_i C(x) = \operatorname{diag}(A_i/R, -A_i/R),$$

so that $\|E_i\| \leq 1/R$ by the normalization $\|A_i\| \leq 1$, and

$$\partial_i F_\nu(x) = x_i - \xi_i + \nu a \operatorname{tr}[P_x E_i].$$

The oracle estimates this derivative for one uniformly random coordinate i from a dense snapshot at a point $w \in Q$, whose full gradient $\nu a \operatorname{tr}[P_w E_i]$, $i \in S$, has been formed, and from a sketch of the increment

$$E := P_x - P_w.$$

Choose integers $k, s \geq 2$. Draw a standard Gaussian matrix $\Omega \in \mathbb{R}^{2n \times 2k}$, let Y have orthonormal columns spanning the range of $E\Omega$, and put $\Pi := YY^\top$. Define

$$E_{\text{low}} := \Pi E + E\Pi - \Pi E\Pi, \quad E_{\text{res}} := (I - \Pi)E(I - \Pi),$$

so that $E = E_{\text{low}} + E_{\text{res}}$ exactly. For fresh independent standard Gaussian vectors $z_1, \dots, z_s \in \mathbb{R}^{2n}$, define

$$\hat{t}_i := \text{tr}[E_i E_{\text{low}}] + \frac{1}{2s} \sum_{j=1}^s z_j^\top (E_i E_{\text{res}} + E_{\text{res}} E_i) z_j$$

and

$$\hat{g}_i := x_i - \xi_i + \nu a(\text{tr}[P_w E_i] + \hat{t}_i).$$

Conditional on x, w, i and Ω , Gaussian isotropy $\mathbb{E}[zz^\top] = I$ gives

$$\mathbb{E}[\hat{t}_i] = \text{tr}[E_i E_{\text{low}}] + \text{tr}[E_i E_{\text{res}}] = \text{tr}[E_i E],$$

so the estimator is conditionally unbiased when the exponential actions are exact.

For every matrix M and a standard Gaussian vector z ,

$$\mathbb{V}[z^\top M z] = 2\|\frac{1}{2}(M + M^\top)\|_F^2.$$

The matrix $E_i E_{\text{res}} + E_{\text{res}} E_i$ is symmetric with Frobenius norm at most $2\|E_i\|\|E_{\text{res}}\|_F$, so the s independent probes give

$$\mathbb{V}[\hat{g}_i \mid x, w, i, \Omega] \leq \frac{2\nu^2 a^2}{sR^2} \|E_{\text{res}}\|_F^2.$$

The residual is controlled by the Gaussian range-finder estimate underlying Hutch++ of Meyer, Musco, Musco, and Woodruff [MMM21]:

$$\mathbb{E}_\Omega[\|E_{\text{res}}\|_F^2] \leq \frac{3}{k} \|E\|_1^2.$$

The constant 3 is quoted in Section 10, so here is its derivation from the deterministic projector bound of Halko, Martinsson, and Tropp [HMT11, Theorem 9.1]. Let $E = W\Sigma V^\top$ be a singular value decomposition with nonincreasing singular values, partition $\Sigma = \text{diag}(\Sigma_1, \Sigma_2)$ with Σ_1 of size $k \times k$, partition $V = [V_1 \ V_2]$ conformally, and put $\Omega_1 := V_1^\top \Omega \in \mathbb{R}^{k \times 2k}$ and $\Omega_2 := V_2^\top \Omega$. Since $\|(I - \Pi)E(I - \Pi)\|_F \leq \|(I - \Pi)E\|_F$, the projector bound gives

$$\|E_{\text{res}}\|_F^2 \leq \|\Sigma_2\|_F^2 + \|\Sigma_2 \Omega_2 \Omega_1^\dagger\|_F^2.$$

Here Ω_1 is a $k \times 2k$ standard Gaussian matrix independent of Ω_2 . Conditioning on Ω_1 , the identity $\mathbb{E}[\Omega_2^\top \Sigma_2^\dagger \Omega_2] = \|\Sigma_2\|_F^2 I$ [HMT11, Proposition A.1] bounds the expectation of the second term by $\|\Sigma_2\|_F^2 \mathbb{E}[\|\Omega_1^\dagger\|_F^2]$, and the Gaussian pseudoinverse identity $\mathbb{E}[\|\Omega_1^\dagger\|_F^2] = k/(k-1)$ [HMT11, Appendix A] makes the whole expectation at most $(1 + k/(k-1))\|\Sigma_2\|_F^2 \leq 3\|\Sigma_2\|_F^2$ for $k \geq 2$. Finally,

$$\|\Sigma_2\|_F^2 \leq \sigma_{k+1}(E) \|E\|_1 \leq \frac{1}{k} \|E\|_1^2,$$

because the singular values are nonincreasing and $\sigma_{k+1}(E)$ is at most the average of the k larger ones, which proves the stated estimate.

From trace norm to objective gaps. For positive definite matrices P, P' of possibly different traces define

$$\mathcal{D}(P\|P') := \text{tr}[P(\log P - \log P') - P + P'].$$

Bansal, George, Ghosh, Sikora, and Zheng [BGG⁺25, Theorem 1.3] prove the generalized Pinsker inequality

$$\|P - P'\|_1^2 \leq 4 \max\{\text{tr}[P], \text{tr}[P']\} \mathcal{D}(P\|P').$$

Let $P_\nu := P_{x_\nu}$. Since $\log P_x = D_x$ is affine in x , the divergence $\mathcal{D}(P_\nu\|P_x)$ equals $\Phi(x) - \Phi(x_\nu) - \langle \nabla \Phi(x_\nu), x - x_\nu \rangle$, the Bregman divergence of Φ , and adding the nonnegative Bregman divergence of q gives

$$\nu \mathcal{D}(P_\nu\|P_x) \leq D_{F_\nu}(x, x_\nu),$$

where the right side is the Bregman divergence of F_ν . The box-constrained optimality condition

$$\langle \nabla F_\nu(x_\nu), x - x_\nu \rangle \geq 0 \quad (x \in Q)$$

shows $D_{F_\nu}(x, x_\nu) = \Delta_x - \langle \nabla F_\nu(x_\nu), x - x_\nu \rangle \leq \Delta_x$.

Suppose $F_\nu(x), F_\nu(w), F_\nu(x_\nu) \leq H$ for the level H fixed above. Then $\nu \text{tr}[P_x], \nu \text{tr}[P_w], \nu \text{tr}[P_\nu] \leq H$, and the Pinsker inequality for the pairs (P_ν, P_x) and (P_ν, P_w) , the trace-norm triangle inequality, and $(u + v)^2 \leq 2u^2 + 2v^2$ give

$$\nu^2 \|P_x - P_w\|_1^2 \leq 8H(\Delta_x + \Delta_w).$$

Combining the variance bound, the range-finder estimate, and this display, and writing $\epsilon_i := \hat{g}_i - \partial_i F_\nu(x)$ for the oracle error, proves

$$\mathbb{E}[|\epsilon_i|^2 \mid x, w] \leq \frac{\eta}{m}(\Delta_x + \Delta_w)$$

whenever

$$ks \geq C \frac{a^2 m H}{R^2 \eta}.$$

A loopless coordinate method. Set

$$\bar{L} := \max\{1 + ea^2 H/R^2, 2a/R\}, \quad \alpha := \frac{1}{m\bar{L}}.$$

The first term in $\bar{L}$ is the coordinate smoothness constant on the level set, proved below; the second is a trust-step correction that makes the comparison displacement admissible. Given the selected coordinate i and the estimate $\hat{g}_i$, minimize

$$\hat{g}_i \gamma + \frac{\bar{L}}{2} \gamma^2$$

over all displacements γ with $x_i + \gamma \in Q_i$ and $|\gamma| \leq R/a$, and put $y := x + \gamma e_i$. If $F_\nu(x) \leq H$, then for every ϑ between 0 and γ ,

$$\|a(C(x + \vartheta e_i) - C(x))\| = a|\vartheta| \|E_i\| \leq 1,$$

so $D_{x+\vartheta e_i} \preceq D_x + I$ and $\Phi(x + \vartheta e_i) \leq e\Phi(x)$ by monotonicity of tr exp . Differentiating twice under the trace (Duhamel's formula) gives

$$\frac{d^2}{d\vartheta^2} \text{tr}[\exp(D_{x+\vartheta e_i})] = a^2 \int_0^1 \text{tr}[E_i \exp((1-\theta)D_{x+\vartheta e_i}) E_i \exp(\theta D_{x+\vartheta e_i})] d\theta.$$

To bound the integrand, diagonalize $D_{x+\vartheta e_i}$ with eigenvalues λ_j and write E_i in the eigenbasis; the weighted arithmetic–geometric mean inequality $e^{(1-\theta)\lambda_j+\theta\lambda_l} \leq (1-\theta)e^{\lambda_j} + \theta e^{\lambda_l}$ and $\sum_l |(E_i)_{jl}|^2 \leq \|E_i\|^2$ give

$$\sum_{j,l} |(E_i)_{jl}|^2 \exp((1-\theta)\lambda_j + \theta\lambda_l) \leq \|E_i\|^2 \sum_j \exp(\lambda_j).$$

Therefore the second derivative is at most $a^2\|E_i\|^2\Phi(x+\vartheta e_i) \leq a^2\Phi(x+\vartheta e_i)/R^2$, and with $\nu\Phi(x) \leq F_\nu(x) \leq H$,

$$\partial_{ii}^2 F_\nu(x + \vartheta e_i) \leq 1 + \frac{ea^2 H}{R^2} \leq \bar{L}.$$

This proves coordinate smoothness along the proposed segment before the proposed point has passed any level test.

For every admissible comparison displacement γ' (that is, $x_i + \gamma' \in Q_i$ and $|\gamma'| \leq R/a$), the one-dimensional proximal three-point inequality for the minimized quadratic, the smoothness bound along the segment, and Young's inequality on the error term $\epsilon_i(\gamma' - \gamma)$ give

$$F_\nu(y) \leq F_\nu(x) + \partial_i F_\nu(x)\gamma' + \frac{\bar{L}}{2}(\gamma')^2 + \frac{\epsilon_i^2}{2\bar{L}}.$$

The choice

$$\gamma' := \frac{x_{\nu,i} - x_i}{\bar{L}}$$

is admissible: it is a convex-combination displacement inside Q_i , as $\bar{L} \geq 1$, and $|\gamma'| \leq 2/\bar{L} \leq R/a$, as the box has width below 2 and $\bar{L} \geq 2a/R$. Averaging over the uniform coordinate i and using the 1-strong convexity of F_ν in the form $\sum_i (\partial_i F_\nu(x)(x_{\nu,i} - x_i) + \frac{1}{2}(x_{\nu,i} - x_i)^2) \leq -\Delta_x$ yields

$$\mathbb{E}[\Delta_y \mid x, w] \leq (1 - \alpha)\Delta_x + \frac{1}{2\bar{L}}\mathbb{E}[|\epsilon_i|^2 \mid x, w].$$

With the oracle bound and $\frac{1}{2\bar{L}} \cdot \frac{\eta}{m} = \frac{\eta\alpha}{2}$ this reads

$$\mathbb{E}[\Delta_y \mid x, w] \leq A_0\Delta_x + B_0\Delta_w, \quad A_0 := 1 - (1 - \eta/2)\alpha, \quad B_0 := \eta\alpha/2.$$

The unguarded contraction. After forming y , set $x^+ := y$ and, independently, set

$$w^+ := \begin{cases} y, & \text{with probability } \alpha, \\ w, & \text{with probability } 1 - \alpha. \end{cases}$$

When a refresh occurs, form the dense snapshot and all its coordinate derivatives. For

$$V(x, w) := \Delta_x + \frac{1}{2}\Delta_w$$

direct substitution gives

$$\mathbb{E}[V(x^+, w^+) \mid x, w] \leq (1 - \alpha/4)V(x, w) \quad (\eta \leq 1/4).$$

Indeed, $\mathbb{E}[\Delta_{w^+} \mid x, w] \leq \alpha\mathbb{E}[\Delta_y \mid x, w] + (1 - \alpha)\Delta_w$, so the coefficients of Δ_x and Δ_w on the right are $(1 + \alpha/2)A_0$ and $(1 + \alpha/2)B_0 + (1 - \alpha)/2$, and for $\eta \leq 1/4$ and $\alpha \in (0, 1]$,

$$(1 + \alpha/2)A_0 \leq 1 - \alpha/4, \quad (1 + \alpha/2)B_0 + \frac{1 - \alpha}{2} \leq \frac{1 - \alpha/4}{2}.$$

Consequently, $T := \lceil 16m\bar{L} \rceil$ coordinate queries reduce the potential by the factor $(1 - \alpha/4)^T \leq e^{-4}$, and the expected number of dense snapshot refreshes is at most $T\alpha \leq 17$. The candidate phase of Section 7.3 uses this update only when its level guard accepts the proposal; if the guard rejects, the phase terminates and returns its starting point.

Remark 7.3 (What the sketch buys). The oracle's mean-square error is $(\eta/m)(\Delta_x + \Delta_w)$, a $1/m$ fraction of the objective gaps, at sketch size $ks = \tilde{O}(m)$, that is $k, s = \tilde{\Theta}(\sqrt{m})$. The factor $1/m$ is what a coordinate method needs: its step $\alpha = 1/(m\bar{L})$ contracts the gap by $1 - \Theta(\alpha)$ per query, so an error of order $\alpha\Delta$ per query is absorbed and $\tilde{O}(m)$ queries reduce the gap by a constant factor. The same argument without the range finder, with Hutchinson probes on $P_x - P_w$ alone, needs $s = \tilde{\Theta}(m)$ probes for the same variance, and a query then costs $\mathcal{T}_{\text{mat}}(n, n, m) = n^{\omega+o(1)}$, the dense cost of Section 6; the range finder of rank $k = \tilde{\Theta}(\sqrt{m})$ removes the k leading singular directions of $P_x - P_w$ and leaves a residual of squared Frobenius norm at most $3/k$ times $\|P_x - P_w\|_1^2$, so $s = \tilde{\Theta}(\sqrt{m})$ probes suffice. The speedup comes from this cheaper certified oracle, not from a new acceleration theorem.

7.3 The level guard, the crossing bound, and localization

A fixed-multiplier solve starts at $z_0 := 0$, so $F_0 = F_\nu(z_0)$ and $H = 16F_0$ are the level constants fixed above. Each constant-reduction phase starts from $x = w = z$ with $F_\nu(z) \leq H/16$ and proposes T coordinate steps, every one of which is guarded.

Localization of the spectrum. On the level $F_\nu(x) \leq H$, the bound $\nu e^{\lambda_{\max}(D_x)} \leq \nu\Phi(x) \leq H$ and the $\pm$ symmetry of the spectrum of $C(x)$ give

$$\lambda_{\max}(D_x) \leq \log(H/\nu), \quad \lambda_{\min}(D_x) \geq -2ab - \log(H/\nu).$$

A trust step $|\gamma| \leq R/a$ moves D_x by at most $\|a\gamma E_i\| \leq 1$ in operator norm, so at a proposed point the spectrum lies in the interval enlarged by one at each end. Choose a polynomial q_r uniformly approximating $\lambda \mapsto e^{\lambda/2}$ on this enlarged interval $[\lambda_-, \lambda_+]$ and put

$$\tilde{P}_x := q_r(D_x)^2.$$

The square makes $\tilde{P}_x$ positive semidefinite. An explicit degree bound: put $c := (\lambda_- + \lambda_+)/2$ and truncate the Taylor series,

$$q_r(\lambda) := e^{c/2} \sum_{j=0}^r \frac{((\lambda - c)/2)^j}{j!}.$$

The Taylor remainder and $(r+1)! \geq ((r+1)/e)^{r+1}$ show that its uniform error on the interval is at most

$$e^{\lambda_+/2} \left(\frac{e(\lambda_+ - \lambda_-)}{4(r+1)} \right)^{r+1}.$$

Choose r so that this error is at most $\varepsilon_{\text{op}}/(3(1 + e^{\lambda_+/2}))$; then $|q_r(\lambda)^2 - e^\lambda| \leq \varepsilon_{\text{op}}$ throughout the interval, since $|q_r^2 - e^\lambda| \leq |q_r - e^{\lambda/2}|(2e^{\lambda/2} + |q_r - e^{\lambda/2}|)$. As $\lambda_+ - \lambda_- = O(ab + |\log(H/\nu)|)$, it is enough to take

$$r = O(ab + |\log(H/\nu)| + \log(1/\varepsilon_{\text{op}})),$$

and the spectral theorem gives $\|\tilde{P}_x - P_x\| \leq \varepsilon_{\text{op}}$.

The trace guard. For every positive semidefinite M and $z \sim \mathcal{N}(0, I)$,

$$\mathbb{E}[z^\top M z] = \text{tr}[M], \quad \mathbb{V}[z^\top M z] = 2\|M\|_F^2 \leq 2(\text{tr}[M])^2.$$

Set

$$s_{\text{tr}} := \lceil C \log(TJR_{\text{rep}}/p_0) \rceil.$$

Split s_{tr} fresh probes into batches of a fixed size, apply Chebyshev's inequality to each batch mean, and take the median of the batch means; the resulting estimate $\hat{\tau}_y$ obeys

$$|\hat{\tau}_y - \text{tr}[\tilde{P}_y]| \leq \text{tr}[\tilde{P}_y]/4$$

except with probability $p_0/(100TJR_{\text{rep}})$. Choose ε_{op} with $d_0\varepsilon_{\text{op}} \leq \phi_0/8$ (the choice of Section 7.4 is stronger) and define

$$U(y) := \frac{4}{3}\hat{\tau}_y + d_0\varepsilon_{\text{op}}.$$

Since $\Phi(y) \geq \phi_0$, $|\text{tr}[\tilde{P}_y] - \Phi(y)| \leq d_0\varepsilon_{\text{op}} \leq \Phi(y)/8$, and the successful certificate satisfies

$$\Phi(y) \leq U(y) \leq 2\Phi(y) :$$

the lower bound is $\frac{4}{3} \cdot \frac{3}{4} \text{tr}[\tilde{P}_y] + d_0\varepsilon_{\text{op}} \geq \Phi(y)$, and the upper bound is $\frac{4}{3} \cdot \frac{5}{4} \cdot \frac{9}{8}\Phi(y) + \frac{1}{8}\Phi(y) = 2\Phi(y)$. Accept the proposal only when

$$q(y) + \nu U(y) \leq H.$$

Every accepted point is in the H -level set, while every proposal with $F_\nu(y) \leq H/3$ is accepted. Because $\Phi(y) \geq \phi_0 = 2n \exp(-ab) = n^{-O(1)}$, the required additive trace accuracy is inverse polynomial. If the test rejects, terminate the candidate phase and return z . A conditional union bound is valid for the adaptive queries because every guard uses fresh probes; the T guards of one candidate phase are all correct with probability at least 0.99, since $T \cdot p_0/(100TJR_{\text{rep}}) \leq 0.01$.

The crossing bound. A crossing means that the first proposed point y with $F_\nu(y) > H/3$ has been formed. Taking $\gamma' = 0$ in the pathwise coordinate inequality gives

$$F_\nu(y) \leq F_\nu(x) + \frac{\epsilon_i^2}{2L}.$$

Before the first crossing of $H/3$, both x and w lie below $H/3$, so $\Delta_x + \Delta_w \leq 2H/3$ and the oracle bound gives

$$\mathbb{E}[|\epsilon_i|^2] \leq \frac{2\eta H}{3m}.$$

Summing the pathwise inequality up to the crossing, a crossing from the initial level $H/16$ within T steps requires

$$\sum_{t < T} \epsilon_{it}^2 \geq 2\bar{L} \left(\frac{H}{3} - \frac{H}{16} \right) = \frac{13}{24} \bar{L} H.$$

Since $T \leq 17m\bar{L}$, the expected stopped sum is at most $17m\bar{L} \cdot 2\eta H/(3m) = 34\eta\bar{L}H/3$, and Markov's inequality gives

$$\Pr[\text{a crossing before time } T] \leq \frac{34/3}{13/24} \eta = \frac{272}{13} \eta < 0.021 \quad (\eta = 10^{-3}).$$

On the event of no crossing and correct guards, every proposal is accepted, and the guarded process coincides with the unguarded one with x and w below $H/3 \leq H$ at every step. Therefore the killed-process version of the unguarded contraction, started from $V(z, z) = \frac{3}{2}\Delta_z$, gives

$$\mathbb{E}[\mathbb{1}[\text{safe}]V_T] \leq \frac{3}{2}e^{-4}\Delta_z.$$

This is the exact-oracle contribution to the phase estimate; the next subsection adds the error of finite exponential actions.

7.4 Finite exponential actions

The gradient oracle is now run on the surrogates. Apply the deflated construction of Section 7.2 to

$$\tilde{E} := \tilde{P}_x - \tilde{P}_w$$

in place of E , with the snapshot baseline $\text{tr}[\tilde{P}_w E_i]$, so that the estimator has conditional expectation $x_i - \xi_i + \nu a \text{tr}[\tilde{P}_x E_i]$. With $d_0 = 2n$ the order of the matrices,

$$\|\tilde{E} - (P_x - P_w)\|_1 \leq 2d_0\varepsilon_{\text{op}} \quad \text{and} \quad \|\tilde{P}_x - P_x\|_1 \leq d_0\varepsilon_{\text{op}}.$$

The bias of the estimator is therefore $\nu a |\text{tr}[(\tilde{P}_x - P_x)E_i]| \leq \nu a d_0 \varepsilon_{\text{op}}/R$, and the range-finder estimate applied to $\tilde{E}$ gives

$$\mathbb{E}[|\mathbf{e}_i|^2 \mid x, w] \leq \frac{C\nu^2 a^2}{ksR^2} \|\tilde{E}\|_1^2 + \left(\frac{\nu a d_0}{R} \varepsilon_{\text{op}}\right)^2.$$

Moreover, by the triangle inequality and $(u + v)^2 \leq 2u^2 + 2v^2$,

$$\|\tilde{E}\|_1^2 \leq 2\|P_x - P_w\|_1^2 + 8d_0^2 \varepsilon_{\text{op}}^2.$$

After increasing ks by a universal constant, the $\|P_x - P_w\|_1^2$ term is again at most $\frac{\eta}{m}(\Delta_x + \Delta_w)$. Put

$$\delta_g^2 := \min\{\eta H/(100m), \varepsilon_{\text{opt}}/(C_0 J m)\},$$

where C_0 is a sufficiently large universal constant, and choose

$$\varepsilon_{\text{op}} \leq \min\{\phi_0/(16d_0), R\delta_g/(4\nu a d_0), R\sqrt{ks}\delta_g/(C\nu a d_0)\}.$$

The first constraint is the guard's requirement with room, the second bounds the squared bias by $\delta_g^2/16$, and the third bounds the $\varepsilon_{\text{op}}^2$ term of the variance by $\delta_g^2/2$ for a suitable universal C . The resulting total mean-square error satisfies

$$\mathbb{E}[|\mathbf{e}_i|^2 \mid x, w] \leq \frac{\eta}{m}(\Delta_x + \Delta_w) + \delta_g^2.$$

The one-step Lyapunov calculation then acquires only the additive term $C\delta_g^2/\bar{L}$, and over $T \leq 17m\bar{L}$ iterations its total contribution is $O(m\delta_g^2)$. Choose C_0 so that the noisy killed-process estimate is

$$\mathbb{E}[\mathbb{1}[\text{safe}]V_T] \leq \frac{3}{2}e^{-4}\Delta_z + \frac{\varepsilon_{\text{opt}}}{1024J}.$$

The first term in the definition of δ_g changes the crossing bound by less than 0.005: the expected stopped sum gains at most $T\delta_g^2 \leq 17\eta\bar{L}H/100$, which raises the Markov bound by $\frac{17/100}{13/24}\eta < 0.005$.

Markov's inequality applied to the noisy estimate, with $\Delta_{x_T} \leq V_T$ and $\max\{3e^{-4}, 32/1024\} < 0.055$, now gives

$$\Pr[\mathbb{1}[\text{safe}] = 1 \text{ and } \Delta_{x_T} > \Delta_z/2 + \varepsilon_{\text{opt}}/(32J)] < 0.055.$$

Together with the crossing probability (below $0.021 + 0.005$) and the guard failure (0.01), one candidate phase returns a point with gap at most $\Delta_z/2 + \varepsilon_{\text{opt}}/(32J)$ with probability greater than $1 - 0.026 - 0.01 - 0.055 > 0.9$. Since all parameters multiplying ε_{opt} are polynomially bounded, the displayed degree r is polylogarithmic. A constant-relative trace guard uses only $O(\log(TJR_{\text{rep}}/p_0))$ fresh Gaussian probes, so its work is absorbed by the polylogarithmic factors.

7.5 Candidate selection and the certified wrapper

Candidate selection. Repeat each candidate phase independently R_{rep} times and pass the returned points, together with the old point z , to COMPAREVALUEEXACT with width $\varepsilon_{\text{opt}}/(128J)$. The procedure encloses every objective in an outward-rounded interval of width at most $\varepsilon_{\text{opt}}/(128J)$ using one dense, inverse-polynomial-accuracy matrix-function evaluation per point: $q(x)$ is evaluated exactly in the real-arithmetic model and $\Phi(x)$ is enclosed to absolute error at most $\varepsilon_{\text{opt}}/(256\nu J)$, and multiplying that interval by ν gives the stated width. These evaluations are deterministic given their accuracy; if a randomized trace certificate is used instead, allocate failure probability $p_0/(100J)$ among them. Among the candidates whose certified upper endpoint lies below the old point's certified lower endpoint, the procedure retains the one with the least upper endpoint; if there is none, it retains the old point. Thus the accepted phase-start objectives never increase. If $\Delta_z > \varepsilon_{\text{opt}}$, a successful candidate, with gap at most $\Delta_z/2 + \varepsilon_{\text{opt}}/(32J)$, is separated from the old point by more than $\Delta_z/2 - \varepsilon_{\text{opt}}/(32J) > 15\varepsilon_{\text{opt}}/32$, which exceeds the two certificate widths, and the retained point pays at most one further width. Hence, except with probability p_0/J (all R_{rep} candidates fail with probability $0.1^{R_{\text{rep}}} \leq p_0/(2J)$ for C_{rep} large, and the internal budgets add at most $p_0/(2J)$), either the old point already has gap at most ε_{opt} or the selected point satisfies

$$\Delta_{j+1} \leq \Delta_j/2 + \varepsilon_{\text{opt}}/(16J),$$

as $\frac{1}{32} + \frac{1}{128} < \frac{1}{16}$. Unrolling the recurrence over the J phases, with $2^{-J}F_0 \leq \varepsilon_{\text{opt}}/4$, shows that the returned point has objective gap at most ε_{opt} , except with probability $J \cdot p_0/J = p_0$. Snapshot refreshes are capped at a logarithmic multiple of their mean; a binomial tail bound absorbs the cap failure into p_0 .

The multiplier wrapper. Let ξ_Q be the coordinatewise box projection of ξ onto Q . Certify an interval of length at most $\tau/4$ for $\Phi(\xi_Q)$ (τ is fixed below). If it lies below 1, the box projection ξ_Q is feasible and is the required exact projection. Otherwise $\Phi(\xi_Q) \geq 1 - \tau/4$, and Slater's condition at the origin gives a multiplier $\nu_* \geq 0$ such that x_{ν_*} is the exact projection x_* , with $\Phi(x_*) \leq 1$ and $\nu_* > 0$ only if $\Phi(x_*) = 1$; the estimates below use only $\nu_* \geq 0$ and $\Phi(x_*) \leq 1$. Define

$$\Lambda := \frac{2q(0)}{1 - \phi_0}.$$

Lemma 7.4 (Lower bound on the accepted draw). *On the multiplier branch, $\|\xi\|_2 \geq \|\xi_Q\|_2 \geq 184$. Consequently $\Lambda = \|\xi\|_2^2/(1 - \phi_0) = \Omega(1)$, and $\Lambda = O(m)$ on the accepted draw.*

Proof. On this branch $\Phi(\xi_Q) \geq 1 - \tau/4$, so $h(\xi_Q) = b + a^{-1} \log \Phi(\xi_Q) \geq b - \tau/(2a)$, using $\log(1 - u) \geq -2u$ for $u \leq 1/2$. The retained bound $h(x) \leq \|B(x)\|/R + \chi$ with $\chi = 5 \cdot 10^{-7}$, the estimate

$\|B(x)\| \leq \sum_{i \in S} |x_i| =: \|x\|_1 \leq \sqrt{m} \|x\|_2$, and $|(\xi_Q)_i| \leq |\xi_i|$ (because $0 \in Q$) give

$$\|\xi\|_2 \geq \|\xi_Q\|_2 \geq \frac{(b - \chi - \tau/(2a))R}{\sqrt{m}} = (b - \chi - \tau/(2a))c_{\text{rad}}(\ell_{\text{asp}})\ell_{\text{asp}} \geq 184,$$

where the last step uses $c_{\text{rad}}(\ell_{\text{asp}})\ell_{\text{asp}} \geq 279$ for every $\ell_{\text{asp}} \geq 1$ by Table 2, $b - \chi = 1 - 10^{-6}$, and $\tau/(2a) \leq 10^{-9}$ (as $\tau \leq d^2 \leq 1/1024$ and $a \geq 2 \cdot 10^6$). Hence $\Lambda = \|\xi\|_2^2/(1 - \phi_0) \geq 184^2$, and on the accepted draw $\Lambda \leq \frac{4}{3}\|\xi\|_2^2 \leq \frac{4}{3}C_g m$, as $\phi_0 < 1/4$. $\square$

By Lemma 7.4, every tolerance below that involves Λ or $1/\underline{\nu}$ is inverse polynomial. Comparing x_* and x_Λ with the origin in F_{ν_*} and F_Λ respectively, using $q \geq 0$ and $\Phi(x_*) = 1$ when $\nu_* > 0$, gives

$$\nu_* \leq \frac{q(0)}{1 - \phi_0} = \Lambda/2, \quad \Phi(x_\Lambda) \leq \frac{1 + \phi_0}{2} < 1.$$

For $\nu > \mu > 0$, optimality of x_ν and x_μ gives

$$F_\nu(x_\nu) \leq F_\nu(x_\mu), \quad F_\mu(x_\mu) \leq F_\mu(x_\nu),$$

and adding the two inequalities yields $(\nu - \mu)(\Phi(x_\nu) - \Phi(x_\mu)) \leq 0$. Hence $\nu \mapsto \Phi(x_\nu)$ is nonincreasing. Adding the strong-convexity inequalities $F_\nu(x_*) \geq F_\nu(x_\nu) + \frac{1}{2}\|x_\nu - x_*\|_2^2$ and $F_{\nu_*}(x_\nu) \geq F_{\nu_*}(x_*) + \frac{1}{2}\|x_\nu - x_*\|_2^2$ gives $\|x_\nu - x_*\|_2^2 \leq (\nu - \nu_*)(\Phi(x_*) - \Phi(x_\nu))$, and the right side is at most $|\nu - \nu_*||\Phi(x_\nu) - 1|$ in both cases: for $\nu \geq \nu_*$ by monotonicity and $\Phi(x_*) \leq 1$, for $\nu < \nu_*$ because then $\Phi(x_*) = 1$. Thus

$$\|x_\nu - x_*\|_2^2 \leq |\nu - \nu_*||\Phi(x_\nu) - 1|.$$

For $\nu \geq \nu_*$, convexity of Φ and $\Phi(x_*) \leq 1$ further give, through $\Phi(x_*) - \Phi(x_\nu) \leq \|\nabla\Phi(x_*)\|_2\|x_* - x_\nu\|_2$ and $|\partial_i\Phi(x)| = a|\text{tr}[P_x E_i]| \leq a\Phi(x)/R$ for every i ,

$$\|x_\nu - x_*\|_2 \leq (\nu - \nu_*)G_*, \quad G_* := a\sqrt{m}/R.$$

Put

$$\nu_{\min} := \frac{\rho}{8G_*}, \quad \underline{\nu} := \min\{\nu_{\min}, \Lambda/2\}.$$

Solve first at $\underline{\nu}$ and certify an interval of length at most $\tau/4$ for $\Phi(x_{\underline{\nu}})$ as below. If the interval lies below 1, then $\Phi(x_{\underline{\nu}}) < 1$, which forces $\underline{\nu} \geq \nu_*$ (otherwise $\nu_* > 0$, $\Phi(x_*) = 1$, and monotonicity would give $\Phi(x_{\underline{\nu}}) \geq 1$), and the path estimate makes $x_{\underline{\nu}}$ $\rho/8$ -close to x_* . If the interval meets 1, the residual estimate Eq. (18) below gives distance at most $d/2$ directly. Otherwise $\Phi(x_{\underline{\nu}}) > 1 \geq \Phi(x_*)$, so $\underline{\nu} < \nu_* \leq \Lambda/2$ and $[\underline{\nu}, \Lambda]$ brackets ν_* .

Certified bisection. Put

$$H_{\max} := 16(q(0) + \Lambda\phi_0 + 1), \quad M_{\text{lev}} := H_{\max}/\underline{\nu}, \quad G_{\text{lev}} := G_*M_{\text{lev}},$$

$$\tau := \min\{d^2/(1 + \Lambda), d(1 - \phi_0)/(16\sqrt{m})\}.$$

At a tested multiplier $\nu \in [\underline{\nu}, \Lambda]$, solve to Euclidean error at most

$$\sigma := \min\{d/8, \tau/(16G_{\text{lev}})\}$$

by invoking the fixed-multiplier solver with

$$\varepsilon_{\text{opt}} \leq \sigma^2/2;$$

the 1-strong convexity of F_ν then gives $\|\hat{x} - x_\nu\|_2 \leq \sigma$ for the returned point $\hat{x}$. Evaluate $\Phi(\hat{x})$ with an outward-rounded absolute error at most $\tau/16$. The wrapper requires of the fixed-multiplier solver the interface hypothesis

$$F_\nu(\hat{x}) \leq H_{\max}.$$

The solver of this section meets it: $\hat{x}$ is an accepted guarded iterate or the old point, so $F_\nu(\hat{x}) \leq H = 16F_\nu(0) = 16(q(0) + \nu\phi_0) \leq H_{\max}$ as $\nu \leq \Lambda$; Sections 8 and 9 verify it for their own solvers. Since also $F_\nu(x_\nu) \leq F_\nu(0) \leq H_{\max}$, both endpoints of the segment from $\hat{x}$ to x_ν lie in the $H_{\max}$ -level set of F_ν , and convexity puts the entire segment there. Every point v of the segment therefore has $\nu\Phi(v) \leq H_{\max}$, so $\Phi(v) \leq M_{\text{lev}}$ and

$$\|\nabla\Phi(v)\|_2 \leq \frac{a\sqrt{m}}{R}\Phi(v) \leq G_*M_{\text{lev}} = G_{\text{lev}},$$

whence

$$|\Phi(\hat{x}) - \Phi(x_\nu)| \leq G_{\text{lev}}\sigma \leq \tau/16.$$

The evaluation error $\tau/16$ on each side and the transfer error $\tau/16$ on each side produce a certified interval for $\Phi(x_\nu)$ of length at most $\tau/4$. If the interval lies above or below 1, monotonicity chooses the correct half of the bracket. If it meets 1, then $|\Phi(x_\nu) - 1| \leq \tau/4$ and $|\nu - \nu_*| \leq \Lambda$, so the residual estimate gives

$$\|x_\nu - x_*\|_2^2 \leq |\nu - \nu_*| |\Phi(x_\nu) - 1| \leq \Lambda\tau/4 \leq d^2/4, \quad (18)$$

distance at most $d/2$, and $\hat{x}$ is retained. Otherwise stop when the multiplier bracket has length at most $d/(4G_*)$ and use its feasible upper endpoint: the path estimate gives distance at most $d/4$ from x_* , and the fixed-multiplier optimization error adds at most $\sigma \leq d/8$. The number of tested multipliers is $O(\log(\Lambda G_*/d)) = O(\log(n/\rho))$.

Exact feasibility. Let $\hat{x}$ be the approximate output of the wrapper and let $\bar{\Phi} \geq \Phi(\hat{x})$ be its certified upper bound. Define

$$\beta_0 := \begin{cases} 1, & \bar{\Phi} \leq 1, \\ (1 - \phi_0)/(\bar{\Phi} - \phi_0), & \bar{\Phi} > 1, \end{cases}$$

and compute a downward-rounded number $\beta \leq \beta_0$ with $0 \leq \beta_0 - \beta \leq d/(32\sqrt{m})$. Since $0, \hat{x} \in Q$ and Φ is convex with $\Phi(0) = \phi_0$, $\Phi(\beta\hat{x}) \leq (1 - \beta)\phi_0 + \beta\bar{\Phi} \leq 1$ for $\beta \leq \beta_0$. Moreover, $1 - \beta_0 = (\bar{\Phi} - 1)/(\bar{\Phi} - \phi_0) \leq (\bar{\Phi} - 1)/(1 - \phi_0)$ when $\bar{\Phi} > 1$, and $\|\hat{x}\|_2 \leq \sqrt{m}$ because every coordinate of Q has modulus below 1, so

$$(1 - \beta_0)\|\hat{x}\|_2 \leq \frac{\bar{\Phi} - 1}{1 - \phi_0}\sqrt{m}.$$

The accuracy choices above ensure $\bar{\Phi} - 1 \leq \tau/2$ (in the retained case $\Phi(x_\nu) \leq 1 + \tau/4$ and the two evaluation errors add $\tau/8$; in every other case $\bar{\Phi} \leq 1$), and the second term in the definition of τ makes the preceding display at most $d/16$. Downward rounding moves the point by at most $(\beta_0 - \beta)\|\hat{x}\|_2 \leq d/32$. The point $\beta\hat{x}$ is therefore exactly feasible for $\Phi \leq 1$, equivalently for $h \leq b$, and its distance to the exact projection is at most

$$\frac{d}{2} + \frac{d}{8} + \frac{d}{16} + \frac{d}{32} = \frac{23d}{32} < d = \frac{\rho}{4} \leq \rho,$$

the four terms being the residual or path bound, the fixed-multiplier error, the radial correction, and the rounding.

7.6 Correctness transfer, failure budget, and running time

Proof of Theorem 7.1. Step 1: correctness transfer. Every projection accepted by the wrapper is exactly feasible for $\Phi \leq 1$, equivalently for $h \leq b$, and lies within the retained accuracy ρ of the exact projection. Therefore the snapping proof, the success probability of each refreshed projection, the discrepancy constant 156000, and the cleanup of the proof of Theorem 6.1 are unchanged.

Step 2: the failure budget. Every randomized ingredient is capped and returns FAIL when its cap is reached, so an uncertified signing is never output, and the theorem's p is the sum of the cap failures. Allocate it as in the proof of Theorem 6.1: one fixed-multiplier solve fails with probability at most p_0 (Section 7.5), one projection call performs $O(\log(n/\rho))$ solves and one capped rejection sampler, and p_0 is chosen so that these budgets together stay within the 0.02 numerical allowance per attempt, so that an attempt succeeds with probability greater than 0.1; the per-phase trial caps $O(\log \log n + \log(1/p))$ over the $O(\log n)$ outer phases and the cleanup cap $O(\log(1/p))$ then bound the total failure probability by p . Fresh randomness at every call makes the conditional union bound valid under adaptivity, and p_0 enters the running time only through $\log(1/p_0)$, so the caps contribute the factor $\text{polylog}(1/p)$.

Step 3: cost of one selected-coordinate query. On every fixed-multiplier level, $H = O(m)$, $R^2 = \tilde{\Omega}(m)$, and $a = O(\log n)$, so $\bar{L} = \text{polylog}(n)$ and the sketch requirement $ks \geq Ca^2mH/(R^2\eta)$ is met with $ks = \tilde{O}(m)$. Take integers

$$k, s = \tilde{\Theta}(\sqrt{m}),$$

both at least 2 as the range-finder estimate requires, which is automatic since $m \geq N_0$ throughout the outer loop. One query applies polynomial functions of the two dense $n \times n$ matrices $B(x)$ and $B(w)$, through which the surrogates $\tilde{P}_x$ and $\tilde{P}_w$ act blockwise, to $\tilde{O}(\sqrt{m})$ vectors (the columns of Ω and Y , the probes z_j , and the guard probes), at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil))$; it then materializes the two dense matrices $\Pi\tilde{E} + \tilde{E}\Pi - \Pi\tilde{E}\Pi$ and $\sum_{j=1}^s (\tilde{E}_{\text{res}} z_j) z_j^\top$ from their rank- $O(k+s)$ factors, at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n))$, and reads the requested coordinate derivative as a Frobenius contraction of E_i with these two matrices, at cost $O(n^2)$. The cost of one query is

$$\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + n^2).$$

QR factorizations, Gaussian generation, projections onto the range of Y , and the maintenance of $B(x)$ after a coordinate update are covered by the n^2 term. A snapshot refresh forms $\tilde{P}_w$ densely by the retained approximate Hermitian diagonalization primitive and then all m coordinate derivatives, at cost

$$n^{\omega+o(1)} + O(mn^2).$$

Refreshes, dense objective evaluations for COMPAREVALUEEXACT, multiplier tests, and candidate repetitions are polylogarithmically many, and the dense inverse-polynomial-accuracy evaluations of Φ use the same primitive at cost $n^{\omega+o(1)}$ each. A fixed-multiplier solve makes $TJR_{\text{rep}} = \tilde{O}(m)$ queries and a projection call makes $O(\log(n/\rho))$ solves, so a projection at active size m costs

$$\tilde{O}(m\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m} \rceil) + m\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m} \rceil, n) + mn^2 + n^\omega).$$

Step 4: the sum over rounds. The active dimensions in the outer partial-coloring algorithm decrease geometrically, so $\sum_t m_t = O(n)$. Monotonicity of $\mathcal{T}_{\text{mat}}$ in each argument gives

$$\sum_t m_t \mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m_t} \rceil) \leq O(n\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil)),$$

and the transposed orientation satisfies the analogous bound. Also,

$$\sum_t m_t n^2 = O(n^3),$$

while all dense snapshots and certified value evaluations together cost $n^{\omega+o(1)}$. Input construction, snapping, the final signed sum, and the retained bounded-dimension cleanup cost at most $n^{3+o(1)}$.

Producing the dense $n \times n$ output gives $\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n) = \Omega(n^2)$, so $n\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n)$ absorbs the n^3 term, and since $\omega < 3$ it also absorbs n^ω . Thus the global cost is

$$n(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n))n^{o(1)} \text{polylog}(1/p).$$

By the definition of the rectangular exponent and the permutation invariance of $\omega(\cdot, \cdot, \cdot)$ recorded in Section 3, both $\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n} \rceil)$ and $\mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n} \rceil, n)$ are $n^{\omega(1, 1/2, 1)+o(1)}$, which is the second bound of the theorem, and the numerical exponent follows from $\omega(1, 1/2, 1) < 2.042776$. $\square$

8 The accelerated block algorithm

Theorem 7.1 spends one sketch of width $\tilde{\Theta}(\sqrt{m})$ on every coordinate query. This section lets ℓ coordinates share one sketch of width $\tilde{\Theta}(\sqrt{m/\ell})$, pays for the sharing with a block curvature of order ℓ , and recovers that price by Catalyst acceleration, which needs only $\tilde{O}(\sqrt{\ell})$ regularized subproblems. The speedup comes from a cheaper certified Gibbs oracle and from a safe-level invariant for the extrapolated centers, not from a new acceleration theorem: the Catalyst centers can leave the Gibbs level set on which the deflated oracle is analyzed, and an energy argument (Lemma 8.6) keeps every regularized inner solve on that level.

Theorem 8.1 (Accelerated block-coordinate dense implementation). *For dense input matrices and every $p \in (0, 1/2)$, a signing satisfying the conclusion of Theorem 5.1 can be found with failure probability at most p using*

$$n^{3+\gamma_\star+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model, where

$$\gamma_\star := \min_{0 \leq b \leq 1/2} \max\{\omega(1, b, 1) + b - 5/2, 1/2 - b\}.$$

Algorithms 4 and 5 record the implementation; their subroutines GUARDEDBLOCKTRIAL and COMPAREVALUE are Algorithms 6 and 7. The rest of the section proves Theorem 8.1.

Remark 8.2 (Reading the exponent). The dual matrix-multiplication exponent α_{dual} is defined in Section 3.2. If $\alpha_{\text{dual}} \geq 1/2$, then $b = 1/2$ gives $\gamma_\star = 0$ and the running time $n^{3+o(1)}$; this holds in particular if $\omega = 2$. The rectangular matrix-multiplication bounds of Alman, Duan, Vassilevska Williams, Xu, Xu and Zhou [ADVW⁺25, Table 1] give $\gamma_\star \leq 0.0183197999\dots$, so the exponent $3.018320 + o(1)$ improves on the exponent $3.042776 + o(1)$ of Theorem 7.1 and on the exponent $2 + \omega/2 + o(1) < 3.185670$ of Theorem 6.1, the latter with $\omega < 2.371339$ from the same source. The comparison with Theorem 6.1 holds for every value of ω : convexity of $b \mapsto \omega(1, b, 1)$, proved in the proof of Theorem 8.1 below, gives $\omega(1, 1/2, 1) \leq (\omega(1, 0, 1) + \omega(1, 1, 1))/2 = 1 + \omega/2$, hence $\gamma_\star \leq \omega(1, 1/2, 1) - 2 \leq (\omega - 2)/2$.

Algorithm 4 $n^{3+\gamma_\star+o(1)}$ polylog($1/p$) time algorithm for Matrix Spencer signing

- 1: **procedure** MAINBLOCK($A_1, \dots, A_n, n, p$) ▷ Theorem 8.1
 - 2: Choose $b \in [0, 1/2]$ attaining the exponent in Theorem 8.1 up to $o(1)$, and set $\ell \leftarrow \max\{2, \lceil n^{1-2b} \rceil\}$
 - 3: Run Algorithm 2 with the same fixed failure-budget allocation, multiplier certificates, snapping rule, and cleanup
 - 4: Replace every fixed-multiplier solve by ACCELERATEDMULTIPLIER; return its signing or FAIL
 - 5: **end procedure**
-

Algorithm 5 Accelerated fixed-multiplier solver

- 1: **procedure** ACCELERATEDMULTIPLIER($F_\nu, Q, m, \ell, a, R, \varepsilon_{\text{opt}}, \delta_\nu$) ▷ Lemmas 8.5 and 8.6
 - 2: **if** $m < Ca^2\ell^2$ **then**
 - 3: **return** COORDINATEMULTIPLIER($F_\nu, Q, m, a, R, \varepsilon_{\text{opt}}, \delta_\nu$) ▷ Algorithm 2
 - 4: **end if**
 - 5: $U \leftarrow F_\nu(0)$, $H \leftarrow 64U$, $\kappa \leftarrow \ell$, $\mu \leftarrow 1 + \kappa$, $\alpha \leftarrow 1/\sqrt{1 + \kappa}$, and $\vartheta \leftarrow (1 - \alpha)/(1 + \alpha)$
 - 6: $K \leftarrow \lceil C\alpha^{-1} \log(U/(\alpha^2\varepsilon_{\text{opt}})) \rceil$, $x_0 \leftarrow 0$, $y_0 \leftarrow 0$, and $h_0 \leftarrow 0$
 - 7: Partition $[m]$ into $q \leftarrow \lceil m/\ell \rceil$ fixed blocks of size at most ℓ
 - 8: $\widehat{L} \leftarrow \max\{1 + \kappa + ea^2H\ell/R^2, 2a(1 + \kappa)\ell/R\}$, $\gamma \leftarrow \mu/(q\widehat{L})$, and $T \leftarrow \lceil 16/\gamma \rceil$
 - 9: Choose $k, s \geq 2$ with $ks \geq Ca^2H\ell q/(\eta\mu R^2)$ and $k, s = \Theta(\sqrt{m/\ell})$
 - 10: **for** $j = 1, \dots, K$ **do**
 - 11: $G_j(x) \leftarrow F_\nu(x) + \kappa\|x - y_{j-1}\|_2^2/2$ and $\varepsilon_{\text{ph},j} \leftarrow c_0\alpha^3U(1 - \alpha/2)^j$
 - 12: $J_j \leftarrow \lceil \log_2(4H/\varepsilon_{\text{ph},j}) \rceil$, $\delta_{\text{ph}} \leftarrow \delta_\nu/(KJ_j)$, $R_{\text{rep}} \leftarrow \lceil C_{\text{rep}} \log(1/\delta_{\text{ph}}) \rceil$, and $\delta_{\text{tr}} \leftarrow \delta_{\text{ph}}/(3R_{\text{rep}})$
 - 13: $\delta_{g,j}^2 \leftarrow \min\{\eta\mu H/(Cm), \mu\varepsilon_{\text{ph},j}/(CJ_jm)\}$, and choose ε_{op} by the three-term bound of Section 8.1 with $\delta_g = \delta_{g,j}$
 - 14: $\mathcal{P}_j \leftarrow (H, a, R, \ell, q, \widehat{L}, \gamma, T, k, s, \delta_{g,j}, \varepsilon_{\text{op}}, c_{\text{prox}}\varepsilon_{\text{ph},j}/J_j, \delta_{\text{tr}})$ and $z \leftarrow x_{j-1}$
 - 15: **for** $r = 1, \dots, J_j$ **do**
 - 16: $\mathcal{C} \leftarrow \{z\}$
 - 17: **for** R_{rep} independent repetitions **do**
 - 18: $x \leftarrow \text{GUARDEDBLOCKTRIAL}(G_j, z, \mathcal{P}_j)$; if the trial did not abort, set $\mathcal{C} \leftarrow \mathcal{C} \cup \{x\}$
 - 19: **end for**
 - 20: $z \leftarrow \text{COMPAREVALUE}(\mathcal{C}, \varepsilon_{\text{ph},j}/(128J_j), \delta_{\text{ph}}/3)$
 - 21: **end for**
 - 22: $x_j \leftarrow z$, $h_j \leftarrow x_j - x_{j-1}$, and $y_j \leftarrow x_j + \vartheta h_j$
 - 23: **end for**
 - 24: **return** x_K
 - 25: **end procedure**
-

8.1 Setup and constants

Fix an active set of size m , a multiplier $\nu > 0$, and one projection subproblem, and retain the notation of the proof of Theorem 7.1: the box Q , whose coordinate intervals have length ε_{box} , the matrices $B(x)$, $C(x)$, D_x and E_i , the parameters R , a and b , the Gibbs trace Φ , its value $\phi_0 = \Phi(0)$,

the dimension $d_0 = 2n$, and the accepted Gaussian draw ξ . Put

$$F(x) := \frac{1}{2}\|x - \xi\|_2^2 + \nu\Phi(x), \quad \Phi(x) := \text{tr}[\exp(D_x)], \quad x \in Q.$$

The function F is 1-strongly convex; write x_* for its unique minimizer over Q and $F_* := F(x_*)$. The solver starts at $x_0 := 0$. Put

$$U := F(x_0) = O(m), \quad H := 64U,$$

where the bound $U = O(m)$ holds uniformly over every multiplier tested by the certified search of Section 7. The level H is four times the level $16F_\nu(0) = 16U$ of Section 7: Catalyst subproblems start at objective values up to $3U$ (Lemma 8.6), and every phase must start below $H/16$. The unsubscripted letter C denotes a sufficiently large universal constant whose value may change from line to line.

Blocks and the regularized objective. Fix a block size $\ell \geq 2$. The accelerated solver runs only at active sizes $m \geq Ca^2\ell^2$; smaller active sizes use the selected-coordinate solver of Theorem 7.1. Section 6 fixes $R = c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ with $\ell_{\text{asp}} \geq 1$ and $c_{\text{rad}}(\ell_{\text{asp}})$ bounded below by a universal constant, so $R^2 \geq c_R m$ for a universal constant $c_R > 0$, and $m \geq Ca^2\ell^2$ gives

$$\ell \leq \min\{m, R/(4a)\}$$

once $C \geq 16/c_R$. Partition $[m]$ into $q := \lceil m/\ell \rceil$ fixed blocks of size at most ℓ ; then $q\ell \leq m + \ell \leq 2m$. Put

$$\kappa := \ell, \quad \mu := 1 + \kappa, \quad \alpha := \frac{1}{\sqrt{1 + \kappa}}, \quad \vartheta := \frac{1 - \alpha}{1 + \alpha}.$$

For a center $y \in \mathbb{R}^m$ define on Q the regularized objective

$$G(x) := F(x) + \frac{\kappa}{2}\|x - y\|_2^2.$$

It is μ -strongly convex; write x_G for its minimizer over Q and $\Delta_v := G(v) - G(x_G)$ for the gap at $v \in Q$. The center enters only the quadratic part of G , which is evaluated exactly; every matrix-function calculation below is one for F .

Block curvature, step, and phase length. Put

$$\widehat{L} := \max\{1 + \kappa + ea^2H\ell/R^2, 2a(1 + \kappa)\ell/R\}, \quad \gamma := \frac{\mu}{q\widehat{L}}, \quad T := \lceil 16/\gamma \rceil.$$

Since $\widehat{L} \geq \mu$ and $q \geq 1$, $\gamma \leq 1$ and $T\gamma \leq 17$. Because $\ell \leq R/(4a)$, the second term of $\widehat{L}$ is at most $\mu/2$; because $H = O(m)$, $R^2 \geq c_R m$ and $a = O(\log n)$, the first term is $\widetilde{O}(1 + \kappa)$. Hence $\widehat{L}/(1 + \kappa) = \widetilde{O}(1)$.

Accuracy, bias floor, sketch width, and proximal tolerance. A regularized subproblem is solved to a target accuracy $\varepsilon_{\text{ph}} \in (0, U]$ in

$$J := \lceil \log_2(4H/\varepsilon_{\text{ph}}) \rceil$$

phases. Let $\eta \leq 1/4$ be a sufficiently small universal constant. The oracle carries a bias floor $\delta_g > 0$ with

$$\delta_g^2 \leq \min\{\eta\mu H/(Cm), \mu\varepsilon_{\text{ph}}/(CJm)\},$$

a range sketch of width k and s residual probes, $k, s \geq 2$, with

$$ks \geq C \frac{a^2 H \ell q}{\eta \mu R^2}, \quad k = s = \tilde{\Theta}(\sqrt{m/\ell}),$$

and the polynomial precision

$$\varepsilon_{\text{op}} \leq \min\{\phi_0/(16d_0), R\delta_g/(4\nu ad_0), R\sqrt{ks}\delta_g/(C\nu ad_0)\}$$

of Section 7, whose first term also forces $\nu d_0 \varepsilon_{\text{op}} \leq \nu \phi_0/16 \leq U/16 = H/1024$. The scalar searches inside the proximal steps are stopped at residuals ρ_t with

$$\sum_{t < T} \rho_t \leq c_{\text{prox}} \min\{H, \varepsilon_{\text{ph}}/J\}$$

in every phase, where $c_{\text{prox}} \leq 1/8$ is a sufficiently small universal constant.

Catalyst schedule and failure budgets. Catalyst runs K regularized subproblems with centers y_{j-1} and target accuracies

$$\varepsilon_{\text{ph},j} := c_0 \alpha^3 U (1 - \alpha/2)^j, \quad K := \lceil C \alpha^{-1} \log(U/(\alpha^2 \varepsilon_{\text{opt}})) \rceil,$$

where $c_0 \leq 1/4$ is a sufficiently small universal constant and ε_{opt} is the accuracy requested by the multiplier wrapper of Section 7; subproblem j uses $\varepsilon_{\text{ph}} = \varepsilon_{\text{ph},j}$, $J = J_j$ and $\delta_g = \delta_{g,j}$ in the displays above. The wrapper gives the fixed-multiplier solve a failure budget δ_ν . Each of the J_j phases of subproblem j receives

$$\delta_{\text{ph}} := \frac{\delta_\nu}{K J_j},$$

so that the phase budgets sum to δ_ν . A phase runs $R_{\text{rep}} := \lceil C_{\text{rep}} \log(1/\delta_{\text{ph}}) \rceil$ independent trials, gives each trial the budget

$$\delta_{\text{tr}} := \frac{\delta_{\text{ph}}}{3 R_{\text{rep}}},$$

spends $\delta_{\text{ph}}/3$ on its objective comparison, and keeps $\delta_{\text{ph}}/3$ for the event that no trial improves. A trial spends $\delta_{\text{tr}}/2$ on its snapshot cap and $\delta_{\text{tr}}/(2T)$ on each of its T guard certificates.

8.2 Block oracle variance

Conditional on the history, one block query selects a block uniformly, draws one fresh range sketch and one fresh family of residual probes, and evaluates the finite-action oracle of Section 7 simultaneously for all coordinates of the block. The sketch and the probes are shared only among the coordinates of that query.

Lemma 8.3 (Block oracle variance). *Let G be a regularized objective as in Section 8.1, let $\mathcal{F}_t$ contain the history before query t , and let the iterate x and the snapshot w be $\mathcal{F}_t$ -measurable points of Q with $G(x) \leq H$ and $G(w) \leq H$. Conditional on $\mathcal{F}_t$, select the block I_t uniformly from the fixed partition, draw a fresh range sketch of width k and s fresh residual probes, and let $\mathbf{e}_{I_t}$ be the error vector of the finite-action oracle with snapshot w and polynomial precision ε_{op} on the coordinates of I_t . Then*

$$\mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t] \leq \frac{\eta \mu}{q} (\Delta_x + \Delta_w) + \ell \delta_g^2. \quad (19)$$

Proof. Put $P_v := \exp(D_v)$ for $v \in Q$. The proof of Theorem 7.1 bounds the mean-square error of one coordinate before converting the trace norm into optimizer gaps; under the snapshot convention, that raw estimate, applied to the deflated difference $\tilde{P}_x - \tilde{P}_w$ of the polynomial operators, and its finite-action perturbation bound give, uniformly for every fixed coordinate i ,

$$\mathbb{E}[|\mathbf{e}_i|^2 \mid \mathcal{F}_t] \leq \frac{C\nu^2 a^2}{ksR^2} \|P_x - P_w\|_1^2 + \delta_g^2.$$

The bound does not depend on which other coordinates share the sketch. Since the squared Euclidean norm is the sum of the coordinate squares, correlations between the coordinates of a block do not enter, and summing over all m coordinates gives

$$\sum_{i=1}^m \mathbb{E}[|\mathbf{e}_i|^2 \mid \mathcal{F}_t] \leq \frac{C\nu^2 a^2 m}{ksR^2} \|P_x - P_w\|_1^2 + m\delta_g^2.$$

Step 1: the trace norm is controlled by the gaps. Write

$$\mathcal{D}(P\|P') := \text{tr}[P(\log P - \log P') - P + P']$$

for the extended matrix relative entropy of positive definite P and P' . The generalized Pinsker inequality of Bansal, George, Ghosh, Sikora and Zheng [BGG⁺25, Theorem 1.3], in the form used in Section 7, bounds $\|P - P'\|_1^2$ by $4 \max\{\text{tr}[P], \text{tr}[P']\} \mathcal{D}(P\|P')$. Since $x \mapsto \log P_x = D_x$ is affine, $\mathcal{D}(P_{x_G}\|P_v)$ is the Bregman divergence of Φ between v and its base point x_G ; the quadratic parts of F and G add nonnegative Bregman terms, and the constrained optimality condition $\langle \nabla G(x_G), v - x_G \rangle \geq 0$ removes the linear term of D_G . Hence, for every $v \in Q$,

$$\nu \mathcal{D}(P_{x_G}\|P_v) \leq D_F(v, x_G) \leq D_G(v, x_G) \leq \Delta_v.$$

The level hypothesis gives $\nu \text{tr}[P_v] = \nu \Phi(v) \leq G(v) \leq H$ for $v = x$ and $v = w$, and also for the reference point $v = x_G$, because $G(x_G) \leq G(w) \leq H$. The trace-norm triangle inequality through P_{x_G} , the bound $(\alpha_1 + \alpha_2)^2 \leq 2\alpha_1^2 + 2\alpha_2^2$, and Pinsker at the trace level H/ν therefore yield

$$\nu^2 \|P_x - P_w\|_1^2 \leq 8H(\Delta_x + \Delta_w).$$

Step 2: averaging over the block. Every coordinate lies in exactly one block, and each block is selected with probability $1/q$, so $\mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t]$ is $1/q$ times the coordinate sum. Combining the coordinate sum with Step 1 and $m/q \leq \ell$ gives

$$\mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t] \leq \frac{Ca^2 \ell H}{ksR^2} (\Delta_x + \Delta_w) + \ell \delta_g^2.$$

The sketch condition $ks \geq Ca^2 H \ell q / (\eta \mu R^2)$ makes the first coefficient at most $\eta \mu / q$, which is Eq. (19). The finite-action term δ_g^2 per coordinate comes from the fixed polynomial operator $\tilde{P}_v = q_r(D_v)^2$ of Section 7 under the three-term bound on ε_{op} ; the Catalyst center changes only the exactly evaluated quadratic part of G and does not enter this calculation. $\square$

8.3 The guarded block step

For a block I and a point $x \in Q$, the trust ball is the set of directions d with $\text{supp}(d) \subseteq I$ and $\|d\|_2 \leq R/(a\sqrt{\ell})$, and $\mathcal{D}_I(x)$ is the set of such d with $x + d \in Q$. Given the oracle output $\widehat{\nabla}_I G(x)$, put

$$\phi_I(u) := \langle \widehat{\nabla}_I G(x), u \rangle + \frac{\widehat{L}}{2} \|u\|_2^2, \quad u \in \mathcal{D}_I(x),$$

and write $\mathbf{e}_I := \widehat{\nabla}_I G(x) - \nabla_I G(x)$ for the oracle error. A displacement $d \in \mathcal{D}_I(x)$ has three-point residual $\rho \geq 0$ if

$$\phi_I(u) \geq \phi_I(d) + \frac{\widehat{L}}{2} \|u - d\|_2^2 - \rho \quad \text{for every } u \in \mathcal{D}_I(x);$$

the exact minimizer of ϕ_I over the convex set $\mathcal{D}_I(x)$ has residual 0, which is the strong three-point inequality.

Lemma 8.4 (Guarded block step). *Let G , $\widehat{L}$ and γ be as in Section 8.1, let $x \in Q$ with $G(x) \leq H$, and let I be a block.*

- (i) *For every d in the trust ball of I at x with $x + d \in Q$ and every $t \in [0, 1]$, $\Phi(x + td) \leq e\Phi(x)$ and*

$$\frac{d^2}{dt^2} G(x + td) \leq (1 + \kappa + ea^2 H \ell / R^2) \|d\|_2^2 \leq \widehat{L} \|d\|_2^2.$$

- (ii) *Let $d \in \mathcal{D}_I(x)$ have three-point residual ρ_t and put $x^+ := x + d$. For every $d' \in \mathcal{D}_I(x)$,*

$$G(x^+) \leq G(x) + \langle \nabla_I G(x), d' \rangle + \frac{\widehat{L}}{2} \|d'\|_2^2 + \frac{1}{2\widehat{L}} \|\mathbf{e}_I\|_2^2 + \rho_t,$$

and, when $I = I_t$ is selected uniformly conditional on a history $\mathcal{F}_t$ containing x ,

$$\mathbb{E}[\Delta_{x^+} \mid \mathcal{F}_t] \leq (1 - \gamma) \Delta_x + \frac{1}{2\widehat{L}} \mathbb{E}[\|\mathbf{e}_{I_t}\|_2^2 \mid \mathcal{F}_t] \quad (20)$$

holds with the additive term ρ_t on the right. Such a d is computed by a scalar bisection stopped at the tolerance ρ_t in $O(\ell \log(1/\rho_t))$ arithmetic operations, all scales being polynomial in n .

- (iii) *Suppose $G(x) \leq H/3$ and let $x^+ = x + d$ with d in the trust ball of I at x . Given $\delta \in (0, 1)$, a certificate $\widehat{G}(x^+)$ computed from $\lceil 8 \log(1/\delta) \rceil$ fresh batches of a universal constant number of Gaussian probes satisfies, except with probability δ ,*

$$G(x^+) \leq \widehat{G}(x^+) \leq G(x^+) + \frac{9H}{512},$$

and $9H/512 < H/48$. Its cost is $O(\log(1/\delta))$ polynomial actions of degree r on vectors, that is, $\widetilde{O}(n^2 \log(1/\delta))$ arithmetic operations.

Proof. Step 1: block smoothness on the trust ball. For d in the trust ball, $\|E_i\| \leq 1/R$ and $\sum_{i \in I} |d_i| \leq \sqrt{\ell} \|d\|_2$ give

$$\|D_{x+d} - D_x\| = a \left\| \sum_{i \in I} d_i E_i \right\| \leq a \sqrt{\ell} \|d\|_2 / R \leq 1.$$

Hence along the segment the matrix exponent moves by at most one in operator norm, $D_{x+td} \preceq D_x + I_{2n}$, and monotonicity of the trace exponential gives $\Phi(x + td) \leq e\Phi(x)$; in particular $\nu\Phi(x + td) \leq eG(x) \leq eH$. Write $E_d := \sum_{i \in I} d_i E_i$, so that $\|E_d\| \leq \sqrt{\ell} \|d\|_2 / R$. The integral formula for the second derivative of $t \mapsto \text{tr}[\exp(D_{x+td})]$ and the weighted arithmetic-geometric mean bound in the proof of Theorem 7.1, with E_d in place of E_i , give

$$\frac{d^2}{dt^2} \Phi(x + td) \leq a^2 \|E_d\|^2 \Phi(x + td) \leq \frac{a^2 \ell \|d\|_2^2}{R^2} e\Phi(x).$$

Multiplying by ν , using $\nu\Phi(x) \leq H$, and adding the second derivative $(1 + \kappa)\|d\|_2^2$ of the two quadratic terms of G gives the first inequality of (i); the second inequality is the first term in the definition of $\widehat{L}$.

Step 2: the proximal step and the comparison displacement. A displacement with the required residual is inexpensive: for a scalar multiplier $\lambda \geq 0$, clip each coordinate of $-\widehat{\nabla}_I G(x)/(\widehat{L}(1 + \lambda))$ to the corresponding box interval and bisect on λ until the ball constraint is met. The residual ρ_t is certified by the same bisection at logarithmic additional cost, and every scale involved is polynomial in n , so the cost is $O(\ell \log(1/\rho_t))$ at tolerance ρ_t . Let first d be the exact minimizer of ϕ_I over $\mathcal{D}_I(x)$ and $d' \in \mathcal{D}_I(x)$. By (i), $G(x + d) - G(x) \leq \langle \nabla_I G(x), d \rangle + \frac{\widehat{L}}{2}\|d\|_2^2 = \phi_I(d) - \langle \mathbf{e}_I, d \rangle$; the strong three-point inequality at $u = d'$ and Young's inequality $\langle \mathbf{e}_I, d' - d \rangle \leq \frac{\widehat{L}}{2}\|d' - d\|_2^2 + \frac{1}{2\widehat{L}}\|\mathbf{e}_I\|_2^2$ then give

$$\begin{aligned} G(x + d) - G(x) &\leq \phi_I(d) - \langle \mathbf{e}_I, d \rangle \\ &\leq \langle \nabla_I G(x), d' \rangle + \frac{\widehat{L}}{2}\|d'\|_2^2 + \langle \mathbf{e}_I, d' - d \rangle - \frac{\widehat{L}}{2}\|d' - d\|_2^2 \\ &\leq \langle \nabla_I G(x), d' \rangle + \frac{\widehat{L}}{2}\|d'\|_2^2 + \frac{1}{2\widehat{L}}\|\mathbf{e}_I\|_2^2. \end{aligned}$$

If d has residual ρ_t instead, the residual inequality replaces the strong three-point inequality, and the last two lines acquire the additive term ρ_t ; this is the pathwise inequality of (ii). The calculation uses only the second moment of $\mathbf{e}_I$; it does not assume that the finite-action error is unbiased.

The comparison displacement

$$d' := \frac{\mu}{\widehat{L}}(x_G - x)_I$$

is admissible. It stays in the box because $\mu/\widehat{L} \leq 1$, so each coordinate of $x + d'$ is a convex combination of the corresponding coordinates of x and x_G . Every coordinate interval of Q has length $\varepsilon_{\text{box}} < 2$, so

$$\|d'\|_2 \leq \frac{2\mu\sqrt{\ell}}{\widehat{L}} \leq \frac{R}{a\sqrt{\ell}}$$

by the second term in the definition of $\widehat{L}$. Each block is selected with probability $1/q$, and the blockwise pieces of d' assemble to $\frac{\mu}{\widehat{L}}(x_G - x)$, so the conditional expectation of the two comparison terms is

$$\frac{\mu}{q\widehat{L}}(\langle \nabla G(x), x_G - x \rangle + \frac{\mu}{2}\|x_G - x\|_2^2) \leq -\gamma\Delta_x,$$

where the inequality is μ -strong convexity of G between x and x_G . Taking conditional expectations in the pathwise inequality and subtracting $G(x_G)$ gives Eq. (20), with the additive term ρ_t when the search is stopped early.

Step 3: the additive guard. Suppose $G(x) \leq 5H/16$ and let $x^+ = x + d$ with d in the trust ball. The quadratic part $\frac{1}{2}\|x^+ - \xi\|_2^2 + \frac{\kappa}{2}\|x^+ - y\|_2^2$ of $G(x^+)$ is computed exactly, so only $\nu\Phi(x^+)$ needs a certificate. By (i), $\nu\Phi(x^+) \leq e\nu\Phi(x) \leq eG(x) \leq eH/3$. Let $\tilde{P}_{x^+} = q_r(D_{x^+})^2$ be the polynomial operator of Section 7 on the interval enlarged by one; then $\|\tilde{P}_{x^+} - P_{x^+}\| \leq \varepsilon_{\text{op}}$, so $|\text{tr}[\tilde{P}_{x^+}] - \Phi(x^+)| \leq d_0\varepsilon_{\text{op}}$, and Section 8.1 gives $\nu d_0\varepsilon_{\text{op}} \leq H/1024$. Hence

$$\nu \text{tr}[\tilde{P}_{x^+}] \leq \frac{e}{3}H + \frac{H}{1024} < 0.91H.$$

For a standard Gaussian vector z and the positive semidefinite matrix $M := \tilde{P}_{x^+}$, $\mathbb{E}[z^\top M z] = \text{tr}[M]$ and $\mathbb{V}[z^\top M z] = 2\|M\|_F^2 \leq 2(\text{tr}[M])^2$, and $z^\top M z = \|q_r(D_{x^+})z\|_2^2$ is one polynomial action. Average

$z^\top M z$ over a batch of 2^{17} fresh probes: by Chebyshev's inequality the batch mean lies within $\text{tr}[M]/128$ of $\text{tr}[M]$ with probability at least $3/4$, because $2 \cdot 128^2/2^{17} = 1/4$. Let $\hat{\tau}$ be the median of $\lceil 8 \log(1/\delta) \rceil$ independent batch means; Hoeffding's inequality for the number of batches outside this window gives

$$|\hat{\tau} - \text{tr}[\tilde{P}_{x^+}]| \leq \text{tr}[\tilde{P}_{x^+}]/128$$

except with probability δ . Define

$$\hat{G}(x^+) := \frac{1}{2}\|x^+ - \xi\|_2^2 + \frac{\kappa}{2}\|x^+ - y\|_2^2 + \nu\hat{\tau} + \frac{9H}{1024}.$$

On the good event the statistical error is $\nu|\hat{\tau} - \text{tr}[\tilde{P}_{x^+}]| \leq \nu \text{tr}[\tilde{P}_{x^+}]/128 \leq H/128$ and the bias is $\nu|\text{tr}[\tilde{P}_{x^+}] - \Phi(x^+)| \leq H/1024$; their sum $9H/1024$ is the shift built into $\hat{G}$. Hence $\nu\Phi(x^+) \leq \nu\hat{\tau} + 9H/1024$, which is $G(x^+) \leq \hat{G}(x^+)$, and $\nu\hat{\tau} \leq \nu\Phi(x^+) + 9H/1024$, which is $\hat{G}(x^+) \leq G(x^+) + 18H/1024 = G(x^+) + 9H/512$. Finally $9/512 < 1/48$ because $9 \cdot 48 = 432 < 512$. The certificate uses $2^{17}\lceil 8 \log(1/\delta) \rceil$ polynomial actions, each $O(rn^2)$ arithmetic operations with r polylogarithmic, since $D_{x^+} = aC(x^+) - abI_{2n}$ is available from the maintained $B(x^+)$. $\square$

8.4 One guarded trial and the regularized subproblem

A trial (Algorithm 6) starts at a point z with $G(z) \leq H/16$, sets $x := w := z$, and makes T block queries. Each query selects a block, evaluates the oracle of Lemma 8.3, computes the proximal step of Lemma 8.4(ii), and guards the proposal with the certificate of Lemma 8.4(iii); the trial aborts if $\hat{G}(x^+) > H/3$. Otherwise it moves to x^+ and, with probability γ and independently of the oracle and guard randomness, refreshes the snapshot to $w := x^+$; it also aborts once the number of refreshes exceeds $C_{\text{snap}} \log(2/\delta_{\text{tr}})$. Every refresh rebuilds the dense snapshot data. The ideal trial coupled to it uses the same block selections, oracle randomness and refresh coins, never aborts, and is killed at the first crossing of the level $5H/16$.

Lemma 8.5 (One guarded trial and the regularized subproblem). *Let $G, \varepsilon_{\text{ph}} \in (0, U]$, $J, \delta_g, k, s, \varepsilon_{\text{op}}$ and the residuals ρ_t be as in Section 8.1.*

- (i) *Let $z \in Q$ with $G(z) \leq H/16$ and $\delta_{\text{tr}} \in (0, 1)$. GUARDEDBLOCKTRIAL($G, z, \mathcal{P}$) makes T block queries and at most $C_{\text{snap}} \log(2/\delta_{\text{tr}}) + 1$ dense snapshot constructions, and every point it accepts satisfies $G \leq H/3$. Except on an event of probability at most δ_{tr} (a wrong guard certificate or a reached snapshot cap), it agrees with the coupled ideal trial, which, with probability at least a universal constant $c_\star > 0$, is not killed and returns a point x_T with*

$$\Delta_{x_T} \leq \frac{\Delta_z}{2} + \frac{\varepsilon_{\text{ph}}}{32J}.$$

- (ii) *Let z be as in (i) and $\delta_{\text{ph}} \in (0, 1)$. Run $R_{\text{rep}} = \lceil C_{\text{rep}} \log(1/\delta_{\text{ph}}) \rceil$ independent trials from z with budgets $\delta_{\text{tr}} = \delta_{\text{ph}}/(3R_{\text{rep}})$, and let z^+ be the output of COMPAREVALUE on z and the returned points, with width $\varepsilon_{\text{ph}}/(128J)$ and budget $\delta_{\text{ph}}/3$. Except with probability δ_{ph} ,*

$$\Delta_{z^+} \leq \frac{\Delta_z}{2} + \frac{\varepsilon_{\text{ph}}}{16J}, \quad G(z^+) \leq G(z) + \frac{\varepsilon_{\text{ph}}}{128J}.$$

- (iii) *Let $z_0 \in Q$ with $G(z_0) \leq 3U$. Starting from z_0 , J phases as in (ii) return an ε_{ph} -minimizer of G over Q except with probability $J\delta_{\text{ph}}$, using $\tilde{O}(q\hat{L}/\mu)$ block queries and $\tilde{O}(1)$ dense snapshot constructions, both counts capped before the phases start; the hidden factors are logarithmic in $1/\varepsilon_{\text{ph}}$ and $1/\delta_{\text{ph}}$.*

Proof. Step 1: the Lyapunov potential and the snapshot cap. Put

$$V(x, w) := \Delta_x + \frac{1}{2}\Delta_w.$$

Suppose the level hypothesis of Lemma 8.3 holds for x and w . Eqs. (19) and (20) give $\mathbb{E}[\Delta_{x^+} | \mathcal{F}_t] \leq A_0\Delta_x + B_0\Delta_w + \ell\delta_g^2/(2\widehat{L}) + \rho_t$ with

$$A_0 := 1 - (1 - \eta/2)\gamma, \quad B_0 := \eta\gamma/2,$$

because $\eta\mu/(2q\widehat{L}) = \eta\gamma/2$. The refresh rule gives $\mathbb{E}[\Delta_{w^+} | \mathcal{F}_t, x^+] = \gamma\Delta_{x^+} + (1 - \gamma)\Delta_w$. Hence the coefficient of Δ_x in $\mathbb{E}[V(x^+, w^+) | \mathcal{F}_t]$ is $(1 + \gamma/2)A_0 \leq 1 - \gamma/4$ and that of Δ_w is $(1 + \gamma/2)B_0 + (1 - \gamma)/2 \leq (1 - \gamma/4)/2$, both because $\eta \leq 1/4$ and $\gamma \leq 1$. Therefore

$$\mathbb{E}[V(x^+, w^+) | \mathcal{F}_t] \leq (1 - \gamma/4)V(x, w) + C\ell\delta_g^2/\widehat{L} + C\rho_t.$$

Thus $T = \lceil 16/\gamma \rceil$ block queries reduce the potential by a constant factor, up to $O(m\delta_g^2/\mu) + O(\sum_{t < T} \rho_t)$ additive error, since $q\ell \leq 2m$. The expected number of snapshot refreshes in these T queries is at most $T\gamma \leq 17$. The refresh count is stochastically dominated by a binomial random variable with mean at most 17, so a Chernoff bound makes the probability of exceeding $C_{\text{snap}} \log(2/\delta_{\text{tr}})$ refreshes at most $\delta_{\text{tr}}/2$ after increasing C_{snap} . Each guard certificate uses the budget $\delta_{\text{tr}}/(2T)$ in Lemma 8.4(iii), so all T certificates are correct except with probability $\delta_{\text{tr}}/2$. The two failure sources together have probability at most δ_{tr} , as claimed in (i).

Step 2: the coupling and the crossing estimate. Index the queries by $t = 0, \dots, T - 1$ and define

$$\tau := \min\{t + 1 : G(x_{t+1}) > 5H/16\},$$

with $\tau := T + 1$ if there is no crossing; the ideal trial is killed at τ . Before τ , both the iterate and the snapshot, which is a past iterate, are below $5H/16$, so the level hypotheses of Lemmas 8.3 and 8.4 hold, and Lemma 8.4(i) keeps every intervening matrix-exponential path below the H trace level because $5e/16 < 1$. On the event that no crossing occurs and all certificates are correct, every proposal has true value at most $5H/16$, so its certificate is at most $5H/16 + 9H/512 = 169H/512 < H/3$, the implemented trial never aborts on a guard, and the two trials agree. Conversely, an accepted proposal satisfies $G(x^+) \leq \widehat{G}(x^+) \leq H/3$, which is the level claim of (i). The Lyapunov calculation is not conditioned on the global guard-success event; certificate failures are added afterwards by the union bound.

Taking $d' = 0$ in the pathwise inequality of Lemma 8.4(ii) gives

$$G(x^+) \leq G(x) + \frac{1}{2\widehat{L}}\|\mathbf{e}_I\|_2^2 + \rho_t.$$

Since the allocated proximal errors sum to at most $c_{\text{prox}}H \leq H/8$, a crossing from $H/16$ to $5H/16$ requires

$$\sum_{t < T} \mathbb{1}[\tau > t] \|\mathbf{e}_{I_t}\|_2^2 \geq \frac{1}{4}\widehat{L}H.$$

The event $\{\tau > t\}$ is $\mathcal{F}_t$ -measurable, and on it $\Delta_{x_t}, \Delta_{w_t} \leq 5H/16$. By Tonelli's theorem and Eq. (19), with $T \leq 17q\widehat{L}/\mu$ and $q\ell \leq 2m$, the expectation of the stopped nonnegative sum is at most

$$C\eta\widehat{L}H + Cm\widehat{L}\delta_g^2/\mu.$$

Markov's inequality and the first term in the bound on δ_g^2 make the crossing probability at most $C\eta$, a small universal constant once η is small enough.

Step 3: the killed potential. Put $A_t := \{\tau > t\}$ for $0 \leq t \leq T$ and define

$$W_t := \mathbb{1}[A_t]V(x_t, w_t), \quad B := C\ell\delta_g^2/\widehat{L}.$$

The event A_t is $\mathcal{F}_t$ -measurable, $A_{t+1} \subseteq A_t$, and $V \geq 0$, so $W_{t+1} \leq \mathbb{1}[A_t]V(x_{t+1}, w_{t+1})$ and the one-step inequality of Step 1 applies on A_t :

$$\mathbb{E}[W_{t+1} \mid \mathcal{F}_t] \leq \mathbb{1}[A_t]((1 - \gamma/4)V(x_t, w_t) + B + C\rho_t).$$

Iterating, with $(1 - \gamma/4)^T \leq e^{-4}$ because $T\gamma \geq 16$ and a geometric sum for the bias terms,

$$\mathbb{E}[W_T] \leq e^{-4}V(z, z) + 4B/\gamma + C \sum_{t < T} \rho_t.$$

Since $q\ell \leq 2m$, the bias floor and the proximal tolerance make $4B/\gamma + C \sum_{t < T} \rho_t \leq \varepsilon_{\text{ph}}/(512J)$ after increasing the universal constant in the upper bound for δ_g and decreasing c_{prox} . Markov's inequality therefore bounds the probability of the event

$$A_T \cap \{V(x_T, w_T) > V(z, z)/3 + \varepsilon_{\text{ph}}/(32J)\}$$

by $\max\{3e^{-4}, 1/16\} = 1/16$. The stopped noise-energy estimate of Step 2 separately bounds $\Pr[A_T^c]$. Hence the ideal trial has no crossing and satisfies $V(x_T, w_T) \leq V(z, z)/3 + \varepsilon_{\text{ph}}/(32J)$ with probability at least $c_\star := 1 - 1/16 - C\eta > 0$. Since $V(z, z) = 3\Delta_z/2$ and $\Delta_{x_T} \leq V(x_T, w_T)$, this is the improvement in (i); the additive term $\varepsilon_{\text{ph}}/(32J)$ covers the near-target case.

Step 4: one phase. The R_{rep} trials are independent. COMPAREVALUE encloses G at each of the at most $R_{\text{rep}} + 1$ candidates in a fresh outward-rounded interval of width at most $\varepsilon_{\text{ph}}/(128J)$, computed as in Section 7: the quadratic part exactly, Φ by one dense inverse-polynomial-accuracy matrix-function evaluation with its certificate, each interval correct except with probability $\delta_{\text{ph}}/(3(R_{\text{rep}} + 1))$; it returns a candidate z^+ with the smallest upper endpoint. Three events cover a failure of the phase: some trial has a wrong certificate or reaches its cap, of probability at most $R_{\text{rep}}\delta_{\text{tr}} = \delta_{\text{ph}}/3$ by (i); no ideal trial succeeds, of probability at most $(1 - c_\star)^{R_{\text{rep}}} \leq \delta_{\text{ph}}/3$ once C_{rep} is large; some comparison interval is wrong, of probability at most $\delta_{\text{ph}}/3$. Off these events the implemented trials agree with their ideal versions, some returned point x^{best} has $\Delta_{x^{\text{best}}} \leq \Delta_z/2 + \varepsilon_{\text{ph}}/(32J)$, and every interval is correct, so $G(z^+)$ is at most the upper endpoint of x^{best} , hence at most $G(x^{\text{best}}) + \varepsilon_{\text{ph}}/(128J)$, which gives $\Delta_{z^+} \leq \Delta_z/2 + \varepsilon_{\text{ph}}/(32J) + \varepsilon_{\text{ph}}/(128J) \leq \Delta_z/2 + \varepsilon_{\text{ph}}/(16J)$. Because z is among the candidates, $G(z^+)$ is also at most the upper endpoint of z , hence at most $G(z) + \varepsilon_{\text{ph}}/(128J)$.

Step 5: the phases. Let $z_0, \dots, z_J$ be the phase starts and put $\Delta_r := \Delta_{z_r}$. By the second bound of (ii), $G(z_r) \leq G(z_0) + r\varepsilon_{\text{ph}}/(128J) \leq 3U + \varepsilon_{\text{ph}}/128 \leq 3U + U/128 < 4U = H/16$ for $r \leq J$, so every phase starts below $H/16$ and (ii) applies to it; the phase failures add to $J\delta_{\text{ph}}$. On the complementary event $\Delta_{r+1} \leq \Delta_r/2 + \varepsilon_{\text{ph}}/(16J)$ for every $r < J$, and unrolling the recurrence gives

$$\Delta_J \leq 2^{-J}\Delta_0 + \frac{\varepsilon_{\text{ph}}}{8J} \leq \frac{\varepsilon_{\text{ph}}}{4} + \frac{\varepsilon_{\text{ph}}}{8} < \varepsilon_{\text{ph}},$$

because $\Delta_0 \leq G(z_0) \leq 3U \leq H$ and $2^{-J} \leq \varepsilon_{\text{ph}}/(4H)$. The block queries number $JR_{\text{rep}}T = \widetilde{O}(q\widehat{L}/\mu)$, and the dense snapshot constructions number $JR_{\text{rep}}(C_{\text{snap}} \log(2/\delta_{\text{tr}}) + 1) + J(R_{\text{rep}} + 1) = \widetilde{O}(1)$, with logarithms in $1/\varepsilon_{\text{ph}}$ and $1/\delta_{\text{ph}}$; both are fixed before the phases start. All polynomial actions use one fixed positive semidefinite operator per oracle call, and every guard and every comparison uses fresh probes, so the conditional calculations above remain valid under adaptivity. $\square$

Algorithm 6 One guarded block trial

```
1: procedure GUARDEDBLOCKTRIAL( $G, z, \mathcal{P}$ ) ▷ Lemma 8.5(i);  
    $\mathcal{P} = (H, a, R, \ell, q, \widehat{L}, \gamma, T, k, s, \delta_g, \varepsilon_{\text{op}}, \rho_{\text{tot}}, \delta_{\text{tr}})$   
2:    $x \leftarrow z, w \leftarrow z$ , build the dense snapshot data at  $w$ , and set the refresh counter to 0  
3:   Fix residual tolerances  $\rho_0, \dots, \rho_{T-1} \geq 0$  with  $\sum_{t < T} \rho_t \leq \rho_{\text{tot}}$   
4:   for  $t = 0, \dots, T - 1$  do  
5:     Select a block  $I_t$  uniformly from the fixed partition, independently of the past  
6:     Draw a fresh range sketch of width  $k$  and  $s$  fresh residual probes, and compute the  
       finite-action oracle  $\widehat{\nabla}_{I_t} G(x)$  with snapshot  $w$  at polynomial precision  $\varepsilon_{\text{op}}$   
7:     Compute  $d \in \mathcal{D}_{I_t}(x)$  with three-point residual  $\rho_t$  by scalar bisection, and put  $x^+ \leftarrow x + d$   
8:     Compute the additive guard certificate  $\widehat{G}(x^+)$  with fresh probes and budget  $\delta_{\text{tr}}/(2T)$   
9:     if  $\widehat{G}(x^+) > H/3$  then  
10:      return ABORT  
11:     end if  
12:      $x \leftarrow x^+$ ; with probability  $\gamma$ , independently of everything else, set  $w \leftarrow x$ , rebuild the  
       snapshot data, and increment the refresh counter  
13:     if the refresh counter exceeds  $C_{\text{snap}} \log(2/\delta_{\text{tr}})$  then  
14:       return ABORT  
15:     end if  
16:   end for  
17:   return  $x$   
18: end procedure
```

Algorithm 7 Certified objective comparison

```
1: procedure COMPAREVALUE( $\mathcal{C}, \theta, \delta$ ) ▷ Lemma 8.5(ii);  $\mathcal{C}$  contains the phase start  $z$ ,  $\theta$  is the  
   interval width,  $\delta$  the failure budget  
2:   for each  $u \in \mathcal{C}$  do  
3:     Evaluate the quadratic part of  $G$  at  $u$  exactly, and enclose  $\Phi(u)$  by one dense inverse-  
       polynomial-accuracy matrix-function evaluation with certified absolute error at most  $\theta/(2\nu)$ ,  
       failing with probability at most  $\delta/|\mathcal{C}|$   
4:     Record the outward-rounded interval  $[\underline{G}(u), \overline{G}(u)]$  of width at most  $\theta$   
5:   end for  
6:   return a  $u \in \mathcal{C}$  with the smallest upper endpoint  $\overline{G}(u)$   
7: end procedure
```

8.5 Catalyst acceleration and the safe-level invariant

Start with $y_0 := x_0 = 0$ and $h_0 := 0$. For $j \geq 1$, let x_j be an $\varepsilon_{\text{ph},j}$ -minimizer over Q of

$$G_j(x) := F(x) + \frac{\kappa}{2} \|x - y_{j-1}\|_2^2,$$

computed by the phases of Lemma 8.5 started at x_{j-1} , and set

$$h_j := x_j - x_{j-1}, \quad y_j := x_j + \vartheta h_j.$$

These are the strongly convex Catalyst parameters of Lin, Mairal and Harchaoui [LMH18]: the usual quantity $q_{\text{cat}} := 1/(1 + \kappa)$ satisfies $\alpha = \sqrt{q_{\text{cat}}}$, and the momentum coefficient is ϑ .

Lemma 8.6 (Catalyst acceleration and the safe-level invariant). *Let $\kappa, \alpha, \vartheta, \varepsilon_{\text{ph},j}$ and K be as in Section 8.1, and suppose that $G_j(x_j) - \min_{x \in Q} G_j(x) \leq \varepsilon_{\text{ph},j}$ for $j = 1, \dots, K$.*

(i) *For every $j \geq 0$, $G_{j+1}(x_j) \leq 3U < 4U = H/16$: every regularized subproblem starts below $H/16$, and every accepted inner iterate satisfies $\nu\Phi \leq H$.*

(ii) *With $\varrho := \alpha/2$,*

$$F(x_K) - F_* \leq \frac{8}{(\alpha - \varrho)^2} (1 - \varrho)^{K+1} U;$$

hence the choice of K in Section 8.1, which is $O(\alpha^{-1} \log(U/(\alpha^2 \varepsilon_{\text{opt}}))) = \tilde{O}(\sqrt{1 + \kappa})$, gives $F(x_K) - F_ \leq \varepsilon_{\text{opt}}$.*

(iii) *If $\nu \leq \Lambda$ is a multiplier tested by the wrapper of Section 7 and $\varepsilon_{\text{opt}} \leq \sigma^2/2$ is the accuracy it requests, then*

$$F(x_K) \leq F_* + \varepsilon_{\text{opt}} \leq U + \varepsilon_{\text{opt}} < H_{\max},$$

where $H_{\max} = 16(q(0) + \Lambda\phi_0 + 1)$ is the level of that wrapper.

Proof. Step 1: the rate. Catalyst is applied with the box constraint Q as the composite term. Theorem 3 and the proof of Proposition 5 of Lin, Mairal and Harchaoui [LMH18, Theorem 3 and Proposition 5] are homogeneous in an upper bound for the initial gap and in the prescribed absolute-error sequence. Rerunning that proof with $U \geq F(x_0) - F_*$ in place of the unknown initial gap, and using $c_0 \alpha^3 U \leq 2U/9$ for sufficiently small c_0 , gives the bound in (ii). Since $8/(\alpha - \varrho)^2 = 32\alpha^{-2}$ and $(1 - \varrho)^{K+1} \leq \exp(-(K+1)\alpha/2)$, an objective error at most ε_{opt} follows once $K \geq 2\alpha^{-1} \log(32U/(\alpha^2 \varepsilon_{\text{opt}}))$, which the choice $K = \lceil C\alpha^{-1} \log(U/(\alpha^2 \varepsilon_{\text{opt}})) \rceil$ of Section 8.1 satisfies for a large enough C , because $U \geq 1$ and $\varepsilon_{\text{opt}} < 1$ on every solve the wrapper requests. Catalyst is invoked only after each regularized subproblem has been solved to its prescribed absolute accuracy, so its outer theorem needs the pathwise inexactness event of Lemma 8.5, not an unconditional linear-rate recurrence for the killed inner process.

Step 2: the energy. The generic theorem does not supply the localization. Define

$$\mathcal{E}_j := F(x_j) - F_* + \frac{\kappa\vartheta}{2} \|h_j\|_2^2.$$

Let x_j^* be the exact minimizer of G_j over Q and put $e_j := x_j - x_j^*$. Since G_j is $(1 + \kappa)$ -strongly convex and x_j^* is its constrained minimizer, $G_j(x_j) \geq G_j(x_j^*) + \frac{1+\kappa}{2} \|e_j\|_2^2$, and $G_j(x_j) - G_j(x_j^*) \leq \varepsilon_{\text{ph},j}$ gives

$$\|e_j\|_2^2 \leq \frac{2\varepsilon_{\text{ph},j}}{1 + \kappa}.$$

The same strong convexity between x_j^* and x_{j-1} , followed by $G_j(x_j) \leq G_j(x_j^*) + \varepsilon_{\text{ph},j}$, gives

$$G_j(x_j) + \frac{1 + \kappa}{2} \|x_j^* - x_{j-1}\|_2^2 \leq G_j(x_{j-1}) + \varepsilon_{\text{ph},j}.$$

Substitute $y_{j-1} = x_{j-1} + \vartheta h_{j-1}$, so that $G_j(x_j) = F(x_j) + \frac{\kappa}{2} \|h_j - \vartheta h_{j-1}\|_2^2$ and $G_j(x_{j-1}) = F(x_{j-1}) + \frac{\kappa\vartheta^2}{2} \|h_{j-1}\|_2^2$, write $x_j^* - x_{j-1} = h_j - e_j$, expand the squared norms, and bound the cross term by

$$\kappa\vartheta \langle h_j, h_{j-1} \rangle \leq \frac{\kappa\vartheta}{2} (\|h_j\|_2^2 + \|h_{j-1}\|_2^2).$$

Subtracting F_* from both sides yields, with $M := 2\kappa(1 - \vartheta) + 1$,

$$\mathcal{E}_j + \frac{M}{2} \|h_j\|_2^2 - (1 + \kappa) \langle h_j, e_j \rangle + \frac{1 + \kappa}{2} \|e_j\|_2^2 \leq \mathcal{E}_{j-1} + \varepsilon_{\text{ph},j}.$$

Completing the square in h_j gives

$$\frac{M}{2} \|h_j\|_2^2 - (1 + \kappa) \langle h_j, e_j \rangle \geq -\frac{(1 + \kappa)^2}{2M} \|e_j\|_2^2.$$

Dropping the nonnegative term $\frac{1+\kappa}{2} \|e_j\|_2^2$ and inserting the bound on $\|e_j\|_2^2$ proves

$$\mathcal{E}_j \leq \mathcal{E}_{j-1} + \left(1 + \frac{1 + \kappa}{2\kappa(1 - \vartheta) + 1}\right) \varepsilon_{\text{ph},j}. \quad (21)$$

Here $1 + \kappa = \alpha^{-2}$ and, since $1 - \vartheta = 2\alpha/(1 + \alpha)$ and $\kappa = (1 - \alpha)(1 + \alpha)/\alpha^2$, $M = 4/\alpha - 3$; the coefficient of $\varepsilon_{\text{ph},j}$ in Eq. (21) is therefore $1 + 1/(4\alpha - 3\alpha^2) \leq 2/\alpha$, because $3\alpha^2 - 10\alpha + 7 = (3\alpha - 7)(\alpha - 1) \geq 0$ for $\alpha \leq 1$. Since $\mathcal{E}_0 = F(x_0) - F_* \leq U$ and $\sum_{r \geq 1} \varepsilon_{\text{ph},r} \leq c_0 \alpha^3 U \cdot 2/\alpha = 2c_0 \alpha^2 U$,

$$\mathcal{E}_j \leq \mathcal{E}_0 + \frac{2}{\alpha} \sum_{r=1}^j \varepsilon_{\text{ph},r} \leq 2U$$

when $c_0 \leq 1/4$. Finally, $\vartheta^2 \leq \vartheta$ gives

$$G_{j+1}(x_j) - F_* = F(x_j) - F_* + \frac{\kappa \vartheta^2}{2} \|h_j\|_2^2 \leq \mathcal{E}_j \leq 2U,$$

and $F_* \leq F(x_0) = U$ gives $G_{j+1}(x_j) \leq 3U < 4U = H/16$, the invariant of (i). Because $G_j \geq F \geq \nu\Phi$ and every accepted inner iterate has $G_j \leq H/3$ (Lemma 8.5(i)), every accepted inner iterate satisfies $\nu\Phi \leq H$; the sketch width and the matrix-polynomial degree therefore remain those of the $O(m)$ level, rather than growing with κ .

Step 3: the hand-off. By (ii), $F(x_K) \leq F_* + \varepsilon_{\text{opt}} \leq F(x_0) + \varepsilon_{\text{opt}} = U + \varepsilon_{\text{opt}}$. The wrapper of Section 7 tests multipliers $\nu \leq \Lambda$ only, since its first solve is at $\underline{\nu} \leq \Lambda/2$ and its bisection stays inside $[\underline{\nu}, \Lambda]$; hence $U = q(0) + \nu\phi_0 \leq q(0) + \Lambda\phi_0$. Its accuracy satisfies $\varepsilon_{\text{opt}} \leq \sigma^2/2 \leq 2^{-17} < 1$, because $\sigma \leq d/8$ and $d \leq 1/32$ there. Therefore $U + \varepsilon_{\text{opt}} < q(0) + \Lambda\phi_0 + 1 = H_{\text{max}}/16 \leq H_{\text{max}}$, which is (iii). $\square$

8.6 Arithmetic cost

Sharing a sketch reduces the work per block query, while Catalyst increases the number of dense snapshot constructions. We count both contributions before summing over the geometrically decreasing active dimensions.

Lemma 8.7 (Arithmetic cost). *Let $\ell \geq 2$ and let C be the universal constant of Section 8.1.*

(i) *At an active size $m \geq Ca^2\ell^2$, one selected-block query costs*

$$\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m/\ell} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m/\ell} \rceil, n) + \ell n^2)$$

arithmetic operations, one dense snapshot construction costs $n^{\omega+o(1)} + O(mn^2)$, one fixed-multiplier solve uses $\tilde{O}(m/\sqrt{\ell})$ selected-block queries and $\tilde{O}(\sqrt{\ell})$ dense snapshot constructions, and, with

$$M_{m,\ell} := \mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{m/\ell} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{m/\ell} \rceil, n),$$

one projection at active size m costs

$$\tilde{O}\left(\frac{m}{\sqrt{\ell}}(M_{m,\ell} + \ell n^2) + \sqrt{\ell}(n^{\omega} + mn^2)\right). \quad (22)$$

(ii) Using the accelerated solver at active sizes $m \geq Ca^2\ell^2$ and the selected-coordinate solver of Theorem 7.1 below that size, the accelerated rounds of the outer algorithm cost

$$\tilde{O}\left(\frac{n}{\sqrt{\ell}}(\mathcal{T}_{\text{mat}}(n, n, \lceil \sqrt{n/\ell} \rceil) + \mathcal{T}_{\text{mat}}(n, \lceil \sqrt{n/\ell} \rceil, n)) + n^3\sqrt{\ell} + n^\omega\sqrt{\ell}\right) \quad (23)$$

and the late rounds cost $\tilde{O}(n^2\ell^3 + n^\omega)$.

(iii) For $b \in [1/3, 1/2]$ and $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$, the projections of all rounds together cost

$$n^{3+\max\{\omega(1,b,1)+b-5/2, 1/2-b\}+o(1)}$$

arithmetic operations.

Proof. Step 1: one block query and one snapshot. By Section 8.1, $H = O(m)$, $R^2 \geq c_R m$ with $R^2 = \tilde{\Theta}(m)$, and $\ell q/\mu = \Theta(m/\ell)$, so the sketch condition is met by

$$ks = \tilde{\Theta}(m/\ell), \quad k = s = \tilde{\Theta}(\sqrt{m/\ell}), \quad k, s \geq 2,$$

with integer ceilings understood. Stack the range vectors and residual probes into a batch of width

$$r_{\text{query}} := \tilde{O}(\sqrt{m/\ell})$$

and apply every matrix polynomial to the whole batch, at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, n, r_{\text{query}}))$. One rectangular product materializes the dense low-rank correction and another the symmetrized residual moment matrix, each at cost $\tilde{O}(\mathcal{T}_{\text{mat}}(n, r_{\text{query}}, n))$. Thereafter every requested coordinate is one Frobenius contraction with these two dense matrices, costing $O(n^2)$ per coordinate, so sharing the sketch across the block introduces neither an extra probe factor nor an extra block factor. The ℓn^2 term covers the block contractions, the update of $B(x)$, the proximal step of Lemma 8.4(ii), and the guard of Lemma 8.4(iii). Although the Catalyst center y_j is generally dense, it occurs only in the quadratic term of G_j , whose gradient is evaluated coordinatewise, so no matrix image of y_j is needed and no $O(mn^2)$ matrix reconstruction is incurred at a block query. This is the query cost in (i). A dense snapshot construction evaluates the matrix exponential at the snapshot to inverse-polynomial accuracy with the approximate Hermitian-diagonalization primitive of Theorem 6.1 and forms the full gradient, at cost

$$n^{\omega+o(1)} + O(mn^2).$$

Step 2: one projection. By Lemma 8.5(iii) and $\hat{L}/(1+\kappa) = \tilde{O}(1)$, each regularized subproblem uses $\tilde{O}(q) = \tilde{O}(m/\ell)$ block queries and $\tilde{O}(1)$ dense snapshot constructions; by Lemma 8.6(ii), Catalyst uses $\tilde{O}(\sqrt{\ell})$ subproblems. Hence one fixed-multiplier solve uses $\tilde{O}(m/\sqrt{\ell})$ selected-block queries and $\tilde{O}(\sqrt{\ell})$ dense snapshot constructions, and its cost is $\frac{m}{\sqrt{\ell}}(M_{m,\ell} + \ell n^2) + \sqrt{\ell}(n^\omega + mn^2)$ up to polylogarithmic factors. The certified multiplier search, the polynomial precision, the guard amplification, the objective comparisons, and the allocation of the failure budget over all inner solves multiply this by polylogarithmic factors only, which gives Eq. (22).

Step 3: all rounds. For a fixed block size ℓ , the accelerated solver is used while $m \geq Ca^2\ell^2$, where $\ell \leq R/(4a)$ by Section 8.1, and the selected-coordinate solver of Theorem 7.1 once $m < Ca^2\ell^2$. The active dimensions decrease geometrically and have sum $O(n)$. Monotonicity of $\mathcal{T}_{\text{mat}}$ and Eq. (22) give Eq. (23) for the accelerated rounds: the query terms sum to $\tilde{O}(\frac{n}{\sqrt{\ell}}(M_{n,\ell} + \ell n^2))$, and the $O(\log n)$ rounds contribute $\tilde{O}(\sqrt{\ell}(n^\omega + n^3))$ through their snapshots. The last term of Eq. (23) is dominated by $n^3\sqrt{\ell}$ because $\omega < 3$. In the late rounds the largest active dimension is $\tilde{O}(\ell^2)$ and

the width in the rectangular products is $\tilde{O}(\sqrt{m}) = \tilde{O}(\ell)$; using classical multiplication for these thin products, the geometric sum of their cost is

$$\tilde{O}(n^2\ell^3 + n^\omega).$$

Step 4: the exponent. Put $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$ for $b \in [1/3, 1/2]$; then $n/\sqrt{\ell} = O(n^{1/2+b})$ and $\lceil \sqrt{n/\ell} \rceil = O(n^b)$. Permutation invariance of $\omega(\cdot, \cdot, \cdot)$, recorded in Section 3, makes both rectangular products in Eq. (23) cost $n^{\omega(1,b,1)+o(1)}$, so the accelerated rounds cost $n^{3+\max\{\omega(1,b,1)+b-5/2, 1/2-b\}+o(1)}$. The late rounds cost $n^{5-6b+o(1)} + n^{\omega+o(1)} \leq n^{3+o(1)}$ because $b \geq 1/3$ and $\omega < 3$, and $3 \leq 3 + (1/2 - b)$, so they do not add a term to the maximum. $\square$

8.7 Proof of Theorem 8.1

Proof. Correctness. Algorithm 4 runs the body, projection accuracy, certified multiplier search, snapping rule, and outer partial-coloring construction of Theorems 6.1 and 7.1, and replaces every fixed-multiplier solve by Algorithm 5, which at active sizes $m < Ca^2\ell^2$ is the solver of Theorem 7.1. Consider one accelerated solve at a tested multiplier $\nu \leq \Lambda$, with the accuracy $\varepsilon_{\text{opt}} \leq \sigma^2/2$ and the budget δ_ν of the wrapper. By Lemma 8.6(i), every regularized subproblem starts at $G_j(x_{j-1}) \leq 3U$, and $\varepsilon_{\text{ph},j} \leq c_0U \leq U$, so Lemma 8.5(iii) applies to subproblem j with $\varepsilon_{\text{ph}} = \varepsilon_{\text{ph},j}$ and the phase budget $\delta_{\text{ph}} = \delta_\nu/(KJ_j)$: the subproblem fails with probability at most $J_j\delta_{\text{ph}} = \delta_\nu/K$, and the K subproblems fail with probability at most δ_ν in total. On the complementary event every inexactness condition $G_j(x_j) - \min_Q G_j \leq \varepsilon_{\text{ph},j}$ holds pathwise, and Lemma 8.6(ii) and (iii) give

$$F(x_K) \leq F_* + \varepsilon_{\text{opt}} \leq U + \varepsilon_{\text{opt}} < H_{\max}.$$

This is the interface hypothesis stated by the multiplier wrapper of Section 7: an ε_{opt} -accurate fixed-multiplier solution in the $H_{\max}$ -level set. Hence the wrapper's certified bisection, residual estimate, and exact feasibility correction apply as written, every accepted projection is exactly feasible and within the same distance of the exact projection as in Section 7, and the partial-coloring success probability, the snapping argument, and the discrepancy constant of Theorem 5.1 are unchanged. Distributing the failure budget over all active rounds, multiplier tests, fixed-multiplier solves, and cleanup trials, with fresh randomness at every call, contributes the factor $\text{polylog}(1/p)$.

Running time. By Lemma 8.7(iii), for every $b \in [1/3, 1/2]$ the block size $\ell := \max\{2, \lceil n^{1-2b} \rceil\}$ makes all projections cost $n^{3+\max\{\omega(1,b,1)+b-5/2, 1/2-b\}+o(1)}$; input construction, the bounded-dimension cleanup, and all remaining work cost at most $n^{3+o(1)}$, and the failure allocation contributes $\text{polylog}(1/p)$. The choice $b = 1/2$ and the known bound $\omega(1, 1/2, 1) < 2.042776$ first show that $\gamma_* \leq 0.042776$. Since the second term in the defining maximum is $1/2 - b$, a minimizer may therefore be taken with $b \geq 0.457224$, which lies in $[1/3, 1/2]$. Minimizing over such b gives the exponent $3 + \gamma_* + o(1)$ of the theorem.

The numerical value. The function $b \mapsto \omega(1, b, 1)$ is convex on $[0, 1]$: it is nondecreasing, because $\mathcal{T}_{\text{mat}}(n, \lceil n^b \rceil, n)$ is nondecreasing in b , and it is midpoint convex, because algorithms for $\langle n, n^{b_1}, n \rangle$ and $\langle n, n^{b_2}, n \rangle$ compose by tensor product into an algorithm for $\langle n^2, n^{b_1+b_2}, n^2 \rangle$, so that $2\omega(1, (b_1 + b_2)/2, 1) \leq \omega(1, b_1, 1) + \omega(1, b_2, 1)$; a nondecreasing midpoint-convex function is convex. With the bounds

$$\omega(1, 0.4, 1) < 2.009280, \quad \omega(1, 0.5, 1) < 2.042776$$

of Alman, Duan, Vassilevska Williams, Xu, Xu and Zhou [ADVW⁺25, Table 1], convex interpolation gives

$$\omega(1, b, 1) \leq 2.009280 + 0.33496(b - 0.4)$$

for $0.4 \leq b \leq 0.5$. Thus the first term in the defining maximum is at most

$$1.33496b - 0.624704,$$

whereas the second term is $1/2 - b$. Balancing these terms gives

$$b_\star := \frac{1.124704}{2.33496} = 0.4816802001\dots$$

and therefore

$$\gamma_\star \leq \frac{1}{2} - b_\star = 0.0183197999\dots,$$

the value quoted in Remark 8.2. □

9 The cubic-time algorithm

This section removes the exponent loss in Theorem 8.1 by tracking the dense Gibbs matrix recursively: a random refresh keeps the tracking error from accumulating, and the variance of every nonrefresh increment is proportional to the squared length of the coordinate step, which the negative step-energy term of coordinate descent pays for. The change is in the fixed-multiplier solver, not in the partial-coloring procedure around it, which is that of Section 7.

Theorem 9.1 (Recursive Gibbs-tracking implementation). *There is an algorithm, given by procedure MAINCUBIC in Algorithm 8, that for every $p \in (0, 1/2)$ finds a signing satisfying the conclusion of Theorem 5.1 with failure probability at most p using*

$$n^{3+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model.

The algorithm uses no structure of the A_i beyond the standing hypotheses (symmetric, $\|A_i\| \leq 1$) and reads them entry by entry; for dense input, whose size is n^3 (Section 3.2), its running time is therefore within an $n^{o(1)} \text{polylog}(1/p)$ factor of the time needed to read the input.

Proof. Lemma 9.4 and independent repetition give Lemma 9.3, including its accuracy and certificate claims. Lemma 9.2 then gives the required signing and failure probability, and Lemma 9.9 gives the claimed running time. □

The rest of the section states the three procedures (Algorithms 8–10) and proves the four lemmas the proof composes.

9.1 The signing algorithm

Algorithm 8 records the outer signing procedure. It changes only the fixed-multiplier solver in the partial-coloring procedure of Section 7 (Algorithm 2).

Algorithm 8 $n^{3+o(1)}$ polylog($1/p$) time algorithm for Matrix Spencer signing

```

1: procedure MAINCUBIC( $A_1, \dots, A_n, n, p$ ) ▷ Theorem 9.1
2:    $y \leftarrow 0$ ,  $S \leftarrow [n]$ , and  $N_0 \leftarrow 3 \cdot 10^8$ 
3:   Allocate disjoint randomized-call, certification, and cleanup budgets from  $p$ ; denote the
   cleanup budget by  $p_{\text{clean}}$ 
4:   while  $|S| \geq N_0$  do
5:      $m \leftarrow |S|$ ,  $u \leftarrow \log(n/m)$ ,  $\ell_{\text{asp}} \leftarrow \max\{1, u\}$ , and  $R \leftarrow c_{\text{rad}}(\ell_{\text{asp}})\sqrt{m}\ell_{\text{asp}}$ 
6:     Form  $Q$ ,  $a$ ,  $b$ ,  $\rho$ ,  $G_*$ , and  $\nu_{\min}$  as in Section 7
7:     repeat
8:       Sample an accepted  $\xi \sim \mathcal{N}(0, I_m)$  and keep it fixed through all multiplier tests
9:       Form the multiplier bracket  $[\underline{\nu}, \Lambda]$  and all certified tolerances from the accepted  $\xi$  as
       in Section 7
10:      if the coordinatewise box projection of  $\xi$  is certifiably feasible then
11:         $x \leftarrow$  the box projection
12:      else
13:        while the certified multiplier stopping rule of Section 7 has not fired do
14:          Choose the next tested  $\nu \in [\underline{\nu}, \Lambda]$  and construct  $F_\nu$ 
15:           $\hat{x}_\nu \leftarrow \text{RECURSIVEMULTIPLIER}(F_\nu, Q, m, n, a, R, \varepsilon_{\text{opt}}, \delta_\nu)$  ▷ Algorithm 9
16:          Update the bracket only from the certified interval for  $\Phi(\hat{x}_\nu)$ , or retain  $\hat{x}_\nu$  if
          that interval meets 1
17:        end while
18:         $x \leftarrow$  the retained candidate or the certified feasible bracket endpoint
19:      end if
20:      Apply the downward-rounded radial correction and snap within  $2\rho$ ; call the result  $\bar{x}$ 
21:      until more than  $\delta m$  coordinates of  $\bar{x}$  are snapped, or the prescribed trial cap is reached
22:      if the trial cap is reached without success then
23:        return FAIL
24:      end if
25:       $y_i \leftarrow y_i + 2\bar{x}_i/\varepsilon_{\text{box}}$  for every  $i \in S$ 
26:       $S \leftarrow \{i \in [n] : |y_i| < 1\}$ 
27:    end while
28:    return CERTIFIEDCLEANUP( $A_1, \dots, A_n, y, S, p_{\text{clean}}$ ) ▷ Algorithm 3
29: end procedure

```

The outer algorithm uses each fixed-multiplier solution to certify a projection before snapping coordinates. Its discrepancy and progress estimates therefore depend on the accuracy and level certificates supplied by the inner solver.

Lemma 9.2 (Outer-wrapper reduction). *Suppose Algorithm 9 returns each fixed-multiplier solution with the accuracy, certificate, and failure probability required by the certified multiplier wrapper of Section 7. Then Algorithm 8 returns a signing satisfying Theorem 5.1, with discrepancy constant 156000 and failure probability at most p .*

Proof. The smoothed body, certified multiplier search, feasibility correction, snapping rule, and outer partial-coloring construction are exactly those of Sections 6, 7, and 8. The hypothesis supplies the only interface used there: an accurately solved fixed-multiplier problem, returned with a valid certificate inside the level set on which the wrapper works. Hence the snapping argument and the progress in every outer round are unchanged. The same terminal cleanup contributes at most

$6.5\sqrt{n}$, and the same union-bound allocation over the prescribed caps gives total failure probability at most p , so the discrepancy constant remains 156000. $\square$

9.2 The recursive fixed-multiplier solver

Algorithm 9 uses certified halving phases. Each candidate phase starts from the currently certified point, whereas its random probes, refresh coins, and guard certificates are fresh.

Algorithm 9 Recursive fixed-multiplier solver

```

1: procedure RECURSIVEMULTIPLIER( $F_\nu, Q, m, n, a, R, \varepsilon_{\text{opt}}, \delta_\nu$ ) ▷ Lemma 9.3
2:    $U \leftarrow F_\nu(0), H \leftarrow 64U, \varepsilon_{\text{opt}} \leftarrow \min\{\varepsilon_{\text{opt}}, H\}$ , and  $z \leftarrow 0$ 
3:    $J \leftarrow \lceil \log_2(\max\{2, 4H/\varepsilon_{\text{opt}}\}) \rceil$ 
4:    $L \leftarrow 4 \max\{1 + ea^2H/R^2, 2a/R, 1\}$  and  $\alpha \leftarrow 1/(mL)$ 
5:    $\pi \leftarrow \min\{1, \max\{n^{-1/2}, 4\alpha\}\}$ 
6:    $k, s \leftarrow \lceil C\pi^{-1/2} \rceil$  and  $T \leftarrow \lceil 16/\alpha \rceil$ 
7:   for  $j = 1, \dots, J$  do
8:      $\varepsilon_{\text{ph},j} \leftarrow \max\{\varepsilon_{\text{opt}}, 2^{-j}H\}$  and  $\delta_{g,j}^2 \leftarrow c \min\{H/m, \varepsilon_{\text{ph},j}/(Jm)\}$ 
9:     Choose the finite-action and proximal tolerances from Lemma 9.8
10:     $\mathcal{C} \leftarrow \{z\}$  and  $R_{\text{rep}} \leftarrow \lceil C \log(J/\delta_\nu) \rceil$ 
11:     $\delta_{\text{ph}} \leftarrow \delta_\nu/(8JR_{\text{rep}})$ 
12:    for  $h = 1, \dots, R_{\text{rep}}$  do
13:       $x \leftarrow \text{TRACKEDPHASE}(F_\nu, z, Q, H, L, \pi, k, s, T, \delta_{g,j}, \delta_{\text{ph}}, \varepsilon_{\text{ph},j}, J)$  ▷ Algorithm 10
14:      If  $x \neq \text{FAIL}$ , set  $\mathcal{C} \leftarrow \mathcal{C} \cup \{x\}$ 
15:    end for
16:     $z \leftarrow \text{COMPAREVALUE}(\mathcal{C}, \varepsilon_{\text{ph},j}/(128J), \delta_\nu/(8J))$  using fresh outward intervals and the
      smallest upper endpoint ▷ Algorithm 7
17:  end for
18:  return  $z$  with fresh outward-rounded intervals for  $F_\nu(z)$  and  $\Phi(z)$ , using failure budget
       $\delta_\nu/8$ 
19: end procedure

```

Certified comparison lets independent phase trials be combined while controlling the starting level for the next phase. This converts a constant-probability phase improvement into the accuracy and interval certificates required by the multiplier search.

Lemma 9.3 (Fixed-multiplier reduction). *Assume that every independent call to Algorithm 10 satisfies Lemma 9.4. Then Algorithm 9 returns an ε_{opt} -accurate fixed-multiplier solution z with $F_\nu(z) < H/16$, with probability at least $1 - \delta_\nu$. It also returns outward-rounded intervals for its objective value and for Φ at the accuracy requested by the certified multiplier wrapper. Its number of phases and candidate repetitions is polylogarithmic in $n/\varepsilon_{\text{opt}}$ and $1/\delta_\nu$.*

Proof. Put $\Delta_j := F_\nu(z_j) - F_\nu(x_\nu)$, where z_j is the point retained after phase j and x_ν is the exact minimizer. Lemma 9.4 and the choice of $\delta_{g,j}$ show that each candidate satisfies

$$\Delta(x) \leq \frac{1}{2}\Delta_{j-1} + \frac{\varepsilon_{\text{ph},j}}{64J}$$

with a universal positive probability. Hence R_{rep} independent candidates make the probability that no such candidate is returned at most $\delta_\nu/(4J)$ after increasing the universal constant. The internal failure probabilities sum to at most $\delta_\nu/(8J)$.

The old point belongs to $\mathcal{C}$, and COMPAREVALUE chooses the smallest certified upper endpoint. The interval widths therefore give, on the certificate-success event,

$$\Delta_j \leq \frac{1}{2}\Delta_{j-1} + \frac{\varepsilon_{\text{ph},j}}{16J}. \quad (24)$$

Since $\Delta_0 \leq U = H/64$ (as $F_\nu \geq 0$) and $H2^{-J} \leq \varepsilon_{\text{opt}}/4$ by the choice of J , unrolling Eq. (24) gives

$$\Delta_J \leq 2^{-J}\Delta_0 + \frac{1}{16J} \sum_{j=1}^J 2^{-(J-j)} \max\{\varepsilon_{\text{opt}}, H2^{-j}\} \leq \varepsilon_{\text{opt}}.$$

The same comparison rule proves the guard invariant needed by the next phase and the level bound in the statement. Indeed, the retained objective can exceed the preceding objective by at most $\varepsilon_{\text{ph},j}/(64J)$. In all uses by the multiplier wrapper, $\varepsilon_{\text{opt}} \leq H$, and hence

$$F_\nu(z_j) \leq U + \frac{1}{64J} \sum_{r=1}^j \varepsilon_{\text{ph},r} \leq \frac{H}{64} + \frac{H}{32} < \frac{H}{16}.$$

Thus Lemma 9.4 applies at every phase, and the returned point $z = z_J$ satisfies $F_\nu(z) < H/16$. Give COMPAREVALUE failure budget $\delta_\nu/(8J)$ in phase j , and give the final objective and Φ evaluations total budget $\delta_\nu/8$. Lemma 9.8 supplies their requested widths. Together with the candidate and internal phase budgets above, a final union bound gives failure probability at most δ_ν . The displayed choices of J and R_{rep} give the claimed polylogarithmic counts. $\square$

9.3 The recursive Gibbs-tracking phase

The phase called by Algorithm 9 is analyzed first with exact matrix functions; Section 9.6 supplies their finite implementations.

Fix an active set of size m , a multiplier $\nu > 0$, and one projection subproblem, and retain from Section 7 the box Q of Eq. (15), the accepted draw ξ , the parameters a , b , and R , the matrices $B(x)$, $C(x)$, D_x , $P_x = \exp(D_x)$, and $E_i = \partial_i C(x) = \text{diag}(A_i/R, -A_i/R)$, and the Gibbs trace $\Phi(x) = \text{tr}[P_x]$. The fixed-multiplier objective is written without its subscript,

$$F(x) := \frac{1}{2}\|x - \xi\|_2^2 + \nu\Phi(x), \quad x \in Q;$$

x_* is its minimizer on Q , and

$$\Delta(x) := F(x) - F(x_*)$$

is the objective gap. The normalization $\|A_i\| \leq 1$ of the input matrices gives

$$\|E_i\| \leq \frac{1}{R}, \quad \sum_{i=1}^m E_i^2 \preceq \frac{m}{R^2} I_{2n}, \quad (25)$$

and differentiating F in the i th coordinate gives

$$\partial_i F(x) = x_i - \xi_i + \nu a \text{tr}[P_x E_i]. \quad (26)$$

Fix the level $H := 64U$, $U := F(0)$, as in the accelerated solver of Section 8; Lemma 9.3 needs the factor 64, since its recurrence keeps every retained point below $3H/64 < H/16$, the level at

which Lemma 9.4 starts a phase, and the level $16F(0)$ of Section 7 would not. Define the coordinate curvature parameter

$$L := 4 \max\{1 + ea^2H/R^2, 2a/R, 1\}. \quad (27)$$

On every fixed-multiplier level used by the wrapper, $H = O(m)$, $R^2 = \tilde{\Omega}(m)$, and $a = O(\log n)$, so $L = n^{o(1)}$.

Algorithm 10 One guarded recursive Gibbs-tracking phase

```

1: procedure TRACKEDPHASE( $F_\nu, z, Q, H, L, \pi, k, s, T, \delta_g, \delta_{\text{ph}}, \varepsilon_{\text{ph}}, J$ ) ▷ Lemma 9.4
2:   Choose the polynomial and proximal accuracies from Lemma 9.8
3:    $M_{\text{ref}} \leftarrow \lceil C(T\pi + \log(1/\delta_{\text{ph}})) \rceil$ 
4:    $x_0 \leftarrow z$ ,  $\hat{P}_0 \leftarrow \tilde{P}_z$ , and  $N_{\text{ref}} \leftarrow 0$ 
5:   for  $t = 0, \dots, T - 1$  do
6:     Draw  $i_t$  uniformly from  $[m]$  and set  $\hat{g}_{t,i_t} \leftarrow x_{t,i_t} - \xi_{i_t} + \nu a \text{tr}[\hat{P}_t E_{i_t}]$ 
7:     Compute the certified trust-capped proximal step  $d_t$  and set  $x' \leftarrow x_t + d_t e_{i_t}$ 
8:     Compute a fresh certificate  $F_\nu(x') \leq \bar{F}_\nu(x') \leq F_\nu(x') + H/48$ 
9:     if  $\bar{F}_\nu(x') > H/3$  then
10:       return FAIL
11:     end if
12:     if an independent  $\pi$ -refresh coin succeeds then
13:        $\hat{P}_{t+1} \leftarrow \tilde{P}_{x'}$  and  $N_{\text{ref}} \leftarrow N_{\text{ref}} + 1$ 
14:     else
15:       Construct  $\hat{\Delta}_t$  from Eq. (29) for  $\tilde{P}_{x'} - \tilde{P}_{x_t}$ 
16:        $\hat{P}_{t+1} \leftarrow \hat{P}_t + \hat{\Delta}_t$ 
17:     end if
18:      $x_{t+1} \leftarrow x'$ 
19:     if  $N_{\text{ref}} > M_{\text{ref}}$  then
20:       return FAIL
21:     end if
22:   end for
23:   return  $x_T$  together with a fresh outward-rounded objective interval
24: end procedure

```

The phase analysis couples descent with control of the tracking error, while the guard preserves the spectral level needed for each matrix-increment estimate. Its additive error allowance accommodates finite matrix actions and inexact proximal steps.

Lemma 9.4 (Tracked-phase guarantee). *Suppose $F(z) \leq H/16$, $\pi \geq 4/(mL)$, and $ks \geq K_0/\pi$, where K_0 is a sufficiently large universal constant. With the accuracies in Lemma 9.8, one call to Algorithm 10 returns a candidate x satisfying*

$$\Delta(x) \leq \frac{1}{2}\Delta(z) + C_3 m \delta_g^2 \quad (28)$$

with a universal positive probability. Except on the assigned event of probability at most δ_{ph} , every returned candidate satisfies the true objective guard and has a valid outward-rounded objective interval.

Proof. The step-sensitive variance bound is Lemma 9.5. The coupled estimate in Lemma 9.6 then gives constant-factor contraction and bounds the probability of leaving the guarded level.

By Lemma 9.8, polynomial actions and inexact scalar minimization add at most $C_3 m \delta_g^2$ to that contraction. The guard has an $H/48$ margin, and the refresh cap has failure probability at most $\delta_{\text{ph}}/4$ by a binomial tail bound. The remaining certificate failures receive the other parts of δ_{ph} . Markov's inequality and a union bound therefore give Eq. (28) with a universal positive probability and give the asserted conditional guard and certificate guarantee. The universal constant c in the definition of $\delta_{g,j}$ is chosen so that $C_3 m \delta_{g,j}^2 \leq \varepsilon_{\text{ph},j}/(64J)$. $\square$

9.4 The local matrix-increment estimator

Let $x, y \in Q$ with $y = x + de_i$, $F(x) \leq H$, and $|d| \leq R/a$, and put $E := P_y - P_x$. For integers $k, s \geq 2$, draw a standard Gaussian matrix $\Omega \in \mathbb{R}^{2n \times 2k}$. Let Y have orthonormal columns spanning $E\Omega$, set $\Pi := YY^\top$, and define

$$E_{\text{low}} := \Pi E + E \Pi - \Pi E \Pi, \quad E_{\text{res}} := (I_{2n} - \Pi)E(I_{2n} - \Pi).$$

Expanding the second product shows $E = E_{\text{low}} + E_{\text{res}}$. For independent standard Gaussian vectors $z_1, \dots, z_s \in \mathbb{R}^{2n}$, set

$$\widehat{E} := E_{\text{low}} + \frac{1}{2s} \sum_{j=1}^s \{(E_{\text{res}} z_j) z_j^\top + z_j (E_{\text{res}} z_j)^\top\}. \quad (29)$$

Conditional on Π , Gaussian isotropy $\mathbb{E}[z_j z_j^\top] = I_{2n}$ gives $\mathbb{E}[\widehat{E} \mid \Pi] = E$.

For a symmetric matrix Z , define the seminorm

$$\|Z\|_{\mathcal{G}}^2 := \frac{\nu^2 a^2}{m} \sum_{r=1}^m \text{tr}[E_r Z]^2. \quad (30)$$

It is the mean squared coordinate-gradient perturbation induced by Z : by Eq. (26), replacing P_x by $P_x + Z$ changes $\partial_r F(x)$ by $\nu a \text{tr}[E_r Z]$, and the average of the squares over a uniform $r \in [m]$ is the right-hand side.

Lemma 9.5 (Local increment variance). *Let $x, y \in Q$ with $y = x + de_i$, $F(x) \leq H$, and $|d| \leq R/a$, put $E := P_y - P_x$, and let $\widehat{E}$ be the estimator of Eq. (29) with integers $k, s \geq 2$. Then*

$$\mathbb{E}[\|\widehat{E} - E\|_{\mathcal{G}}^2] \leq \frac{CL^2}{ks} d^2$$

for a universal constant C .

Proof. Conditional on Π , the Gaussian quadratic-form variance identity and Eq. (25) give

$$\begin{aligned} \mathbb{E}[\|\widehat{E} - E\|_{\mathcal{G}}^2 \mid \Pi] &\leq \frac{2\nu^2 a^2}{ms} \sum_{r=1}^m \|E_{\text{res}} E_r\|_F^2 \\ &= \frac{2\nu^2 a^2}{ms} \text{tr}[E_{\text{res}}^2 \sum_{r=1}^m E_r^2] \\ &\leq \frac{2\nu^2 a^2}{sR^2} \|E_{\text{res}}\|_F^2. \end{aligned} \quad (31)$$

Here the first inequality applies the variance identity to the symmetric part of $E_r E_{\text{res}}$, whose Frobenius norm is at most $\|E_{\text{res}} E_r\|_F$; the equality is cyclicity of the trace; and the last inequality

is the second bound of Eq. (25) against $E_{\text{res}}^2 \succeq 0$. The Gaussian range-finder estimate proved in Section 7 yields

$$\mathbb{E}_{\Omega}[\|E_{\text{res}}\|_F^2] \leq \frac{C}{k} \|E\|_1^2.$$

For positive definite matrices, let

$$\mathcal{D}(P\|P') := \text{tr}[P(\log P - \log P') - P + P'].$$

Since $\log P_x = D_x$ is affine in x , this is the Bregman divergence of the convex function Φ at the base point x :

$$\mathcal{D}(P_x\|P_y) = \Phi(y) - \Phi(x) - \langle \nabla \Phi(x), y - x \rangle.$$

The trust condition $|d| \leq R/a$ gives $\|D_y - D_x\| = a|d| \|E_i\| \leq 1$, so monotonicity of trexp and $\nu\Phi(x) \leq F(x) \leq H$ give $\Phi(y) \leq e\Phi(x) \leq eH/\nu$. The generalized Pinsker inequality used in Section 7 therefore gives

$$\nu^2 \|P_y - P_x\|_1^2 \leq 4eH\nu\mathcal{D}(P_x\|P_y). \quad (32)$$

Along the segment from x to y , the coordinate second derivative of F is at most $1 + ea^2H/R^2 \leq L/4$ by Eq. (27), and the quadratic part of F contributes $d^2/2 \geq 0$ to the Bregman remainder. Consequently,

$$\nu\mathcal{D}(P_x\|P_y) \leq F(y) - F(x) - \partial_i F(x)d \leq \frac{L}{8}d^2.$$

Combining this estimate with Eq. (31), the range-finder bound, and Eq. (32), and using $a^2H/R^2 \leq L$ from Eq. (27) in the last step, gives

$$\mathbb{E}[\|\hat{E} - E\|_{\mathcal{G}}^2] \leq \frac{C}{ks} \frac{a^2H}{R^2} Ld^2 \leq \frac{CL^2}{ks} d^2. \quad \square$$

9.5 Coupled contraction and localization

This subsection couples the recursive tracking error to the progress of randomized coordinate descent.

At time t , maintain a symmetric dense matrix $\hat{P}_t$ and an iterate x_t . Select i_t uniformly from $[m]$, independently of the past, and use

$$\hat{g}_{t,i_t} := x_{t,i_t} - \xi_{i_t} + \nu a \text{tr}[\hat{P}_t E_{i_t}] \quad (33)$$

as the selected derivative. After the coordinate step produces x_{t+1} , independently refresh with probability π by setting $\hat{P}_{t+1} := P_{x_{t+1}}$. Otherwise, construct an independent copy $\hat{\Delta}_t$ of Eq. (29) for $P_{x_{t+1}} - P_{x_t}$ and set

$$\hat{P}_{t+1} := \hat{P}_t + \hat{\Delta}_t.$$

The matrix $\hat{P}_t$ need not be positive semidefinite; it is used only in Eq. (33). The objective guards continue to use fresh positive-semidefinite surrogates.

Put

$$Z_t := \hat{P}_t - P_{x_t}, \quad \mathcal{E}_t := \|Z_t\|_{\mathcal{G}}^2.$$

At a nonrefresh step, the increment error has conditional matrix expectation zero, so its cross term with Z_t vanishes in the seminorm. Lemma 9.5, applied at the base point x_t when $F(x_t) \leq H$, gives

$$\mathbb{E}_t[\mathcal{E}_{t+1}] \leq (1 - \pi)\{\mathcal{E}_t + CL^2\mathbb{E}_t[d_t^2]/(ks)\}. \quad (34)$$

The coordinate step and its energy. For the selected coordinate $i := i_t$, define

$$\mathbf{e}_{t,i} := \nu a \operatorname{tr}[E_i Z_t].$$

By Eq. (30) and the independent uniform choice of i_t ,

$$\mathbb{E}_{i_t}[\mathbf{e}_{t,i_t}^2 \mid \mathcal{F}_t] = \mathcal{E}_t. \quad (35)$$

Choose d_t to minimize

$$\widehat{g}_{t,i} d + \frac{L}{2} d^2$$

over $x_{t,i} + d \in Q_i$ and $|d| \leq R/a$, and put $x_{t+1} := x_t + d_t e_i$. The comparison displacement

$$d'_i := (x_{*,i} - x_{t,i})/L$$

is feasible. Indeed, every coordinate interval of Q has length $\varepsilon_{\text{box}} < 1$, while Eq. (27) gives $L \geq 1$ and $L \geq 2a/R$.

The strong three-point inequality for the one-dimensional proximal step, the coordinate Hessian bound $L/4$, and Young's inequality give

$$F(x_{t+1}) - F(x_t) \leq \partial_i F(x_t) d'_i + \frac{L}{2} (d'_i)^2 + \frac{\mathbf{e}_{t,i}^2}{2L} - \frac{L}{4} d_t^2. \quad (36)$$

The negative term arises because the model has curvature L , whereas the true coordinate curvature is at most $L/4$; Young's inequality uses at most half of the remaining model curvature.

Averaging the comparison terms over the uniform coordinate, and using the 1-strong convexity of F in the last step, gives

$$\begin{aligned} \mathbb{E}_i[\partial_i F(x) d'_i + L(d'_i)^2/2] &= \frac{1}{mL} \{ \langle \nabla F(x), x_* - x \rangle + \|x - x_*\|_2^2/2 \} \\ &\leq -\frac{\Delta(x)}{mL}. \end{aligned}$$

With

$$\alpha := \frac{1}{mL}, \quad (37)$$

Eqs. (35) and (36) imply

$$\mathbb{E}_t[\Delta(x_{t+1})] \leq (1 - \alpha) \Delta(x_t) + \frac{\mathcal{E}_t}{2L} - \frac{L}{4} \mathbb{E}_t[d_t^2]. \quad (38)$$

Weighting the tracking error by $1/(\pi L)$ allows the decrease from refreshes to absorb its contribution to the objective gap. The negative step-energy term then absorbs the variance of the matrix increments and controls the probability of leaving the analytical level.

Lemma 9.6 (Coupled contraction). *Suppose*

$$\pi \geq 4\alpha, \quad ks \geq K_0/\pi,$$

where K_0 is a sufficiently large universal constant. Then, at every step t with $F(x_t) \leq H$, the potential

$$V_t := \Delta(x_t) + \frac{\mathcal{E}_t}{\pi L}$$

satisfies

$$\mathbb{E}_t[V_{t+1}] \leq (1 - \alpha)V_t - \frac{L}{8}\mathbb{E}_t[d_t^2]. \quad (39)$$

Moreover, if $x_0 := z$, $\hat{P}_0 := P_z$, $F(z) \leq H/16$, and $T := \lceil 16/\alpha \rceil$, then the stopped ideal phase crosses the $5H/16$ analytical level with probability at most $1/20$. A candidate phase therefore halves the objective gap with a universal positive probability.

Proof. Step 1: the two coefficients. Combining Eqs. (34) and (38), the coefficient of $\mathcal{E}_t$ is

$$\frac{1}{2L} + \frac{1 - \pi}{\pi L} = \frac{1 - \pi/2}{\pi L} \leq \frac{1 - \alpha}{\pi L}.$$

The coefficient of $\mathbb{E}_t[d_t^2]$ is at most

$$-\frac{L}{4} + \frac{CL(1 - \pi)}{\pi ks} \leq -\frac{L}{8}$$

after increasing K_0 . These are exactly the two coefficients in Eq. (39).

Step 2: contraction and step energy over a phase. Initialize a reduction phase at z by setting $x_0 := z$ and $\hat{P}_0 := P_z$. Thus $\mathcal{E}_0 = 0$. For

$$T := \lceil 16/\alpha \rceil = O(mL),$$

iterating Eq. (39) and using $(1 - \alpha)^T \leq e^{-16}$ gives $\mathbb{E}[V_T] \leq e^{-16}V_0$ as long as the process stays on the guarded level. Summing Eq. (39) over $t < T$ and dropping $V_T \geq 0$ also gives the step-energy estimate

$$L \sum_{t=0}^{T-1} \mathbb{E}[d_t^2] \leq 8V_0. \quad (40)$$

Step 3: localization. This step justifies the qualification in Step 2. Start a phase at z with $F(z) \leq H/16$, and define

$$\tau := \min\{t + 1 : F(x_t + d_t e_{i_t}) > 5H/16\}.$$

Stop the ideal process after the crossing proposal. Before this time, every current iterate is below $5H/16$. The trust condition changes the matrix exponential by at most a factor e , so the entire proposed segment remains in the H -trace level because $5e/16 < 1$. Thus all preceding local estimates apply through the crossing proposal.

Taking the comparison displacement to be zero in the proximal inequality gives, before stopping,

$$F(x_{t+1}) - F(x_t) \leq \mathfrak{e}_{t,i_t}^2/L. \quad (41)$$

Therefore a crossing requires

$$\sum_{t=0}^{\tau-1} \mathfrak{e}_{t,i_t}^2/L \geq cH$$

for a universal constant $c > 0$. Summing the stopped version of Eq. (34), using $\mathcal{E}_0 = 0$, gives

$$\pi \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] \mathcal{E}_t] \leq \frac{CL^2}{ks} \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] d_t^2].$$

The stopped version of Eq. (40), together with $V_0 \leq H/16$, now yields

$$\begin{aligned} \mathbb{E}\left[\sum_{t=0}^{T-1} \mathbb{1}[t < \tau] \mathbf{e}_{t,i_t}^2 / L\right] &= \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] \mathcal{E}_t / L] \\ &\leq \frac{CL}{\pi k s} \sum_{t=0}^{T-1} \mathbb{E}[\mathbb{1}[t < \tau] d_t^2] \\ &\leq \frac{CH}{K_0}. \end{aligned}$$

Markov's inequality shows that $\Pr[\tau \leq T] \leq C/K_0$. Choose K_0 so that this probability is at most $1/20$.

Step 4: agreement with the implemented process. On the event of no analytical crossing, every proposal has true objective at most $5H/16$. A correct $H/48$ -accurate upper certificate is therefore at most $H/3$, so every proposal is retained and the implemented process agrees with the ideal process. The killed-process argument of Section 7, applied to Eq. (39), then shows that one candidate phase halves the gap with a universal positive probability. Independent candidate repetitions and outward-rounded objective comparisons amplify this to the failure budget assigned to the phase. $\square$

Remark 9.7 (Positivity of the tracker is not used). Positivity of $\hat{P}_t$ plays no role in the proof of Lemma 9.6: the tracker enters only through the selected derivative of Eq. (33), and the one-sided guard uses a fresh positive-semidefinite approximation of $P_{x_{t+1}}$ at the proposed point, not the tracker.

9.6 Finite matrix-function implementation

For finite matrix actions, the recursive tracker is centered on a common polynomial surrogate, so its stochastic increment error remains conditionally unbiased. The difference between that surrogate and the exact exponential contributes a deterministic bias that must fit within the phase's additive error allowance.

Lemma 9.8 (Finite-surrogate stability). *There are inverse-polynomial action accuracies and certified scalar minimization tolerances, computable from the phase parameters, with the following properties.*

- (i) *The finite implementation of Algorithm 10 satisfies Eq. (28) with a universal positive probability (the killed-process bound of Lemma 9.6).*
- (ii) *Except on an event of probability at most δ_{ph} , the fresh guard of Algorithm 10 obeys $F_\nu(x') \leq \bar{F}_\nu(x') \leq F_\nu(x') + H/48$ and its phase-end objective interval has its requested outward-rounded width.*
- (iii) *Outside the event of (ii), the fixed-multiplier Φ -interval returned by Algorithm 9 has its requested outward-rounded width.*

The required polynomial degrees and scalar iteration counts are polylogarithmic in the inverse target accuracy.

Proof. Step 1: the polynomial surrogate and its bias. Replace every exact exponential by the same finite polynomial surrogate used in Section 7. On the guarded spectral interval, let q_r approximate $t \mapsto \exp(t/2)$ and define

$$\tilde{P}_x := q_r(D_x)^2. \tag{42}$$

The degree r is polylogarithmic in the inverse target accuracy. The square in Eq. (42) ensures $\tilde{P}_x \succeq 0$, as required by the one-sided trace certificates.

The recursive tracker is now defined relative to $\tilde{P}_x$: a refresh sets $\hat{P}_{t+1} := \tilde{P}_{x_{t+1}}$, and a nonrefresh step adds an unbiased copy of the estimator in Eq. (29) with E replaced by $\tilde{E} := \tilde{P}_{x_{t+1}} - \tilde{P}_{x_t}$. This convention is important: it preserves exact conditional unbiasedness for the finite algorithm. The difference between $\tilde{P}_x$ and P_x is treated separately as deterministic gradient bias.

Let $d_0 := 2n$ and suppose uniformly on the guarded interval that

$$\|\tilde{P}_x - P_x\| \leq \varepsilon_{\text{op}}.$$

Since the trace norm of a $d_0 \times d_0$ matrix is at most d_0 times its operator norm and $\|E_i\| \leq 1/R$,

$$\|\tilde{P}_x - P_x\|_1 \leq d_0 \varepsilon_{\text{op}}, \quad |\nu a \text{tr}[E_i(\tilde{P}_x - P_x)]| \leq \frac{\nu a d_0}{R} \varepsilon_{\text{op}}, \quad (43)$$

and by the triangle inequality

$$\|\tilde{P}_y - \tilde{P}_x - (P_y - P_x)\|_1 \leq 2d_0 \varepsilon_{\text{op}}. \quad (44)$$

For the phase accuracy ε_{ph} , the argument of Algorithm 10 (equal to $\varepsilon_{\text{ph},j}$ in phase j of Algorithm 9), and the number J of halving phases, choose an inverse-polynomial δ_g with

$$\delta_g^2 \leq c \min\{H/m, \varepsilon_{\text{ph}}/(Jm)\}, \quad (45)$$

and take

$$\varepsilon_{\text{op}} \leq \min\{\phi_0/(16d_0), H/(192\nu d_0), R\delta_g/(C\nu a d_0), R\sqrt{ks}\delta_g/(C\nu a d_0)\}. \quad (46)$$

The certified wrapper in Section 7 has $\nu \geq \underline{\nu} = n^{-O(1)}$, and all other quantities in Eq. (46) are polynomially bounded. The guarded spectral interval and the required polynomial degree are therefore polylogarithmic in the inverse target accuracy.

Step 2: the finite recurrences. Eqs. (43) and (44), the range-finder estimate, and Young's inequality change the two exact recurrences only to

$$\mathbb{E}_t[\Delta(x_{t+1})] \leq (1 - \alpha)\Delta(x_t) + \frac{C_1 \mathcal{E}_t}{L} - \frac{L}{4} \mathbb{E}_t[d_t^2] + \frac{C_1 \delta_g^2}{L}, \quad (47)$$

$$\mathbb{E}_t[\mathcal{E}_{t+1}] \leq (1 - \pi)\{\mathcal{E}_t + CL^2 \mathbb{E}_t[d_t^2]/(ks) + C\delta_g^2/(ks)\}. \quad (48)$$

Here $\mathcal{E}_t$ measures the stochastic tracking error relative to $\tilde{P}_{x_t}$, and C_1 is universal. To see the only new term in Eq. (48), use

$$\|\tilde{P}_y - \tilde{P}_x\|_1^2 \leq 2\|P_y - P_x\|_1^2 + 8d_0^2 \varepsilon_{\text{op}}^2$$

in the proof of Lemma 9.5. The bias in Eq. (47) follows directly from Eq. (43).

Step 3: the finite potential. Choose a sufficiently large universal c_1 and use the finite potential

$$V_t^{\text{fin}} := \Delta(x_t) + \frac{c_1 \mathcal{E}_t}{\pi L}.$$

Because $\pi \geq 4\alpha$, increasing c_1 absorbs the $C_1 \mathcal{E}_t/L$ term. Increasing K_0 in $ks \geq K_0/\pi$ absorbs the step-variance term. Thus

$$\mathbb{E}_t[V_{t+1}^{\text{fin}}] \leq (1 - \alpha)V_t^{\text{fin}} - \frac{L}{8} \mathbb{E}_t[d_t^2] + \frac{C_2 \delta_g^2}{L}. \quad (49)$$

Over $T = O(mL)$ queries, the total additive contribution is $O(m\delta_g^2)$. Equation (45) makes this smaller than the allocated phase error and a sufficiently small fraction of the crossing budget. The exact killed-process argument therefore gives Eq. (28), rather than a pure multiplicative halving statement.

Step 4: the scalar minimization. The scalar coordinate minimization can be performed to a certified variational residual ρ_t . As in Section 8, choose the bisection tolerances so that

$$\sum_{t=0}^{T-1} \rho_t \leq c \min\{H, \varepsilon_{\text{ph}}/J\}.$$

This contributes another admissible additive term to Eq. (49).

Step 5: the certificates. It remains to record the certificates. For every proposal, use fresh probes of the positive-semidefinite surrogate and a median-of-means trace estimate, with enough samples that its contribution to the objective is at most $H/192$. The polynomial bias is at most $H/192$ by Eq. (46), so the two-sided width is $H/48$ and

$$F_\nu(x') \leq \overline{F}_\nu(x') \leq F_\nu(x') + H/48.$$

Assign conditional failure probability at most $\delta_{\text{ph}}/(4T)$ to each guard. Fresh randomness makes a union bound valid under adaptivity. The cap $M_{\text{ref}} = \lceil C(T\pi + \log(1/\delta_{\text{ph}})) \rceil$ fails with probability at most $\delta_{\text{ph}}/4$ by a binomial tail bound.

The much narrower phase-end objective intervals and the multiplier Φ -interval of Algorithm 9 are computed separately, exactly as in Section 7: use one fresh dense polynomial evaluation per candidate, choose its operator error so that the trace error is at most $\varepsilon_{\text{ph}}/(256\nu J)$ or the smaller accuracy requested by the multiplier wrapper, and round the resulting interval outward. This requires only polylogarithmic polynomial degree. Allocate the remaining parts of δ_{ph} to these evaluations and to the finite proximal calls. The finite phase therefore satisfies Eq. (28) with the probability of Lemma 9.4, and the certificate claims (ii) and (iii) hold as stated; no additional pure-halving step is asserted. $\square$

9.7 Running time

The refresh probability trades sketch width against the frequency of dense matrix reconstructions. At the chosen scale, thin rectangular products control the cost of each coordinate query, and dense refreshes contribute a lower-order term after summing over the active dimensions.

Lemma 9.9 (Global running time). *With the parameters π , k , s , and T set by Algorithm 9 at every active dimension, Algorithm 8 uses*

$$n^{3+o(1)} \text{polylog}(1/p)$$

arithmetic operations in the real-arithmetic model.

Proof. Step 1: the parameters. The dual matrix-multiplication exponent α_{dual} is defined in Section 3.2. Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] prove $\alpha_{\text{dual}} > 0.321334$, so $\alpha_{\text{dual}} > 1/4$, and Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25] prove $\omega < 2.371339$, so $\omega < 5/2$; only these two consequences are used. At active dimension m , Algorithm 9 sets

$$\pi_m := \min\{1, \max\{n^{-1/2}, 4/(mL)\}\}, \quad k, s := \lceil C\pi_m^{-1/2} \rceil, \quad (50)$$

where the universal constant is large enough that $ks \geq K_0/\pi_m$. Since $\pi_m \geq n^{-1/2}$, $k+s \leq n^{1/4+o(1)}$.

Step 2: the cost of one query. A nonrefresh query applies $\tilde{P}_x$ and $\tilde{P}_y$ to batches of $O(k + s)$ vectors, together with the polylogarithmically many fresh guard probes, constructs $E\Omega$, EY , and the residual actions, forms the low-rank products in Eq. (29), contracts $\text{tr}[\hat{P}_t E_{it}]$, and updates $B(x_t)$. Polynomial actions cost only a polylogarithmic number of thin matrix products. Since $1/4 < \alpha_{\text{dual}}$, both orientations of an $n \times n$ by $n \times n^{1/4+o(1)}$ product, and the corresponding $n \times n^{1/4+o(1)}$ by $n^{1/4+o(1)} \times n$ product, cost $n^{2+o(1)}$. The QR factorization costs $O(n(k + s)^2) = n^{3/2+o(1)}$, while the contraction and the update of $B(x_t)$ cost $O(n^2)$. Therefore a nonrefresh query costs

$$n^{2+o(1)}. \quad (51)$$

A refresh constructs the full dense matrix $\tilde{P}_{x_{t+1}}$ by a polylogarithmic number of dense matrix products, and hence costs

$$n^{\omega+o(1)}. \quad (52)$$

In contrast with the snapshot implementation in Section 7, a refresh does not compute all m coordinate derivatives. The algorithm stores the dense matrix and computes only the subsequently selected contraction.

Step 3: one fixed-multiplier solve. A constant-reduction phase uses $T = O(mL) = mn^{o(1)}$ coordinate queries. Its nonrefresh work is therefore

$$mn^{2+o(1)}.$$

The expected number of refreshes is $O(mL\pi_m + 1)$, and a standard binomial tail bound caps it at a logarithmic multiple of this quantity with the assigned failure probability. If $\pi_m = n^{-1/2}$, the refresh work is

$$mn^{\omega-1/2+o(1)}.$$

If the term $4/(mL)$ determines π_m , there are only $\text{polylog}(n)$ refreshes, whose total cost is $n^{\omega+o(1)}$. Consequently, including the polylogarithmically many halving phases and accuracy repetitions, one fixed-multiplier solve costs

$$mn^{2+o(1)} + mn^{\omega-1/2+o(1)} + n^{\omega+o(1)}. \quad (53)$$

The phase-end objective comparisons and the certified Φ -intervals use only a polylogarithmic number of fresh dense polynomial evaluations. Their cost is absorbed by the $n^{\omega+o(1)}$ term in Eq. (53); their probe counts and outward rounding contribute only the displayed polylogarithmic factors. Certified multiplier bisection contributes only another polylogarithmic factor. Its finite objective intervals, exact-feasibility radial correction, and snapping step are unchanged.

Step 4: all outer rounds. Let N_{out} be the number of outer rounds. Denote their active dimensions by $m_0, m_1, \dots, m_{N_{\text{out}}-1}$. These dimensions decrease geometrically, so $N_{\text{out}} = O(\log n)$ and

$$\sum_{t=0}^{N_{\text{out}}-1} m_t = O(n).$$

Summing Eq. (53) over the outer rounds gives

$$\begin{aligned} \sum_{t=0}^{N_{\text{out}}-1} m_t n^{2+o(1)} &= n^{3+o(1)}, \\ \sum_{t=0}^{N_{\text{out}}-1} m_t n^{\omega-1/2+o(1)} &= n^{\omega+1/2+o(1)}, \end{aligned}$$

$$\sum_{t=0}^{N_{\text{out}}-1} n^{\omega+o(1)} = n^{\omega+o(1)}.$$

The input construction, final signed sum, snapping, and bounded-dimension cleanup already cost at most $n^{3+o(1)}$. Since $\omega < 5/2$, $\omega + 1/2 < 3$. The total arithmetic cost is therefore

$$(n^3 + n^{\omega+1/2} + n^\omega) n^{o(1)} \text{polylog}(1/p) = n^{3+o(1)} \text{polylog}(1/p).$$

Step 5: the hand-off to the wrapper. Lemma 9.3 supplies the objective accuracy and outward-rounded Φ -interval required by the certified multiplier wrapper, and its returned point satisfies $F_\nu(z) < H/16 = 4U \leq H_{\max}$, because $U = F_\nu(0) = \frac{1}{2}\|\xi\|_2^2 + \nu\phi_0$ with $\nu \leq \Lambda$ while $H_{\max} = 16(\frac{1}{2}\|\xi\|_2^2 + \Lambda\phi_0 + 1)$; this is the interface hypothesis of the wrapper of Section 7, so its segment estimate applies to the returned point. The wrapper therefore chooses the correct bracket half or stops with the prescribed residual, and its downward-rounded radial correction makes the selected projection exactly feasible. Lemma 9.2 then preserves the snapping argument, the outer partial-coloring success probability, the terminal cleanup, and the discrepancy constant 156000. Allocating p over all outer rounds, multiplier tests, candidate phases, guards, refresh-count caps, value certificates, and cleanup trials contributes only the displayed polylogarithmic factor. $\square$

10 Bit complexity of the cubic-time algorithm

This section proves that the algorithm of Theorem 9.1 keeps its exponent when every number is stored in a fixed number of bits. The real-arithmetic proof never controls three things that a finite word forces one to control: decisions taken on rounded quantities, random objects that cannot be drawn exactly, and the size of the tracked matrix, which the seminorm of Eq. (30) does not see. We specify a finite version of the algorithm, prove one lemma per primitive, and assemble them into Theorem 10.1. Theorem 6.1 is not covered: its diagonalization step is analyzed by Sobczyk [Sob25, Corollary 2.4] only in the real-arithmetic model, and it is not on the cubic path.

Computational model. A *fixed-point number* with f fractional bits and I integer bits is an integer multiple of $u := 2^{-f}$ of modulus below 2^I ; a *word* holds one such number in $W := I + f$ bits. Addition, subtraction, comparison, and negation of words are exact whenever the result has modulus below 2^{I-1} . The product of two words is a multiple of u^2 ; the routine $\text{fl}(xy)$ rounds it to the grid $u\mathbb{Z}$ with absolute error at most u , and multiplication by an integer constant is exact. Inner products and matrix products are formed exactly in a temporary of $2W + \lceil \log_2(\text{number of terms}) \rceil$ bits and rounded once. Numbers on the coordinate grid Γ of Lemma 10.5, and on the grid $2^{-b_{\text{in}}}\Gamma$ of the entries of $B(x)$, are stored as integer numerators over a common denominator, which costs at most $b_{\text{in}} + 12$ further bits and changes nothing below; a constant number of scalars of the solve ($q(x)$ and $\nu\tilde{c}$) are kept in $2W$ -bit words, and their products with a word are formed in $3W$ -bit temporaries and rounded once. Costs are counted in *bit operations*. Addition or comparison of W -bit words costs $O(W)$ bit operations and multiplication costs $M(W) = O(W \log W)$ (Harvey and van der Hoeven [HvdH21, Theorem 1.1]). Arithmetic saturates: a result of modulus at least 2^{I-1} is replaced by the largest representable value of the same sign, and every loop of the algorithm has a fixed cap, so the operation count below holds on every path; the correctness statements are proved on the event that no cap and no saturation occurs, which the failure budget covers: no saturation occurs while every guard is correct and no cap fires, because then every base point satisfies $F(x_t) \leq H$ and the magnitude bounds of Lemma 10.13(iv) and of Step 1 of the proof of Theorem 10.1 hold deterministically. Randomness is a source of independent fair coins. The model

is the additive reading of the standard floating-point convention; Banks, Garza-Vargas, Kulkarni, and Srivastava [BGVKS23, Remark 1.3] note that the two are interchangeable for algorithms whose magnitudes stay within a fixed polynomial range, which Lemma 10.13 establishes here.

Notation. Throughout, $d_0 := 2n$ is the dimension of D_x , L_{in} is the bit length of the input entries, and the parameters $a, R, b, s_{\text{sm}}, \chi, \varepsilon_{\text{box}}, \delta, N_0, c_{\text{rad}}(\cdot), \rho, \Lambda, \underline{\nu}, H, H_{\text{max}}, d_{\text{w}}, \tau_{\text{w}}, \sigma, \varepsilon_{\text{opt}}, J, R_{\text{rep}}, \delta_g, L, \pi_m, k, K_0, T$ carry the meanings fixed in Sections 5, 6, 7, and 9; where the finite algorithm replaces one of them by a rounded value we write a tilde, and Lemma 10.5 lists every such replacement with the direction of its rounding. Where two of those sections fix the same letter differently, the value of Section 7 is meant, as (F12) states: $\Lambda = 2q(0)/(1 - \phi_0)$ is the multiplier bracket of Section 7, not the bracket $2f_\xi(0)/(b - \chi)$ of Section 6; $H = 16F_\nu(0)$ is the level of Section 7, not the level $64U$ that Sections 8 and 9 set; J is the phase count of Section 7; and L is the curvature constant of Section 9, while the aspect-ratio quantity $\max\{1, \log(n/m)\}$ of Sections 5 and 6 is written $\max\{1, \ell\}$ here. Where a letter of those sections collides with one of Section 9 or with the fixed-point vocabulary, this section renames it as follows; the curvature L and the integer-bit count I keep their letters.

object	letter in Sections 5–9	letter here
grid unit 2^{-f}	—	$\mathbf{u}$
$\log(n/m)$ (Sections 5 and 6)	u	ℓ
smoothing parameter (Section 6)	s	s_{sm}
probe count (Section 9)	s	s
wrapper radius and tolerance (Section 7)	d, τ	$d_{\text{w}}, \tau_{\text{w}}$
coordinate step and crossing time (Section 9)	d_t, τ	d_t, τ
squaring depth of Lemma 10.17	—	J_{sq} (J is the phase count)
radial factor of Lemma 10.18	—	β_{r}
Bini’s error degree	—	h_{B}

Unsubscripted constants C, c are universal and may change from line to line; subscripted constants are fixed where they are introduced.

Spectral bounds. The norm-rejection constant of Section 6 is fixed as $C_g := 2$, and the finite algorithm also rejects draws with $\|\xi'\|_2^2 < m/2$: a standard Gaussian vector in $\mathbb{R}^m$ has $\|\xi\|_2^2 > 2m$ with probability at most $(2/e)^{m/2} \leq e^{-0.1534m}$ and $\|\xi\|_2^2 < m/2$ with probability at most $(e^{1/2}/2)^{m/2} \leq e^{-0.096m}$, and $e^{-0.09m}$ bounds both events for the rounded ξ' of Lemma 10.7(i), whose norm differs from $\|\xi\|_2$ by at most $\sqrt{m}2^{-17}$; the lower rejection is charged to the 0.02 numerical allowance of Section 6. On the accepted event $\Lambda \leq \frac{4}{3}\|\xi'\|_2^2 \leq \frac{8}{3}m$ and $H \geq 16q(0) = 8\|\xi'\|_2^2 \geq 4m$. We write $\lambda_{\text{top}} := \log(H_{\text{max}}/\underline{\nu}) + 1 + \log(2n)$ for an upper bound on the largest eigenvalue of D_x at every point where a matrix function is evaluated (guarded points, their proposals, and the box projection of ξ' after the pre-test of Lemma 10.18(v)) and $M_{\text{sp}} := 2ab + \lambda_{\text{top}} + 1$ for a bound on the modulus of the smallest eigenvalue, so that every $\widehat{D}_x$ that is exponentiated has spectrum in $[-M_{\text{sp}}, \lambda_{\text{top}}]$ (Section 7 gives $[-2ab - \log(H/\nu) - 1, \log(H/\nu) + 1]$ at proposals).

Theorem 10.1 (Bit complexity of the cubic-time signing). *Let $A_1, \dots, A_n$ be symmetric $n \times n$ matrices with $\|A_i\| \leq 1$ whose entries are rationals of bit length at most L_{in} , and let $p \in (0, 1/2)$.*

Assume $\alpha_{\text{dual}} > 1/4$ and $\omega < 5/2$ in the asymptotic-rank sense of Definition 10.2. Put

$$f := \left\lceil \log_2 \frac{1}{\varepsilon_{\text{opt}}} \right\rceil + 3\lceil \lambda_{\text{top}} \log_2 e \rceil + 2\lceil \log_2 M_{\text{sp}} \rceil + 40\lceil \log_2(2n) \rceil + 200, \quad I := \left\lceil \log_2 \frac{H_{\text{max}}}{\underline{\nu}} \right\rceil + 7\lceil \log_2(2n) \rceil + 12, \quad (54)$$

where ε_{opt} and $H_{\text{max}}/\underline{\nu}$ are evaluated at their extreme over the run, which is the round with $m = n$ (every parameter is monotone in m) and the smallest multiplier $\underline{\nu}$ (so with $\tilde{\Lambda} \leq 3n$), so that $W := I + f = O(\log n)$, independent of p . For every fixed $\eta > 0$ there is a randomized algorithm, using fair coins and fixed-point arithmetic on W -bit words (with multiplication temporaries of $2W + c_\eta \lceil \log_2(2n) \rceil$ bits, c_η the coefficient constant of Lemma 10.3, and $3W$ -bit temporaries for a constant number of scalar products), that outputs $\varepsilon \in \{-1, 1\}^n$ with

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| < 156000\sqrt{n}$$

with probability at least $1 - p$, using at most

$$\{n^3 + n^{\omega+1/2} + n^\omega\} n^{2\eta} \text{polylog}(n/p) \cdot M(2W + c_\eta \lceil \log_2(2n) \rceil) + O(n^3(L_{\text{in}} + W) \log(L_{\text{in}} + W)) \quad (55)$$

bit operations; since η is arbitrary, this is $n^{3+o(1)} \text{polylog}(1/p) + \tilde{O}(n^3 L_{\text{in}})$ bit operations. The input bit length enters only through the second term, which reads and rounds the input.

The rest of the section proves Theorem 10.1. We first fix the finite algorithm as a list of replacements of primitives of Sections 6, 7, and 9; each replacement is then justified by the lemma named beside it, and the closing proof assembles the cost.

The finite algorithm. The finite algorithm is the algorithm of Theorem 9.1 (procedures MAIN-CUBIC, RECURSIVEMULTIPLIER, and TRACKEDPHASE of Algorithms 8–10) with the following changes.

- (F1) *Input.* Each A_i is replaced by a b_{in} -bit dyadic contraction $\tilde{A}_i$, $b_{\text{in}} := 2\lceil \log_2 n \rceil + 10$ (Lemma 10.4).
- (F2) *State and parameters.* Iterates, fractional colorings, the matrix $B(x)$, and D_x are exact rationals on fixed grids; every analysis parameter is replaced by a dyadic value rounded in the direction that preserves the inequality it enters (Lemma 10.5).
- (F3) *Randomness.* The Gaussian ξ and the test matrix Ω are exact samples of rounded Gaussians (Lemma 10.7); all Hutchinson probes are Rademacher vectors, coordinate choices and refresh coins are exact (Lemma 10.6); the norm rejection of Section 6 also rejects $\|\xi'\|_2^2 < m/2$.
- (F4) *Matrix exponentials.* The surrogate $q_r(D_x)^2$ of Eq. (42) is replaced by the scaled-Taylor power $\mathcal{Q}(D_x)$ of Eq. (58), evaluated by repeated squaring for dense matrices (Lemmas 10.8 and 10.9) and by repeated application for vector actions (Lemma 10.10).
- (F5) *Fast products.* Every dense or thin matrix product is computed exactly by a recursive bilinear algorithm and rounded once (Lemma 10.3).
- (F6) *Range finder and increment.* The sketch, its Householder basis, and the increment estimator are computed with the rounded test matrix and the action surrogate (Lemmas 10.11 and 10.12).

- (F7) *Tracker.* A refresh also occurs whenever $\lceil 1/\pi' \rceil$ steps have passed since the last one, π' being the dyadic refresh probability fixed with the parameters after Lemma 10.5 (Lemma 10.13).
- (F8) *Coordinate step.* The one-dimensional step is rounded to the coordinate grid before clipping (Lemma 10.14).
- (F9) *Guard.* The trace certificate is additive: a proposal y is accepted when $q(y) + \nu U(y) \leq H$ with an upper estimate $U(y) \geq \Phi(y)$ that satisfies $U(y) \leq \frac{5}{3}\Phi(y) + H/(9\nu)$, in place of the certificate $F_\nu(x') \leq \bar{F}_\nu(x') \leq F_\nu(x') + H/48$ with rejection above $H/3$ of Algorithm 10, itself a replacement of the relative certificate $\Phi(y) \leq U(y) \leq 2\Phi(y)$ of Section 7 (Lemma 10.15; Remark 10.21 records the word length with the relative certificate).
- (F10) *Comparisons.* Every objective comparison, bisection decision, and feasibility correction acts on outward-rounded enclosures (Lemma 10.18).
- (F11) *Cleanup.* The biased coins are exact and the spectral certificate U is computed by normalized repeated squaring (Lemma 10.19).
- (F12) *Wrapper parameters.* The level $H = 16F_\nu(0)$, the phase count J (rounded up as in Lemma 10.5), the repetition count R_{rep} , the candidate-selection rule (a candidate is retained only if its certified upper endpoint lies below the old point's certified lower endpoint), and the failure budgets $p_0/(100TJR_{\text{rep}})$, p_0/J , p_0 are those of Section 7, in place of $H = 64U$, COMPAREVALUE, and the δ_ν -budgets of Algorithm 9; one phase-independent δ_g replaces the per-phase $\delta_{g,j}$ (Lemmas 10.18(ii) and 10.16). Under this selection rule phase-start objectives never increase, so $F(z) \leq F_\nu(0) \leq H/16$ at every phase start, which is the hypothesis of Lemma 10.16; with the explicit constants of Lemma 10.16, $32m\delta_g^2 \leq \varepsilon_{\text{opt}}/(6250J) \leq \varepsilon_{\text{ph},j}/(64J)$ for every phase j , which is the role played by the per-phase requirement $C_3m\delta_{g,j}^2 \leq \varepsilon_{\text{ph},j}/(64J)$ of Lemma 9.4.

10.1 Exact products and the input

The bit count uses bilinear algorithms for matrix multiplication with rational coefficients. We fix the exponent convention that supplies these exact algorithms.

Definition 10.2 (Asymptotic-rank exponents). For a tensor $\mathcal{T}$ over the rationals let $R(\mathcal{T})$ be its rank and $\tilde{R}(\mathcal{T}) := \lim_{k \rightarrow \infty} R(\mathcal{T}^{\otimes k})^{1/k}$ its asymptotic rank. For an integer $q \geq 2$ and a rational $\beta \in [0, 1]$ with q^β an integer, put $\omega(1, \beta, 1) := \log_q \tilde{R}(\langle q, q^\beta, q \rangle)$, where $\langle k_1, k_2, k_3 \rangle$ is the matrix multiplication tensor, and $\alpha_{\text{dual}} := \sup\{\beta : \omega(1, \beta, 1) = 2\}$; write $\omega := \omega(1, 1, 1)$. These are the definitions under which Vassilevska Williams, Xu, Xu, and Zhou [VWXXZ24, Section 3.4] prove $\alpha_{\text{dual}} = \alpha > 0.321334$ and Alman, Duan, Vassilevska Williams, Xu, Xu, and Zhou [ADVW⁺25] prove $\omega < 2.371339$ (their main theorem); both are laser-method degenerations of Coppersmith–Winograd tensors with integer coefficients, so the bounds hold for the rank over $\mathbb{Q}$, which is the field used here.

The asymptotic rank is defined through exact ranks of tensor powers, so no bilinear algorithm has to be extracted from an arithmetic circuit and no border-rank approximation has to be made exact: by Fekete's lemma $\tilde{R}(\mathcal{T}) = \inf_k R(\mathcal{T}^{\otimes k})^{1/k}$, and $\langle q, q^\beta, q \rangle^{\otimes k} = \langle q^k, q^{\beta k}, q^k \rangle$. Where a border-rank bound is converted into a rank bound inside the laser method, the conversion is Bini's interpolation [Bin80], in the form $R(\mathcal{T}) \leq (2h_B + 1)R_{h_B}(\mathcal{T})$ of Bläser [Blä13, Lemma 6.4 and Remark 6.5]; its rational Lagrange weights are the source of the denominators below.

Lemma 10.3 (Exact fast multiplication of fixed-point matrices). *Fix $\eta > 0$ and a rational $\beta \in \{1\} \cup (1/4, \alpha_{\text{dual}})$. There are integers $\varkappa \geq 2$, $t \leq \varkappa^{\omega(1,\beta,1)+\eta}$, and an exact bilinear algorithm $(U, V, W) \in \mathbb{Q}^{\varkappa^{1+\beta} \times t} \times \mathbb{Q}^{\varkappa^{1+\beta} \times t} \times \mathbb{Q}^{\varkappa^2 \times t}$ for $\langle \varkappa, \varkappa^\beta, \varkappa \rangle$ with $\varkappa^\beta$ an integer. Let $\mathfrak{d}$ be a common denominator of its entries, $\alpha_U := \max_s \sum_i |u_{is}|$, $\alpha_V := \max_s \sum_j |v_{js}|$, $\alpha_W := \max_r \sum_s |w_{rs}|$, each taken as the maximum over the three permuted decompositions used for the permuted shapes, and*

$$c_\eta := 2 \left\lceil \frac{\log_2(\mathfrak{d}^3 \alpha_U \alpha_V \alpha_W \varkappa)}{\log_2 \varkappa} \right\rceil + 2.$$

Let $X \in \mathbb{R}^{n_1 \times n_2}$ and $Y \in \mathbb{R}^{n_2 \times n_3}$ be fixed-point matrices with W -bit entries, (n_1, n_2, n_3) at most $(N_{\text{mult}}, N_{\text{mult}}^\beta, N_{\text{mult}})$ or $(N_{\text{mult}}, N_{\text{mult}}, N_{\text{mult}}^\beta)$ entrywise up to a permutation (smaller shapes are padded with zeros), $N_{\text{mult}} \leq 2n$ and $n \geq \varkappa$; recall $\omega(1, 1, \beta) = \omega(1, \beta, 1)$ [VWXXZ24, Section 3.4]. Then the recursive application of (U, V, W) on temporaries of $2W + c_\eta \lceil \log_2(2n) \rceil$ bits computes the exact product XY (an integer multiple of $\mathfrak{u}^2$) and rounds it once to the $\mathfrak{u}$ -grid, so that the returned matrix $\hat{C}$ satisfies $\|\hat{C} - XY\|_{\max} \leq \mathfrak{u}$, using

$$O(N_{\text{mult}}^{\omega(1,\beta,1)+2\eta}) \cdot M(2W + c_\eta \lceil \log_2(2n) \rceil)$$

bit operations. For $\beta = 1$ the exponent is $\omega + 2\eta$; for $\beta \in (1/4, \alpha_{\text{dual}})$ it is $2 + 2\eta$.

Proof. Step 1: the base algorithm. By Definition 10.2 and Fekete's lemma there is k_η with $R(\langle q, q^\beta, q \rangle^{\otimes k_\eta}) \leq q^{k_\eta(\omega(1,\beta,1)+\eta)}$; put $\varkappa := q^{k_\eta}$ and let (U, V, W) be a rank decomposition of $\langle \varkappa, \varkappa^\beta, \varkappa \rangle$ over $\mathbb{Q}$ with t terms. Exactness means that for all $X \in \mathbb{R}^{\varkappa \times \varkappa^\beta}$, $Y \in \mathbb{R}^{\varkappa^\beta \times \varkappa}$ and every output index r ,

$$(XY)_r = \sum_{s=1}^t w_{rs} \left(\sum_i u_{is} x_i \right) \left(\sum_j v_{js} y_j \right), \quad (56)$$

and because Eq. (56) never commutes an x_i past a y_j it holds verbatim when the x_i, y_j are matrix blocks.

Step 2: the recursion is exact. Pad X and Y with zeros to sizes $\varkappa^\varrho \times \varkappa^{\beta\varrho}$ and $\varkappa^{\beta\varrho} \times \varkappa^\varrho$ with $\varrho := \lceil \log_\varkappa N_{\text{mult}} \rceil$; the padding multiplies every size by at most $\varkappa$. Multiply all entries of U, V, W by $\mathfrak{d}$ so that they become integers; then Eq. (56) returns $\mathfrak{d}^3(XY)_r$. The recursive algorithm at depth ϱ forms the t linear forms $X_s := \sum_i \mathfrak{d} u_{is} X_i$ and $Y_s := \sum_j \mathfrak{d} v_{js} Y_j$ of the $\varkappa \times \varkappa^\beta$ resp. $\varkappa^\beta \times \varkappa$ block partitions, calls itself on (X_s, Y_s) , and returns $C_r := \sum_s \mathfrak{d} w_{rs} P_s$. Every operation is an addition or a multiplication by an integer constant of integers (the grid multiples of $\mathfrak{u}$, scaled by 2^f), hence exact, and the base products are exact products of integers. By induction on the depth the output equals $\mathfrak{d}^{3\varrho} XY$ exactly, and dividing by $\mathfrak{d}^{3\varrho}$ and rounding to the $\mathfrak{u}$ -grid gives $\|\hat{C} - XY\|_{\max} \leq \mathfrak{u}$.

Step 3: magnitudes. We claim that every value formed by the recursion has modulus at most $G_\varrho := (\mathfrak{d}^3 \alpha_U \alpha_V \alpha_W \varkappa)^{\varrho} 2^{2I}$ (here $\alpha_U \alpha_V \alpha_W \geq 1$ and each $\alpha_\bullet \geq 1/\mathfrak{d}$, both forced by the exactness of Eq. (56) and the integrality of the scaled coefficients). At depth ϱ the linear forms have entries of modulus at most $\mathfrak{d} \alpha_U 2^I$ and $\mathfrak{d} \alpha_V 2^I$; the exact product returned by a call at depth $\varrho - j$ on operands of moduli M_X, M_Y equals $\mathfrak{d}^{3(\varrho-j)}$ times a product with inner dimension at most $\varkappa^{\varrho-j}$, hence has modulus at most $\mathfrak{d}^{3(\varrho-j)} \varkappa^{\varrho-j} M_X M_Y$; and each assembly multiplies by at most $\mathfrak{d} \alpha_W$. Unrolling, the operands at depth $\varrho - j$ have moduli at most $(\mathfrak{d} \alpha_U)^j 2^I$, $(\mathfrak{d} \alpha_V)^j 2^I$, the returned products at most $\mathfrak{d}^{3(\varrho-j)} \varkappa^{\varrho-j} (\mathfrak{d}^2 \alpha_U \alpha_V)^j 2^{2I}$, and the partial sums of an assembly at most $\mathfrak{d} \alpha_W$ times that, all of which are at most G_ϱ . Since $\varrho \log_2 \varkappa \leq \log_2 N_{\text{mult}} + \log_2 \varkappa \leq 2 \log_2(2n)$ for $\varkappa \leq n$, we have $\log_2 G_\varrho \leq 2I + (c_\eta - 2) \lceil \log_2(2n) \rceil$, and a temporary of $2W + c_\eta \lceil \log_2(2n) \rceil$ bits holds every intermediate value together with its fractional part.

Step 4: cost. There are t^ϱ base products and $O(\varrho t^\varrho)$ additions and constant multiplications, each on the temporaries, so the cost is $O(\varrho t^\varrho M(2W + c_\eta \lceil \log_2(2n) \rceil))$ bit operations (the factor $\varrho = O(\log N_{\text{mult}})$ is absorbed in N_{mult}^η), and $t^\varrho \leq \varkappa^{\varrho(\omega(1,\beta,1)+\eta)} \leq (\varkappa N_{\text{mult}})^{\omega(1,\beta,1)+\eta} = O(N_{\text{mult}}^{\omega(1,\beta,1)+2\eta})$ for $N_{\text{mult}} \geq \varkappa^{(\omega(1,\beta,1)+\eta)/\eta}$; smaller N_{mult} are absorbed in the constant. For $\beta \in (1/4, \alpha_{\text{dual}})$ the exponent is $\omega(1, \beta, 1) = 2$ by definition of α_{dual} . $\square$

The constant c_η is finite for every fixed η and is not quantified by any source: the Lagrange weights of Bini's interpolation at $2N_\eta h_B + 1$ nodes have numerators and denominators of size $\exp(O(N_\eta \log N_\eta))$, so $c_\eta \rightarrow \infty$ as $\eta \rightarrow 0$. This is the same unquantified constant as in the stability exponent of Demmel, Dumitriu, Holtz, and Kleinberg [DDHK07, Theorem 3.3], whose floating-point analysis covers square products only; the fixed-point route above needs no stability theorem because every product is exact.

Rounding the input first reduces the subsequent arithmetic to dyadic matrices; a small contraction preserves the operator-norm normalization.

Lemma 10.4 (Input rounding). *Let $A_1, \dots, A_n$ be symmetric with $\|A_i\| \leq 1$, let $b_{\text{in}} := 2\lceil \log_2 n \rceil + 10$, and put $\eta_{\text{in}} := n2^{-b_{\text{in}}}$ and $\tilde{A}_i := \text{round}_{b_{\text{in}}}((1 - \eta_{\text{in}})A_i)$, the entrywise rounding to the nearest multiple of $2^{-b_{\text{in}}}$, applied identically to the two symmetric positions. Then every $\tilde{A}_i$ is a symmetric contraction with b_{in} -bit dyadic entries, and for every signing $\varepsilon \in \{-1, 1\}^n$,*

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| \leq \left\| \sum_{i=1}^n \varepsilon_i \tilde{A}_i \right\| + 2^{-9}.$$

Forming the $\tilde{A}_i$ from rational entries of bit length L_{in} costs $O(n^3(L_{\text{in}} + b_{\text{in}}) \log(L_{\text{in}} + b_{\text{in}}))$ bit operations.

Proof. The rounding error matrix $F_i := \tilde{A}_i - (1 - \eta_{\text{in}})A_i$ is symmetric with entries of modulus at most $2^{-b_{\text{in}}-1}$, so $\|F_i\| \leq \|F_i\|_F \leq n2^{-b_{\text{in}}-1} = \eta_{\text{in}}/2$. Hence $\|\tilde{A}_i\| \leq (1 - \eta_{\text{in}}) + \eta_{\text{in}}/2 < 1$ and $\|\tilde{A}_i - A_i\| \leq \eta_{\text{in}}\|A_i\| + \eta_{\text{in}}/2 \leq 2\eta_{\text{in}}$. Summing over i with signs, $\|\sum \varepsilon_i (A_i - \tilde{A}_i)\| \leq 2n\eta_{\text{in}} = 2n^2 2^{-b_{\text{in}}} \leq 2^{-9}$, because $2^{b_{\text{in}}} \geq 2^{10} n^2$. Each entry is rounded by one integer division of an $(L_{\text{in}} + b_{\text{in}})$ -bit numerator, at cost $O(M(L_{\text{in}} + b_{\text{in}}))$, and there are n^3 entries. $\square$

Theorem 5.1 is applied to the $\tilde{A}_i$: its hypotheses are only symmetry and $\|\tilde{A}_i\| \leq 1$, its proof certifies $(154709.3 + 6.5)\sqrt{n}$, and $2^{-9} < 0.002\sqrt{n}$ for all $n \geq 1$, so the constant 156000 absorbs the rounding. Below, A_i denotes $\tilde{A}_i$.

10.2 State, parameters, and randomness

The finite algorithm keeps every quantity that the real-arithmetic algorithm updates incrementally on a fixed rational grid, so that these updates are exact, and it replaces each analysis parameter by a nearby dyadic number rounded in the direction that keeps the inequality it enters. The next lemma lists the grids, the rounded parameters, and the three consequences the later sections use: the rounded body still satisfies the hypotheses of Lemma 5.4, the snapping step still lands in the body, and the discrepancy accounting of Theorem 5.1 changes by less than $0.1\sqrt{n}$.

Lemma 10.5 (Exact state and rounded parameters). *Fix an outer round with active set S , $|S| = m \geq N_0$, a fractional coloring $y \in (\mathbb{U}\mathbb{Z})^S \cap (-1, 1)^S$, and $f \geq 40 + \lceil \log_2(2n) \rceil$. Let $\underline{\ell}$ and $\bar{\ell}$ be dyadic numbers with $\underline{\ell} \leq \log(n/m) \leq \bar{\ell} \leq \underline{\ell} + 10^{-9}$, computed by a certified evaluation of the logarithm to 40 bits. Define*

$$\tilde{a} := \lceil a \rceil_{\mathbb{U}}, \quad R_{\text{alg}} := \lceil c_{\text{rad}}(\underline{\ell})\sqrt{m} \max\{1, \bar{\ell}\} \rceil_{\mathbb{U}}, \quad \tilde{c} := \lfloor \tilde{a}/R_{\text{alg}} \rfloor_{\mathbb{U}}, \quad \tilde{R} := \tilde{a}/\tilde{c},$$

$$\tilde{b} := \lfloor b \rfloor_{\mathbf{u}}, \quad \tilde{\theta} := \lfloor \tilde{a}\tilde{b} \rfloor_{\mathbf{u}}, \quad b' := \tilde{\theta}/\tilde{a}, \quad \tilde{\chi} := \log(2n)/\tilde{a},$$

where $\lceil \cdot \rceil_{\mathbf{u}}$ and $\lfloor \cdot \rfloor_{\mathbf{u}}$ round up and down to the grid $\mathbf{u}\mathbb{Z}$ and $c_{\text{rad}}(\cdot)$ is the step function of Table 2, extended by $c_{\text{rad}}(v) := c_{\text{rad}}(1) = 279$ for $v < 1$ so that it is nonincreasing on $\mathbb{R}$. Let $\Gamma := (\mathbf{u}/4000)\mathbb{Z}$, and for $x \in \Gamma^S$ put

$$B(x) := \sum_{i \in S} x_i A_i, \quad D_x := \tilde{c} \text{diag}(B(x), -B(x)) - \tilde{\theta} I_{d_0}, \quad \hat{D}_x := \text{round}_{\mathbf{u}}(D_x),$$

$$\tilde{K} := \{x \in \mathbb{R}^S : \|B(x)\| \leq \tilde{R}\}, \quad \tilde{\mathcal{D}} := \{x \in \mathbb{R}^S : \text{tr}[\exp(D_x)] \leq 1\},$$

$$\tilde{Q} := \prod_{i \in S} \left[-\frac{\varepsilon_{\text{box}}}{2}(1 + y_i), \frac{\varepsilon_{\text{box}}}{2}(1 - y_i) \right].$$

Then the following hold.

- (i) *Grids.* The endpoints of $\tilde{Q}$ lie in Γ ; for $x \in \Gamma^S$ the entries of $B(x)$ lie in $2^{-b_{\text{in}}}\Gamma$ and those of D_x in $\mathbf{u}2^{-b_{\text{in}}}\Gamma$; the update $x \mapsto x + d e_i$ with $d \in \Gamma$, the update of $B(x)$, and $q(x) := \frac{1}{2}\|x - \xi'\|_2^2$ for $\xi' \in (\mathbf{u}\mathbb{Z})^S$ are exact rational operations, and every comparison among these quantities is exact. Moreover $\|\hat{D}_x - D_x\| \leq d_0 \mathbf{u}$; the algorithm recomputes $\hat{D}_x$ from the exact D_x whenever it is needed, so rounding errors in $\hat{D}_x$ never accumulate across steps.
- (ii) *Parameter directions.* $\tilde{a} \geq a$, $\tilde{\chi} \leq \chi$, $R \leq R_{\text{alg}} \leq \tilde{R} \leq (1 + 2^{-30})R_{\text{alg}}$, and $b - 2\mathbf{u} \leq b' \leq \tilde{b} \leq b$; the matrices $E_i := \partial_i D_x = \tilde{c} \text{diag}(A_i, -A_i)$ satisfy $\|E_i\| \leq \tilde{c}$ and $\sum_{i \in S} E_i^2 \preceq m\tilde{c}^2 I_{d_0}$, and $a/(2R) \leq \tilde{c} \leq (a + \mathbf{u})/R \leq 2a/R$.
- (iii) *The rounded body.* $(b' - \tilde{\chi})\tilde{K} \subseteq \tilde{\mathcal{D}} \subseteq b'\tilde{K} \subseteq (1 - s_{\text{sm}})\tilde{K}$, $\tilde{\mathcal{D}}$ is closed, convex, and symmetric, and

$$\gamma_m(\tilde{\mathcal{D}}) \geq \exp(-0.00372m).$$

- (iv) *Snapping.* Let $\tilde{\rho} := \lfloor \min\{\frac{1}{8}, s_{\text{sm}}R_{\text{alg}}/(8(2m + \sqrt{m}))\} \rfloor_{\mathbf{u}}$. If $z \in \tilde{Q}$ satisfies $\|z - x_*\|_2 \leq \tilde{\rho}$ for the exact minimizer x_* of Eq. (15) over $\tilde{Q} \cap \tilde{\mathcal{D}}$, and $\bar{x}$ is obtained from z by moving every coordinate within $2\tilde{\rho}$ of a facet of $\tilde{Q}$ onto that facet, then every coordinate of x_* that lies on a facet is moved, and $\|B(\bar{x})\| \leq (1 - s_{\text{sm}}/2)\tilde{R} < \tilde{R}$.
- (v) *Accounting.* A successful round increases $\ell := \log(n/m)$ by at least $\lambda = -\log(1 - \delta)$ and contributes at most $(1 + 2^{-30})(1 + 10^{-9}) \cdot \frac{2c_{\text{rad}}(\ell - 10^{-9})}{\varepsilon_{\text{box}}} e^{-\ell/2} \max\{1, \ell\} \sqrt{n}$ to the discrepancy, and the sum over all rounds is less than $154710\sqrt{n}$; the rounding of y in Lemma 10.18(iv) is included in this bound.

Proof. (i), *grids.* Since $\varepsilon_{\text{box}}/2 = 847/4000$ and $1 \mp y_i \in \mathbf{u}\mathbb{Z}$, the endpoints $\frac{\varepsilon_{\text{box}}}{2}(1 \mp y_i)$ are multiples of $847\mathbf{u}/4000 \in \Gamma$. The entries of A_i are multiples of $2^{-b_{\text{in}}}$, so $B(x)$ has entries in $2^{-b_{\text{in}}}\Gamma$ for $x \in \Gamma^S$, and $\tilde{c}, \tilde{\theta} \in \mathbf{u}\mathbb{Z}$ give entries of D_x in $\mathbf{u}2^{-b_{\text{in}}}\Gamma$. Additions of numbers on a common grid are exact, $q(x)$ is a sum of squares of numbers in Γ , hence a multiple of $\mathbf{u}^2/(2 \cdot 4000^2)$, and comparisons of rationals with a common denominator are integer comparisons. Rounding each entry of D_x to the nearest multiple of $\mathbf{u}$ changes it by at most $\mathbf{u}/2$, so $\|\hat{D}_x - D_x\| \leq \|\hat{D}_x - D_x\|_F \leq d_0 \mathbf{u}/2$.

(ii), *directions.* Rounding up gives $\tilde{a} \geq a$ and hence $\tilde{\chi} \leq \chi$. Because c_{rad} is nonincreasing and $\underline{\ell} \leq \log(n/m) \leq \bar{\ell}$, $R_{\text{alg}} \geq c_{\text{rad}}(\log(n/m))\sqrt{m} \max\{1, \log(n/m)\} = R$. Rounding down gives $\tilde{c} \leq \tilde{a}/R_{\text{alg}}$, so $\tilde{R} \geq R_{\text{alg}}$, and $\tilde{c} \geq \tilde{a}/R_{\text{alg}} - \mathbf{u}$ gives $\tilde{R} \leq R_{\text{alg}}/(1 - \mathbf{u}R_{\text{alg}}/\tilde{a}) \leq (1 + 2^{-30})R_{\text{alg}}$ because $R_{\text{alg}} \leq 2c_{\text{rad}}(1)\sqrt{n} \log(2n) \leq 2^{f-31}\tilde{a}$ for $f \geq 40 + \lceil \log_2(2n) \rceil$ and $a \geq 2 \cdot 10^6$. Rounding down twice gives $\tilde{\theta} \leq \tilde{a}\tilde{b}$, so $b' \leq \tilde{b} \leq b$, and $\tilde{\theta} \geq \tilde{a}\tilde{b} - \mathbf{u} \geq \tilde{a}(b - \mathbf{u}) - \mathbf{u}$ gives $b' \geq b - 2\mathbf{u}$. The

bounds on E_i follow from $\|A_i\| \leq 1$ and $\sum_i A_i^2 \preceq mI$ exactly as in Eq. (25), with $\tilde{c}$ in place of a/R ; and $\tilde{c} \leq \tilde{a}/R_{\text{alg}} \leq \lceil a \rceil_{\text{u}}/R \leq (a + \text{u})/R$, while $\tilde{c} \geq \tilde{a}/R_{\text{alg}} - \text{u} \geq a/(1.21R) - \text{u} \geq a/(2R)$, because $R_{\text{alg}} \leq (1 + 10^{-9})c_{\text{rad}}(\ell)\sqrt{m} \max\{1, \bar{\ell}\} + \text{u} \leq 1.21R$, the largest ratio of consecutive values of c_{rad} being 240/200.

(iii), *the rounded body*. Write $\tilde{h}(x) := \tilde{a}^{-1} \log \text{tr}[\exp(\tilde{a} \text{diag}(B(x), -B(x))/\tilde{R})]$, so that $D_x = \tilde{a} \text{diag}(B(x), -B(x))/\tilde{R} - \tilde{a}b'I$ and $\text{tr}[\exp(D_x)] \leq 1$ if and only if $h(x) \leq b'$. The log-sum-exp bounds $\|B(x)\|/\tilde{R} \leq \tilde{h}(x) \leq \|B(x)\|/\tilde{R} + \tilde{\chi}$ give the two inclusions, and $b' \leq b = 1 - s_{\text{sm}}$ gives the third. Convexity of h and the symmetry $\tilde{h}(-x) = \tilde{h}(x)$ give the shape statements. For the measure, $\tilde{K} \supseteq K$ by $\tilde{R} \geq R$, so Lemma 4.1 gives $\gamma_m(\tilde{K}) \geq \exp(-0.003712m)$, and the dilation bound $\gamma_m(\theta\tilde{K}) \geq \theta^m \gamma_m(\tilde{K})$ for $\theta \in (0, 1)$, which follows from the change of variables $x \mapsto \theta x$ and the monotonicity of the Gaussian density along rays, gives

$$\begin{aligned} \gamma_m(\tilde{\mathcal{D}}) &\geq (b' - \tilde{\chi})^m \exp(-0.003712m) \\ &\geq \exp(-(0.003712 - \log(1 - s_{\text{sm}} - \chi - 2\text{u}))m) \geq \exp(-0.00372m), \end{aligned}$$

because $-\log(1 - 10^{-6} - 2^{-39}) < 1.0001 \cdot 10^{-6}$ and $0.003712 + 1.0001 \cdot 10^{-6} < 0.00372$.

(iv), *snapping*. A coordinate of x_* on a facet has $|z_i - \text{facet}| \leq \|z - x_*\|_2 \leq \tilde{\rho} < 2\tilde{\rho}$ and is moved. Each move changes one coordinate by at most $2\tilde{\rho}$, and $\|A_i\| \leq 1$, so $\|B(\bar{x}) - B(z)\| \leq 2m\tilde{\rho} \leq s_{\text{sm}}R_{\text{alg}}/4 \leq s_{\text{sm}}\tilde{R}/4$. Since $z \in \tilde{\mathcal{D}} \subseteq (1 - s_{\text{sm}})\tilde{K}$ was not assumed, we use instead $\|B(z)\| \leq \|B(x_*)\| + \|B(z - x_*)\| \leq (1 - s_{\text{sm}})\tilde{R} + \sqrt{m}\tilde{\rho}$ (from $x_* \in \tilde{\mathcal{D}}$ and $\|B(v)\| \leq \|v\|_1 \leq \sqrt{m}\|v\|_2$), so $\|B(\bar{x})\| \leq (1 - s_{\text{sm}})\tilde{R} + (2m + \sqrt{m})\tilde{\rho} \leq (1 - s_{\text{sm}})\tilde{R} + s_{\text{sm}}\tilde{R}/8 < (1 - s_{\text{sm}}/2)\tilde{R}$.

(v), *accounting*. The increment of a round is $2\bar{x}/\varepsilon_{\text{box}}$ with $\|B(\bar{x})\| < \tilde{R} \leq (1 + 2^{-30})c_{\text{rad}}(\ell)\sqrt{m} \max\{1, \bar{\ell}\}$, and $c_{\text{rad}}(\ell) \leq c_{\text{rad}}(\ell - 10^{-9})$, $\max\{1, \bar{\ell}\} \leq (1 + 10^{-9}) \max\{1, \ell\}$ for $\ell = \log(n/m)$, which gives the displayed bound with $\sqrt{m} = e^{-\ell/2}\sqrt{n}$. The summation in the proof of Theorem 5.1 bounds $\sum_{\text{rounds}} c_{\text{rad}}(\ell)g(\ell)$, $g(\ell) = \ell e^{-\ell/2}$, by the bin suprema $\sum_k M_k(a)$ of Lemma 5.5 for $a \in \{1, 35\lambda\}$ and by the geometric count of phases starting below $\ell = 1$. Replacing $c_{\text{rad}}(\ell)$ by $c_{\text{rad}}(\ell - 10^{-9})$ leaves the six values and the count below 1 unchanged (both functions equal 279 for $\ell < 1 + 10^{-9}$) and moves the five jump points t_r right by 10^{-9} . No bin endpoint $a + k\lambda$ lies within 10^{-9} of a jump point (the least distance exceeds $1.6 \cdot 10^{-3}$ for both values of a and every bin before the geometric tail, by the enclosure of λ), so each window $(t_r, t_r + 10^{-9}]$ lies in the interior of one bin together with the left limit $c_{\text{rad}, r-1}g(t_r)$, which that bin's supremum already dominates; hence each of the at most five affected bin suprema grows by at most $279 \sup_{\ell \geq 1} |g'(\ell)| \cdot 10^{-9} < 10^{-7}$, each sum $\sum_k M_k(a)$ by less than 10^{-6} , and the total after the factor $2/\varepsilon_{\text{box}}$ by less than $10^{-5} < 10^{-3}$. Multiplying the constant 154709.3 of the proof of Theorem 5.1 by $(1 + 2^{-30})(1 + 10^{-9})$ and adding 10^{-3} gives less than 154710. $\square$

The remaining parameters of the wrapper and the solver are rounded as follows, each in the direction that preserves the inequality it enters, and each computable in $O(\text{polylog}(n))$ bit operations from $n, m, \|\xi'\|_2^2$, and ν : $\tilde{\Lambda} := \lceil (8/3)q(0) \rceil_{\text{u}} \geq \Lambda$ (using $\phi_0 < 1/4$); $\tilde{H} := \lceil 16(q(0) + \nu/4) \rceil_{\text{u}} \geq 16F_{\nu}(0)$ and $\tilde{H}_{\text{max}} := \lceil 16(q(0) + \tilde{\Lambda}/4 + 1) \rceil_{\text{u}} \geq H_{\text{max}}$; $\tilde{G}_* := \lceil \tilde{c}\sqrt{m} \rceil_{\text{u}} \geq \tilde{c}\sqrt{m}$, the Lipschitz constant of Φ for the finite objective; $\tilde{\nu}_{\min} := \lfloor \tilde{\rho}/(8\tilde{G}_*) \rfloor_{\text{u}}$ and $\tilde{\nu} := \min\{\tilde{\nu}_{\min}, \tilde{\Lambda}/2\}$ (rounded down to $\text{u}\mathbb{Z}$); $\tilde{d}_{\text{w}} := \lfloor \tilde{\rho}/4 \rfloor_{\text{u}}$, $\tilde{\tau}_{\text{w}} := \lfloor \min\{\tilde{d}_{\text{w}}^2/(1 + \tilde{\Lambda}), 3\tilde{d}_{\text{w}}/(64\sqrt{m})\} \rfloor_{\text{u}}$, $\tilde{G}_{\text{lev}} := \lceil \tilde{G}_*\tilde{H}_{\text{max}}/\tilde{\nu} \rceil_{\text{u}}$, $\tilde{\sigma} := \lfloor \min\{\tilde{d}_{\text{w}}/8, \tilde{\tau}_{\text{w}}/(16\tilde{G}_{\text{lev}})\} \rfloor_{\text{u}}$, $\tilde{\varepsilon}_{\text{opt}} := \lfloor \tilde{\sigma}^2/2 \rfloor_{\text{u}}$, $\tilde{J} := \lceil \log_2(4\tilde{H}/\tilde{\varepsilon}_{\text{opt}}) \rceil$; $\tilde{L} := 4 \cdot 2^{\lceil \log_2 \max\{1 + 3\tilde{c}^2\tilde{H}, 2\tilde{c}, 1\} \rceil}$, $\tilde{\alpha} := 1/(m\tilde{L})$, $\tilde{T} := 16m\tilde{L}$; $\tilde{\delta}_g := \lfloor \min\{\tilde{H}/(10^5\tilde{m}), \tilde{\varepsilon}_{\text{opt}}/(C_0\tilde{J}m)\}^{1/2} \rfloor_{\text{u}}$ with $C_0 := 2 \cdot 10^5$; $\pi' :=$ the least power of two with $\pi' \geq \max\{n^{-1/2}, 4\tilde{\alpha}\}$, so that $\pi' \in [\pi_m, 2\pi_m]$ and $\pi' \geq 4\tilde{\alpha}$; $k = s := \lceil C_{ks}\pi'^{-1/2} \rceil$ with $C_{ks} := 45$, so that $ks \geq C_{ks}^2/\pi' \geq K_0/\pi'$ for $K_0 = 2000$. The tolerances ε_{op} and ε_{act} of the matrix functions are fixed in Lemmas 10.9 and 10.10. Every inequality of Sections 7 and 9 that involves $a, R, b, H, L, \Lambda, \underline{\nu}, \tau_{\text{w}}, \sigma, \varepsilon_{\text{opt}}, \delta_g$, or π_m is re-verified below for the rounded values:

two-sidedly in (ii)–(v) for the parameters of the body, and through the ratio bounds recorded after the requirement table in the proof of Theorem 10.1 for the wrapper parameters; no monotonicity in the rounding direction is claimed. In the rest of the section we drop the tildes on these parameters and on Φ , F , and Δ , which now refer to the finite objective $F(x) := q(x) + \nu \operatorname{tr}[\exp(D_x)]$ on $\tilde{Q}$ with ξ' in place of ξ ; F is 1-strongly convex, and the coordinate curvature bound $\partial_{ii}^2 F \leq 1 + e\tilde{c}^2 H \leq L/4$ and the trust radius $1/\tilde{c}$ of Section 9 hold with $\tilde{c}$ in place of a/R by Lemma 10.5(ii).

Rademacher vectors supply bounded probes with the isotropy and variance estimates used by the tracker and its guards. The remaining random choices have exact implementations from fair coins.

Lemma 10.6 (Discrete probes and exact coins). *Let $z \in \{-1, 1\}^{d_0}$ have independent fair-coin entries and let M be a real symmetric $d_0 \times d_0$ matrix. Then*

$$\mathbb{E}[zz^\top] = I_{d_0}, \quad \mathbb{E}[(Mz)z^\top] = M, \quad \mathbb{V}[z^\top Mz] = 2\|M\|_F^2 - 2 \sum_j M_{jj}^2 \leq 2\|M\|_F^2, \quad \|z\|_2^2 = d_0.$$

Moreover, a uniform index in $[m]$, a Bernoulli coin of dyadic bias π' , and a Bernoulli coin of bias $(1 + y_i)/2$ for $y_i \in \mathbb{u}\mathbb{Z}$ can each be drawn exactly with an expected $O(\log m)$ (capped as described in the proof), exactly $\log_2(1/\pi')$, and exactly $f + 1$ fair coins.

Proof. The first two identities follow from $\mathbb{E}[z_j z_l] = \delta_{jl}$. For the variance, $z^\top Mz = \sum_j M_{jj} + 2 \sum_{j < l} M_{jl} z_j z_l$, and the products $z_j z_l$, $j < l$, are pairwise uncorrelated fair signs, so $\mathbb{V}[z^\top Mz] = 4 \sum_{j < l} M_{jl}^2 = 2\|M\|_F^2 - 2 \sum_j M_{jj}^2$. The norm identity is $z_j^2 = 1$. A uniform index is drawn by rejection from $\lceil \log_2 m \rceil$ coins, with acceptance probability at least $1/2$ per round; the rounds are capped at $\lceil \log_2(N_s/p_s) \rceil$ for a procedure drawing N_s indices, the cap being a declared failure of probability at most p_s in total; a dyadic bias $\pi' = 2^{-j}$ is the event that j coins are all heads; and $(1 + y_i)/2 \in 2^{-f-1}\mathbb{Z}$ is compared with a uniform $(f + 1)$ -bit integer. $\square$

With Rademacher probes the Hutchinson estimates of Sections 7 and 9 keep their means, and their variance bounds $2\|M\|_F^2$ hold with $\leq$ in place of $=$; the range-finder test matrix is the one place where a Gaussian is needed, and it is treated in Lemmas 10.7 and 10.11.

Lemma 10.7 (Rounded Gaussians and the projection lemma). *Let $h_g := 2^{-16}$ and $h_\Omega := 2^{-b_\Omega}$ with $b_\Omega := \lceil 4.5 \log_2 n \rceil + 7$.*

- (i) *Exact rounded Gaussians. There is an algorithm that, from fair coins, outputs a number $\xi' \in h\mathbb{Z}$ whose law is exactly that of $\operatorname{round}_h(G)$ for $G \sim \mathcal{N}(0, 1)$, for any $h = 2^{-b_h}$, using an expected $O(b_h)$ coins and $O(b_h)$ bit operations; on the same coin space there is a $G \sim \mathcal{N}(0, 1)$ with $|\xi' - G| \leq h/2$ surely. Capping the coins per sample at $C_{\text{cap}}(b_h + \log_2(N_s/p_s))$ for a universal C_{cap} and declaring failure at the cap adds at most p_s to the failure probability of any procedure that draws N_s samples.*
- (ii) *Lipschitz transfer. If $F : \mathbb{R}^m \rightarrow \mathbb{R}$ is 1-Lipschitz for the Euclidean norm and $\xi' = \operatorname{round}_h(G)$ coordinatewise, then $|F(\xi') - F(G)| \leq \sqrt{m} h/2$ surely.*
- (iii) *The projection lemma survives. Let $m \geq N_0$, let $K \subseteq \mathbb{R}^m$ be a symmetric closed convex set with $\gamma_m(K) \geq \exp(-0.00372m)$, let $y \in (-1, 1)^m$, and let $\xi' = \operatorname{round}_{h_g}(G)$ with $G \sim \mathcal{N}(0, I_m)$. Then the Euclidean projection of ξ' onto $K \cap Q_y$ has more than δm coordinates on facets of Q_y with probability greater than 0.126.*

(iv) Rounded test matrix. Let $\Omega_{\max} := 1 + \sqrt{2 \log(4nk/u^2)}$. The matrix $\hat{\Omega} \in (h_\Omega \mathbb{Z})^{d_0 \times 2k}$ is drawn as independent samples of $\text{round}_{h_\Omega}(\mathcal{N}(0, 1))$; if $\|\hat{\Omega}\|_{\max} > \Omega_{\max}$ it is redrawn once, and a second oversized draw is a declared failure. One draw satisfies $\hat{\Omega} = \Omega + E_\Omega$ for a standard Gaussian Ω on the same coin space with $\|E_\Omega\|_{\max} \leq h_\Omega/2$ and is oversized with probability at most u^2 ; the declared failure has probability at most u^4 per test matrix, and on its complement the accepted matrix has exactly the law of one draw conditioned on the event $\{\|\hat{\Omega}\|_{\max} \leq \Omega_{\max}\}$, of probability at least $1 - u^2$; $\Omega_{\max} = O(\sqrt{\log n})$ for $f = O(\log n)$ as in Eq. (54).

Proof. (i), the sampler. This is Karney's algorithm [Kar16, Section 3]: it generates the binary digits of a normal deviate one at a time, using only integer comparisons of partially generated uniform deviates, so that the digits generated are those of an exactly normal G ; stopping after the digit that determines the rounding to $h\mathbb{Z}$ yields $\xi' = \text{round}_h(G)$ with G the deviate that the continued digit stream defines. The number of coins consumed is a stopping time with an exponential tail, because each of the $O(1)$ expected rejection rounds terminates a comparison with probability at least $1/2$ per digit, so a cap of $C_{\text{cap}}(b_h + \log_2(N_s/p_s))$ coins fails with probability at most p_s/N_s per sample, and a union bound over N_s samples gives (i). (ii), transfer. This is $|F(\xi') - F(G)| \leq \|\xi' - G\|_2 \leq \sqrt{m} h/2$.

(iii), transfer of the projection lemma. The proof of Lemma 5.4 has two parts. The deterministic part shows that for every point $\xi \in \mathbb{R}^m$ with

$$d(\xi, Q_y) > d(\xi, K \cap H(\mathcal{I})) \quad \text{for all } \mathcal{I} \subseteq [m] \text{ with } |\mathcal{I}| \leq \delta m, \quad (57)$$

the projection of ξ onto $K \cap Q_y$ has more than δm tight facets. The probabilistic part shows that a standard Gaussian G satisfies Eq. (57) with margin: on an event of probability greater than 0.126 the left side exceeds $(\sqrt{\psi(\varepsilon_{\text{box}})} - 1/N_0 - 3 \cdot 10^{-5})\sqrt{m}$ and every right side is below $(\sqrt{\delta} + 2(0.00372) + 1/N_0 + \sqrt{2H_2(\delta)} + 3 \cdot 10^{-5})\sqrt{m}$ (H_2 the binary entropy, written h in Section 5), and the difference of the two thresholds is $(r_{\text{sep}} - 6 \cdot 10^{-5})\sqrt{m} > 1.72 \cdot 10^{-5}\sqrt{m}$ by the enclosure $10^6(r_{\text{sep}} - 6 \cdot 10^{-5}) > 17.2$ displayed there (r_{sep} is the separation constant r of that proof). Both distance functions are 1-Lipschitz in ξ , so by (ii) with $h = h_g$ each moves by at most $\sqrt{m} 2^{-17}$ when G is replaced by ξ' , and Eq. (57) holds for ξ' on the same event because $2 \cdot 2^{-17} < 1.53 \cdot 10^{-5} < 1.72 \cdot 10^{-5}$. The deterministic part applied to ξ' gives (iii).

(iv), the test matrix. Apply (i) with $h = h_\Omega$ to each of the $2d_0k$ entries. For a standard normal G and $t \geq 1$, $\Pr[|G| > t] \leq e^{-t^2/2}$, and $|\hat{\Omega}_{jl}| > \Omega_{\max}$ forces $|\Omega_{jl}| > \Omega_{\max} - 1$, an event of probability at most $u^2/(4nk)$; the union bound over $4nk$ entries gives u^2 , and the two draws are independent, so both are oversized with probability at most u^4 . Writing $\mathcal{A}$ for the event that a draw is not oversized, the accepted matrix lies in a set S and no failure occurs with probability $\Pr[\mathcal{A} \cap S](1 + \Pr[\mathcal{A}^c])$, while no failure occurs with probability $1 - \Pr[\mathcal{A}^c]^2 = (1 + \Pr[\mathcal{A}^c]) \Pr[\mathcal{A}]$; the ratio is $\Pr[S \mid \mathcal{A}]$, the conditioned law of one draw. $\square$

The rejection test $\|\xi'\|_2^2 \leq 2m$ ($C_g = 2$) of Section 6 is an exact comparison on ξ' , and $\tilde{\Lambda} \leq \frac{8}{3}m + u$ on the accepted event. In Lemma 10.7(iii) the body is $\tilde{\mathcal{D}}$ of Lemma 10.5(iii), and the exact projection onto $\tilde{\mathcal{D}} \cap Q_y$ is the point x_* of Lemma 10.5(iv), so a computed point within $\tilde{\rho}$ of x_* snaps to more than δm signs on the event of Lemma 10.7(iii). The cube deviation $3 \cdot 10^{-5}$ and section deviation $3 \cdot 10^{-5}$ are unchanged, so the failure probabilities $e^{-0.135} + e^{-4558}$ and the success probability 0.126 are those of Lemma 5.4; the coupling consumes $1.53 \cdot 10^{-5}$ of the $1.72 \cdot 10^{-5}$ margin and nothing else.

10.3 Matrix exponentials in fixed point

The real-arithmetic algorithm approximates e^{D_x} by the centered Taylor square $q_r(D_x)^2$ of Section 7. Its coefficients $e^{c/2}/j!$ with $c \approx -ab$ are of size $(2n)^{-10^6}$ and its partial sums cancel by a factor e^{ab} below the midpoint of the spectral interval, so evaluating it needs about $2 \cdot 10^6 \log_2(2n)$ bits per number. We use instead the scaled Taylor power

$$\mathcal{Q}(D) := T_r(D/2^{\kappa_{\text{sc}}})^{2^{\kappa_{\text{sc}}}}, \quad T_r(y) := \sum_{j=0}^r \frac{y^j}{j!}, \quad (58)$$

whose factors have spectrum in $[1/3, e^{(\lambda_{\text{sp}}+1)/\sigma_{\text{sc}}}]$, evaluated by repeated squaring for dense matrices and by repeated application for vectors. It is one exact matrix, a deterministic function of x , positive definite, and it serves every place where Sections 7 and 9 use $\tilde{P}_x$.

Lemma 10.8 (Scaled Taylor surrogate). *Let D be symmetric with spectrum in $[-M_{\text{sp}}, \lambda_{\text{sp}}]$, $\lambda_{\text{sp}} \geq 0$, let $2^{\kappa_{\text{sc}}} =: \sigma_{\text{sc}} \geq M_{\text{sp}} + \lambda_{\text{sp}}$, let $r \geq 2$ satisfy $\sigma_{\text{sc}}\eta_r \leq 1$ with $\eta_r := e^2/(r+1)!$, and put $X := D/\sigma_{\text{sc}}$. Then:*

- (i) *for $y \in [-1, 1]$, $T_r(y) = e^y(1 + \vartheta)$ with $|\vartheta| \leq \eta_r$, T_r is nondecreasing on $[-1, 1]$, and $T_r(y) \geq T_r(-1) \geq 1/3$;*
- (ii) *$\|T_r(X)^j\| \leq e^{(\lambda_{\text{sp}}+1)j/\sigma_{\text{sc}}} \leq e^{\lambda_{\text{sp}}+1}$ for $0 \leq j \leq \sigma_{\text{sc}}$, and $\mathcal{Q}(D) = T_r(X)^{\sigma_{\text{sc}}} \succeq 3^{-\sigma_{\text{sc}}} I \succ 0$;*
- (iii) *$\|\mathcal{Q}(D) - e^D\| \leq 3\sigma_{\text{sc}}e^{\lambda_{\text{sp}}+2}/(r+1)!;$*
- (iv) *for symmetric D' with $\|D' - D\| \leq \zeta \leq 1$, $\|e^{D'} - e^D\| \leq \zeta e^{\lambda_{\text{sp}}+1}$.*

Proof. (i) The Lagrange remainder gives $|T_r(y) - e^y| \leq e^{\max\{y, 0\}} |y|^{r+1}/(r+1)! \leq e/(r+1)!$ on $[-1, 1]$, and dividing by $e^y \geq e^{-1}$ gives $|\vartheta| \leq e^2/(r+1)! = \eta_r$. Since $T'_r = T_{r-1}$ and $T_1(y) = 1 + y \geq 0$ on $[-1, 1]$, induction gives $T'_r \geq 0$ there, so $T_r(y) \geq T_r(-1)$; and $T_r(-1) = \sum_{j \leq r} (-1)^j/j!$ is an alternating partial sum of e^{-1} , so $|T_r(-1) - e^{-1}| \leq 1/(r+1)! \leq 1/120$ for $r \geq 4$, giving $T_r(-1) > 0.359$, while $T_2(-1) = 1/2$ and $T_3(-1) = 1/3$; hence $T_r(-1) \geq 1/3$ for all $r \geq 2$.

(ii) The spectrum of X lies in $[-1, \lambda_{\text{sp}}/\sigma_{\text{sc}}] \subseteq [-1, 1]$, so by (i) and functional calculus $\|T_r(X)^j\| = \max_y |T_r(y)|^j \leq (e^{\lambda_{\text{sp}}/\sigma_{\text{sc}}}(1 + \eta_r))^j \leq e^{(\lambda_{\text{sp}}+\sigma_{\text{sc}}\eta_r)j/\sigma_{\text{sc}}} \leq e^{(\lambda_{\text{sp}}+1)j/\sigma_{\text{sc}}}$, and $T_r(X) \succeq (1/3)I$ gives the lower bound.

(iii) Diagonalize $D = \sum_i \mu_i v_i v_i^\top$; then $\|\mathcal{Q}(D) - e^D\| = \max_i |T_r(\mu_i/\sigma_{\text{sc}})^{\sigma_{\text{sc}}} - e^{\mu_i}|$. For $y = \mu_i/\sigma_{\text{sc}} \in [-1, 1]$, (i) gives $T_r(y)^{\sigma_{\text{sc}}} - e^{\sigma_{\text{sc}}y} = e^{\sigma_{\text{sc}}y}((1 + \vartheta)^{\sigma_{\text{sc}}} - 1)$, and $|(1 + \vartheta)^{\sigma_{\text{sc}}} - 1| \leq e^{\sigma_{\text{sc}}\eta_r} - 1 \leq (e - 1)\sigma_{\text{sc}}\eta_r \leq 3\sigma_{\text{sc}}\eta_r$ because $\sigma_{\text{sc}}\eta_r \leq 1$ and $e^t - 1 \leq (e - 1)t$ on $[0, 1]$; finally $e^{\sigma_{\text{sc}}y} = e^{\mu_i} \leq e^{\lambda_{\text{sp}}}$.

(iv) Duhamel's formula $e^{D'} - e^D = \int_0^1 e^{(1-t)D'}(D' - D)e^{tD} dt$ and $\|e^{tB}\| = e^{t\lambda_{\text{max}}(B)}$ for symmetric B give $\|e^{D'} - e^D\| \leq \zeta e^{\max\{\lambda_{\text{max}}(D), \lambda_{\text{max}}(D')\}} \leq \zeta e^{\lambda_{\text{sp}}+\zeta}$. $\square$

On the cubic path $D = \hat{D}_x$ with x a guarded point or a proposal from one, so by Section 7 and Lemma 10.5(i) the spectrum of $\hat{D}_x$ lies in $[-M_{\text{sp}}, \lambda_{\text{top}}]$ once $d_0 u \leq \log(2n)$, the rounding shift $d_0 u$ being covered by the $\log(2n)$ term of λ_{top} . We fix

$$\sigma_{\text{sc}} := 2^{\kappa_{\text{sc}}}, \quad \kappa_{\text{sc}} := \lceil \log_2(M_{\text{sp}} + \lambda_{\text{top}}) \rceil, \quad r := \min\{r \geq 2 : (r+1)! \geq 24\sigma_{\text{sc}}e^{\lambda_{\text{top}}+2}/\varepsilon_{\text{op}}\}$$

so that $\sigma_{\text{sc}} = O(ab + \lambda_{\text{top}}) = O(\log n)$, $r = O(\log(\sigma_{\text{sc}}e^{\lambda_{\text{top}}}/\varepsilon_{\text{op}})) = O(\log n)$, $\sigma_{\text{sc}}\eta_r \leq 1$ (as $\varepsilon_{\text{op}} \leq 1$), $\kappa_{\text{sc}} \geq 1$, and Lemma 10.8(iii) gives $\|\mathcal{Q}(\hat{D}_x) - e^{\hat{D}_x}\| \leq \varepsilon_{\text{op}}/8$; here ε_{op} is the operator tolerance fixed

in Eq. (61) below. With $d_0 \mathbf{u} e^{\lambda_{\text{top}}+1} \leq \varepsilon_{\text{op}}/8$, which is one of the requirements collected in the proof of Theorem 10.1, Lemma 10.8(iv) and Lemma 10.5(i) give

$$\|\mathcal{Q}(\widehat{D}_x) - P_x\| \leq \varepsilon_{\text{op}}/4, \quad P_x = e^{D_x}. \quad (59)$$

Repeated squaring evaluates the surrogate as a dense matrix, and its error propagation is governed by the largest eigenvalue of the exponent.

Lemma 10.9 (Dense evaluation of the surrogate). *Under the hypotheses of Lemma 10.8 with $\kappa_{\text{sc}} \geq 1$, let $\widehat{X}_0$ be the Horner evaluation $T_r(X) = I + X(I + \frac{X}{2}(I + \cdots (I + \frac{X}{r}) \cdots))$ with each matrix product computed exactly by Lemma 10.3 and rounded once, let $\widehat{X}_{i+1} := \text{round}_{\mathbf{u}}(\widehat{X}_i \widehat{X}_i)$ for $0 \leq i \leq \kappa_{\text{sc}} - 2$, and let $\widetilde{P} := \text{round}_{\mathbf{u}}(\widehat{X}_{\kappa_{\text{sc}}-1} \widehat{X}_{\kappa_{\text{sc}}-1})$. If $2e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}e^{\lambda_{\text{sp}}+1}d_0\mathbf{u} \leq 1$, then*

$$\|\widetilde{P} - \mathcal{Q}(D)\| \leq \varepsilon_{\text{dense}} := 2e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}e^{\lambda_{\text{sp}}+1}d_0\mathbf{u},$$

every stored entry has modulus at most $e^{\lambda_{\text{sp}}+1} + 1$, and the computation costs $r + \kappa_{\text{sc}}$ dense products of $d_0 \times d_0$ fixed-point matrices.

Proof. Write $X_i := T_r(X)^{2^i}$ and $f_i := \|\widehat{X}_i - X_i\|$. Each Horner step forms $I + \text{round}_{\mathbf{u}}(X\widehat{Z})/j$ from the previous $\widehat{Z}$: the product and then the division by j each round once, so the step contributes at most $2d_0\mathbf{u}$ in operator norm (the entrywise error $\mathbf{u}$ of a $d_0 \times d_0$ matrix has Frobenius, hence operator, norm at most $d_0\mathbf{u}$), and it is multiplied downstream by $\|X\|/(j-1)\cdots$, so that $f_0 \leq 2d_0\mathbf{u} \sum_{j \geq 0} 1/j! \leq 2ed_0\mathbf{u} \leq 6d_0\mathbf{u}$. For the squarings, $\widehat{X}_{i+1} - X_{i+1} = X_i F_i + F_i X_i + F_i^2 + E_{i+1}$ with $F_i := \widehat{X}_i - X_i$ and $\|E_{i+1}\| \leq d_0\mathbf{u}$, so with $N_i := e^{2^i(\lambda_{\text{sp}}+1)/\sigma_{\text{sc}}} \geq \|X_i\|$ from Lemma 10.8(ii),

$$f_{i+1} \leq 2N_i f_i \left(1 + \frac{f_i}{2N_i}\right) + d_0\mathbf{u}.$$

Unrolling to $i = \kappa_{\text{sc}} - 1$ and using $\prod_{i=j}^{\kappa_{\text{sc}}-2} N_i = \exp((\lambda_{\text{sp}} + 1)(2^{\kappa_{\text{sc}}-1} - 2^j)/\sigma_{\text{sc}}) \leq e^{(\lambda_{\text{sp}}+1)/2}$ and $\prod_i (1 + f_i/(2N_i)) \leq e$ (the latter because $\sum_i f_i/(2N_i) \leq 1$ under the hypothesis, as one checks inductively from the bound being proved), we get

$$f_{\kappa_{\text{sc}}-1} \leq e 2^{\kappa_{\text{sc}}-1} e^{(\lambda_{\text{sp}}+1)/2} (f_0 + (\kappa_{\text{sc}} - 1)d_0\mathbf{u}) \leq e \sigma_{\text{sc}} e^{(\lambda_{\text{sp}}+1)/2} (\kappa_{\text{sc}} + 5)d_0\mathbf{u}/2.$$

Finally $\|\widetilde{P} - \mathcal{Q}(D)\| \leq \|\widehat{X}_{\kappa_{\text{sc}}-1}^2 - X_{\kappa_{\text{sc}}-1}^2\| + d_0\mathbf{u} \leq f_{\kappa_{\text{sc}}-1}(2\|X_{\kappa_{\text{sc}}-1}\| + f_{\kappa_{\text{sc}}-1}) + d_0\mathbf{u} \leq 3e^{(\lambda_{\text{sp}}+1)/2} f_{\kappa_{\text{sc}}-1} + d_0\mathbf{u}$, using $\|X_{\kappa_{\text{sc}}-1}\| \leq e^{(\lambda_{\text{sp}}+1)/2}$ and $f_{\kappa_{\text{sc}}-1} \leq e^{(\lambda_{\text{sp}}+1)/2}$, which gives the stated $\varepsilon_{\text{dense}}$ after $3e/2 + 1 \leq 2e$. The magnitude bound follows from $\|X_i\| \leq e^{\lambda_{\text{sp}}+1}$ and $f_i \leq 1$. The count is r products in the Horner scheme, $\kappa_{\text{sc}} - 1$ squarings, and one final product. $\square$

For the range finder and trace guards, the surrogate is applied to thin blocks of vectors. Repeated application of the scaled Taylor polynomial gives the same exact surrogate as the dense evaluation.

Lemma 10.10 (Vector actions of the surrogate). *Under the hypotheses of Lemma 10.8, let X be stored exactly with κ_{sc} additional fractional bits, and for a vector v let $\text{Horner}_r(X, v)$ be the vector obtained from $y_r := v$, $y_{j-1} := v + \text{round}_{\mathbf{u}}(X y_j)/j$ ($j = r, \dots, 1$, the division rounded once) as y_0 . For a fixed-point vector z put $w_0 := z$ and $w_i := \text{Horner}_r(X, w_{i-1})$, $1 \leq i \leq \sigma_{\text{sc}}$. Then for every $1 \leq J_{\text{ap}} \leq \sigma_{\text{sc}}$,*

$$\|w_{J_{\text{ap}}} - T_r(X)^{J_{\text{ap}}} z\|_2 \leq \varepsilon_{\text{act}}(J_{\text{ap}}) := 6J_{\text{ap}} e^{(\lambda_{\text{sp}}+1)J_{\text{ap}}/\sigma_{\text{sc}}} d_0^{3/2} \mathbf{u}, \quad \|w_{J_{\text{ap}}}\|_2 \leq e^{(\lambda_{\text{sp}}+1)J_{\text{ap}}/\sigma_{\text{sc}}} \|z\|_2 + \varepsilon_{\text{act}}(J_{\text{ap}}),$$

in particular $\|w_{\sigma_{\text{sc}}} - \mathcal{Q}(D)z\|_2 \leq \varepsilon_{\text{act}} := 6\sigma_{\text{sc}} e^{\lambda_{\text{sp}}+1} d_0^{3/2} \mathbf{u}$ and $\|w_{\sigma_{\text{sc}}/2} - \mathcal{Q}(D)^{1/2} z\|_2 \leq \varepsilon_{\text{act}}/2$, where $\mathcal{Q}(D)^{1/2} = T_r(X)^{\sigma_{\text{sc}}/2}$. Applying the scheme to a block of ℓ_{bl} vectors costs $\sigma_{\text{sc}} r$ products of a $d_0 \times d_0$ matrix by a $d_0 \times \ell_{\text{bl}}$ matrix.

Proof. One Horner step computes $v + Xy_j/j$ with a matrix–vector product rounded entrywise (Euclidean error at most $d_0^{3/2}u$) and a division rounded entrywise (at most $\sqrt{d_0}u$); the addition is exact. The error e_{j-1} in y_{j-1} therefore obeys $\|e_{j-1}\|_2 \leq \|X\| \|e_j\|_2/j + 2d_0^{3/2}u \leq \|e_j\|_2/j + 2d_0^{3/2}u$, so $\|e_0\|_2 \leq 2d_0^{3/2}u \sum_{j \geq 0} 1/j! \leq 6d_0^{3/2}u$: one application of the scheme returns $T_r(X)w + \phi$ with $\|\phi\|_2 \leq 6d_0^{3/2}u$, independently of r and of $\|w\|_2$. Hence $w_{J_{\text{ap}}} - T_r(X)^{J_{\text{ap}}}z = \sum_{i=1}^{J_{\text{ap}}} T_r(X)^{J_{\text{ap}}-i} \phi_i$, and Lemma 10.8(ii) bounds each term by $e^{(\lambda_{\text{sp}}+1)(J_{\text{ap}}-i)/\sigma_{\text{sc}}} 6d_0^{3/2}u$; summing the J_{ap} terms gives $\varepsilon_{\text{act}}(J_{\text{ap}})$, and the same bound with $\|T_r(X)^{J_{\text{ap}}}z\|_2 \leq e^{(\lambda_{\text{sp}}+1)J_{\text{ap}}/\sigma_{\text{sc}}} \|z\|_2$ gives the magnitude. The half-power statement is the case $J_{\text{ap}} = \sigma_{\text{sc}}/2$ with $e^{(\lambda_{\text{sp}}+1)/2} \leq e^{\lambda_{\text{sp}}+1}$. $\square$

Lemmas 10.9 and 10.10 use the same exact matrix $\mathcal{Q}(\hat{D}_x)$, so a refreshed tracker and a tracked increment refer to one surrogate; this is the property Section 9 calls the convention that “preserves exact conditional unbiasedness”. The error made at stage i of the repeated application is amplified by $\|T_r(X)^{J_{\text{ap}}-i}\| \leq e^{\lambda_{\text{sp}}+1}$ and never by $e^{M_{\text{sp}}}$, which is the reason the scheme is stable for symmetric D although the spectral interval has width $M_{\text{sp}} \approx 4 \cdot 10^6 \log(2n)$; Al-Mohy and Higham [AMH11, Section 4] make the same observation. The count $\sigma_{\text{sc}}r = O(\log^2 n)$ of matrix–vector products per action is polylogarithmic but large (about 10^{10} at $n = 3 \cdot 10^8$); a Chebyshev expansion of $e^{t/2}$ on the guarded interval evaluated by the three-term recurrence needs only $O(\sqrt{M_{\text{sp}}} \log n) \approx 10^5$ products with the same error structure, and an implementation would use it. We keep the scaled Taylor power in the proof because its error analysis is four lines.

10.4 The range finder and the increment estimator

For a symmetric Z we use the seminorm of Eq. (30) in the form

$$\|Z\|_{\mathcal{G}}^2 := \frac{\nu^2}{m} \sum_{i \in S} \text{tr}[E_i Z]^2, \quad E_i = \tilde{c} \text{diag}(A_i, -A_i), \quad (60)$$

which is the mean squared perturbation of the coordinate gradient $\partial_i F(x) = x_i - \xi'_i + \nu \text{tr}[P_x E_i]$ induced by replacing P_x by $P_x + Z$.

The range finder receives a rounded Gaussian test matrix and an approximate matrix action. Its residual estimate must therefore account for errors in both the test matrix and the sketch.

Lemma 10.11 (Range finder with a rounded test matrix and an inexact sketch). *Let A be a symmetric $d_0 \times d_0$ matrix, $2 \leq k \leq n$, and let $\hat{\Omega} = \Omega + E_{\Omega} \in \mathbb{R}^{d_0 \times 2k}$ with Ω standard Gaussian and $E_{\Omega} = E_{\Omega}(\Omega)$ measurable with $\|E_{\Omega}\|_{\max} \leq 2^{-b_{\Omega}}$, $b_{\Omega} \geq 4.5 \log_2 n + 7$. Let $\hat{Y} \in \mathbb{R}^{d_0 \times 2k}$ satisfy $\|\hat{Y} - A\hat{\Omega}\|_F \leq \theta$ for a deterministic θ , and let $\mathcal{P}$ be an orthogonal projector with $\text{range}(\mathcal{P}) \supseteq \text{range}(\hat{Y})$. Then*

$$\mathbb{E}_{\Omega} [\|(I - \mathcal{P})A\|_F^2] \leq \frac{64}{k} \|A\|_1^2 + 1200 kn^2 \theta^2.$$

Proof. Write $A = U\Sigma V^{\top}$ with $\sigma_1 \geq \sigma_2 \geq \dots$, let Σ_1 be the top $k \times k$ block and Σ_2 the rest, and put $\Omega_1 := V_1^{\top} \Omega$ ($k \times 2k$), $\Omega_2 := V_2^{\top} \Omega$, and likewise $\hat{\Omega}_1, \hat{\Omega}_2, E_1, E_2$. By rotation invariance Ω_1 and Ω_2 are independent standard Gaussian matrices. Put $\tau_{\Omega} := 1/(2n)$ and $\eta_E := \|E_{\Omega}\|_2 \leq \|E_{\Omega}\|_F \leq \sqrt{4nk} 2^{-b_{\Omega}} \leq 2n 2^{-b_{\Omega}}$.

Step 1: the good event. Let $\mathcal{G}_1 := \{\sigma_k(\Omega_1) \geq \tau_{\Omega}\}$ and $\mathcal{G}_2 := \{\|\Sigma_2 \Omega_2\|_F^2 \leq 8nk \|\Sigma_2\|_F^2\}$. For a $k \times 2k$ standard Gaussian matrix, Halko, Martinsson, and Tropp [HMT11, Proposition A.3] (after Chen and Dongarra [CD05, Lemma 4.1]), whose prefactor is at most 1.6 for the $k \times 2k$ shape and all $k \geq 2$, give $\Pr[\|\Omega_1^{\dagger}\| \geq t] \leq 2t^{-(k+1)} \leq 2t^{-3}$ for $t \geq 1$ and $k \geq 2$, so $\Pr[\mathcal{G}_1^c] \leq 2(2n)^{-3} = 1/(4n^3)$. Since $\mathbb{E}[\|\Sigma_2 \Omega_2\|_F^2] = 2k \|\Sigma_2\|_F^2$ [HMT11, Proposition A.1], Markov’s inequality gives $\Pr[\mathcal{G}_2^c] \leq 1/(4n)$. On

$\mathcal{G} := \mathcal{G}_1 \cap \mathcal{G}_2$, Weyl's inequality and $\eta_E \leq 2n2^{-b_\Omega} \leq \tau_\Omega/2$ give $\sigma_k(\widehat{\Omega}_1) \geq \tau_\Omega/2$, so $\widehat{\Omega}_1$ has full row rank, $\widehat{\Omega}_1 \widehat{\Omega}_1^\dagger = I_k$, and $\|\widehat{\Omega}_1^\dagger\| \leq 2/\tau_\Omega = 4n$. Off $\mathcal{G}$ we use $\|(I - \mathcal{P})A\|_F \leq \|A\|_F \leq \|A\|_1$, so the complement contributes at most $\|A\|_1^2 \Pr[\mathcal{G}^c] \leq \|A\|_1^2/(2n) \leq \|A\|_1^2/k$ to the expectation.

Step 2: a deterministic bound on $\mathcal{G}$. Write $\widehat{Y} = A\widehat{\Omega} + G_0$ with $\|G_0\|_F \leq \theta$, and put $\mathcal{W} := \widehat{Y}\widehat{\Omega}_1^\dagger$, whose range lies in $\text{range}(\widehat{Y}) \subseteq \text{range}(\mathcal{P})$, so $(I - \mathcal{P})\mathcal{W} = 0$. Since $A\widehat{\Omega}\widehat{\Omega}_1^\dagger = U\Sigma V^\top \widehat{\Omega}\widehat{\Omega}_1^\dagger$ and $V^\top \widehat{\Omega} = [\widehat{\Omega}_1; \widehat{\Omega}_2]$,

$$\mathcal{W} = U \begin{bmatrix} \Sigma_1 + G_1 \\ \Sigma_2 T' + G_2 \end{bmatrix}, \quad T' := \widehat{\Omega}_2 \widehat{\Omega}_1^\dagger, \quad \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} := U^\top G_0 \widehat{\Omega}_1^\dagger, \quad \left\| \begin{bmatrix} G_1 \\ G_2 \end{bmatrix} \right\|_F \leq 4n\theta =: g_\theta.$$

Split $[k]$ into $\mathbf{a} := \{j : \sigma_j > 2g_\theta\}$ and $\mathbf{b} := [k] \setminus \mathbf{a}$, and let $U_{\mathbf{a}}, U_{\mathbf{b}}, U_2$ be the corresponding column blocks of U . The columns of $\mathcal{W}$ indexed by $\mathbf{a}$ are $\mathcal{W}_{\mathbf{a}} = U_{\mathbf{a}}M + U_{\mathbf{b}}G_{1,\mathbf{ba}} + U_2(\Sigma_2 T' + G_2)_{:, \mathbf{a}}$ with $M := \Sigma_{1\mathbf{a}} + G_{1,\mathbf{aa}}$. Because $\|G_{1,\mathbf{aa}}\| \leq g_\theta < \sigma_{\min}(\Sigma_{1\mathbf{a}})/2$, M is invertible and $\|M^{-1}\Sigma_{1\mathbf{a}}\| = \|(I + \Sigma_{1\mathbf{a}}^{-1}G_{1,\mathbf{aa}})^{-1}\| \leq 2$. Since $(I - \mathcal{P})\mathcal{W}_{\mathbf{a}} = 0$, we have $(I - \mathcal{P})U_{\mathbf{a}}M = -(I - \mathcal{P})[U_{\mathbf{b}}G_{1,\mathbf{ba}} + U_2(\Sigma_2 T' + G_2)_{:, \mathbf{a}}]$; multiplying by $M^{-1}\Sigma_{1\mathbf{a}}$ on the right and using $\|I - \mathcal{P}\| \leq 1$,

$$\|(I - \mathcal{P})U_{\mathbf{a}}\Sigma_{1\mathbf{a}}\|_F \leq 2(\|G_{1,\mathbf{ba}}\|_F + \|\Sigma_2 T'\|_F + \|G_2\|_F) \leq 2\|\Sigma_2 T'\|_F + 3g_\theta,$$

using $\|G_{1,\mathbf{ba}}\|_F + \|G_2\|_F \leq \sqrt{2}g_\theta$. Since $A = U_{\mathbf{a}}\Sigma_{1\mathbf{a}}V_{\mathbf{a}}^\top + U_{\mathbf{b}}\Sigma_{1\mathbf{b}}V_{\mathbf{b}}^\top + U_2\Sigma_2V_2^\top$ and $\|\Sigma_{1\mathbf{b}}\|_F \leq 2g_\theta\sqrt{k}$,

$$\|(I - \mathcal{P})A\|_F \leq 2\|\Sigma_2 T'\|_F + (3 + 2\sqrt{k})g_\theta + \|\Sigma_2\|_F, \quad \|(I - \mathcal{P})A\|_F^2 \leq 12\|\Sigma_2 T'\|_F^2 + 3(2\sqrt{k} + 3)^2 g_\theta^2 + 3\|\Sigma_2\|_F^2.$$

Step 3: the perturbed pseudo-inverse. Decompose $\Sigma_2 T' = \Sigma_2 \Omega_2 \Omega_1^\dagger + \Sigma_2 \Omega_2 (\widehat{\Omega}_1^\dagger - \Omega_1^\dagger) + \Sigma_2 E_2 \widehat{\Omega}_1^\dagger$. For two full-row-rank $k \times 2k$ matrices P, Q the identity $Q^\dagger - P^\dagger = -Q^\dagger(Q - P)P^\dagger + (I - Q^\dagger Q)(Q - P)^\top (P^\dagger)^\top P^\dagger$ (Stewart [Ste77, Section 3]) gives $\|Q^\dagger - P^\dagger\| \leq \|Q - P\| \|P^\dagger\| (\|Q^\dagger\| + \|P^\dagger\|)$; on $\mathcal{G}$ with $P = \Omega_1$, $Q = \widehat{\Omega}_1$ this is at most $\eta_E \cdot 2n \cdot 6n = 12n^2 \eta_E$. Hence the second and third terms have Frobenius norm at most $\sqrt{8nk} \|\Sigma_2\|_F \cdot 12n^2 \eta_E + \|\Sigma_2\|_F \eta_E \cdot 4n \leq 38n^3 \eta_E \|\Sigma_2\|_F \leq 76n^4 2^{-b_\Omega} \|\Sigma_2\|_F \leq \|\Sigma_2\|_F / \sqrt{2n}$, the last step by $b_\Omega \geq 4.5 \log_2 n + 7$. By $(x + y)^2 \leq 2x^2 + 2y^2$, the independence of Ω_2 from Ω_1 (rotation invariance) with the Frobenius identity [HMT11, Proposition A.1], and $\mathbb{E}[\|\Omega_1^\dagger\|_F^2] = k/(k - 1) \leq 2$ [HMT11, Proposition A.5],

$$\mathbb{E}[\|\Sigma_2 T'\|_F^2; \mathcal{G}] \leq 2\|\Sigma_2\|_F^2 \mathbb{E}[\|\Omega_1^\dagger\|_F^2] + 2\|\Sigma_2\|_F^2/(2n) \leq 5\|\Sigma_2\|_F^2.$$

Step 4: assembly. Combining Steps 1–3 with $\|\Sigma_2\|_F^2 \leq \sigma_{k+1}\|A\|_1 \leq \|A\|_1^2/k$, $g_\theta^2 = 16n^2\theta^2$, and $3(2\sqrt{k} + 3)^2 \leq 75k$,

$$\mathbb{E}[\|(I - \mathcal{P})A\|_F^2] \leq (60 + 3) \frac{\|A\|_1^2}{k} + 75k \cdot 16n^2\theta^2 + \frac{\|A\|_1^2}{k} = \frac{64}{k} \|A\|_1^2 + 1200kn^2\theta^2.$$

If $\widehat{\Omega}$ is the conditioned draw of Lemma 10.7(iv) instead of one draw, and the hypothesis $\|\widehat{Y} - A\widehat{\Omega}\|_F \leq \theta$ is available only on the acceptance event $\mathcal{A}$, the bound holds with the right side multiplied by $(1 - u^2)^{-1}$: on $\mathcal{A}^c$ replace $\widehat{Y}$ by $A\widehat{\Omega}$ and $\mathcal{P}$ by the projector onto its range, so that the hypothesis holds surely and $X := \|(I - \mathcal{P})A\|_F^2$ is unchanged on $\mathcal{A}$; then $X \geq 0$ gives $\mathbb{E}[X] \geq \Pr[\mathcal{A}] \mathbb{E}[X | \mathcal{A}]$. $\square$

The constant $C = 3$ of Section 7 in $\mathbb{E}[\|A_{\text{res}}\|_F^2] \leq (C/k)\|A\|_1^2$ becomes 64 here and 129 in Lemma 10.12(ii); it enters only K_0 . The hypothesis $k \leq n$ is always met: the choice after Lemma 10.5 gives $k = \lceil C_{ks}\pi'^{-1/2} \rceil \leq 45n^{1/4} + 1 \leq n$ for $n \geq N_0$, and $k \geq C_{ks} \geq 2$.

The increment analysis separates the exactly unbiased estimator defined by the computed basis from the deterministic errors in its actions and matrix products.

Lemma 10.12 (The computed increment estimator). *Fix a nonrefresh step from $x = x_t$ to $y = x_{t+1}$ and put $A := \mathcal{Q}(\widehat{D}_y) - \mathcal{Q}(\widehat{D}_x)$. The algorithm draws $\widehat{\Omega}$ as in Lemma 10.7(iv) and Rademacher probes $z_1, \dots, z_s$, and computes*

- (a) $\widehat{Y} := \text{act}_y(\widehat{\Omega}) - \text{act}_x(\widehat{\Omega})$, where $\text{act}_x(\cdot)$ is the columnwise scheme of Lemma 10.10 for $\mathcal{Q}(\widehat{D}_x)$;
- (b) a Householder QR factorization of $\widehat{Y}$ in fixed point, returning $\widehat{U} \in \mathbb{R}^{d_0 \times 2k}$;
- (c) $\widehat{Z} := \text{act}_y(\widehat{U}) - \text{act}_x(\widehat{U})$ and the dense matrix $\widehat{A}_{\text{low}} := \widehat{U}\widehat{Z}^\top + \widehat{Z}\widehat{U}^\top - \widehat{U}(\widehat{U}^\top \widehat{Z})\widehat{U}^\top$ by three exact-then-rounded products of Lemma 10.3;
- (d) for each j , $v_j := z_j - \widehat{U}(\widehat{U}^\top z_j)$, $w_j := \text{act}_y(v_j) - \text{act}_x(v_j)$, $r_j := w_j - \widehat{U}(\widehat{U}^\top w_j)$, and the dense matrix $\widehat{A}_{\text{hut}} := \frac{1}{2s} \sum_j (r_j z_j^\top + z_j r_j^\top)$ by one exact-then-rounded thin product;
- (e) $\widehat{A}^{\text{comp}} := \widehat{A}_{\text{low}} + \widehat{A}_{\text{hut}}$, symmetrized exactly.

Let $\widehat{\Pi} := \widehat{U}\widehat{U}^\top$, $A_{\text{res}} := (I - \widehat{\Pi})A(I - \widehat{\Pi})$, and let $\widehat{A}^{\text{ideal}} := \widehat{\Pi}A + A\widehat{\Pi} - \widehat{\Pi}A\widehat{\Pi} + \frac{1}{2s} \sum_j \{(A_{\text{res}} z_j) z_j^\top + z_j (A_{\text{res}} z_j)^\top\}$ be the exact-arithmetic estimator built from the same $\widehat{\Pi}$ and z_j . Then:

- (i) $\mathbb{E}[\widehat{A}^{\text{ideal}} | \widehat{\Omega}] = A$ and $\mathbb{E}[\|\widehat{A}^{\text{ideal}} - A\|_G^2 | \widehat{\Omega}] \leq \frac{2\nu^2 \tilde{c}^2}{s} \|A_{\text{res}}\|_F^2$;
- (ii) with $Y_{\text{max}} := \Omega_{\text{max}} \sqrt{2kd_0} (e^{\lambda_{\text{top}}} + 1) + 2\sqrt{2k} \varepsilon_{\text{act}}$, which bounds $\|\widehat{Y}\|_F$ for every accepted $\widehat{\Omega}$, $\varepsilon_{\text{qr}} := 16k^{3/2} d_0^{3/2} u (1 + Y_{\text{max}})$, and $\theta := 2\sqrt{2k} \varepsilon_{\text{act}} + \varepsilon_{\text{qr}}$ (all deterministic), if $3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$ then

$$\mathbb{E}_\Omega[\|A_{\text{res}}\|_F^2] \leq \frac{129}{k} \|A\|_1^2 + 2401 kn^2 \theta^2;$$

- (iii) $\|\widehat{A}^{\text{comp}} - \widehat{A}^{\text{ideal}}\| \leq \varepsilon_{\text{step}} := 40 d_0^{3/2} (1 + \|A\|) (\varepsilon_{\text{act}} + d_0 u)$, deterministically, under the hypothesis of (ii);
- (iv) the step costs $6\sigma_{\text{sc}} r + O(1)$ products of $d_0 \times d_0$ by $d_0 \times O(k+s)$ matrices, four exact-then-rounded products of shape $\langle d_0, O(k+s), d_0 \rangle$, and $O(d_0 k^2 + d_0^2)$ further operations.

Proof. (i), unbiasedness and variance. Conditional on $\widehat{\Omega}$ the matrix $\widehat{\Pi}$ is fixed and the z_j are independent of it, so Lemma 10.6 gives $\mathbb{E}[(A_{\text{res}} z_j) z_j^\top] = A_{\text{res}}$ and hence $\mathbb{E}[\widehat{A}^{\text{ideal}} | \widehat{\Omega}] = A_{\text{low}} + A_{\text{res}} = A$, where the identity $\widehat{\Pi}A + A\widehat{\Pi} - \widehat{\Pi}A\widehat{\Pi} + (I - \widehat{\Pi})A(I - \widehat{\Pi}) = A$ holds for every matrix $\widehat{\Pi}$. For the variance, by symmetry of E_i , $\text{tr}[E_i(\widehat{A}^{\text{ideal}} - A)] = \frac{1}{s} \sum_j z_j^\top A_{\text{res}} E_i z_j - \text{tr}[A_{\text{res}} E_i]$, and Lemma 10.6, applied to the symmetric part of $A_{\text{res}} E_i$ (whose Frobenius norm is at most $\|A_{\text{res}} E_i\|_F$), gives $\mathbb{V}[z^\top A_{\text{res}} E_i z] \leq 2\|A_{\text{res}} E_i\|_F^2$, so

$$\mathbb{E}[\|\widehat{A}^{\text{ideal}} - A\|_G^2] \leq \frac{\nu^2}{m} \sum_{i \in S} \frac{2}{s} \|A_{\text{res}} E_i\|_F^2 = \frac{2\nu^2}{ms} \text{tr} \left[A_{\text{res}}^2 \sum_{i \in S} E_i^2 \right] \leq \frac{2\nu^2 \tilde{c}^2}{s} \|A_{\text{res}}\|_F^2$$

by Lemma 10.5(ii).

(ii), the range finder with computed data. *The sketch.* Each column of $\widehat{Y}$ is a difference of two computed actions on the same column of $\widehat{\Omega}$, so Lemma 10.10 gives $\|\widehat{Y} - A\widehat{\Omega}\|_F \leq \sqrt{2k} \cdot 2\varepsilon_{\text{act}}$. *The factorization.* Householder QR processes the columns of $\widehat{Y}$ in turn. At step j it computes a vector v_j from the current j -th column (its entries and the norm rounded once each) and defines the reflector $H_j := I - 2v_j v_j^\top / \|v_j\|_2^2$, which is an exactly orthogonal matrix for the computed v_j , whatever rounding produced it; a column whose remaining part rounds to zero is skipped (its norm, at most $\sqrt{d_0} u$, is charged to $\Delta \widehat{Y}$), and each v_j is rescaled by a power of two to norm in $[1/2, 2]$

before use, which leaves H_j unchanged (scaling up is exact; scaling down drops low bits of v_j , a perturbation of at most $\sqrt{d_0}u$ of the order already charged, and H_j is exactly orthogonal for the vector actually used). Applying H_j to a column w costs the inner product $v_j^\top w$ (exact accumulation, one rounding), the scalar $2v_j^\top w/\|v_j\|_2^2$ (one rounding), and the update $w - (\cdot)v_j$ (one rounding per entry), so the computed image differs from $H_j w$ by at most $4\sqrt{d_0}u(1 + \|w\|_2)$ in Euclidean norm; setting the computed entries below the diagonal to zero, which the exact H_j would have done only for the exact Householder vector, costs at most another $2\sqrt{d_0}u(1 + \|w\|_2)$. Because every H_j is exactly orthogonal, an error made at step j is carried unchanged in norm to the end, and every intermediate column has norm at most $2Y_{\max} + 1$, where $\|\hat{Y}\|_F \leq \|A\|\|\hat{\Omega}\|_F + 2\sqrt{2k}\varepsilon_{\text{act}} \leq Y_{\max}$ by Lemma 10.10 and $\|A\| \leq e^{\lambda_{\text{top}}} + 1$, the latter from Lemma 10.8(iv) with $\lambda_{\text{sp}} = \log(H/\nu)$ at the guarded x and $\zeta = \|D_y - D_x\| \leq 1$ ($\|P_y - P_x\| \leq eH/\nu \leq e^{\lambda_{\text{top}}}/(2n)$) together with Eq. (59) ($\varepsilon_{\text{op}}/2 \leq 1$). Hence the computed upper triangular $\hat{R}$ satisfies $\hat{Y} + \Delta\hat{Y} = Q_H \hat{R}$ with $Q_H := H_1 \cdots H_{2k}[I_{2k}; 0]$ exactly orthonormal and $\|\Delta\hat{Y}\|_F \leq 2k \cdot \sqrt{2k} \cdot 6\sqrt{d_0}u(2Y_{\max} + 2) \leq \varepsilon_{\text{qr}}$; this is the fixed-point form of the backward error bound of Higham [Hig02, Theorem 19.4]. The returned $\hat{U}$ is the computed product of the same reflectors applied to the first $2k$ unit vectors, so $\|\hat{U} - Q_H\|_F \leq \varepsilon_{\text{qr}}$ by the same count with $\|w\|_2 \leq 1$. Apply Lemma 10.11 with the sketch $\hat{Y} + \Delta\hat{Y}$, whose distance to $A\hat{\Omega}$ is at most θ , and $\mathcal{P} := Q_H Q_H^\top$, whose range contains $\text{range}(\hat{Y} + \Delta\hat{Y})$: $\mathbb{E}[\|(I - \mathcal{P})A\|_F^2] \leq (1 - u^2)^{-1}\{(64/k)\|A\|_1^2 + 1200kn^2\theta^2\}$, the factor coming from the conditioned draw. *The near-projector.* $\|\hat{\Pi} - \mathcal{P}\| \leq 2\|\hat{U} - Q_H\| + \|\hat{U} - Q_H\|^2 \leq 3\varepsilon_{\text{qr}} =: \varepsilon_{\text{proj}}$, so $\|A_{\text{res}}\|_F \leq \|(I - \mathcal{P})A(I - \mathcal{P})\|_F + 3\varepsilon_{\text{proj}}\|A\|_F \leq \|(I - \mathcal{P})A\|_F + 3\varepsilon_{\text{proj}}\|A\|_1$, and squaring with $(x+y)^2 \leq 2x^2 + 2y^2$, $2(1 - u^2)^{-1} \cdot 64 \leq 128 + 1/(2k)$ and $2(1 - u^2)^{-1} \cdot 1200 \leq 2401$ (both because $u \leq 1/(240k^2 d_0^{3/2})$ under the hypothesis), and $18\varepsilon_{\text{proj}}^2 \leq 18/(25k)$ gives (ii).

(iii), *computed versus ideal.* Every stage is a deterministic function of the same inputs. In (c), $\hat{Z}$ differs from $A\hat{U}$ by at most $2\sqrt{2k}\varepsilon_{\text{act}}$ in Frobenius norm, and $\|\hat{U}\| \leq 1 + \varepsilon_{\text{qr}} \leq 2$, so the three terms of $\hat{A}_{\text{low}}$ differ from those of $\hat{\Pi}A + A\hat{\Pi} - \hat{\Pi}A\hat{\Pi}$ by at most $12 \cdot 2\sqrt{2k}\varepsilon_{\text{act}}$ plus three roundings of at most d_0u each, the intermediate products of the triple product being formed exactly in wider temporaries. In (d), v_j differs from $(I - \hat{\Pi})z_j$ by at most $3\sqrt{d_0}u$ (inner products and the product with $\hat{U}$ rounded entrywise), w_j from Av_j by at most $2\varepsilon_{\text{act}} + 3\sqrt{d_0}u\|A\|$, and r_j from $A_{\text{res}}z_j$ by at most $2(2\varepsilon_{\text{act}} + 3\sqrt{d_0}u\|A\|) + 3\sqrt{d_0}u(1 + 2\sqrt{d_0}\|A\|) =: \varepsilon_{\text{rd}}$, using $\|w_j\|_2 \leq 2\sqrt{d_0}\|A\|$; hence $\hat{A}_{\text{hut}}$ differs from its ideal counterpart by at most $\varepsilon_{\text{rd}}\|z_j\|_2 = \varepsilon_{\text{rd}}\sqrt{d_0}$ plus one rounding d_0u . Summing the stages and using $k \leq n \leq d_0$ gives $\varepsilon_{\text{step}} \leq 38\sqrt{d_0}\varepsilon_{\text{act}} + 4d_0u + \sqrt{d_0}\varepsilon_{\text{rd}} \leq 40d_0^{3/2}(1 + \|A\|)(\varepsilon_{\text{act}} + d_0u)$.

(iv), *cost.* The actions in (a), (c), (d) apply the scheme of Lemma 10.10 to blocks of $2k$, $2k$, and s vectors for each of the two matrices, i.e. $2\sigma_{\text{scr}}$ thin products per block; the products in (c) and (d) have the stated shapes; the Householder factorization costs $O(d_0k^2)$ and the symmetrization $O(d_0^2)$. $\square$

10.5 The tracker, the step, the guard, and the phase

A deterministic upper bound on the time between refreshes controls both accumulated rounding error and the components of the tracker invisible to the gradient seminorm.

Lemma 10.13 (Tracker with bounded drift). *Fix a candidate phase at multiplier ν with level H , $T = 16mL$ steps, refresh probability π' , and $L_{\max} := \lceil 1/\pi' \rceil$. The finite tracker starts from $\hat{P}_0 := \tilde{P}_z$, the dense surrogate of Lemma 10.9 at the phase start z ; after x_{t+1} is formed it refreshes, setting $\hat{P}_{t+1} := \tilde{P}_{x_{t+1}}$, if an independent coin of bias π' shows heads or if $L_{\max}$ steps have passed since the last refresh, and otherwise sets $\hat{P}_{t+1} := \hat{P}_t + \hat{A}_t^{\text{comp}}$ with the estimator of Lemma 10.12 for $A_t := Q(\hat{D}_{x_{t+1}}) - Q(\hat{D}_{x_t})$. Put $Z_t := \hat{P}_t - P_{x_t}$ and split $Z_t = Z_t^{\text{rand}} + Z_t^{\text{det}}$, where at a refresh*

$Z_{t+1}^{\text{rand}} := 0$ and $Z_{t+1}^{\text{det}} := \tilde{P}_{x_{t+1}} - P_{x_{t+1}}$, and at a nonrefresh step

$$Z_{t+1}^{\text{rand}} := Z_t^{\text{rand}} + (\hat{A}_t^{\text{ideal}} - A_t), \quad Z_{t+1}^{\text{det}} := Z_t^{\text{det}} + (\hat{A}_t^{\text{comp}} - \hat{A}_t^{\text{ideal}}) + (A_t - (P_{x_{t+1}} - P_{x_t})).$$

At $t = 0$, $Z_0^{\text{rand}} := 0$ and $Z_0^{\text{det}} := \tilde{P}_z - P_z$. Assume $\varepsilon_{\text{step}}, \varepsilon_{\text{dense}} \leq 1$, $\varepsilon_{\text{op}} \leq 2$, and $3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$, where $\varepsilon_{\text{step}}$ is evaluated at $\|A\| = e^{\lambda_{\text{top}}} + 1$; the accuracy choices of Eq. (61) meet these requirements, since they give $\varepsilon_{\text{dense}}, \varepsilon_{\text{step}} \leq \varepsilon_{\text{det}} \leq \delta_g/(2\nu\tilde{c}d_0) \leq 1$ (with ε_{det} the deterministic error total defined in (i) below) and $\varepsilon_{\text{op}} \leq 1/6$, and $3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$ is one of the four requirements listed there. Let $\mathcal{E}_t := \|Z_t^{\text{rand}}\|_{\mathcal{G}}^2$, let $\mathcal{F}_t$ be the history up to and including x_t and $\hat{P}_t$, and suppose every base point satisfies $F(x_t) \leq H$ and every step $|d_t| \leq 1/\tilde{c}$. Then:

- (i) $\|Z_t^{\text{det}}\| \leq \varepsilon_{\text{det}} := \varepsilon_{\text{dense}} + 3\varepsilon_{\text{op}}/4 + L_{\text{max}}\varepsilon_{\text{step}}$ for every t , deterministically;
- (ii) $\mathbb{E}[\mathcal{E}_{t+1} \mid \mathcal{F}_t] \leq (1 - \pi')\{\mathcal{E}_t + \frac{65L^2}{ks}\mathbb{E}[d_t^2 \mid \mathcal{F}_t] + \frac{2\delta_g^2}{ks}\}$, provided $\varepsilon_{\text{op}} \leq \delta_g/(12\nu\tilde{c}d_0)$ and $\theta \leq \delta_g/(70\nu\tilde{c}kn)$ with θ as in Lemma 10.12(ii);
- (iii) the selected gradient $\hat{g}_{t,i} := x_{t,i} - \xi'_i + \text{round}_u(\nu \text{tr}[\hat{P}_t E_i])$ satisfies $\hat{g}_{t,i} = \partial_i F(x_t) + \mathbf{e}_{t,i} + \beta_{t,i}$ with $\mathbf{e}_{t,i} := \nu \text{tr}[E_i Z_t^{\text{rand}}]$, $\mathbb{E}[\mathbf{e}_{t,i}^2 \mid \mathcal{F}_t] = \mathcal{E}_t$ for the uniform i_t , and $|\beta_{t,i}| \leq \nu\tilde{c}d_0\varepsilon_{\text{det}} + u$;
- (iv) $\|\hat{P}_t\| \leq e^{\lambda_{\text{top}}} + 1 + L_{\text{max}}(25d_0 + 30)(e^{\lambda_{\text{top}}} + 1)$ for every t , so every entry of $\hat{P}_t$ fits in I integer bits with I as in Eq. (54);
- (v) the number of refreshes in the phase is at most $T\pi' + 1$ forced ones plus a $\text{Binomial}(T, \pi')$ number of coin refreshes; the refresh-count cap M_{ref} of Algorithm 10 is applied to the coin refreshes only.

The forced-refresh decision is a deterministic function of the step count, hence $\mathcal{F}_t$ -measurable, and the coin, the test matrix $\hat{\Omega}_t$, the probes z_j , and the guard's probes are fresh coins independent of $\mathcal{F}_t$ and of each other.

Proof. The identity $Z_t = Z_t^{\text{rand}} + Z_t^{\text{det}}$ holds by induction: at a nonrefresh step $Z_{t+1} = Z_t + \hat{A}_t^{\text{comp}} - (P_{x_{t+1}} - P_{x_t})$, which is the sum of the three increments.

(i), the drift telescopes. Let $t_0 \leq t$ be the last refresh time. The surrogate increments telescope, $\sum_{t_0 \leq \tau < t} (A_\tau - (P_{x_{\tau+1}} - P_{x_\tau})) = (\mathcal{Q}(\hat{D}_{x_t}) - P_{x_t}) - (\mathcal{Q}(\hat{D}_{x_{t_0}}) - P_{x_{t_0}})$, of norm at most $\varepsilon_{\text{op}}/2$ by Eq. (59); the refresh term has norm at most $\varepsilon_{\text{dense}} + \varepsilon_{\text{op}}/4$ by Lemma 10.9 and Eq. (59); and there are at most L_{max} computation errors of norm at most $\varepsilon_{\text{step}}$ each by Lemma 10.12(iii).

(ii), the zero-mean recurrence. On a refresh (forced or by coin) $\mathcal{E}_{t+1} = 0$. On a nonrefresh step, which given $\mathcal{F}_t$ has probability at most $1 - \pi'$ because the coin is independent of everything else, $Z_{t+1}^{\text{rand}} = Z_t^{\text{rand}} + \Delta_t$ with $\mathbb{E}[\Delta_t \mid \mathcal{F}_t, i_t, \hat{\Omega}_t] = 0$ by Lemma 10.12(i); since $\|\cdot\|_{\mathcal{G}}^2$ is a quadratic form and Z_t^{rand} is $\mathcal{F}_t$ -measurable, the cross term vanishes and $\mathbb{E}[\mathcal{E}_{t+1} \mid \mathcal{F}_t] \leq (1 - \pi')\{\mathcal{E}_t + \mathbb{E}[\|\Delta_t\|_{\mathcal{G}}^2 \mid \mathcal{F}_t]\}$. By Lemma 10.12(i)–(ii), $\mathbb{E}[\|\Delta_t\|_{\mathcal{G}}^2 \mid \mathcal{F}_t, i_t, d_t] \leq \frac{2\nu^2\tilde{c}^2}{s}\{\frac{129}{k}\|A_t\|_1^2 + 2401kn^2\theta^2\}$. By Eq. (59), $\|A_t\|_1 \leq \|P_{x_{t+1}} - P_{x_t}\|_1 + d_0\varepsilon_{\text{op}}/2$, and the argument of Lemma 9.5, whose hypotheses $F(x_t) \leq H$ and $|d_t| \leq 1/\tilde{c}$ hold by assumption, gives $\nu^2\|P_{x_{t+1}} - P_{x_t}\|_1^2 \leq \frac{\varepsilon}{2}HLd_t^2$. Hence

$$\mathbb{E}[\|\Delta_t\|_{\mathcal{G}}^2 \mid \mathcal{F}_t, i_t, d_t] \leq \frac{258\tilde{c}^2}{ks} \left(eHLd_t^2 + \frac{\nu^2 d_0^2 \varepsilon_{\text{op}}^2}{2} \right) + \frac{4802\nu^2\tilde{c}^2 kn^2\theta^2}{s} \leq \frac{65L^2}{ks} d_t^2 + \frac{\delta_g^2}{ks} + \frac{\delta_g^2}{ks},$$

where the first term uses $e\tilde{c}^2H \leq L/4$ from the definition of L , the second uses $129\nu^2\tilde{c}^2 d_0^2 \varepsilon_{\text{op}}^2 \leq \delta_g^2$, and the third uses the hypothesis on θ . Taking $\mathbb{E}[\cdot \mid \mathcal{F}_t]$ gives (ii).

(iii), the used gradient. The inner product $\langle \hat{P}_{t,11} - \hat{P}_{t,22}, A_i \rangle$ is formed exactly, the factor $\nu\tilde{c}$ is a number of the solve stored exactly with $2f$ fractional bits, and their product is formed in a

3W-bit temporary and rounded once, so $\hat{g}_{t,i} = \partial_i F(x_t) + \nu \text{tr}[E_i Z_t] + \vartheta$ with $|\vartheta| \leq u$; split Z_t and use $|\text{tr}[E_i Z_t^{\text{det}}]| \leq \|E_i\| \|Z_t^{\text{det}}\|_1 \leq \tilde{c} d_0 \varepsilon_{\text{det}}$. The identity for $\mathbb{E}[\mathfrak{e}^2]$ is Eq. (60) with i_t uniform and independent of $\mathcal{F}_t$.

(iv), *magnitude*. By Lemma 10.9 the refreshed matrix has norm at most $e^{\lambda_{\text{top}}+1} + 1$. Between refreshes at most L_{max} increments are added. Each satisfies $\|\hat{A}_t^{\text{comp}}\| \leq \|\hat{A}_t^{\text{ideal}}\| + \varepsilon_{\text{step}}$ and $\|\hat{A}_t^{\text{ideal}}\| \leq 24\|A_t\| + \frac{1}{2s} \sum_j 2\|A_{\text{res}} z_j\|_2 \|z_j\|_2 \leq (24 + 25d_0)\|A_t\|$, using $\|\hat{\Pi}\| \leq 4$ and $\|z_j\|_2^2 = d_0$; and $\|A_t\| \leq \|P_{x_{t+1}} - P_{x_t}\| + \varepsilon_{\text{op}}/2 \leq e^{\lambda_{\text{top}}} + 1$ by Lemma 10.8(iv) with $\lambda_{\text{sp}} = \log(H/\nu)$ at the base point x_t and $\zeta = \|D_{x_{t+1}} - D_{x_t}\| \leq 1$, which gives $\|P_{x_{t+1}} - P_{x_t}\| \leq eH/\nu \leq e^{\lambda_{\text{top}}}/(2n)$, and $\varepsilon_{\text{op}}/2 \leq 1$. Summing, and absorbing $\varepsilon_{\text{step}} \leq 1$ into the constant 30, gives the bound; with $L_{\text{max}} \leq 2\sqrt{n} + 1$ and $e^{\lambda_{\text{top}}} = 2neH_{\text{max}}/\nu$ its base-two logarithm is at most $\log_2(H_{\text{max}}/\nu) + 2\log_2(2n) + \log_2(50d_0) + 5 \leq I - 2 - 3\lceil \log_2(2n) \rceil - 4$ for $n \geq N_0$; the spare $3\lceil \log_2(2n) \rceil + 4$ bits cover the selected-gradient scalar $|\nu \text{tr}[\hat{P}_t E_i]| \leq \Lambda \tilde{c} d_0 \|\hat{P}_t\|$, since $\log_2(\Lambda \tilde{c} d_0) \leq 3\log_2(2n) + 4$ for $n \geq N_0$ (using $\tilde{\Lambda} \leq 3n$ and $\tilde{c} \leq 4 \cdot 10^6 \log(2n)$).

(v), *the refresh count*. Forced refreshes occur at most once per $L_{\text{max}} \geq 1/\pi'$ steps; coin refreshes are independent Bernoulli(π') trials, one per step. $\square$

Without the forced refresh no bound on $\|\hat{P}_t\|$ is available: the zero-mean increments accumulate in the kernel of $\|\cdot\|_G$ over an inter-refresh gap that is only geometrically distributed, and the real-arithmetic proof never needs to control them. The forced refresh is free for the analysis, since a refresh resets $\mathcal{E}$, and it at most doubles the expected refresh count.

Rounding the scalar proximal minimizer can be absorbed into a bounded perturbation of the selected gradient, so the descent estimate applies to the step on the coordinate grid.

Lemma 10.14 (Finite coordinate step). *Let $\hat{g}$ be the selected gradient of Lemma 10.13(iii) for coordinate i , let $[d_i^-, d_i^+]$ be the feasible displacement interval $\{d \in \mathbb{R} : x_{t,i} + d \in \tilde{Q}_i, |d| \leq R_a\}$, whose endpoints lie in Γ , with the trust radius $R_a := \lfloor 1/\tilde{c} \rfloor_\Gamma$, and let $d_t := \text{clip}_{[d_i^-, d_i^+]}(\text{round}_\Gamma(-\hat{g}/L))$. Then d_t is the exact minimizer of $(\hat{g} + L\vartheta)d + \frac{L}{2}d^2$ over $d \in [d_i^-, d_i^+]$ for some $|\vartheta| \leq u/8000$; consequently the step is the exact proximal coordinate step of Section 9 for the gradient value $\partial_i F(x_t) + \mathfrak{e}_{t,i} + \beta_{t,i}$ with $|\beta_{t,i}| \leq \nu \tilde{c} d_0 \varepsilon_{\text{det}} + (L+1)u$, and $x_{t+1} = x_t + d_t e_i$ lies in $\Gamma^S \cap \tilde{Q}$ and satisfies $|d_t| \leq 1/\tilde{c}$. The comparison displacement $d'_i := (x_{*,i} - x_{t,i})/L$ of Eq. (36) lies in the trust interval because $|d'_i| \leq \varepsilon_{\text{box}}/L \leq 1/(2\tilde{c}) \leq R_a$.*

Proof. Put $\vartheta := -\hat{g}/L - \text{round}_\Gamma(-\hat{g}/L)$, so $|\vartheta| \leq u/8000$. The unconstrained minimizer of $(\hat{g} + L\vartheta)d + \frac{L}{2}d^2$ is $-\hat{g}/L - \vartheta = \text{round}_\Gamma(-\hat{g}/L)$, and clipping a one-dimensional convex quadratic's minimizer to an interval gives the constrained minimizer. The bias bound adds $L|\vartheta| \leq Lu$ to Lemma 10.13(iii). The endpoints of $\tilde{Q}_i$ and R_a lie in Γ , so $d_t \in \Gamma$ and $x_{t+1} \in \Gamma^S$; $R_a \leq 1/\tilde{c}$ gives the trust bound; and $R_a \geq 1/\tilde{c} - u/4000 \geq 1/(2\tilde{c})$ together with $L \geq 2\tilde{c}$ and $\varepsilon_{\text{box}} < 1$ gives the last claim. $\square$

We now fix the tolerances. Put $C_{\text{op}} := 12$ and

$$\varepsilon_{\text{op}} := \left\lfloor \frac{\delta_g}{C_{\text{op}} \nu \tilde{c} d_0} \right\rfloor_u, \quad \text{and require} \quad \nu \tilde{c} d_0 \varepsilon_{\text{det}} \leq \frac{\delta_g}{2}, \quad (L+1)u \leq \frac{\delta_g}{2}, \quad \theta \leq \frac{\delta_g}{70\nu \tilde{c} kn}, \quad 3\varepsilon_{\text{qr}} \leq \frac{1}{5\sqrt{k}}, \quad (61)$$

where $\theta = 2\sqrt{2k}\varepsilon_{\text{act}} + \varepsilon_{\text{qr}}$. Then $|\bar{\beta}_{t,i}| \leq \delta_g$ in Lemma 10.14, and the hypotheses of Lemma 10.13(ii) hold. Moreover $\delta_g \leq 2\nu \tilde{c} d_0$: indeed $\delta_g \leq \tilde{\sigma} \leq \tilde{\tau}_w \tilde{\nu}/(16\tilde{c}\sqrt{m}\tilde{H}_{\text{max}}) \leq \tilde{\tau}_w \nu/(16\tilde{c}\sqrt{m}\tilde{H}_{\text{max}})$ by the definitions after Lemma 10.5 and $\nu \geq \tilde{\nu}$ for every tested multiplier, and $\tilde{\tau}_w \leq \tilde{d}_w^2 \leq 2^{-10} \leq 32\tilde{c}^2 \sqrt{m} d_0 \tilde{H}_{\text{max}}$ since $\tilde{\rho} \leq 1/8$, $\tilde{c} \geq a/(2R)$, $d_0 \geq 2m$, and $\tilde{H}_{\text{max}} \geq 16$. Hence $\varepsilon_{\text{op}} \leq \delta_g/(12\nu \tilde{c} d_0) \leq 1/6$, and the first requirement gives $\varepsilon_{\text{dense}}, \varepsilon_{\text{step}} \leq \varepsilon_{\text{det}} \leq \delta_g/(2\nu \tilde{c} d_0) \leq 1$: the standing assumptions

of Lemma 10.13 hold. Every quantity in Eq. (61) is inverse polynomial in n , and the requirements are met once u is small enough; the proof of Theorem 10.1 lists the resulting bit counts.

The trace guard needs an upper certificate for accepted proposals and enough slack to accept every proposal below the analytical crossing level.

Lemma 10.15 (Additive trace guard). *Let y be a proposal from a base point x_t with $F(x_t) \leq H$, let $s_{\text{tr}} := 128 \lceil 8 \log(100TJR_{\text{rep}}/p_0) \rceil$, and let $z_1, \dots, z_{s_{\text{tr}}}$ be fresh Rademacher vectors, and assume $\varepsilon_{\text{act}} \leq 1$. Compute $\hat{y}_j$, the half-action of Lemma 10.10 for $\mathcal{Q}(\hat{D}_y)^{1/2}$ on z_j , the statistics $\hat{s}_j := \|\hat{y}_j\|_2^2$ (exact sums of squares), the means of $\lceil 8 \log(100TJR_{\text{rep}}/p_0) \rceil$ batches of 128 statistics, and their median $\hat{\tau}_y$, and put*

$$U(y) := \left\lceil \frac{4}{3} \hat{\tau}_y + \frac{4}{3} (\varepsilon_{\text{trace}} + 2u) + \frac{d_0 \varepsilon_{\text{op}}}{4} \right\rceil_u,$$

where

$$\varepsilon_{\text{trace}} \in u\mathbb{Z} \text{ with } 2e^{(\lambda_{\text{top}}+1)/2} \sqrt{d_0} \varepsilon_{\text{act}} \leq \varepsilon_{\text{trace}} \leq 2e^{(\lambda_{\text{top}}+1)/2} \sqrt{d_0} \varepsilon_{\text{act}} + u$$

(a certified upper enclosure). Accept y if and only if $q(y) + \lceil \nu U(y) \rceil_u \leq H$, an exact comparison. Then, except with probability $p_0/(100TJR_{\text{rep}})$,

(i) if y is accepted then $F(y) \leq H$;

(ii) if $F(y) \leq H/3$ then y is accepted, provided $\nu(2d_0\varepsilon_{\text{op}} + 3\varepsilon_{\text{trace}} + 7u) + u \leq H/9$.

The guard costs $\sigma_{\text{sc}}r/2$ thin products on a block of s_{tr} vectors and never evaluates ϕ_0 .

Proof. Write $\mathcal{Q} := \mathcal{Q}(\hat{D}_y)$, $\mathcal{T}_y := \text{tr}[\mathcal{Q}]$, and $s_j^{\text{ex}} := z_j^\top \mathcal{Q} z_j$. By Lemma 10.6, $\mathbb{E}[s_j^{\text{ex}}] = \mathcal{T}_y$ and $\mathbb{V}[s_j^{\text{ex}}] \leq 2\|\mathcal{Q}\|_F^2 \leq 2\mathcal{T}_y^2$ because $\mathcal{Q} \succeq 0$. Chebyshev's inequality gives $\Pr[|\text{batch mean} - \mathcal{T}_y| > \mathcal{T}_y/4] \leq (2\mathcal{T}_y^2/128)/(\mathcal{T}_y^2/16) = 1/4$ for each batch of 128, and Hoeffding's inequality for the median of n_b independent batches gives $\Pr[|\hat{\tau}_y^{\text{ex}} - \mathcal{T}_y| > \mathcal{T}_y/4] \leq e^{-n_b/8} \leq p_0/(100TJR_{\text{rep}})$ for $n_b = \lceil 8 \log(100TJR_{\text{rep}}/p_0) \rceil$, where $\hat{\tau}_y^{\text{ex}}$ is the median of the exact batch means. By Lemma 10.10 with $J_{\text{ap}} = \sigma_{\text{sc}}/2$, $\|\hat{y}_j - \mathcal{Q}^{1/2} z_j\|_2 \leq \varepsilon_{\text{act}}/2$ and $\|\mathcal{Q}^{1/2} z_j\|_2 \leq e^{(\lambda_{\text{top}}+1)/2} \sqrt{d_0}$, so $|\hat{s}_j - s_j^{\text{ex}}| \leq (2e^{(\lambda_{\text{top}}+1)/2} \sqrt{d_0} + \varepsilon_{\text{act}}/2) \varepsilon_{\text{act}}/2 \leq \varepsilon_{\text{trace}}$ for $\varepsilon_{\text{act}} \leq 1$. Averages and medians move by at most the largest perturbation of their inputs, and the two roundings add $2u$, so $|\hat{\tau}_y - \hat{\tau}_y^{\text{ex}}| \leq \varepsilon_{\text{trace}} + 2u$. On the good event, $\hat{\tau}_y \geq \frac{3}{4} \mathcal{T}_y - \varepsilon_{\text{trace}} - 2u$, hence $U(y) \geq \mathcal{T}_y + d_0 \varepsilon_{\text{op}}/4 \geq \Phi(y)$ by $\|\mathcal{Q} - P_y\|_1 \leq d_0 \varepsilon_{\text{op}}/4$ (Eq. (59)), and acceptance gives $F(y) = q(y) + \nu \Phi(y) \leq q(y) + \lceil \nu U(y) \rceil_u \leq H$, which is (i). Conversely $\hat{\tau}_y \leq \frac{5}{4} \mathcal{T}_y + \varepsilon_{\text{trace}} + 2u$ and $\mathcal{T}_y \leq \Phi(y) + d_0 \varepsilon_{\text{op}}/4$ give $U(y) \leq \frac{5}{3} \Phi(y) + 2d_0 \varepsilon_{\text{op}} + 3\varepsilon_{\text{trace}} + 7u$, so if $F(y) \leq H/3$ then $q(y) + \lceil \nu U(y) \rceil_u \leq \frac{5}{3} F(y) + \nu(2d_0 \varepsilon_{\text{op}} + 3\varepsilon_{\text{trace}} + 7u) + u \leq \frac{5}{9} H + \frac{1}{9} H < H$, which is (ii). The half-actions apply Lemma 10.10 with $J_{\text{ap}} = \sigma_{\text{sc}}/2$ to a block of s_{tr} vectors. $\square$

Lemma 10.15 replaces a relative certificate by an additive one: the relative sandwich $\Phi(y) \leq U(y) \leq 2\Phi(y)$ of Section 7, whose requirement $d_0 \varepsilon_{\text{op}} \leq \phi_0/8$ is the only place where $\phi_0 = (2n)^{-1999998}$ enters the algorithm, is replaced by the two one-sided properties (i) and (ii), which are all that the crossing argument uses. The requirement in (ii) is implied by Eq. (61) and $\delta_g^2 \leq H/(10^5 m)$ with room to spare.

The phase analysis combines the stochastic tracker recurrence with the bounded gradient bias. A stopped potential estimate controls both the objective gap and the probability of crossing the analytical level.

Lemma 10.16 (Finite Lyapunov contraction and localization). *Let one candidate phase of the finite algorithm be run from z with $F(z) \leq H/16$, under $K_0 := 2000$, $c_1 := 4/3$, and Eq. (61) in force, and let $\tau := \min\{t+1 : F(x_t + d_t e_{i_t}) > H/3\}$ be the first crossing of $H/3$ by a proposal. Then, with $V_t := \Delta(x_t) + c_1 \mathcal{E}_t/(\pi' L)$:*

(i) for every $t < \tau$, $\mathbb{E}[V_{t+1} \mid \mathcal{F}_t] \leq (1 - \alpha)V_t - \frac{L}{8}\mathbb{E}[d_t^2 \mid \mathcal{F}_t] + \frac{2\delta_g^2}{L}$;

(ii) $\Pr[\tau \leq T] \leq 1/15$;

(iii) on the event $\{\tau > T\} \cap \{\text{all guards correct}\}$ the guarded and the unguarded finite processes coincide pathwise, and $\mathbb{E}[\mathbb{1}[\tau > T]\Delta(x_T)] \leq e^{-16}\Delta(z) + 32m\delta_g^2$.

Consequently one candidate phase returns a point x_T with $\Delta(x_T) \leq \Delta(z)/2 + \varepsilon_{\text{opt}}/(32J)$ with probability at least 0.9, by the choice $C_0 = 2 \cdot 10^5$ in the definition of δ_g .

Proof. Step 1: the gap recurrence. For $t < \tau$ the base point has $F(x_t) \leq H/3 \leq H$ and the step is the exact proximal step for the gradient value $\partial_i F(x_t) + \mathbf{e}_{t,i} + \bar{\beta}_{t,i}$ with $|\bar{\beta}_{t,i}| \leq \delta_g$ (Lemma 10.14), so the three-point inequality Eq. (36) holds with $\mathbf{e}_{t,i} + \bar{\beta}_{t,i}$ in place of $\mathbf{e}_{t,i}$. Using $(\mathbf{e} + \bar{\beta})^2 \leq 2\mathbf{e}^2 + 2\delta_g^2$, the average over the uniform coordinate as in Eq. (38), and Lemma 10.13(iii),

$$\mathbb{E}[\Delta(x_{t+1}) \mid \mathcal{F}_t] \leq (1 - \alpha)\Delta(x_t) + \frac{\mathcal{E}_t}{L} + \frac{\delta_g^2}{L} - \frac{L}{4}\mathbb{E}[d_t^2 \mid \mathcal{F}_t].$$

Step 2: the potential. Add $c_1/(\pi'L)$ times Lemma 10.13(ii). The coefficient of $\mathcal{E}_t$ is $1/L + c_1(1 - \pi')/(\pi'L) \leq c_1(1 - \alpha)/(\pi'L)$ because $c_1 \geq \pi'/(1 - \alpha)$, which holds for $c_1 = 4/3$ since $\pi' \geq 4\alpha$; the coefficient of $\mathbb{E}[d_t^2 \mid \mathcal{F}_t]$ is $-L/4 + c_1(1 - \pi')65L/(\pi'ks) \leq -L/4 + (4/3) \cdot 65L/K_0 \leq -L/8$ because $ks \geq K_0/\pi'$ and $K_0 \geq 694$; and the additive term is $\delta_g^2/L + c_1 \cdot 2\delta_g^2/(\pi'ksL) \leq (1 + 8/(3K_0))\delta_g^2/L \leq 2\delta_g^2/L$. This is (i).

Step 3: the crossing. With $d' = 0$ in the three-point inequality, $F(x_{t+1}) - F(x_t) \leq (\mathbf{e}_{t,i} + \bar{\beta}_{t,i})^2/(2L) \leq (\mathbf{e}_{t,i}^2 + \delta_g^2)/L$, so a crossing from $F(z) \leq H/16$ to above $H/3$ within T steps forces $\sum_{t < \tau} (\mathbf{e}_{t,i}^2 + \delta_g^2)/L \geq 13H/48$. The event $\{t < \tau\}$ is $\mathcal{F}_t$ -measurable, so $\mathbb{E}[\mathbb{1}[t < \tau]\mathbf{e}_{t,i}^2] = \mathbb{E}[\mathbb{1}[t < \tau]\mathcal{E}_t]$ by Lemma 10.13(iii). Multiplying Lemma 10.13(ii) by $\mathbb{1}[t < \tau]$, using $\mathbb{1}[t + 1 < \tau] \leq \mathbb{1}[t < \tau]$, and summing over $t < T$ with $\mathcal{E}_0 = 0$,

$$\pi' \sum_{t < T} \mathbb{E}[\mathbb{1}[t < \tau]\mathcal{E}_t] \leq \frac{65L^2}{ks} \sum_{t < T} \mathbb{E}[\mathbb{1}[t < \tau]d_t^2] + \frac{2T\delta_g^2}{ks}.$$

Summing (i) over $t < \tau$ likewise gives the stopped step energy $\frac{L}{8} \sum_{t < T} \mathbb{E}[\mathbb{1}[t < \tau]d_t^2] \leq V_0 + 2T\delta_g^2/L \leq H/16 + 32m\delta_g^2$, using $V_0 = \Delta(z) \leq F(z) \leq H/16$ and $T = 16mL$. Hence, with $ks \geq K_0/\pi'$,

$$\mathbb{E}\left[\sum_{t < \tau} \frac{\mathbf{e}_{t,i}^2}{L}\right] \leq \frac{65}{K_0} \left(\frac{H}{2} + 256m\delta_g^2\right) + \frac{32m\delta_g^2}{K_0} \leq \frac{65H}{2K_0} + \frac{16672m\delta_g^2}{K_0},$$

and $\sum_{t < \tau} \delta_g^2/L \leq T\delta_g^2/L = 16m\delta_g^2$. With $K_0 = 2000$ and $m\delta_g^2 \leq 10^{-5}H$ (definition of δ_g), the expected crossing mass is at most $(0.01625 + 0.00009 + 0.00016)H < 0.0166H$, and Markov's inequality gives $\Pr[\tau \leq T] \leq 0.0166 \cdot 48/13 < 1/15$.

Step 4: coupling and the endpoint. Before τ every proposal has $F \leq H/3$, so by Lemma 10.15(ii) a correct guard accepts it; hence on $\{\tau > T\} \cap \{\text{all guards correct}\}$ the guarded process performs exactly the steps of the unguarded process, which are deterministic functions of the same coins. Iterating (i) over $t < T$ on $\{\tau > T\}$ gives $\mathbb{E}[\mathbb{1}[\tau > T]V_T] \leq (1 - \alpha)^T V_0 + 2T\delta_g^2/L \leq e^{-16}\Delta(z) + 32m\delta_g^2$, and $\Delta \leq V$. Markov's inequality with the threshold $\Delta(z)/2 + \varepsilon_{\text{opt}}/(32J)$ and $32m\delta_g^2 \leq 32\varepsilon_{\text{opt}}/(C_0J)$ bounds the probability that the endpoint misses the threshold on this event by $2e^{-16} + 1024/C_0 \leq 0.01$; the guards fail with probability at most $T \cdot p_0/(100TJR_{\text{rep}}) \leq 0.01$; and $\Pr[\tau \leq T] \leq 1/15$. The three failure probabilities sum to less than 0.1. $\square$

10.6 Certified decisions, the cleanup, and the proof of Theorem 10.1

Two decisions of the wrapper concern points that are not guarded: the box projection ξ'_Q of the Gaussian, whose matrix $D_{\xi'_Q}$ can have eigenvalues of order $a\sqrt{m}$, and the cleanup error matrix. Both are handled by a spectral-norm certificate that never forms a matrix exponential.

Lemma 10.17 (Normalized-squaring norm certificate). *Let E be a symmetric $n' \times n'$ fixed-point matrix, $n' \leq d_0$, with $\|E\|_F > 1$, let $\varpi \in (0, 1/2]$, and put $J_{\text{sq}} := \max\{0, \lceil \log_2(\log n' / \log(1 + \varpi/2)) \rceil - 1\}$ ($J_{\text{sq}} := 0$ when $n' = 1$). Set $\hat{X}_0 := E$ and, for $0 \leq j < J_{\text{sq}}$, $\hat{N}_j := \lceil \|\hat{X}_j\|_F \rceil_{\mathbf{u}}$ (exact sum of squares, integer square root), $\hat{Y}_j := \text{round}_{\mathbf{u}}(\hat{X}_j / \hat{N}_j)$, $\hat{X}_{j+1} := \text{round}_{\mathbf{u}}(\hat{Y}_j \hat{Y}_j)$ with the product exact; finally $\hat{N}_{J_{\text{sq}}} := \lceil \|\hat{X}_{J_{\text{sq}}}\|_F \rceil_{\mathbf{u}}$. Let $\hat{\Lambda}$ satisfy $0 \leq \hat{\Lambda} - \sum_{j \leq J_{\text{sq}}} 2^{-j} \log \hat{N}_j \leq \varpi/16$ (range reduction and Taylor series), and let $U \in \mathbf{u}\mathbb{Z}$ satisfy $e^{\hat{\Lambda} + \varpi/16} \leq U \leq e^{\hat{\Lambda} + \varpi/16} + \mathbf{u}$ (a certified upper enclosure). If $10^5 n'^2 \mathbf{u} \leq \varpi$ then*

$$\|E\| \leq U \leq (1 + \varpi)\|E\|.$$

The certificate costs J_{sq} exact dense products and $J_{\text{sq}} + 1$ Frobenius norms, $J_{\text{sq}} = O(\log(\log n' / \varpi))$.

Proof. Write $a_j := \|\hat{X}_j\|$, $F_j := \|\hat{X}_j\|_F$, so $F_j \leq \hat{N}_j \leq F_j + 2\mathbf{u}$, and let $Y_j^{\text{ex}} := \hat{X}_j / \hat{N}_j$, with $\|Y_j^{\text{ex}}\|_F \leq 1$ and $\|Y_j^{\text{ex}}\| = a_j / \hat{N}_j$. The rounding of $\hat{Y}_j$ perturbs it by a matrix G_j of operator norm at most $n'\mathbf{u}/2$ (entrywise error $\mathbf{u}/2$, Frobenius bound), and the rounding of the product by at most $n'\mathbf{u}$, so $\hat{X}_{j+1} = (Y_j^{\text{ex}})^2 + E_{j+1}$ with $\|E_{j+1}\| \leq 2\|Y_j^{\text{ex}}\| \|G_j\| + \|G_j\|^2 + n'\mathbf{u} \leq n'\mathbf{u} + n'^2 \mathbf{u}^2 / 4 + n'\mathbf{u} \leq 4n'\mathbf{u}$, hence $a_{j+1} \in [(a_j / \hat{N}_j)^2 - 4n'\mathbf{u}, (a_j / \hat{N}_j)^2 + 4n'\mathbf{u}]$.

Lower bounds. $F_0 > 1$ by hypothesis, and for $j \geq 1$, $F_j \geq a_j \geq \|Y_{j-1}^{\text{ex}}\|^2 - 4n'\mathbf{u} \geq F_{j-1}^2 / (n' \hat{N}_{j-1}^2) - 4n'\mathbf{u} \geq 1/(2n')$, using $\|Y\| \geq \|Y\|_F / \sqrt{n'}$, $\hat{N}_{j-1} \leq F_{j-1}(1 + 4n'\mathbf{u})$, and induction. Hence $(a_j / \hat{N}_j)^2 \geq 1/(2n')$ for every j , and the additive perturbation $4n'\mathbf{u}$ of a_{j+1} is a relative one of at most $8n'^2 \mathbf{u}$: $a_j = \hat{N}_j a_{j+1}^{1/2} (1 + \vartheta_j)^{1/2}$ with $|\vartheta_j| \leq 8n'^2 \mathbf{u}$.

Unrolling. Iterating from $j = 0$ to $J_{\text{sq}} - 1$ and using $F_{J_{\text{sq}}} / \sqrt{n'} \leq a_{J_{\text{sq}}} \leq F_{J_{\text{sq}}} \leq \hat{N}_{J_{\text{sq}}} \leq F_{J_{\text{sq}}}(1 + 4n'\mathbf{u})$,

$$\|E\| = \prod_{j < J_{\text{sq}}} \hat{N}_j^{2^{-j}} \cdot a_{J_{\text{sq}}}^{2^{-J_{\text{sq}}}} \cdot \prod_{j < J_{\text{sq}}} (1 + \vartheta_j)^{2^{-j-1}} \in \left[\Pi n'^{-2^{-J_{\text{sq}}-1}} \frac{1 - 16n'^2 \mathbf{u}}{1 + 4n'\mathbf{u}}, \Pi(1 + 16n'^2 \mathbf{u}) \right], \quad \Pi := \prod_{j \leq J_{\text{sq}}} \hat{N}_j^{2^{-j}}.$$

The certificate. $\hat{\Lambda} + \varpi/16 \geq \log \Pi + \varpi/16 \geq \log(\Pi(1 + 16n'^2 \mathbf{u})) \geq \log \|E\|$ because $\varpi/16 \geq 16n'^2 \mathbf{u}$, so $U \geq \|E\|$. Conversely $U \leq \Pi e^{\varpi/8} + \mathbf{u} \leq \Pi e^{\varpi/8} (1 + 2\sqrt{n'} \mathbf{u})$, because $\Pi \geq \|E\| / (1 + 16n'^2 \mathbf{u}) \geq 1/(2\sqrt{n'})$ from $\|E\| \geq \|E\|_F / \sqrt{n'} > 1/\sqrt{n'}$; hence $U \leq \|E\| n'^{2^{-J_{\text{sq}}-1}} e^{\varpi/8} (1 + 2\sqrt{n'} \mathbf{u}) (1 + 4n'\mathbf{u}) / (1 - 16n'^2 \mathbf{u})$, and $n'^{2^{-J_{\text{sq}}-1}} \leq 1 + \varpi/2$ by the choice of J_{sq} , $e^{\varpi/8} \leq 1 + 0.134\varpi$ for $\varpi \leq 1/2$, and $(1 + 2\sqrt{n'} \mathbf{u})(1 + 4n'\mathbf{u}) / (1 - 16n'^2 \mathbf{u}) \leq 1 + 40n'^2 \mathbf{u} \leq 1 + \varpi/2500$; the product of the three factors is at most $1 + 0.7\varpi \leq 1 + \varpi$. $\square$

The multiplier search, candidate selection, and radial feasibility correction require outward enclosures at finer accuracy than the step guards.

Lemma 10.18 (Certified enclosures and exact decisions). *Let $x \in \Gamma^S$ be a point of the guarded level set, or the box projection ξ'_Q of ξ' in case (C) of item (v) below, and let $\text{tol} \in \{\varepsilon_{\text{opt}}/(256\nu J), \tau_{\mathbf{w}}/16\}$. Compute $\tilde{P}_x$ by Lemma 10.9 with the degree $r_{\text{tol}} := \min\{r \geq 2 : (r+1)! \geq 9\sigma_{\text{sc}} e^{\lambda_{\text{top}}+2} d_0 / \text{tol}\}$ and put $\hat{\Phi}(x) := \text{round}_{\mathbf{u}}(\text{tr}[\tilde{P}_x])$. Then:*

- (i) if $d_0 \varepsilon_{\text{dense}} \leq \text{tol}/3$ and $d_0^2 u e^{\lambda_{\text{top}}+1} \leq \text{tol}/3$, the interval $[\widehat{\Phi}(x) - \text{tol} - u, \widehat{\Phi}(x) + \text{tol} + u]$ contains $\Phi(x)$, and $[q(x) + \lfloor \nu(\widehat{\Phi}(x) - \text{tol} - u) \rfloor_u, q(x) + \lceil \nu(\widehat{\Phi}(x) + \text{tol} + u) \rceil_u]$ contains $F(x)$ and has length at most $2\nu \text{tol} + 2(\nu + 1)u$;
- (ii) the selection rule of Section 7 (retain the candidate whose upper endpoint lies below the old point's lower endpoint and is least, else the old point), the three-way bisection decision on the enclosure of $\Phi(x_\nu)$, the box shortcut, and the radial correction are exact comparisons of grid numbers (every tested multiplier is first rounded to $u\mathbb{Z}$, a shift of at most u of a bisection midpoint, which the wrapper tolerates because its bracket length never falls below $\tilde{d}_w/(4\tilde{G}_*) \gg u$), and every conclusion Section 7 draws from the corresponding exact comparison holds for them;
- (iii) the radial factor $\beta_r := \lfloor (3/4)/(\bar{\Phi} - 1/4) \rfloor_u$ for $\bar{\Phi} := \widehat{\Phi}(\hat{x}) + \text{tol} + u > 1$ satisfies $\Phi(\beta_r \hat{x}) \leq 1$ and $(1 - \beta_r)\|\hat{x}\|_2 \leq d_w/16$, and $\beta_r \hat{x}$ is formed exactly;
- (iv) after snapping at threshold $2\tilde{\rho}$ (Lemma 10.5(iv)), the update $y_i := \text{round}_u^{\rightarrow 0}(y_i + 2\bar{x}_i/\varepsilon_{\text{box}})$ for unsnapped i and $y_i := \pm 1$ for snapped i keeps $y \in (u\mathbb{Z})^S \cap [-1, 1]^S$, makes $S := \{i : |y_i| < 1\}$ exact, and moves the effective increment by at most $m\varepsilon_{\text{box}}u/2$ in $\|B(\cdot)\|$, which the slack $s_{\text{sm}}\tilde{R}/2$ of Lemma 10.5(iv) absorbs (as $u \leq 2^{-40}/(2n)$ and $\tilde{R} \geq 279\sqrt{m}$); the acceptance test “at least δm snapped coordinates” compares an integer with δm for the exact rational $\delta = 35209/1250000$;
- (v) (pre-test for the box shortcut) let $\tilde{\chi}_{\text{hi}}$ be a dyadic number with $\tilde{\chi} \leq \tilde{\chi}_{\text{hi}} \leq 1.01\tilde{\chi}$ (a certified evaluation of $\log(2n)$ to 40 bits), let ϖ be a dyadic number in $[\tilde{\chi}/16, \tilde{\chi}/8]$, and let U_B be the certificate of Lemma 10.17 for $E := B(\xi'_Q)$ with this ϖ (or $U_B := 1$ if $\|B(\xi'_Q)\|_F \leq 1$; with $U_B = 1$ the test lands in case (A), as $(b' - \tilde{\chi}_{\text{hi}})\tilde{R} > 278\sqrt{m} > 1$). The decisions compare the dyadic numbers $\tilde{c}U_B$ and $(b' \mp \tilde{\chi}_{\text{hi}} \cdot \{1, 1/4\})\tilde{a}$, which is exact. In case (A), i.e. $\tilde{c}U_B \leq (b' - \tilde{\chi}_{\text{hi}})\tilde{a}$, the point ξ'_Q lies in $\tilde{\mathcal{D}}$ and is the exact projection. In case (B), i.e. $\tilde{c}U_B > (b' + \tilde{\chi}_{\text{hi}}/4)\tilde{a}$, the point ξ'_Q is infeasible. In the remaining case (C), $\lambda_{\max}(D_{\xi'_Q}) \leq \log(2n)/3$ and $\lambda_{\min}(D_{\xi'_Q}) \geq -2ab - \log(2n)/3 - 2u \geq -M_{\text{sp}}$, so the enclosure of (i) applies to ξ'_Q and decides its feasibility up to the wrapper's tolerance. In cases (B) and (C), $\|\xi'\|_2 > 184$.

Proof. (i), the enclosure. By Lemmas 10.8(iii),(iv) and 10.9 with the stated degree, $\|\tilde{P}_x - P_x\| \leq \text{tol}/(3d_0) + \varepsilon_{\text{dense}} + d_0 u e^{\lambda_{\text{top}}+1} \leq \text{tol}/d_0$, so $|\text{tr}[\tilde{P}_x] - \Phi(x)| \leq \text{tol}$; the trace is an exact sum of grid entries rounded once, ν is dyadic, $q(x)$ is exact, and the two products are rounded outward.

(ii), exact decisions. Each rule compares numbers on the grid $u\mathbb{Z}$ or Γ -derived rationals with a common denominator, which is an integer comparison. The arguments of Section 7 use only that the intervals contain the true values and have the stated widths: for the candidate selection rule, “accepted phase-start objectives never increase” and the separation $\Delta_z/2 - \varepsilon_{\text{opt}}/(32J) > 2\varepsilon_{\text{opt}}/(64J)$ when $\Delta_z > \varepsilon_{\text{opt}}$; for the bisection, the three cases “interval above 1”, “meets 1”, “below 1” with the residual estimate $\|x_\nu - x_*\|_2^2 \leq |\nu - \nu_*| |\Phi(x_\nu) - 1|$ of Section 7 (the Φ -analog of Eq. (17)) in the middle case (the enclosure of $\Phi(x_\nu)$ has length at most $\tau_w/4 + 2u$, and $2\Lambda u \leq d_w^2/(4(1 + \Lambda))$), the bisection row of Step 1 in the proof of Theorem 10.1, keeps the product of Λ and the length below $d_w^2/4$ as Section 7 requires) and the Lipschitz transfer $G_{\text{lev}}\sigma \leq \tau_w/16$ (which holds for the rounded constants because $\tilde{\sigma} \leq \tilde{\tau}_w/(16\tilde{G}_{\text{lev}})$ by definition and $\tilde{G}_{\text{lev}}$ bounds the Lipschitz constant of the finite Φ). For the box shortcut in case (C) of (v), if the upper endpoint at ξ'_Q is at most 1 the point is exactly feasible and is the projection; otherwise $\Phi(\xi'_Q) > 1 - \tau_w/8 - 2u$ and the wrapper proceeds as written, handling a barely feasible ξ'_Q through its first solve at $\underline{\nu}$.

(iii), *the radial factor*. For $\theta \in (0, 1/4)$ the map $\theta \mapsto (1 - \theta)/(\bar{\Phi} - \theta)$ is decreasing when $\bar{\Phi} > 1$, and $\phi_0 < 1/4$, so $\beta_r \leq (3/4)/(\bar{\Phi} - 1/4) \leq (1 - \phi_0)/(\bar{\Phi} - \phi_0)$ and convexity gives $\Phi(\beta_r \hat{x}) \leq \beta_r \bar{\Phi} + (1 - \beta_r)\phi_0 \leq 1$ as in Section 7. Moreover $1 - \beta_r \leq (\bar{\Phi} - 1)/(\bar{\Phi} - 1/4) + u \leq \frac{4}{3}(\bar{\Phi} - 1) + u \leq \frac{4}{3} \cdot \frac{\tau_w}{2} + u$ by the accuracy choices (and $u \leq \tau_w/(8\Lambda)$, the bisection row of Step 1), so with $\|\hat{x}\|_2 \leq \sqrt{m}$ and $\tau_w \leq 3d_w/(64\sqrt{m})$ we get $(1 - \beta_r)\|\hat{x}\|_2 \leq \frac{2}{3}\tau_w\sqrt{m} + u\sqrt{m} \leq d_w/32 + u\sqrt{m} \leq d_w/16$, the last step by $u\sqrt{m} \leq d_w/32$. The product $\beta_r \hat{x}$ of a u -grid number and a Γ -vector lies on the grid $u\Gamma$ and is exact.

(v), *the pre-test*. By Lemma 10.17, $\|B(\xi'_Q)\| \leq U_B \leq (1 + \tilde{\chi}/8)\|B(\xi'_Q)\|$, and $\tilde{c}U_B \leq (b' - \tilde{\chi}_{hi})\tilde{a}$ means $U_B \leq (b' - \tilde{\chi}_{hi})\tilde{R}$ since $\tilde{R} = \tilde{a}/\tilde{c}$. In case (A), $\|B(\xi'_Q)\| \leq (b' - \tilde{\chi}_{hi})\tilde{R} \leq (b' - \tilde{\chi})\tilde{R}$, so $\tilde{h}(\xi'_Q) \leq b'$ by Lemma 10.5(iii), i.e. $\xi'_Q \in \tilde{\mathcal{D}}$; the box projection of ξ' that lies in the body is the projection onto $\tilde{Q} \cap \tilde{\mathcal{D}}$. In case (B), $\|B(\xi'_Q)\| \geq U_B/(1 + \tilde{\chi}/8) > (b' + \tilde{\chi}_{hi}/4)\tilde{R}/(1 + \tilde{\chi}/8) \geq b'\tilde{R}$ because $b' \leq 1$ and $\tilde{\chi}_{hi} \geq \tilde{\chi}$, so $\tilde{h}(\xi'_Q) > b'$ and $\xi'_Q \notin \tilde{\mathcal{D}}$. In case (C), $(b' - \tilde{\chi}_{hi})\tilde{R}/(1 + \tilde{\chi}/8) \leq \|B(\xi'_Q)\| \leq (b' + \tilde{\chi}_{hi}/4)\tilde{R}$, so the eigenvalues of $D_{\xi'_Q} = \tilde{a} \text{diag}(B, -B)/\tilde{R} - \tilde{a}b'I$ lie in $[-\tilde{a}(2b' + \tilde{\chi}_{hi}/4), \tilde{a}\tilde{\chi}_{hi}/4]$, and $\tilde{a}\tilde{\chi}_{hi} \leq 1.01 \log(2n)$ gives the two spectral bounds (recall $\tilde{a}b' \leq ab + u$); the enclosure of (i) then applies because its hypotheses are exactly these spectral bounds. In cases (B) and (C), $\|\xi'\|_2 \geq \|\xi'_Q\|_2 \geq \|\xi'_Q\|_1/\sqrt{m} \geq \|B(\xi'_Q)\|/\sqrt{m} \geq (b' - \tilde{\chi}_{hi})\tilde{R}/((1 + \tilde{\chi}/8)\sqrt{m}) \geq (1 - s_{sm} - 1.01\chi - 2u)c_{\text{rad}}(\ell) \max\{1, \ell\}/(1 + \chi/8) > 184$, using $c_{\text{rad}}(\ell) \max\{1, \ell\} \geq 279$ for every regime of Table 2 and $\tilde{R} \geq R$; this is the bound $\Lambda = \Omega(1)$ that Section 7 derives from the infeasibility of the box projection.

(iv), *the update of y* . $y_i + 2\bar{x}_i/\varepsilon_{\text{box}} \in [-1, 1]$ because $\bar{x} \in \tilde{Q}$, and rounding toward zero keeps a value in $(-1, 1)$ strictly inside; snapped coordinates are exactly ± 1 . The rounding changes the increment $2\bar{x}_i/\varepsilon_{\text{box}}$ by at most u , i.e. $\bar{x}_i$ by at most $\varepsilon_{\text{box}}u/2$, and $\|A_i\| \leq 1$ gives the bound on $\|B(\cdot)\|$. $\square$

The independent cleanup signs have the stored fractional coloring as their exact mean. The spectral certificate then converts a successful rounding into a certified bound within the cleanup allowance.

Lemma 10.19 (Cleanup: exact coins and the norm certificate). *Let $|S| < N_0$, let ε_i for $i \in S$ be drawn with $\Pr[\varepsilon_i = 1] = (1 + y_i)/2$ exactly (Lemma 10.6), and let $E := \sum_{i \in S} (\varepsilon_i - y_i)A_i$, an exact matrix with entries in $2^{-b_{\text{in}}}u\mathbb{Z}$ of modulus at most $2|S|$. If $\|E\|_F \leq 1$ put $U := 1$; otherwise let U be the certificate of Lemma 10.17 for E with $\varpi := 0.0046$ and $n' := n$. If $10^5 n^2 u \leq 0.0046$ then $\|E\| \leq U \leq 1.0046\|E\|$; hence $U \leq 6.5\sqrt{n}$ whenever $\|E\| \leq \sqrt{2(\log(2N_0) + \log 2)}\sqrt{n}$, the Hoeffding radius of Section 5 (numerically $6.46616\sqrt{n}$), and $U \leq 6.5\sqrt{n}$ implies $\|E\| \leq 6.5\sqrt{n}$. The certificate costs $J_{\text{sq}} = \max\{0, \lceil \log_2(\log n / \log 1.0023) \rceil - 1\} \leq 14$ dense products for $n \leq 10^{18}$.*

Proof. If $\|E\|_F \leq 1$ then $\|E\| \leq 1 = U \leq 6.5\sqrt{n}$. Otherwise Lemma 10.17 gives the two-sided bound, and $6.46616 \cdot 1.0046 < 6.4960 < 6.5$. $\square$

Matrix Hoeffding is applied with the stored y as the mean, so its centering is exact and the Hoeffding radius of Section 5 is unchanged; the additive certification allowance $0.03\sqrt{n}$ of Section 6 becomes the multiplicative 1.0046 here, and the certificate replaces the diagonalization-based certificate of Algorithm 1, a real-arithmetic tool (Remark 10.22), by one that needs only exact products. A direct power $\text{tr}[E^{2^{J_{\text{sq}}}}]$ would overflow every grid, because $\|E\|/(2m)$ can be as small as 10^{-4} ; the per-level normalization keeps every stored matrix at Frobenius norm about one.

Remark 10.20 (The generalized Pinsker inequality needs no precision argument). The inequality $\|P - Q\|_1^2 \leq 4 \max\{\text{tr } P, \text{tr } Q\} \mathcal{D}(P\|Q)$ of Bansal, George, Ghosh, Sikora, and Zheng [BGG⁺25, Theorem 1.3], used in Lemma 10.13(ii) through Lemma 9.5, is a statement about arbitrary positive

semidefinite matrices and is applied here to the exact matrices $P_{x_t}, P_{x_{t+1}}$, never to computed ones. Nothing is computed from it, so it carries no rounding.

Remark 10.21 (The relative certificate). Theorem 10.1 holds for the algorithm with the relative guard $\Phi(y) \leq U(y) \leq 2\Phi(y)$ of Section 7 as well, with the same proof: the requirement $d_0\varepsilon_{\text{op}} \leq \phi_0/8$ of Section 7, met there by the choice $\varepsilon_{\text{op}} \leq \phi_0/(16d_0) = (2n)^{-1999998}/(32n)$, replaces Eq. (61), and every bit count that involves $\log_2(1/\varepsilon_{\text{op}})$ grows by about $1999998\log_2(2n) + \log_2(32n)$, which is $5.8 \cdot 10^7$ at $n = 3 \cdot 10^8$ and still $O(\log n)$. The exponent of the bit complexity is therefore the same, and the additive guard is a matter of the constant in W , not of the theorem. In fixed point it is also a matter of representability: the relative guard resolves Φ at the scale ϕ_0 , which the additive guard never forms.

Remark 10.22 (Theorem 6.1 is not covered). The $n^{2+\omega/2}$ algorithm of Section 6 diagonalizes $B(x)$ at every block query with Sobczyk’s algorithm [Sob25, Corollary 2.4], whose eigenvector computation is analyzed only in the real-arithmetic model; its finite-precision behavior is Sobczyk’s own open problem, and Theorem 6.1 remains a real-arithmetic statement. No diagonalization occurs on the cubic path.

Proof of Theorem 10.1. Step 1: the word length. The lemmas impose the following requirements on $\mathbf{u} = 2^{-f}$, each of the form $f \geq \log_2(1/\text{tolerance}) + \log_2(\text{polynomial factor})$; we list the tolerance and the factor and check that Eq. (54) dominates each.

requirement	tolerance	factor (base-two logarithm at most)
Lemma 10.5	2^{-40}	$\log_2(2n)$
Eq. (59)	$\varepsilon_{\text{op}}/8$	$(\lambda_{\text{top}} + 1)\log_2 e + \log_2 d_0$
Lemma 10.9 ($\varepsilon_{\text{dense}} \leq \varepsilon_{\text{op}}/8$)	ε_{op}	$(\lambda_{\text{top}} + 1)\log_2 e + \log_2(16e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}d_0)$
Eq. (61), ε_{det} term	δ_g	$2\lambda_{\text{top}}\log_2 e + \log_2(\nu\tilde{c}) + 4\log_2 d_0 + \log_2(L_{\text{max}}\sigma_{\text{sc}}) + 12$
Eq. (61), $(L+1)\mathbf{u}$ term	δ_g	$\log_2(2(L+1))$
Eq. (61), θ term	δ_g	$2\lambda_{\text{top}}\log_2 e + \log_2(\nu\tilde{c}kn) + 2\log_2 d_0 + \log_2(k^{3/2}\sigma_{\text{sc}}) + 14$
Lemma 10.12(ii) ($3\varepsilon_{\text{qr}} \leq 1/(5\sqrt{k})$)	1	$\lambda_{\text{top}}\log_2 e + \log_2(k^{5/2}d_0^2\Omega_{\text{max}}) + 10$
Lemma 10.15(ii)	H/ν	$1.5\lambda_{\text{top}}\log_2 e + 2\log_2 d_0 + \log_2 \sigma_{\text{sc}} + 12$
Lemma 10.18(i), candidate selection	$\varepsilon_{\text{opt}}/(256\nu J)$	$(\lambda_{\text{top}} + 1)\log_2 e + \log_2(48e(\kappa_{\text{sc}} + 5)\sigma_{\text{sc}}d_0^2)$
Lemma 10.18(i), rounding	ε_{opt}	$\log_2(512\Lambda J)$
Lemma 10.18(ii), bisection	$d_w^2/(1+\Lambda)$	$\log_2(8\Lambda)$
Lemma 10.18(iii)	d_w	$\log_2(32\sqrt{m})$
Lemma 10.17 (pre-test, $\varpi \geq \tilde{\chi}/16$)	$\tilde{\chi}/16$	$\log_2(10^5 n^2)$
Lemma 10.19 ($\varpi = 0.0046$)	0.0046	$\log_2(10^5 n^2)$

Every tolerance is at least ε_{opt} times an inverse polynomial: indeed $\delta_g^2 \geq \varepsilon_{\text{opt}}/(4C_0Jm)$ (the ε_{opt} term is the smaller one in the minimum: $\tilde{\varepsilon}_{\text{opt}} \leq \tilde{\sigma}^2/2 \leq 2^{-17} < 1 \leq 4m \leq 2JH$, using $\tilde{\sigma} \leq \tilde{d}_w/8 \leq 1/256$ and $H \geq 16q(0) = 8\|\xi'\|_2^2 \geq 4m$ from the norm rejection; the floor costs at most a factor 2) gives $\log_2(1/\delta_g) \leq \frac{1}{2}\log_2(1/\varepsilon_{\text{opt}}) + \frac{1}{2}\log_2(4C_0Jm)$, and $\varepsilon_{\text{op}} \geq \delta_g/(C_{\text{op}}\nu\tilde{c}d_0) - \mathbf{u}$ gives $\log_2(1/\varepsilon_{\text{op}}) \leq \log_2(1/\delta_g) + \log_2(2C_{\text{op}}\Lambda\tilde{c}d_0)$. With $\tilde{\Lambda} \leq \frac{8}{3}m + \mathbf{u} \leq 3n$ ($C_g = 2$), $\Omega_{\text{max}} \leq 2n$ (since $(\Omega_{\text{max}} - 1)^2 \leq 2\log(4n^2) + 4f\log 2$ and $f \leq n^2$ for $n \geq N_0$, every term of Eq. (54) being $O(\log n)$, so $\Omega_{\text{max}} \leq 1 + \sqrt{2\log(4n^2) + 4n^2\log 2} \leq 2n$; the requirement of Lemma 10.12(ii) is monotone in f , so it suffices to check it at the f of Eq. (54)), $\tilde{c} \leq 2a/R \leq 4 \cdot 10^6 \log(2n)$, $k \leq n$, $L_{\text{max}} \leq 2\sqrt{n} + 1$, $\sigma_{\text{sc}} \leq 2(M_{\text{sp}} + \lambda_{\text{top}}) \leq 4M_{\text{sp}}$, $L \leq 8\max\{1 + 3\tilde{c}^2H, 2\tilde{c}\} \leq 2^{61}n\log^2(2n)$, $J \leq 2\log_2(4H/\varepsilon_{\text{opt}})$, and $C_0 = 2 \cdot 10^5$, every row is at most

$$\log_2 \frac{1}{\varepsilon_{\text{opt}}} + 3\lambda_{\text{top}}\log_2 e + 2\log_2 M_{\text{sp}} + 40\log_2(2n) + 200,$$

which is f of Eq. (54). The rows are stated for the rounded parameters of Lemma 10.5, while Eq. (54) uses the parameters of Section 7; the definitions in and after Lemma 10.5 bound the

ratios by constants ($\tilde{\rho} \geq \rho/3$, $\tilde{d}_w \geq d_w/3$, $\tilde{\tau}_w \geq \tau_w/16$, $\tilde{H}_{\max} \leq \frac{5}{3}H_{\max}$, $\tilde{\nu} \geq \nu/4$, $\tilde{G}_{\text{lev}} \leq 7G_{\text{lev}}$, the rounding floors costing at most u each, negligible under Eq. (54)), hence $\tilde{\varepsilon}_{\text{opt}} \geq 2^{-14}\varepsilon_{\text{opt}}$ and $\tilde{H}_{\max}/\tilde{\nu} \leq 2^3 H_{\max}/\nu$ (at $n = 3 \cdot 10^8$ the factors are $2^{8.3}$ and $2^{1.7}$), which the additive 200 in f and 12 in I absorb. The integer bits are dictated by Lemma 10.13(iv), by the action intermediates $e^{\lambda_{\text{top}}+1}\Omega_{\max}\sqrt{d_0}$ of Lemma 10.10 (with $\Omega_{\max} \leq 2n$), by the sketch and Householder columns (norm at most $2Y_{\max}+1$) and the probe images w_j (norm at most $2\sqrt{d_0}(e^{\lambda_{\text{top}}+1})$) of Lemma 10.12, and by the wrapper constants $\tilde{G}_{\text{lev}} \leq \tilde{c}\sqrt{m}\tilde{H}_{\max}/\tilde{\nu}$; each has base-two logarithm at most $\log_2(H_{\max}/\nu) + 3\log_2(2n)+30$, and the selected-gradient scalar has logarithm at most $\log_2(H_{\max}/\nu)+6\log_2(2n)+15$ (Lemma 10.13(iv)); both are below $I-2$ for $n \geq N_0$. Since $\log_2(1/\varepsilon_{\text{opt}})$, λ_{top} , $\log_2 M_{\text{sp}}$, and $\log_2(H_{\max}/\nu)$ are $O(\log n)$ (all wrapper tolerances are inverse polynomial in n by Section 7), $W = O(\log n)$; the failure probability p enters only through counts.

Step 2: correctness. Run the finite algorithm on the $\tilde{A}_i$ of Lemma 10.4. In each outer round, Lemma 10.5 supplies the rounded body with $\gamma_m(\tilde{\mathcal{D}}) \geq \exp(-0.00372m)$ and Lemma 10.7(iii) gives, for the rounded Gaussian ξ' , more than δm tight facets at the exact projection x_* with probability greater than 0.126; with the norm rejection and the sampler caps this is at least 0.1 per attempt after the 0.02 numerical allowance of Section 6. The multiplier wrapper runs $O(\log(n/\tilde{\rho}))$ fixed-multiplier solves on exact enclosures (Lemma 10.18(ii)). Each solve runs J phases of R_{rep} candidates; by Lemma 10.16 a candidate halves the gap up to $\varepsilon_{\text{opt}}/(32J)$ with probability at least 0.9, guards being correct except with probability $p_0/(100TJR_{\text{rep}})$ each (Lemma 10.15) and refresh counts being capped as in Algorithm 10 (Lemma 10.13(v)); the candidate-selection enclosures of width at most $\varepsilon_{\text{opt}}/(128J)+4\Lambda u \leq \varepsilon_{\text{opt}}/(64J)$ (a requirement listed in Step 1) then give $\Delta_{j+1} \leq \Delta_j/2 + \varepsilon_{\text{opt}}/(16J)$ and a final gap at most ε_{opt} except with probability p_0 , exactly as in Section 7. Strong convexity converts the gap into $\|\hat{x} - x_\nu\|_2 \leq \sigma$, the bisection and radial correction return an exactly feasible point within $\tilde{\rho}$ of x_* (Lemma 10.18(iii)), and Lemma 10.5(iv) and Lemma 10.18(iv) put the snapped increment in $\tilde{K}$ with more than δm new signs. Lemma 10.5(v) bounds the sum of the increments by $154710\sqrt{n}$, Lemma 10.19 bounds the cleanup by $6.5\sqrt{n}$ with probability at least 1/2 per trial, and Lemma 10.4 adds 2^{-9} ; the total is below $156000\sqrt{n}$. The failure budget is that of Theorem 9.1 with one additional class of events charged to the numerical allowance, the caps of the samplers (the coin caps of Lemmas 10.7(i) and 10.6, and the one-redraw cap of Lemma 10.7(iv), whose failure probability u^4 per test matrix sums over the steps of one projection attempt to less than 2^{-4000} , charged to that attempt's numerical allowance), and nothing else, since every rounding is deterministic and no saturation occurs while all guards are correct (Lemma 10.13(iv) and Step 1).

Step 3: cost. Write $W' := 2W + c_\eta[\log_2(2n)]$. A nonrefresh step performs $6\sigma_{\text{sc}}r + O(1)$ thin products of shape $\langle d_0, d_0, O(k+s) \rangle$ or $\langle d_0, O(k+s), d_0 \rangle$ (Lemma 10.12(iv)), each costing $O(n^{2+2\eta})M(W')$ by Lemma 10.3 with $\beta \in (1/4, \alpha_{\text{dual}})$ since $k+s = O(n^{1/4+o(1)}) \leq n^\beta$ for large n , plus $O(d_0^2)$ word operations for the contraction, the update of $B(x_t)$, and the exact addition to $\hat{P}_t$, plus the guard's $\sigma_{\text{sc}}r/2$ thin products on a block of s_{tr} vectors (Lemma 10.15); with $\sigma_{\text{sc}}r = O(\log^2 n)$ and $s_{\text{tr}} = O(\log(n/p))$ this is $n^{2+o(1)}\text{polylog}(1/p)M(W')$. A refresh performs $r + \kappa_{\text{sc}} = O(\log n)$ dense products at $O(n^{\omega+2\eta})M(W')$ each (Lemma 10.9). A phase has $T = 16mL = mn^{o(1)}$ steps and, by Lemma 10.13(v) and the binomial tail cap, at most $O(T\pi_m \log(n/p) + 1)$ refreshes, so a phase costs $mn^{2+o(1)} + mn^{\omega-1/2+o(1)} + n^{\omega+o(1)}$ times $\text{polylog}(1/p)M(W')$, as in Eq. (53). Phases per solve (JR_{rep}), solves per projection ($O(\log(n/\tilde{\rho}))$), enclosure evaluations per solve ($O(JR_{\text{rep}})$ dense evaluations of $r_{\text{tol}} + \kappa_{\text{sc}}$ products each), projection attempts per round ($O(\log \log n + \log(1/p))$), and rounds ($O(\log n)$) are polylogarithmic, and $\sum_{\text{rounds}} m = O(n)$. The pre-test of Lemma 10.18(v) performs $O(\log(a)) = O(\log \log n)$ dense products per projection attempt, and the cleanup performs $O(\log(1/p))$ trials of $O(\log \log n)$ dense products (Lemma 10.19); the per-step probes and test matrix cost $O(n^{5/4}\text{polylog}(n/p))$ bit operations per step, below the $O(d_0^2)$ word operations

already counted, and the outer-round coins, the index sampler, and the parameter computations cost $O(n \text{polylog}(n/p))$ bit operations per round. Summing over the rounds as in Section 9 gives the first term of Eq. (55), and Lemma 10.4 gives the second. Finally $M(W') = O(W' \log W') = O(\log n \log \log n)$ for fixed η and $\omega + 1/2 < 3$; for every fixed $\eta > 0$ the count is $O(n^{3+2\eta}) \text{polylog}(1/p) M(W')$, and since η is arbitrary this is $n^{3+o(1)} \text{polylog}(1/p)$ in the sense in which Sections 6–9 use $n^{\omega+o(1)}$; this gives $n^{3+o(1)} \text{polylog}(1/p) + \tilde{O}(n^3 L_{\text{in}})$. $\square$

Remark 10.23 (Reading the word length). Eq. (54) is a bound with room: it charges every row of the table in Step 1 at its worst and adds 200 guard bits. At $n = m = 3 \cdot 10^8$ with the parameters of Sections 5–9, $\log_2(1/\varepsilon_{\text{opt}}) \approx 400$, $\lambda_{\text{top}} \log_2 e \approx 115$, $\log_2 M_{\text{sp}} \approx 27$, and $\log_2(H_{\text{max}}/\underline{\nu}) \approx 84$, so $f \approx 2200$ and $I \approx 310$, about $90 \log_2 n$ bits; the individual rows of the table reach at most about 660 fractional bits (the candidate-selection enclosure and the tracker’s ε_{det} term) and about 160 integer bits (the tracked matrix), so a ledger that tracks each quantity separately gives a word of about 820 bits, about $29 \log_2 n$. Asymptotically the formula is dominated by its $40 \lceil \log_2(2n) \rceil$ term, while a quantity-by-quantity ledger gives about $18 \log_2 n + O(\log \log n)$ bits, the ε_{det} row (about $14 \log_2 n$) plus the tracked matrix (about $4 \log_2 n$), with $\log_2(1/\varepsilon_{\text{opt}}) \approx 7 \log_2 n$ the largest single demand. The quantities that are never resolved are $\phi_0 = (2n)^{-1999998}$, the bottom eigenvalues e^{-2ab} of P_x , and the Taylor prefactor $e^{c/2}$ of the surrogate of Section 7; the only object whose size is set by a count rather than by an accuracy is the tracked matrix, through L_{max} . The bit cost of the fast products carries the unquantified constant c_η of Lemma 10.3, inherited from the coefficient growth of the laser-method algorithms behind $\alpha_{\text{dual}} > 1/4$; it affects the temporaries, not the stored words, and not the exponent.

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A An existence constant below 10

This appendix combines a smooth spectral barrier, covariance-sensitive Gaussian moments, and a lossless conversion from Gaussian measure to partial signs. The scalar inequalities needed by the proof are certified on their entire domains. Every terminating decimal below denotes an exact rational number.

Theorem A.1 (Existence constant below 10). *Let $1 \leq N \leq n$, and let $A_1, \dots, A_n \in \mathbb{R}^{N \times N}$ be symmetric matrices with $\|A_i\| \leq 1$. There are signs $\epsilon_1, \dots, \epsilon_n \in \{-1, 1\}$ such that*

$$\left\| \sum_{i=1}^n \epsilon_i A_i \right\| \leq 9.999565738962\sqrt{n} < 9.9996\sqrt{n}.$$

Proof. Pad the matrices with zero blocks to reduce to $N = n$. Lemma A.33 gives hereditary Gaussian small-ball estimates for the seven stages of Definition A.30. Lemma A.3, with the quartic entropy reserve certified in Lemma A.32, turns these estimates into partial signs. Lemma A.34 sums their increments and the entire remaining tail, giving the displayed bound. $\square$

The following analytic reductions also supply the two-component small-ball families used in Section B. Sections A.8–A.12 refine them for the existence bound, without changing those families or their algorithmic application.

A.1 Lossless Gaussian-to-sign coding

Put $h(x) := -x \log x - (1-x) \log(1-x)$ for $0 < x < 1$, with $h(0) := 0$ and $h(1) := 0$, and fix

$$s_\star := 0.67448, \quad a_\star := 0.635, \quad b_\star := \frac{1 - 2s_\star a_\star}{s_\star^2}.$$

Let Z have the even density

$$f_Z(z) := \begin{cases} a_\star, & |z| \leq s_\star, \\ b_\star(2s_\star - |z|), & s_\star < |z| \leq 2s_\star, \\ 0, & |z| > 2s_\star. \end{cases}$$

The identity $2s_*a_* + s_*^2b_* = 1$ normalizes this density. In particular it is not a Gaussian density.

The conversion to partial signs starts from a compactly supported density whose random sign translates preserve a Gaussian mass lower bound.

Lemma A.2 (A peaked translated density). *If σ is a uniform sign independent of Z , the law P of $Z + s_*\sigma$ has an even density nonincreasing on $[0, \infty)$, and*

$$P([-x, x]) \geq \gamma_1([-x, x]) \quad (x \geq 0).$$

Consequently, for every symmetric convex Borel set $K \subseteq \mathbb{R}^m$, $P^{\otimes m}(K) \geq \gamma_m(K)$.

Proof. For this proof write $s := s_*$, $a := a_*$, $b := b_*$, $F(x) := \gamma_1([-x, x])$ and $g(x) := F'(x) = \sqrt{2/\pi}e^{-x^2/2}$. Let $F_P(x) := P([-x, x])$ for $x \geq 0$. Direct integration gives

$$F_P(x) = \begin{cases} (a + bs)x - bx^2/2, & 0 \leq x \leq s, \\ 1/2 + a(x - s), & s \leq x \leq 2s, \\ 1 - b(3s - x)^2/2, & 2s \leq x \leq 3s, \\ 1, & x \geq 3s. \end{cases}$$

The inequalities $b > 0$ and $a > bs$ show that the density of P is even and nonincreasing, including at its two jumps. Here are explicit sufficient checks for $F_P \geq F$, with $\epsilon := 0.01$:

$$\begin{aligned} a + bs &> g(0), & g(s) &> a > g(s + \epsilon), & s &< 1, \\ \frac{1}{2} - F(s) &> \epsilon(g(s) - a) > 0, \\ 1 - F(5s/2) &> bs^2/2, & 1 - F(3s) &> bs^2/8. \end{aligned}$$

All these strict inequalities are certified in Lemma A.16. To see that they suffice, on $[0, s]$ the derivative of $F_P - F$ is convex, since its second derivative is $(1 - x^2)g(x) > 0$. It is positive at zero and negative at s , so the minimum of $F_P - F$ is attained at an endpoint. Its endpoint values are 0 and $1/2 - F(s) > 0$. On $[s, 2s]$ its derivative $a - g(x)$ increases, becomes positive before $s + \epsilon$, and can lose at most $\epsilon(g(s) - a)$ before doing so. On $[2s, 5s/2]$ use $F_P(x) \geq 1 - bs^2/2$ and $F(x) \leq F(5s/2)$; on $[5s/2, 3s]$ use $F_P(x) \geq 1 - bs^2/8$ and $F(x) \leq F(3s)$. Beyond $3s$ the assertion is immediate.

For completeness, the tensorization does not require a mixture of log-concave densities to remain log-concave. Every even nonincreasing density on the line is a mixture of uniform densities on centered intervals, by its layer-cake representation. Fix the intervals for the other coordinates. Prékopa's marginal theorem shows that integrating $\mathbb{1}[K]$ against their uniform laws and any remaining Gaussian coordinates gives an even log-concave function of the coordinate being replaced. It is therefore nonincreasing in its absolute value. The one-dimensional interval comparison, integrated over its level sets, permits replacement of that Gaussian coordinate by P . Averaging over the centered-interval mixtures preserves the comparison. Replace the coordinates one at a time. Truncation and monotone convergence cover unbounded sets and densities. $\square$

To turn Gaussian mass into a partial signing, we combine the translated density with a diameter bound for subsets of the discrete cube.

Lemma A.3 (Lossless partial signs). *Let $K \subseteq \mathbb{R}^m$ be symmetric and convex, with $\gamma_m(K) \geq e^{-Em}$. If $0 < q < 1$ and*

$$\log 2 - E > h((1 - q)/2),$$

*then there is $y \in \{-1, 0, 1\}^m$ supported on more than $(1 - q)m$ coordinates such that $s_*y \in K$. In particular, the choice $E := q^2/2$ is sufficient for every $m \geq 1$.*

Proof. For independent copies of Z , averaging over $z := (Z_1, \dots, Z_m)$ gives

$$\mathbb{E}_z[|\{\sigma \in \{-1, 1\}^m : z + s_\star \sigma \in K\}|] = 2^m P^{\otimes m}(K) \geq e^{m(\log 2 - E)}.$$

Choose a translate with at least this many signs. Kleitman's cube-diameter theorem says that a family of Hamming diameter $D < m$ has size at most

$$\sum_{i=0}^r \binom{m}{i} \quad (D = 2r), \quad 2 \sum_{i=0}^r \binom{m-1}{i} \quad (D = 2r + 1).$$

These are at most $e^{mh(D/(2m))}$. For even D this is the binomial entropy bound; for odd D use that bound in dimension $m - 1$ and concavity of h with

$$\frac{D}{2m} = \frac{m-1}{m} \frac{r}{m-1} + \frac{1}{m} \frac{1}{2}.$$

The case $m = 1$ is immediate. Thus two members σ, σ' differ on more than $(1 - q)m$ coordinates. Their half-difference $y := (\sigma - \sigma')/2$ has the required support and $s_\star y \in K$, by symmetry and convexity; the translate cancels. Finally

$$\log 2 - h((1 - q)/2) = \sum_{r=1}^{\infty} \frac{q^{2r}}{2r(2r-1)} > q^2/2.$$

The diameter theorem can equivalently be read in Huang–Klurman–Pohoata [HKP20, Theorem 1.1]; this includes both parity cases used here. $\square$

A.2 Smooth barriers and spectral curvature

We use the divided-difference convention of Definition 3.2: $f'[u, v] := (f'(u) - f'(v))/(u - v)$, with value $f''(u)$ at $u = v$. For the two-component families below, the additive barrier parameters are denoted by $\kappa_0 > 0$ and $\kappa_{\text{tail}} > 0$; thus the coefficient κ_1 in the original barrier notation is $\kappa_0 + \kappa_{\text{tail}}$. For an integer $\tau \geq 2$, set

$$\phi(u) := \kappa_0 \phi_0(u) + \kappa_{\text{tail}} \phi_{1,\tau}(u), \quad \phi_0(u) := -\log(1 - u^2), \quad \phi_{1,\tau}(u) := \sum_{j=\tau}^{\infty} \frac{u^{2j}}{j},$$

for $|u| < 1$, and extend ϕ by $+\infty$ outside this interval. Write $\Phi(X) := \text{tr}[\phi(X)]$. Let $X := \sum_{i=1}^m \xi_i N_i$ with symmetric $N_i \in \mathbb{R}^{N \times N}$ and $\sum_{i=1}^m N_i^2 \preceq aI$. We allow the even tilted measure

$$d\mu := Z_\mu^{-1} e^{-\Phi(X) - Q(\xi)} d\gamma_m(\xi),$$

where Q is any nonnegative quadratic form. In the applications below $\kappa_0 + \kappa_{\text{tail}} > 2$. For $\sigma \in \{-1, 1\}$ write $Y_\sigma := I + \sigma X$.

We allocate the scalar barrier curvature across spectral values to control the resolvent weights near the boundary.

Definition A.4 (Spectral allocation profile). For $0 < \zeta, \theta < 1$, put

$$d_\zeta(u) := \min\{1, \sqrt{(1 - u)/\zeta}\}, \\ F_\theta(v) := \max\{2(1 + \theta - v)/(1 + \theta), 3(1 - v) + (1/\theta - 1)(1 - v)^2\}.$$

The all-pairs curvature condition is

$$(1 - u)(1 - v)\phi'[u, v] \geq F_\theta(d_\zeta(u)d_\zeta(v)) \quad (-1 < u, v < 1). \quad (62)$$

Reflection gives the analogous condition at the other edge.

The spectral allocation condition controls the curvature of the resolvent weights, which is the input to the weighted variance estimate.

Lemma A.5 (One- and two-slot weighted Poincaré bounds). *Assume Eq. (62) and $2a < \zeta^2$. Put*

$$p_1 := (1 - a/\zeta^2)^{-1}, \quad p_2 := (1 - 2a/\zeta^2)^{-1}.$$

For a one-slot weight $W := Y_\sigma^{-1}$, or a two-slot weight $W := Y_{\sigma_1}^{-1} \otimes Y_{\sigma_2}^{-1}$, respectively,

$$\inf_v \mathbb{E}_\mu[(f - v)^\top W(f - v)] \leq p_r \sum_{i=1}^m \mathbb{E}_\mu[(\partial_i f)^\top W(\partial_i f)], \quad r = 1, 2. \quad (63)$$

Here v is a constant vector of the appropriate dimension; direct sums of these weights are also allowed.

Proof. We spell out the ordered curvature calculation. For a positive matrix density A , its curvature blocks are

$$\mathcal{K}_{i,j}(A) := (\partial_i A)A^{-1}(\partial_j A) - \partial_{i,j}A.$$

If $A = e^{-V}W$, then

$$e^V \mathcal{K}_{i,j}(A) = W_i W^{-1} W_j - W_{i,j} + V_{i,j} W.$$

For $W = Y^{-1}$, congruence by $Y^{1/2}$ turns this into $V_{i,j}I - C_j C_i$, where $C_i := Y^{-1/2}(\partial_i Y)Y^{-1/2}$. Notice the reversed order $C_j C_i$. For a tensor product of two inverse weights, the mixed derivatives cancel the mixed products, leaving one such negative block in each slot.

Consider $Y := I - X$; reflection treats $I + X$. In its eigenbasis let $D := \min\{I, \sqrt{Y/\zeta}\}$, with eigenvalues d_a , and put $C_{i,1} := DC_i D$. Since $DY^{-1/2} \preceq \zeta^{-1/2}I$,

$$\sum_{i=1}^m C_{i,1}^2 \preceq \frac{a}{\zeta^2} I.$$

The tensor Gram estimate therefore bounds the regular curvature loss by a/ζ^2 in each slot. To estimate the remaining loss, fix test vectors z_i and set $T_{ab,c} := \sum_{i=1}^m (C_i)_{ab}(z_i)_c$, so $T_{ab,c} = T_{ba,c}$. Write

$$Q_0 := \sum_{i=1}^m \sum_{j=1}^m \langle z_i, C_j C_i z_j \rangle, \quad Q_1 := \sum_{i=1}^m \sum_{j=1}^m \langle z_i, C_{j,1} C_{i,1} z_j \rangle.$$

Weighted arithmetic-geometric mean, with weights $(1 + \theta - d_a d_b)^{-1}$ on the cyclic products of T , gives

$$2(Q_0 - Q_1) \leq \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N T_{ab,c}^2 (1 + \theta - v) \left(\frac{1 - v d_b d_c}{1 + \theta - d_b d_c} + \frac{1 - v d_a d_c}{1 + \theta - d_a d_c} \right), \quad v := d_a d_b.$$

Put $C := 1 + \theta$. The derivative of the displayed coefficient in d_c has the sign of $1 - Cv$. We distinguish two cases. If $v \geq 1/C$, the coefficient is nonincreasing in d_c , so its maximum is attained at $d_c = 0$ and equals $2(C - v)/C$, the first branch of $F_\theta(v)$.

If $v \leq 1/C$, the coefficient is nondecreasing in d_c , so take $d_c = 1$. Its value is

$$(C - v)[2v + (1 - Cv)\left(\frac{1}{C - d_a} + \frac{1}{C - d_b}\right)].$$

For $s := d_a + d_b$ and $d_a d_b = v$, the reciprocal sum in brackets is $(2C - s)/(C^2 - Cs + v)$. Its derivative with respect to s is

$$\frac{C^2 - v}{(C^2 - Cs + v)^2} > 0.$$

Since $1 - Cv \geq 0$ and $(1 - d_a)(1 - d_b) \geq 0$ gives $s \leq 1 + v$, the maximum is attained at $\{d_a, d_b\} = \{1, v\}$. Substitution gives $3(1 - v) + (1/\theta - 1)(1 - v)^2$, the second branch of $F_\theta(v)$. At $v = 1/C$ either case applies. Thus the coefficient is at most $F_\theta(v)$ throughout the domain, including repeated indices. The spectral Hessian formula and Eq. (62) imply

$$\sum_{i=1}^m \sum_{j=1}^m (\partial_{i,j} \Phi) \langle z_i, z_j \rangle \geq \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N F_\theta(d_a d_b) T_{ab,c}^2 \geq 2(Q_0 - Q_1).$$

Consequently half of the barrier Hessian absorbs the nonregular loss of each slot. One slot uses at most half, and two slots use at most all, of the available Hessian. The Gaussian Hessian supplies I , and $\nabla^2 Q \succeq 0$. This proves the curvature lower bounds $1 - ra/\zeta^2$.

We recall why these curvature bounds imply Eq. (63), including the boundary issue. On a smooth bounded convex domain, set $B_i := A^{-1} \partial_i A$ and $L := -\sum_{i=1}^m (\partial_{i,i} + B_i \partial_i)$. Integration by parts for the Neumann problem gives

$$\|Lu\|_A^2 = \sum_{i=1}^m \sum_{j=1}^m \|\partial_{i,j} u\|_A^2 + \int \sum_{i=1}^m \sum_{j=1}^m (\partial_i u)^\top \mathcal{K}_{i,j}(A) (\partial_j u) + \text{a nonnegative boundary term}.$$

Indeed $A \partial_j B_i = -\mathcal{K}_{j,i}(A)$; the boundary term is the second fundamental form applied to the tangential derivatives. Solve $Lu = f - v$, where $v := (\int A)^{-1} \int A f$. The ordinary coercive Neumann problem is solvable modulo constant vectors. The curvature lower bound and Cauchy–Schwarz, applied to $\|f - v\|_A^2 = \int \sum_{i=1}^m (\partial_i f)^\top A \partial_i u$, give the desired estimate. Smooth convex exhaustion proves it on the open convex domain $\{\xi : \|X(\xi)\| < 1\}$. Its weighted mean converges because $\int A$ and $\int A f$ converge; truncate first if necessary. Near a spectral boundary, $e^{-\Phi}$ vanishes to order $\kappa_0 + \kappa_{\text{tail}} > 2$, which controls up to two inverse factors. Gaussian decay controls infinity, even if the linear map $\xi \mapsto X(\xi)$ has a kernel. Polynomial fields and the inverse-weighted fields below have finite energies. Local smoothing followed by exhaustion justifies each displayed integration by parts without differentiating the spectral cutoff D . $\square$

A.3 Resolvent means and parity-sensitive polynomial energies

We collect the scalar roots that bound the one-slot and two-slot resolvent means.

Definition A.6 (Scalar root bounds). With p_1, p_2 as above, define

$$\begin{aligned} L_1 &:= \frac{2}{1 + \sqrt{1 - 4p_1 a}}, & c_1 &:= L_1 - 1, \\ A_J &:= 1 - c_1^2, & B_J &:= 1 + 2c_1, \\ \Delta_J &:= A_J^2 - 4p_2 a B_J, & J &:= \frac{2B_J}{A_J + \sqrt{\Delta_J}}, & c &:= 1 - 1/J. \end{aligned}$$

We require $1 - 4p_1 a > 0$, $A_J > 0$, and $\Delta_J > 0$. These requirements will be checked at the largest variance at each stage.

Applying the weighted variance bounds to resolvent fields leads to scalar quadratic inequalities for their means.

Lemma A.7 (Resolvent mean bounds). *Under the preceding assumptions,*

$$I \preceq S_1 := \mathbb{E}_\mu[Y_+^{-1}] \preceq L_1 I, \quad I \preceq S := \mathbb{E}_\mu[Y_+^{-1} \otimes Y_+^{-1}] \preceq JI.$$

The same conclusions hold with both signs negative.

Proof. Apply the one-slot estimate to $(Y_+ - I)z$. Its optimized variance is $z^\top (I - S_1^{-1})z$, and its gradient energy is at most $a\|S_1\|\|z\|^2$. Thus $I - S_1^{-1} \preceq p_1 \sum_{i=1}^m N_i S_1 N_i$. Jensen and evenness give $S_1 \succeq I$. Put $l := \|S_1\|$. Then $p_1 a l^2 - l + 1 \geq 0$; continuity from $a = 0$ selects the smaller root L_1 .

For two slots, put $K_1 := I \otimes S_1$, $K_2 := S_1 \otimes I$. Apply the two-slot estimate separately to $((Y_+ - I) \otimes I)z$ and $(I \otimes (Y_+ - I))z$ and add. The uncentered second moments give $U_1 + U_2 \succeq 2I$, since pointwise $Y_+ \otimes Y_+^{-1} + Y_+^{-1} \otimes Y_+ \succeq 2I$. Hence, with $s := \|S\|$,

$$2I - K_1 S^{-1} K_1 - K_2 S^{-1} K_2 \preceq 2p_2 a s I.$$

Evenness gives $S \succeq I$: in a common eigenbasis of X , the geometric mean of the two entries of $Y_+^{-1} \otimes Y_+^{-1}$ and $Y_-^{-1} \otimes Y_-^{-1}$ is at least one. Put $D_i := K_i - I$, so $0 \preceq D_i \preceq c_1 I$. For a unit top eigenvector z of S ,

$$z^\top K_i S^{-1} K_i z \leq \frac{1 + 2c_1}{s} + c_1^2.$$

It follows that $p_2 a s^2 - A_J s + B_J \geq 0$, whose smaller root is J . To justify both root choices, replace X by rX , $0 \leq r \leq 1$, keeping Q fixed. The strict discriminant conditions persist for smaller variance, and the means vary continuously from I ; they cannot cross the gap between the two roots. Integrability established in the preceding proof justifies this continuity. $\square$

For an array f of vectorized matrix fields, let

$$W_\pm := Y_\pm^{-1} \otimes Y_\pm^{-1}, \quad \mathcal{R}(f) := \mathbb{E}_\mu[f^\top W_+ f], \quad \mathcal{D}(f) := \mathbb{E}_\mu[\|f\|^2],$$

where a block-diagonal copy of W_+ is understood for an array.

Evenness of the tilted measure gives different controls on the projected means of odd and even fields.

Lemma A.8 (Odd and even fields). *For a parity-homogeneous field put $\mathcal{P}(f) := \|\mathbb{E}_\mu[W_+ f]\|_{S^{-1}}^2$, and let $B \geq \mathcal{R}(f) - \mathcal{P}(f)$. One may always take $B = p_2 \sum_i \mathcal{R}(\partial_i f)$. If f is odd, then*

$$(\mathcal{R}(f), \mathcal{P}(f)) \leq (J(1+a)B, (J(1+a)-1)B)$$

componentwise. If f is even and $\|\mathbb{E}_\mu[f]\|^2 \leq M$, then

$$\begin{aligned} (\mathcal{R}(f), \mathcal{P}(f)) &\leq (\mathcal{F}_c(M, B), \mathcal{F}_c(M, B) - B), \\ \mathcal{F}_c(M, B) &:= M + \frac{(\sqrt{cM} + \sqrt{cM + (1-c)B})^2}{(1-c)^2}. \end{aligned} \tag{64}$$

Both pairs are nonnegative and nondecreasing in their nonnegative arguments. Constant fields satisfy $\mathcal{R}(f) = \mathcal{P}(f) \leq J\|f\|^2$.

Proof. Put $W_e := (W_+ + W_-)/2$ and $W_o := (W_+ - W_-)/2$. Parity gives $\mathcal{R}(f) = \mathbb{E}[f^\top W_e f]$, and $\mathbb{E}[W_e] = S$. Pointwise $W_e \succeq I$ and

$$W_o W_e^{-1} W_o = W_e - (I + X \otimes X)^{-1}.$$

Gaussian convex domination gives $\text{Cov}_\mu(\xi) \preceq I$; the tensor Gram inequality then gives $\|\mathbb{E}[X \otimes X]\| \leq a$. Operator convexity of inversion implies

$$\mathbb{E}[W_o W_e^{-1} W_o] \preceq S - (1+a)^{-1} I \preceq (1 - [J(1+a)]^{-1}) S.$$

The first step uses the tensor bound and Jensen's inequality for inversion; the second uses $S \preceq JI$. For odd f , $\mathbb{E}[W_+ f] = \mathbb{E}[W_o f]$, so weighted Cauchy–Schwarz gives $\mathcal{P}(f) \leq (1 - [J(1+a)]^{-1}) \mathcal{R}(f)$. Combine this with $\mathcal{R}(f) - \mathcal{P}(f) \leq B$ to obtain both odd bounds.

For even f , put $b := \mathbb{E}[f]$, $w := \mathbb{E}[(W_e - I)f]$, and $q := b^\top S^{-1} b$. As $S - I \preceq cS$, weighted Cauchy–Schwarz gives

$$w^\top S^{-1} w \leq c(\mathcal{R}(f) - \mathcal{D}(f)), \quad q \leq \|b\|^2 \leq \mathcal{D}(f).$$

Consequently $\mathcal{P}(f) \leq (\sqrt{q} + \sqrt{c(\mathcal{R}(f) - q)})^2$, with $q \leq M$. For $R := \mathcal{R}(f) > (1+c)M$, the last expression increases up to $q = M$. The inequality $R - \mathcal{P}(f) \leq B$, with $u := \sqrt{R - M}$, becomes $(1-c)u^2 - 2\sqrt{cM}u \leq B$. Solving gives $R \leq \mathcal{F}_c(M, B)$; substituting this upper bound at $q = M$ gives $\mathcal{P}(f) \leq \mathcal{F}_c(M, B) - B$. If $R \leq (1+c)M$, both assertions follow from $\mathcal{P}(f) \leq R$ and

$$\mathcal{F}_c(M, B) - B = M + 2\alpha M + \frac{cB}{1-c} + 2\sqrt{\alpha M} \sqrt{\alpha M + B/(1-c)}, \quad \alpha := \frac{c}{(1-c)^2}. \quad (65)$$

This identity also proves monotonicity of the even pair. The odd pair is manifestly monotone. Constants have zero weighted variance, and all arguments apply to direct sums. $\square$

Successive derivatives alternate parity, so we encode their energy bounds in a backward recurrence.

Definition A.9 (Polynomial energy recurrence). Let $\text{df}(r) := (2r-1)!!$ for $r \geq 1$, and $\text{df}(0) := 1$. For an integer $\ell \geq 1$, start with $V_\ell := (\ell!)^2 J$. For $j = \ell-1, \dots, 0$, set $r := \ell - j$, $(\ell)_j := \ell!/r!$, and

$$V_j := \begin{cases} p_2 J(1+a) V_{j+1}, & r \text{ odd}, \\ \mathcal{F}_c(M_j, p_2 V_{j+1}), & r \text{ even}, \end{cases} \quad M_j := (\ell)_j^2 \begin{cases} 1, & r = 2, \\ \text{df}(r), & r \neq 2. \end{cases}$$

Write $C_\ell(a) := V_0$, with the stage parameter ζ held fixed.

We apply the parity bounds to all derivative arrays of a matrix power to bound its energy with two resolvent factors.

Lemma A.10 (Polynomial resolvent energies). *For the preceding measure and variance bound,*

$$\frac{1}{N} \mathbb{E}_\mu[\text{tr}[X^{2\ell} Y_+^{-2}]] \leq a^\ell C_\ell(a). \quad (66)$$

The scalar function $C_\ell(a)$ is nondecreasing while the strict root conditions hold.

Proof. Take the full array of ordered derivatives $\nabla^j(X^\ell)$. Set $r := \ell - j$. Then

$$\frac{1}{N} \mathbb{E}_\mu[\|\nabla^j(X^\ell)\|_{\text{HS}}^2] \leq a^\ell (\ell)_j^2 \text{df}(r). \quad (67)$$

Here is a word-level justification that avoids commuting matrices. Polarize the j derivatives using independent standard Gaussian direction vectors $h_1, \dots, h_j$. Their second moment is exactly the sum over the ordered derivative indices. Each differentiated word contains r copies of X and one

copy of each independent Gaussian direction matrix. The map $T \mapsto \sum_{i=1}^m N_i T N_i$ has norm at most a on every Schatten class: use positivity and $\sum_{i=1}^m N_i^2 \preceq aI$ for operator norm, trace duality for trace norm, and interpolation. Integrating the independent directions one at a time, with Schatten Hölder between the remaining powers, bounds the conditional squared Hilbert–Schmidt norm of one word by $a^j \operatorname{tr}[|X|^{2r}]$. There are $(\ell)_j$ words, so Cauchy–Schwarz supplies $(\ell)_j^2$. Hargé’s Gaussian convex-domination theorem, applied to the centered even convex tilt, and the Gaussian trace moment bound give $\mathbb{E}_\mu[\operatorname{tr}[|X|^{2r}]] \leq N \operatorname{df}(r) a^r$. This proves Eq. (67).

For the mean of a degree-two derivative array there is a sharper bound

$$\frac{1}{N} \|\mathbb{E}_\mu[\nabla^{\ell-2}(X^\ell)]\|_{\text{HS}}^2 \leq a^\ell (\ell)_{\ell-2}^2.$$

Indeed, the two undifferentiated occurrences form $LXUXR$ in a word. Their mean is $L\Psi_\Sigma(U)R$, where $\Sigma := \operatorname{Cov}_\mu(\xi) \preceq I$ and $\Psi_\Sigma(U) := \sum_{i=1}^m \sum_{j=1}^m \Sigma_{i,j} N_i U N_j$. This positive map has Schatten-2 norm at most a . Integrating each independent direction matrix, inside or outside Ψ_Σ , costs another factor a in the squared norm. Thus one word costs at most Na^ℓ ; summing words by Cauchy–Schwarz gives the assertion. Odd derivative arrays have mean zero. Other even arrays use Jensen followed by Eq. (67). At $j = \ell$ the array is constant and its weighted energy is at most $Na^\ell (\ell!)^2 J$. Apply Lemma A.8 backwards through the derivative arrays and divide by Na^ℓ . This is exactly Definition A.9; for $j = 0$, vectorization identifies its energy with the left side of Eq. (66).

Finally p_1, p_2, L_1, c_1 increase with a . Write $J = (B_J/A_J)\mathcal{C}(p_2 a B_J/A_J^2)$, where $\mathcal{C}(x) := 2/(1 + \sqrt{1 - 4x})$ has a power series with nonnegative coefficients. Thus J and c increase too. Every operation in the backwards recurrence is nondecreasing in these nonnegative arguments, proving the last assertion. At $a = 0$ all formulas mean their continuous limits. $\square$

A.4 Covariance-power moments and radial integration

Let $Y := \sum_{i=1}^m \xi_i B_i$ with symmetric B_i and $\sum_{i=1}^m B_i^2 \preceq \eta I$. Define the coefficient Gram matrix $\Gamma_{i,j} := \operatorname{tr}[B_i B_j]$, and put

$$C_\Gamma := (I + 2\kappa_0 \Gamma)^{-1}, \quad \nu := \mathcal{N}(0, C_\Gamma), \quad \omega_j := \frac{\operatorname{tr}[\Gamma C_\Gamma^j]}{N\eta} \quad (j \geq 1).$$

Then $0 \leq \omega_j \leq 1$. Write $\psi(u) := \phi(u) - \kappa_0 u^2$, an even nonnegative convex function, and use the radial measures

$$d\mu_t := Z_t^{-1} e^{-\operatorname{tr}[\psi(tY)]} d\nu, \quad Z_t := \mathbb{E}_\nu[e^{-\operatorname{tr}[\psi(tY)]}], \quad 0 \leq t \leq 1.$$

Here $Z_0 = 1$. Relative to γ_m , the non-Gaussian potential is $\Phi(tY) + \kappa_0(1 - t^2)\xi^\top \Gamma \xi$, so all preceding weighted estimates apply with $a := \eta t^2$ and a nonnegative quadratic Q .

Retaining powers of the preconditioned covariance sharpens the ordinary moments used in the radial integral.

Lemma A.11 (Covariance-power trace bound). *For every integer $j \geq 1$,*

$$\frac{1}{N} \mathbb{E}_{\mu_t}[\operatorname{tr}[(tY)^{2j}]] \leq \operatorname{df}(j)(\eta t^2)^j \omega_j.$$

Proof. Rotate the coefficient coordinates to diagonalize Γ , with eigenvalues λ_i and $c_i := (1 + 2\kappa_0 \lambda_i)^{-1}$. The rotated matrices still satisfy $\sum_{i=1}^m B_i^2 \preceq \eta I$. The variance matrix under ν is

$\Sigma_C := \sum_{i=1}^m c_i B_i^2$. In an eigenbasis u_a of Σ_C , set $w_{a,i} := u_a^\top B_i^2 u_a / \eta$. These weights are nonnegative and sum to at most one. Add the missing mass at zero and apply scalar Jensen to $x \mapsto x^j$:

$$(u_a^\top \Sigma_C u_a)^j \leq \eta^j \sum_{i=1}^m w_{a,i} c_i^j.$$

Summing over $a = 1, \dots, N$ proves

$$\text{tr}[\Sigma_C^j] \leq \eta^{j-1} \sum_{i=1}^m \lambda_i c_i^j = \eta^{j-1} \text{tr}[\Gamma C_\Gamma^j].$$

No operator convexity of x^j is used here. The sharp Gaussian noncommutative Khintchine inequality gives

$$\mathbb{E}_\nu[\text{tr}[Y^{2j}]] \leq \text{df}(j) \text{tr}[\Sigma_C^j].$$

We use the sharp double-factorial constant of Buchholz [Buc01, Theorem 5]; see also Tropp [Tro12, Eq. (4.17)].

Finally $e^{-\text{tr}[\psi(tY)]}$ is log-concave and even. Hargé's centered Gaussian convex-domination theorem [Har04], after whitening ν , applies to the convex function $\text{tr}[(tY)^{2j}]$. Combining the two bounds proves the lemma. $\square$

The radial integration combines the polynomial moment bounds with a resolvent estimate for the untruncated tail.

Lemma A.12 (Radial bound with the covariance credit retained). *Fix integers $K \geq \tau \geq 2$ and $\ell \geq 1$ with $L := 2K - \ell \geq 1$. Assume the preceding curvature and root conditions at variance η . Put $N_{\text{grid}} := 512$, $u_i := (i/N_{\text{grid}})^2$, and*

$$p_j := \frac{\text{df}(j)\eta^j}{j}(\kappa_0 + \kappa_{\text{tail}} \mathbb{1}[j \geq \tau]), \quad 2 \leq j < K,$$

$$T := \frac{(\kappa_0 + \kappa_{\text{tail}})\eta^K \sqrt{\text{df}(L)}}{K} \sum_{i=1}^{N_{\text{grid}}} (u_i^K - u_{i-1}^K) \sqrt{C_\ell(\eta u_i)}.$$

Then

$$\nu[\|Y\| < 1] \geq Z_1 \geq \exp[-N(\sum_{j=2}^{K-1} p_j \omega_j + T \sqrt{\omega_L})]. \quad (68)$$

Proof. Let $X := tY$ and $R_j(t) := N^{-1} \mathbb{E}_{\mu_t}[\text{tr}[X^{2j}(I - X^2)^{-1}]]$. Differentiation of the radial partition function gives

$$-\frac{d}{dt} \log Z_t = \frac{2N}{t} (\kappa_0 R_2(t) + \kappa_{\text{tail}} R_\tau(t)).$$

Expand

$$R_j(t) = \sum_{r=j}^{K-1} N^{-1} \mathbb{E}[\text{tr}[X^{2r}]] + R_K(t).$$

By evenness, $R_K(t) = N^{-1} \mathbb{E}[\text{tr}[X^{2K} Y_+^{-1}]]$. Cauchy-Schwarz with the two matrix factors $X^{2K-\ell}$ and $X^\ell Y_+^{-1}$, followed by Lemmas A.10 and A.11, gives

$$R_K(t) \leq (\eta t^2)^K \sqrt{\text{df}(2K - \ell) C_\ell(\eta t^2)} \sqrt{\omega_{2K-\ell}}.$$

The finite terms integrate exactly to $p_j \omega_j$. Substitute $u := t^2$ in the remainder integral and use monotonicity of C_ℓ on each interval $[u_{i-1}, u_i]$. The exact integral of u^{K-1} on this interval is $(u_i^K - u_{i-1}^K)/K$, giving T . Thus this is an analytic upper sum, not numerical quadrature without an error bound. The boundary vanishing discussed above justifies differentiation; the bound is integrable at $t = 0$. Since $\psi \geq 0$ and is infinite outside the spectral domain, $Z_1 \leq \nu[\|Y\| < 1]$. Integrating from $Z_0 = 1$ proves Eq. (68). $\square$

A.5 Returning to the standard Gaussian and hereditary scaling

A change of Gaussian covariance introduces a determinant factor; a scalar profile bounds this factor together with the radial contribution.

Lemma A.13 (Gaussian transfer with a scalar cap). *Let $\Gamma \succeq 0$ be an $m \times m$ matrix with $\text{tr } \Gamma \leq N\eta$ and $\text{rank } \Gamma \leq wN$, where $\kappa, \eta > 0$. Let ν be the probability measure proportional to $e^{-\kappa \xi^\top \Gamma \xi} d\gamma_m(\xi)$, and put $\omega_j := \text{tr}[\Gamma(I + 2\kappa\Gamma)^{-j}]/(N\eta)$. Suppose a Borel set $\mathcal{K}$ satisfies*

$$-N^{-1} \log \nu(\mathcal{K}) \leq C + \sum_j Q_j \omega_j, \quad C, Q_j \geq 0,$$

with finitely many coefficients. For $H > 1$, set

$$L_H(x) := \frac{H \log(1+x) - \log(1+Hx)}{(H-1)x}, \quad L_H(0) := 0.$$

If $b \geq 0$, $B_{\text{cap}} > 0$, and

$$L_H(x) + \frac{H}{(H-1)\kappa\eta} \sum_j \frac{Q_j}{(1+x)^j} - \frac{b}{x} \leq B_{\text{cap}} \quad (x > 0),$$

then

$$\gamma_m(\mathcal{K}) \geq e^{-Ne}, \quad e := \kappa\eta B_{\text{cap}} + bw/2 + HC/(H-1). \quad (69)$$

For $b = 0$ include the continuous value at zero. The same cap gives exponent at most αe after replacing η, w, C, Q_j by $\alpha\eta, \alpha w, C_\alpha, Q_{j,\alpha}$, provided $0 < \alpha \leq 1$, $C_\alpha \leq \alpha C$ and $Q_{j,\alpha} \leq \alpha Q_j$. When $b = 0$ no rank condition is needed.

Proof. Put $Z_\Gamma := \det(I + 2\kappa\Gamma)^{-1/2}$. Hölder gives

$$(Z_\Gamma \nu(\mathcal{K}))^H \leq \gamma_m(\mathcal{K})^{H-1} \det(I + 2H\kappa\Gamma)^{-1/2}.$$

For each positive Gram eigenvalue λ_i , set $x_i := 2\kappa\lambda_i$ and $\vartheta_i := \lambda_i/(N\eta)$. Taking logarithms and substituting the radial bound gives

$$-N^{-1} \log \gamma_m(\mathcal{K}) \leq \kappa\eta \sum_i \vartheta_i (L_H(x_i) + \frac{H}{(H-1)\kappa\eta} \sum_j \frac{Q_j}{(1+x_i)^j}) + \frac{HC}{H-1}.$$

The determinant terms give L_H , and the definition of ω_j gives the rational terms. Zero eigenvalues contribute zero. Since

$$\sum_i \vartheta_i \leq 1, \quad \sum_i \frac{\vartheta_i}{x_i} = \frac{\text{rank } \Gamma}{2\kappa N\eta},$$

the scalar cap proves Eq. (69). Under the stated scaling, every coefficient $Q_j/(\kappa\eta)$ decreases, so the cap persists and each exponent term is at most α times its original value. $\square$

A.6 All-domain scalar certificates and exact data

For the two-component barriers, Lemma A.12 and Young's inequality give $C = Tv/2$ and $Q_j = p_j + \mathbb{1}[j = L]T/(2v)$, with missing p_j interpreted as zero. Thus the required cap is

$$L_H(x) + \frac{H}{(H-1)\kappa_0\eta} \left(\sum_{j=2}^{K-1} \frac{p_j}{(1+x)^j} + \frac{T}{2v(1+x)^L} \right) \leq B_{\text{cap}} \quad (x \geq 0). \quad (70)$$

It gives $e = \kappa_0\eta B_{\text{cap}} + HTv/(2(H-1))$. At variance $\alpha\eta$, the coefficients p_j decrease by α^j and T by at least α^K , by Lemma A.10. Since $j, K \geq 2$, the scaling criterion applies.

To verify the allocation condition, we reduce pairs of spectral values to finitely many inequalities in one variable.

Lemma A.14 (Finite reductions for the spectral condition). *Fix the spectral parameters and put $p := 2\tau - 2$, $a_0 := 1 - \zeta$ and*

$$G(u) := (1-u)^2 \phi''(u) = \frac{2\kappa_0(1+u^2) + 2\kappa_{\text{tail}}u^p((p+1) - (p-1)u^2)}{(1+u)^2}.$$

The following one-variable polynomial checks suffice for Eq. (62):

- (i) $2\kappa_0 > 2\theta/(1+\theta)$ and $G(a_0t) > 2\theta/(1+\theta)$ for $0 \leq t \leq 1$.
- (ii) For $u := a_0 + \zeta t$, $0 \leq t \leq 1$, both $G(u) > 2(\theta+t)/(1+\theta)$ and $G(u) > 3t + (1/\theta - 1)t^2$.
- (iii) $\kappa_{\text{tail}}a_0^{p-1} > \kappa_0$ and, with $u := 1 - \zeta t^2$,

$$2(\kappa_0 + \kappa_{\text{tail}}u^p) > (1+u)F_i(t), \quad i = 1, 2,$$

where $F_1(t) := 2 - 2t/(1+\theta)$ and $F_2(t) := 1/\theta + 2 - (2/\theta + 1)t + (1/\theta - 1)t^2$.

- (iv) For each $[L_0, R_0] := [j/1024, (j+1)/1024]$ and $i = 1, 2$, the following polynomial in $t \in [0, 1]$, with $u := 1 - \zeta t^2$, is strictly positive:

$$\begin{aligned} & 2\kappa_0(1-uL_0)(u+L_0)(u+R_0) \\ & + 2\kappa_{\text{tail}}u^{p+1}(1+R_0)(1-L_0)(u+L_0) \\ & + 2\kappa_{\text{tail}}L_0^{p+1}(1-u^2)(u+R_0) \\ & - F_i(t)(1+u)(1-L_0)(u+L_0)(u+R_0). \end{aligned}$$

In the first two parts, clear the strictly positive denominator $(1+u)^2$ before checking polynomial positivity.

Proof. Call an endpoint active if $u > 1 - \zeta$. For two regular endpoints, the normalized divided difference is a weighted average of G between them, with probability density proportional to $(1-x)^{-2}$. For $x \leq 0$, $G(x) \geq 2\kappa_0$; part (i) handles the rest. For two active endpoints, write their gaps as $h, k \in (0, \zeta)$, and let H_0 have density proportional to H_0^{-2} on the interval between h and k . Then

$$\mathbb{E}[H_0] \leq \sqrt{hk}, \quad \mathbb{E}[H_0^2] = hk.$$

For the first inequality, after setting $z := \frac{1}{2} \log(h/k)$, use $\sinh z \geq z$; coincident gaps follow by continuity. Part (ii), with $t := 1 - H_0/\zeta$, therefore gives both branches at $d_\zeta(u)d_\zeta(v) = \sqrt{hk}/\zeta$. In the quadratic branch the coefficient of $\mathbb{E}[H_0]$ is negative, so the direction of the bound is preserved.

For an active endpoint u and a regular endpoint $v \geq 0$, use

$$(1-u)(1-v)\phi'[u, v] = \frac{2\kappa_0(1+uv)}{(1+u)(1+v)} + 2\kappa_{\text{tail}} \frac{u^{p+1}(1-v)/(1+u) - v^{p+1}(1-u)/(1+v)}{u-v}.$$

Differentiating in v shows that this is nondecreasing for $0 \leq v \leq a_0 \leq u$ under $\kappa_{\text{tail}}a_0^{p-1} \geq \kappa_0$ (expand the difference of powers into its geometric sum). Thus its value at $v = 0$ is a lower bound, and part (iii) gives both branches, since $d_\zeta(v) = 1$ and $d_\zeta(u) = t$.

For a regular negative endpoint v , put $w := -v \in [0, 1)$. The exact normalized difference is

$$\frac{2\kappa_0(1-uw)}{(1+u)(1-w)} + \frac{2\kappa_{\text{tail}}u^{p+1}(1+w)}{(u+w)(1+u)} + \frac{2\kappa_{\text{tail}}w^{p+1}(1-u)}{(u+w)(1-w)}.$$

The first and third terms increase in w and the second decreases. On $[L_0, R_0]$, use L_0 in the first and third terms and R_0 in the second. Clearing the positive common denominator gives exactly part (iv). The last slab includes the limit $w \uparrow 1$; its left endpoint is still less than one. This covers every pair, including the diagonal by continuity. Reflection supplies the other spectral edge. $\square$

The seven stages are specified by coefficients of explicit polynomials. For a polynomial p , let $[x^k]p(x)$ denote its coefficient of x^k .

Definition A.15 (Stage parameters). For $0 \leq k \leq 6$, define $\vartheta_k := [x^k]p_\vartheta(x)$ using the polynomials below, for each symbol

$$\vartheta \in \{\tau, \ell, K, q, \eta, \zeta, \kappa_0, \kappa_{\text{tail}}, \theta, H, v, B_{\text{cap}}\}.$$

At a fixed stage, the subscript k is suppressed on $\tau, \ell, K, \eta, \zeta, \kappa_0, \kappa_{\text{tail}}, \theta, H, v, B_{\text{cap}}$.

The integer parameters are defined by

$$p_\tau(x) := 6 + 9x + 13x^2 + 18x^3 + 23x^4 + 28x^5 + 33x^6, \quad (71)$$

$$p_\ell(x) := 4 + 5x + 6x^2 + 7x^3 + 7x^4 + 7x^5 + 7x^6, \quad (72)$$

$$p_K(x) := 10 + 17x + 26x^2 + 35x^3 + 45x^4 + 55x^5 + 65x^6, \quad (73)$$

$$p_H(x) := 8 + 6x + 5x^2 + 4x^3 + 4x^4 + 4x^5 + 4x^6. \quad (74)$$

The retention and variance parameters are defined by

$$p_q(x) := 10^{-6}(39250 + 23060x + 12280x^2 + 7450x^3 + 5481x^4 + 5271x^5 + 5292x^6). \quad (75)$$

$$p_\eta(x) := 10^{-8}(3381414 + 1851877x + 1152527x^2 + 804555x^3 + 607866x^4 + 479595x^5 + 352981x^6). \quad (76)$$

The spectral barrier parameters are defined by

$$\begin{aligned} p_\zeta(x) := & 0.4088788392442434 + 0.2838471728369487x \\ & + 0.22164300714004487x^2 + 0.1833030277982336x^3 \\ & + 0.161245154965971x^4 + 0.14696938456699069x^5 \\ & + 0.13564659966250536x^6. \end{aligned} \quad (77)$$

$$\begin{aligned}
p_{\kappa_0}(x) := & 0.03445220702450958 + 0.0009674868358782514x \\
& + 1.073394394862783 \cdot 10^{-5}x^2 + 7.511900481158373 \cdot 10^{-8}x^3 \\
& + 4.445895654029819 \cdot 10^{-10}x^4 + 3.314296695413546 \cdot 10^{-12}x^5 \\
& + 2.416901835462038 \cdot 10^{-14}x^6.
\end{aligned} \tag{78}$$

$$\begin{aligned}
p_{\kappa_{\text{tail}}}(x) := & 43.72169566130472 + 1534.0837593501x \\
& + 216801.6634562664x^2 + 57298296.20492192x^3 \\
& + 18027985272.13702x^4 + 4589775273660.708x^5 \\
& + 1134092398730830x^6.
\end{aligned} \tag{79}$$

$$\begin{aligned}
p_{\theta}(x) := & 0.024023827399826305 + 0.0006553898871937424x \\
& + 7.20183924435989 \cdot 10^{-6}x^2 + 4.946902965041738 \cdot 10^{-8}x^3 \\
& + 2.9003477397537836 \cdot 10^{-10}x^4 + 2.1439516828536333 \cdot 10^{-12}x^5 \\
& + 1.5432100042289588 \cdot 10^{-14}x^6.
\end{aligned} \tag{80}$$

The scalar cap parameters are defined by

$$\begin{aligned}
p_v(x) := & 10^{-8}(74094940 + 67644865x + 65112347x^2 + 57745483x^3 \\
& + 41921619x^4 + 10000542x^5 + 10000542x^6).
\end{aligned} \tag{81}$$

$$\begin{aligned}
p_{B_{\text{cap}}}(x) := & 10^{-5}(51055 + 45411x + 42473x^2 + 39255x^3 \\
& + 39162x^4 + 39101x^5 + 39042x^6).
\end{aligned} \tag{82}$$

The next lemma verifies the spectral, measure and coding conditions for these seven stages.

Lemma A.16 (Certified stage inequalities). *Put $w_0 := 1$, $w_{k+1} := q_k w_k$, and $E_k := q_k^2/2$. Every stage satisfies the hypotheses of Lemmas A.5–A.13 at its stated η , and its exponent satisfies $e_k < w_k E_k$. The coding checks in Lemma A.2 hold.*

Proof. All coefficients in Definition A.15 are exact rationals. We give the finite comparisons and the rational enclosure rules used to verify them.

Rational enclosures. For a nonnegative rational x , a square-root enclosure is $[a/2^d, (a+1)/2^d]$, where the integer $a \geq 0$ satisfies $a^2 \leq 2^{2d}x < (a+1)^2$. The two comparisons use integer arithmetic.

For logarithms, reduce the argument to $y \in [1, 2]$ by powers of two and put $z := (y-1)/(y+1)$. Define

$$\ell_{K_{\text{series}}}(z) := 2 \sum_{j=0}^{K_{\text{series}}} \frac{z^{2j+1}}{2j+1}, \tag{83}$$

$$0 \leq \log y - \ell_{K_{\text{series}}}(z) \leq \frac{2z^{2K_{\text{series}}+3}}{(2K_{\text{series}}+3)(1-z^2)}. \tag{84}$$

The second line bounds the remaining positive series by a geometric series. The same formula at $z = 1/3$ encloses $\log 2$.

For $0 \leq y \leq 1$, define $e_{K_{\text{series}}}(y) := \sum_{j=0}^{K_{\text{series}}} y^j/j!$. The exponential remainder satisfies

$$0 \leq e^y - e_{K_{\text{series}}}(y) \leq \frac{y^{K_{\text{series}}+1}}{(K_{\text{series}}+1)!} \frac{1}{1 - y/(K_{\text{series}}+2)}. \tag{85}$$

For a larger positive argument use $e^x = (e^{x/2^r})^{2^r}$ with $x/2^r \leq 1$; for a negative argument take reciprocals. Finally, the identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and consecutive alternating partial sums of $\arctan y = \sum_{j=0}^{\infty} (-1)^j y^{2j+1}/(2j+1)$ give rational bounds for π . These rules, with $d := 256$ and $K_{\text{series}} := 256$, enclose every elementary function used below. Interval addition, multiplication and division propagate the rational endpoints, with the order of endpoints reversed for reciprocals. Polynomial coefficients and terminating-decimal comparisons use exact rational arithmetic.

Spectral curvature and root conditions. For each stage $0 \leq k \leq 6$, take the exact parameters $\vartheta_k := [x^k]p_{\vartheta}(x)$ from Definition A.15 and substitute them into the four numbered parts of Lemma A.14. In parts (i)–(iii), subtract the right-hand side from the left-hand side of each inequality in t ; in parts (i)–(ii), also multiply by $(1+u)^2$ to clear the denominator. This gives $f_{k,0}$ from part (i), $f_{k,1}, f_{k,2}$ from part (ii), and $f_{k,3}, f_{k,4}$ from part (iii), in the order stated there. In part (iv), use $L_0 := j/1024$ and $R_0 := (j+1)/1024$, for $0 \leq j < 1024$. Its displayed expression with F_1 defines $f_{k,5+2j}$, and the expression with F_2 defines $f_{k,6+2j}$. Thus these formulas specify 2053 polynomials at each stage, indexed by $0 \leq r \leq 2052$, and $7 \cdot 2053 = 14371$ in total.

Expand each $f_{k,r}(t)$ in powers of t to obtain its coefficients a_v . Set $n_B := 4\tau$ and pad every polynomial $f(t) = \sum_{v=0}^{n_B} a_v t^v$ with zero coefficients to this degree. On each of the 32 intervals, substitute $t := (i+s)/32$, $0 \leq i < 32$, $0 \leq s \leq 1$. Its power and Bernstein coefficients are respectively

$$a_j^{(i)} := \sum_{v=j}^{n_B} a_v \binom{v}{j} \frac{i^{v-j}}{32^v}, \quad (86)$$

$$b_{i,j} := \sum_{v=0}^j a_v^{(i)} \frac{\binom{j}{v}}{\binom{n_B}{v}}. \quad (87)$$

The convention $0^0 := 1$ applies in the first formula. The polynomial on this interval equals $\sum_{j=0}^{n_B} b_{i,j} \binom{n_B}{j} s^j (1-s)^{n_B-j}$. Thus a positive lower bound for all its Bernstein coefficients proves positivity on the entire interval.

For every $f_{k,r}$ defined by the four parts of Lemma A.14, exact substitution in Eqs. (86)–(87) gives

$$b_{i,j} > 2 \cdot 10^{-16} \quad (0 \leq i < 32, \quad 0 \leq j \leq 4\tau_k). \quad (88)$$

Thus all 14371 polynomials, constructed from Lemma A.14 and Definition A.15, are positive on $[0, 1]$.

The root expressions of Definition A.6, evaluated at $a = \eta_k$, satisfy

$$1 - 2\eta_k/\zeta_k^2 > 1/2, \quad (89)$$

$$1 - 4p_1\eta_k > 4/5, \quad (90)$$

$$\Delta_J > 7/10. \quad (91)$$

These three comparisons follow by the rational enclosure rules applied to the defining root expressions.

The other scalar conditions follow by the same rational comparisons: at every stage $A_J > 0.997$, $\kappa_0 + \kappa_{\text{tail}} - 2 > 41$, $2\kappa_0 - 2\theta/(1+\theta) > 1.7 \cdot 10^{-14}$, and $\kappa_{\text{tail}}(1-\zeta)^{2\tau-3} - \kappa_0 > 0.350$. This proves the spectral condition and all root hypotheses at the largest variance. Their persistence at smaller variance follows from Lemmas A.7 and A.10.

Radial and scalar-profile bounds. Evaluate the 512 terms in the upper sum of Lemma A.12, using Definition A.9 at each $a = \eta_k(i/512)^2$. Define $\bar{T}_k := [x^k]p_{\text{rem}}(x)$ by

$$\begin{aligned} p_{\text{rem}}(x) := & 3.97224831 \cdot 10^{-4} + 5.376579712 \cdot 10^{-6}x \\ & + 3.626126629 \cdot 10^{-8}x^2 + 1.737114912 \cdot 10^{-10}x^3 \\ & + 6.188830148 \cdot 10^{-13}x^4 + 7.993551039 \cdot 10^{-16}x^5 \\ & + 5.696775742 \cdot 10^{-22}x^6. \end{aligned} \quad (92)$$

The rational enclosure rules give $T_k < \bar{T}_k$ for every $0 \leq k \leq 6$. To verify the scalar cap, let $L_H(x)$ be the logarithmic term in Eq. (70), with $L_H(0) := 0$. Define its rational part with the larger remainder by

$$\bar{Q}_H(x) := \frac{H}{(H-1)\kappa_0\eta} \left(\sum_{j=2}^{K-1} \frac{p_j}{(1+x)^j} + \frac{\bar{T}_k}{2v(1+x)^{2K-\ell}} \right). \quad (93)$$

The identity $L_H(x) = Hx \int_0^1 t/((1+tx)(1+Htx))dt$ gives $|L'_H(x)| \leq H/2$, and $\bar{Q}_H$ is decreasing. Set $h := 10/2^{17}$. On the cell $[ih, (i+1)h]$ the profile is therefore at most

$$u_i := L_H((i+1/2)h) + Hh/4 + \bar{Q}_H(ih), \quad 0 \leq i < 2^{17}. \quad (94)$$

For each of the seven stages, the rational enclosure rules give

$$B_{\text{cap}} - u_i > 2 \cdot 10^{-7} \quad (0 \leq i < 2^{17}), \quad (95)$$

$$B_{\text{cap}} - \bar{Q}_H(0) > 2 \cdot 10^{-7}, \quad (96)$$

$$B_{\text{cap}} - \frac{H}{H-1} \frac{\log 11}{10} - \bar{Q}_H(10) > 2 \cdot 10^{-7}. \quad (97)$$

The first line handles $[0, 10]$, and the second includes the continuous value at zero explicitly. For $x \geq 10$, use $L_H(x) \leq H \log(1+x)/((H-1)x)$; both this upper bound and $\bar{Q}_H$ decrease, so the third line handles the remaining half-line. These finite comparisons also admit dyadic subdivision: the Lipschitz estimate implies that bisecting a cell cannot increase its upper bound in Eq. (94).

The exponent in Lemma A.13 is at most $\kappa_0\eta B_{\text{cap}} + H\bar{T}_k v/(2(H-1))$. The coefficients of the stage polynomials and Eq. (92) satisfy the exact rational inequality

$$\frac{\kappa_0\eta B_{\text{cap}} + H\bar{T}_k v/(2(H-1))}{w_k q_k^2/2} < \frac{199}{200} \quad (0 \leq k \leq 6). \quad (98)$$

Since $199/200 < 1$, this proves $e_k < w_k E_k$.

Coding. Set $s := s_*$, $a := a_*$, $b := b_*$, $\epsilon := 0.01$, $F(x) := \gamma_1([-x, x])$ and $g(x) := \sqrt{2/\pi}e^{-x^2/2}$. The Gaussian bounds follow from the two polynomial families

$$\mathbf{e}_n(x) := \sum_{j=0}^n \frac{(-1)^j x^{2j}}{2^j j!}, \quad (99)$$

$$\mathbf{f}_n(x) := \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2^j j!(2j+1)}. \quad (100)$$

Taylor's integral remainder gives $\mathbf{e}_{17}(x) \leq e^{-x^2/2} \leq \mathbf{e}_{16}(x)$. Integration from 0 to x gives the corresponding bounds with $\mathbf{f}_{17}$ and $\mathbf{f}_{16}$.

The Machin identity and rational alternating-series bounds used earlier give $\alpha_- < \sqrt{2/\pi} < \alpha_+$, where $\alpha_- := 0.7978845608028653$ and $\alpha_+ := 0.7978845608028654$. Define $\underline{g} := \alpha_- \mathbf{e}_{17}$, $\bar{g} := \alpha_+ \mathbf{e}_{16}$, $\underline{F} := \alpha_- \mathbf{f}_{17}$ and $\bar{F} := \alpha_+ \mathbf{f}_{16}$. Thus, on $0 \leq x \leq 3s$,

$$\underline{g}(x) \leq g(x) \leq \bar{g}(x), \quad (101)$$

$$\underline{F}(x) \leq F(x) \leq \bar{F}(x). \quad (102)$$

These bounds have rational coefficients and degree at most 35. Exact substitution at $0, s, s + \epsilon, 5s/2, 3s$ gives

$$\min\{b, a - bs, a + bs - \bar{g}(0), \underline{g}(s) - a, a - \bar{g}(s + \epsilon)\} > 1/2000, \quad (103)$$

$$1/2 - \bar{F}(s) - \epsilon(\bar{g}(s) - a) > 6 \cdot 10^{-7}, \quad (104)$$

$$\min\{1 - \bar{F}(5s/2) - bs^2/2, 1 - \bar{F}(3s) - bs^2/8\} > 1/50. \quad (105)$$

These comparisons verify the slope, middle-interval and tail conditions of Lemma A.2, respectively. Together with $\epsilon < s < 1$, this completes the density checks. Lemma A.3 supplies the entropy reserve for every q_k . □

A.7 Uniform small balls for the tail

Once the active set is sufficiently small, the estimate at large aspect ratios supplies a uniform continuation beyond the seven prescribed stages.

Lemma A.17 (Uniform bound for the infinite tail). *For every family of $m \leq N$ symmetric contractions with $u := \log(N/m) > 43/(256/17)$,*

$$\Pr\left[\left\|\sum_{i=1}^m \xi_i A_i\right\| \leq 185u\sqrt{m}\right] \geq e^{-0.000051714m}.$$

Proof. We use the sharper exponent of the last row of Tables 2 and 3, not the uniform exponent 0.003712. We record its continuation to all larger aspect ratios. Put

$$c_{\text{cut}} := 256/17, \quad c_{\text{rad}} := 185, \quad b := 0.499855, \quad D := 221000, \quad A := 228 \cdot 10^9,$$

and $\tau_0 := 44$, $\ell_0 := 43/c_{\text{cut}}$, $r_0 := 44/c_{\text{cut}}$. Set $\hat{\kappa}_3 := b^2/(1+b)$, $\hat{c}_0 := (1-2b)/2$, and $P_{c_{\text{rad}}}(x) := e^x + e^{2x}/(c_{\text{rad}}^2 x^2)$. The first-cell hypotheses in the proof of Lemma 4.1 are

$$A > 8/\hat{c}_0, \quad D > 32/\hat{c}_0, \quad A^2 > D, \quad (44/(c_{\text{rad}}\ell_0))^2 < \hat{\kappa}_3^2/4, \quad 1/(c_{\text{rad}}^2\ell_0^2) < 1/6400.$$

These follow by rational comparison. The proof of Lemma 4.15, with these parameters, bounds the normalized negative exponent by E_{tail} , where

$$\begin{aligned} I_0 &:= 25A\tau_0 P_{c_{\text{rad}}}(\ell_0)(50\tau_0^2/(c_{\text{rad}}^2\ell_0^2))^{\tau_0}, \\ J_0 &:= (2A/\tau_0)4^{-\tau_0} P_{c_{\text{rad}}}(r_0), \\ E_{\text{tail}} &:= 27D/A + I_0 + J_0 + 4e^{\ell_0 - c_{\text{rad}}^2\ell_0^2/8} < 0.000051714. \end{aligned}$$

Applying the exponential polynomial and remainder bound of Lemma A.16 gives

$$I_0 < 0.000025542317676, \quad (106)$$

$$J_0 < 6.221 \cdot 10^{-16}, \quad (107)$$

$$4e^{\ell_0 - c_{\text{rad}}^2 \ell_0^2/8} < 10^{-100}. \quad (108)$$

The first two lines substitute the rational parameters in the defining expressions for I_0 and J_0 . The third bounds the exponential strip term. Adding these bounds gives

$$E_{\text{tail}} < 0.000051713371 < 0.000051714. \quad (109)$$

For $\tau \geq 44$ and $\ell_{\text{asp}} \in ((\tau - 1)/c_{\text{cut}}, \tau/c_{\text{cut}}]$, the interpolation term

$$25A\tau P_{c_{\text{rad}}}(\ell_{\text{asp}})(50\tau^2/(c_{\text{rad}}^2 \ell_{\text{asp}}^2))^\tau$$

decreases in ℓ_{asp} , since its logarithmic derivative is at most $2 - 2\tau/\ell_{\text{asp}} < 0$. The diagonal term $(2A/\tau)4^{-\tau}P_{c_{\text{rad}}}(\ell_{\text{asp}})$ increases, while the strip term decreases. At successive left endpoints the curvature bound $c_{\text{cut}}\tau/(\tau - 1)$ decreases, as does $1/(c_{\text{rad}}^2 \ell_{\text{asp}}^2)$. Successive interpolation maxima have ratio at most

$$\frac{\tau + 1}{\tau} e^{2/c_{\text{cut}}} \frac{50c_{\text{cut}}^2}{c_{\text{rad}}^2} < 0.387,$$

and successive diagonal maxima have ratio at most $e^{2/c_{\text{cut}}}/4 < 0.286$. For the first comparison use $P_{c_{\text{rad}}}(x + 1/c_{\text{cut}}) \leq e^{2/c_{\text{cut}}}P_{c_{\text{rad}}}(x)$ and the decreasing sequence $(1 + 1/\tau)^{\tau+1}$. The rational enclosure rules of Lemma A.16 bound these ratios at $\tau = 44$ by 0.386945 and 0.285509, respectively; both bounds only improve thereafter. Thus the first cell controls every later cell, including its endpoints. Since $u > \ell_0 > 1$, take $\ell_{\text{asp}} := u$ and $\tau := \lceil c_{\text{cut}}u \rceil \geq 44$ in the same partition-function reduction. This proves the assertion over the entire infinite tail. $\square$

A.8 Projection-aware estimates and coupled spectral curvature

The two-component families of Definition A.15 and their certificates supply the algorithmic small-ball estimates in Section B. We now prove the shared moment estimates and the spectral refinements used by the existence parameters of Definition A.30.

For a positive matrix density A , define

$$\begin{aligned} \mathcal{R}_A(f) &:= \int f^\top A f, & \Pi_A f &:= \left(\int A\right)^{-1} \int A f, \\ \mathcal{P}_A(f) &:= \mathcal{R}_A(\Pi_A f), & \mathcal{D}_A(f) &:= \mathcal{R}_A(f) - \mathcal{P}_A(f). \end{aligned}$$

Here $\mathcal{D}_A$ is the centered weighted energy. These definitions extend to arrays by direct sums.

Keeping the projected derivative means in the variance estimate improves the energy recurrence for the sub-10 construction.

Lemma A.18 (Projection-aware weighted variance). *Under the curvature and exhaustion hypotheses of Lemma A.5, with curvature lower bound $\lambda > 0$ and $p := \lambda^{-1}$,*

$$\mathcal{D}_A(f) \leq \frac{p}{2} \sum_{i=1}^m (\mathcal{R}_A(\partial_i f) + \mathcal{P}_A(\partial_i f)). \quad (110)$$

Proof. On a smooth bounded convex exhaustion domain solve $Lu = f - \Pi_A f$ with the Neumann condition, using the operator from Lemma A.5. The Bochner identity and the weighted Poincaré inequality applied to each $\partial_i u$ give

$$\|Lu\|_A^2 \geq \sum_{i=1}^m \sum_{j=1}^m \|\partial_{i,j} u\|_A^2 + \lambda \sum_{i=1}^m \mathcal{R}_A(\partial_i u) \geq \lambda \sum_{i=1}^m \mathcal{P}_A(\partial_i u) + 2\lambda \sum_{i=1}^m \mathcal{D}_A(\partial_i u).$$

The first step is the curvature lower bound and the nonnegative boundary term. The second applies weighted Poincaré to the derivative fields; these fields need not satisfy a Neumann condition. Integration by parts gives $\mathcal{D}_A(f) = \langle \nabla f, \nabla u \rangle_A$. Splitting the gradients into their constant projections and orthogonal complements, then applying Cauchy–Schwarz with weights λ and 2λ , yields

$$\mathcal{D}_A(f)^2 \leq [p \sum_{i=1}^m \mathcal{P}_A(\partial_i f) + \frac{p}{2} \sum_{i=1}^m \mathcal{D}_A(\partial_i f)] \|Lu\|_A^2.$$

Since $\|Lu\|_A^2 = \mathcal{D}_A(f)$, cancellation proves Eq. (110); the zero case is immediate. On exhaustion, the weighted energies and constant projections converge by finite-energy approximation, as in Lemma A.5. No convergence of the auxiliary Neumann solutions is required. $\square$

Mixed products of linear matrix series control the ordinary means in both appendices. The following lemma also supplies the short-word constants used below.

Lemma A.19 (Mixed linear moments). *Let μ be an even convex tilt of standard Gaussian measure on $\mathbb{R}^m$. Define $b_0 := 1$, $b_1 := 1$ and, for $s \geq 2$,*

$$b_s := \begin{cases} b_{s/2}^2 + s^2 b_{s-1}/2, & s \text{ even}, \\ s^2(b_{s-1} + 2b_{(s-1)/2}^2)/3, & s \text{ odd}. \end{cases}$$

Let $Z_k(\xi) := \sum_{i=1}^m \xi_i N_{k,i}$ be square matrix series, not necessarily symmetric, such that

$$\sum_{i=1}^m N_{k,i}^\top N_{k,i} \preceq aI, \quad \sum_{i=1}^m N_{k,i} N_{k,i}^\top \preceq aI.$$

For deterministic contractions $U_0, \dots, U_{2s}$,

$$\|\mathbb{E}_\mu[U_0 Z_1 U_1 \cdots Z_{2s} U_{2s}]\| \leq b_s a^s.$$

For a single such series Z , both $\mathbb{E}_\mu[(Z^s)^\top Z^s]$ and $\mathbb{E}_\mu[Z^s (Z^s)^\top]$ are at most $b_s a^s I$.

The same bounds hold for compatible rectangular words, transposes and amplifications. In particular, for a word W of degree $r \leq 4$ in linear series of variance at most I , with deterministic contractions between them,

$$\mathbb{E}_\mu[W^\top W], \mathbb{E}_\mu[WW^\top] \preceq c_r I, \quad (c_0, c_1, c_2, c_3, c_4) := (1, 1, 3, 18, 153).$$

For an even word of degree $2s \leq 8$, $\|\mathbb{E}_\mu[W]\| \leq c_s$; odd words have zero mean. A deterministic derivative insertion with row or column norm at most $\sqrt{a}$ contributes one further factor a to a squared-norm bound.

Proof. The scalar Bochner identity and parity give

$$\mathbb{V}_\mu[f] \leq \frac{1}{2}\mathbb{E}_\mu[\|\nabla f\|_2^2] \quad (f \text{ even}), \quad \mathbb{E}_\mu[f^2] \leq \frac{1}{3}\mathbb{E}_\mu[\|\nabla f\|_2^2] + \frac{2}{3}\|\mathbb{E}_\mu[\nabla f]\|_2^2 \quad (f \text{ odd}).$$

Indeed, the derivatives of an even Neumann solution are odd and have zero mean, so curvature-one Poincaré and Bochner give the first bound. For an odd solution, its derivatives are even. Applying the first bound gives the gradient metric $3I - 2\Pi$, whose inverse is $(I + 2\Pi)/3$, proving the second bound. Smooth symmetric convex exhaustion and even convex approximation justify these arguments for an extended-valued tilt; Gaussian polynomial domination passes to the limit. The bounds apply componentwise.

We induct on s , starting with covariance domination at $s = 1$. We have

$$\partial_i Z^s = \sum_{q=0}^{s-1} Z^q N_i Z^{s-1-q}.$$

The lower mixed-order bounds give

$$\sum_{i=1}^m \mathbb{E}_\mu[(\partial_i Z^s)^\top (\partial_i Z^s)] \preceq s^2 b_{s-1} a^s I.$$

To perform the coefficient sum without a dimension factor, use the vacuum and one-letter word space. If L_i creates letter i , then $T := \sum_{i=1}^m N_i \otimes L_i$ has norm at most $\sqrt{a}$; vacuum compression of a word containing $T^\top, T$ performs this sum. Each resulting word has $2s - 2$ random factors. Cauchy–Schwarz over the s derivative positions supplies s^2 .

For even s , the ordinary mean has norm at most $b_{s/2} a^{s/2}$. For odd s , the same coefficient encoding gives $\sum_{i=1}^m (\mathbb{E}_\mu[\partial_i Z^s])^\top \mathbb{E}_\mu[\partial_i Z^s] \preceq s^2 b_{(s-1)/2}^2 a^s I$. Apply the corresponding scalar parity bound to $Z^s x$, for each fixed vector x . This gives the two definitions of b_s . Transposition gives the other power-square bound, and operator Cauchy–Schwarz bounds $\|\mathbb{E}_\mu[Z^{2s}]\|$.

Finally, place $Z_1 U_1, \dots, Z_{2s} U_{2s}$ on the edges of a cyclic block matrix. Its two variance sums are block diagonal and at most aI . One diagonal block of its $2s$ th power is the ordered product. Compression and multiplication by U_0 prove the mixed bound, closing the induction.

Zero padding embeds compatible rectangular factors in a common square space and preserves both variance bounds. Apply the mixed bound to $W^\top W$ and $WW^\top$. Since $(b_0, b_1, b_2, b_3, b_4) = (1, 1, 3, 15, 129)$, the stated c_r bounds follow. Coefficient column maps and their transposes are deterministic contractions after division by $\sqrt{a}$; restoring this normalization gives the insertion factors. The case $a = 0$ is immediate. $\square$

To sharpen the weighted mean estimate, we retain the gap between the arithmetic and harmonic means of the resolvent weight.

Lemma A.20 (Weighted means and the arithmetic–harmonic gap). *Let $W \succeq I$ be a random positive matrix, $S := \mathbb{E}[W] \preceq JI$, $T := \mathbb{E}[W^{-1}]$, and $k := \|S^{1/2} T S^{1/2} - I\|$. If $\|\mathbb{E}[f]\|^2 \leq M$ and $\mathcal{D}_W(f) \leq B$, then*

$$\mathcal{P}_W(f) \leq (\sqrt{JM} + \sqrt{kB})^2, \quad k \leq \min\{J - 1, \|\mathbb{E}[(W - I)W^{-1}(W - I)]\|\}. \quad (111)$$

Proof. Put $v := S^{-1}\mathbb{E}[Wf]$, $g := f - v$, and $b := \mathbb{E}[f]$. Since $\mathbb{E}[Wg] = 0$, weighted Cauchy–Schwarz gives

$$(b - v)(b - v)^\top \preceq \mathcal{D}_W(f)(T - S^{-1}).$$

Congruence by $S^{1/2}$ and the triangle inequality imply $\|S^{1/2}v\| \leq \sqrt{JM} + \sqrt{kB}$. Its square is $\mathcal{P}_W(f)$.

Jensen gives $T \succeq S^{-1}$, while $W \succeq I$ gives $T \preceq I$. Hence $0 \leq k \leq J - 1$. Write $B_W := \mathbb{E}[(W - I)W^{-1}(W - I)]$. The exact identity

$$S - T^{-1} = B_W - (T^{-1} - 2I + T) \preceq B_W$$

uses $T^{-1} - 2I + T \succeq 0$. The matrices $S^{1/2}TS^{1/2}$ and $T^{1/2}ST^{1/2}$ have the same eigenvalues. Therefore

$$k = \|T^{1/2}(S - T^{-1})T^{1/2}\| \leq \|T^{1/2}B_W T^{1/2}\| \leq \|B_W\|.$$

The first step uses the equal eigenvalues, the second uses the preceding positive-semidefinite comparison, and the last uses $T \preceq I$. $\square$

We bound this gap through scalar resolvent inequalities and estimate the remainder of a truncated expansion.

Lemma A.21 (Scalar gap and truncated resolvent series). *For $|u|, |v| < 1$, put $w := (1 + uv)/((1 - u^2)(1 - v^2))$. Then $w \geq 1$ and*

$$\frac{(w - 1)^2}{w} \leq \frac{9}{2}(u^4 + v^4) + \frac{13}{8}(u^6 + v^6)w. \quad (112)$$

Put $H_j(u, v) := \sum_{q=0}^{2j} u^q v^{2j-q}$. For every integer $r \geq 1$,

$$0 \leq w - \sum_{j=0}^{r-1} H_j \leq \frac{2r+1}{2}(u^{2r} + v^{2r})w. \quad (113)$$

Proof. For a negative product, $u^2 + v^2 - |uv| - u^2v^2 \geq |uv|(1 - |uv|) \geq 0$; this also proves $w \geq 1$. To prove Eq. (112), first take $u, v \geq 0$, write $p := uv$, $\delta := (u - v)^2$, and $s := u^2 + v^2 = 2p + \delta$. Here $0 \leq \delta \leq (1 - p)^2$. Clearing the positive denominator $(1 - u^2)(1 - v^2)(1 + p)$ gives

$$\begin{aligned} P &:= \frac{9}{2}(s^2 - 2p^2)(1 - s + p^2)(1 + p) + \frac{13}{8}(s^3 - 3p^2s)(1 + p)^2 - (s + p - p^2)^2 \\ &= \frac{p^3(7p - 1)^2}{4} + \delta(A + B\delta + C\delta^2), \end{aligned}$$

where direct expansion gives

$$A := \frac{p(261p^3 + 18p^2 - 83p + 96)}{8}, \quad B := \frac{57p^3 - 12p^2 - 51p + 14}{4}, \quad C := \frac{(p + 1)(13p - 23)}{8}.$$

On $0 \leq p \leq 1$, $A \geq 0$ since $96 - 83p \geq 13$, and $C < 0$. The bracket is concave in δ . At $\delta = (1 - p)^2$ its value is

$$(13p^6 + 52p^5 + 104p^4 + 58p^3 + 40p^2 + 20p + 5)/8 > 0.$$

It is thus nonnegative throughout the interval. For a negative product, replacing u, v by their moduli increases w and preserves the even powers. The function $1 - (2 + b)/w + 1/w^2$ is increasing for $w \geq 1$ and fixed $b \geq 0$. Divide the desired inequality by w and use $b := 9(u^4 + v^4)/2$ to transfer it to the signed pair.

For Eq. (113), each H_j is nonnegative: it is the divided difference of x^{2j+1} , with diagonal value $(2j + 1)u^{2j}$. The convergent identity $w = \sum_{j \geq 0} H_j$ proves the lower bound. For equal signs, summing the geometric series gives

$$\frac{w - \sum_{j < r} H_j}{w} = \frac{H_r - u^2v^2H_{r-1}}{1 + uv} \leq H_r \leq \frac{2r+1}{2}(u^{2r} + v^{2r}).$$

The first step sums the tail, the second drops a nonpositive term and uses $1 + uv \geq 1$, and the last integrates the convexity bound for x^{2r} in the divided-difference formula. For opposite signs put $z := -v$, with $u, z \geq 0$. The same ratio is a weighted average of u^{2r} and z^{2r} , with weights $u(1 - z^2)$ and $z(1 - u^2)$. It is at most their maximum, which gives the asserted bound. Zero and diagonal cases follow by continuity. $\square$

A condition on triples of spectral values controls the coupled curvature terms in the one- and two-slot resolvent weights.

Lemma A.22 (Coupled spectral triples). *Let ϕ be even and convex on $(-1, 1)$, extended by $+\infty$ outside, with logarithmic boundary coefficient greater than two. Put $G(u, v) := (1 - u)(1 - v)\phi'[u, v]$. Choose a nonnegative function d satisfying $d(u)^2/(1 - u) \leq \zeta^{-1}$. For a triple (u, v, w) set*

$$l_u := 1 - d(u)^2 d(v) d(w), \quad l_v := 1 - d(u) d(v)^2 d(w), \quad l_w := 1 - d(u) d(v) d(w)^2.$$

Suppose

$$M(u, v, w) := \begin{pmatrix} G(u, v) & -l_u & -l_v \\ -l_u & G(u, w) & -l_w \\ -l_v & -l_w & G(v, w) \end{pmatrix} \succeq 0 \quad (-1 < u, v, w < 1). \quad (114)$$

Then the conclusions of Lemma A.5 hold with $p_r = (1 - ra/\zeta^2)^{-1}$, $r = 1, 2$, for $2a < \zeta^2$. The same bounds hold with an additional nonnegative quadratic potential.

Proof. Use $Y := I - X$, $C_i := Y^{-1/2} N_i Y^{-1/2}$, and $D := \text{diag}(d(u_b))$ in an eigenbasis of X . Put $C_{i,1} := DC_i D$ and $R := DY^{-1/2}$. Then $R^2 \preceq \zeta^{-1} I$, so

$$\sum_i C_{i,1}^2 = R \left(\sum_i N_i R^2 N_i \right) R \preceq \zeta^{-1} R \left(\sum_i N_i^2 \right) R \preceq a \zeta^{-2} I.$$

The equality is expansion; the first comparison uses $R^2 \preceq \zeta^{-1} I$, and the second uses the variance bound. The regular curvature loss is therefore at most a/ζ^2 per slot.

For test vectors z_i , set $T_{ab,c} := \sum_i (C_i)_{ab} (z_i)_c$, so $T_{ab,c} = T_{ba,c}$. Let Q_0, Q_1 be the ordered curvature forms in the proof of Lemma A.5. Direct expansion gives

$$Q_0 - Q_1 = \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N (1 - d_a d_b^2 d_c) T_{ab,c} T_{bc,a}.$$

Write $\mathcal{H} := \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N G(u_a, u_b) T_{ab,c}^2$ and $t_{abc} := (T_{ab,c}, T_{ac,b}, T_{bc,a})^\top$. Relabeling the ordered sums gives the exact identity

$$\mathcal{H} - 2(Q_0 - Q_1) = \frac{1}{3} \sum_{a=1}^N \sum_{b=1}^N \sum_{c=1}^N t_{abc}^\top M(u_a, u_b, u_c) t_{abc} \geq 0. \quad (115)$$

Each diagonal sum occurs three times, and each of the three cross-term sums equals $Q_0 - Q_1$. This includes repeated indices and does not assume that l_u, l_v, l_w are nonnegative. Spectator indices in an amplified slot can be summed separately. Half the barrier Hessian absorbs the nonregular loss in one slot; two slots use at most the whole Hessian. The Gaussian Hessian and the nonnegative quadratic Hessian leave curvature $1 - ra/\zeta^2$. The Bochner and exhaustion argument of Lemma A.5 now applies. Reflection treats the other edge. No derivative of D is used. $\square$

A.9 Finite resolvent refinement and covariance-sensitive means

In this subsection the barrier may be any of those in Definition A.30. The new parameters are local to this construction. Write $Y = \sum_i \xi_i B_i$, $\sum_i B_i^2 \preceq \eta I_N$, and let κ denote the quadratic coefficient of the barrier. Retain

$$\Gamma_{i,j} := \text{tr}[B_i B_j], \quad C_\Gamma := (I + 2\kappa\Gamma)^{-1}, \quad \nu := \mathcal{N}(0, C_\Gamma), \quad \omega_j := \frac{\text{tr}[\Gamma C_\Gamma^j]}{N\eta}.$$

Put $\psi(u) := \phi(u) - \kappa u^2$ and $d\mu_t := Z_t^{-1} e^{-\text{tr}[\psi(tY)]} d\nu$, with $X := tY$, $a := \eta t^2$, and $Y_\pm := I \pm X$. Relative to the standard Gaussian the additional potential is $\text{tr}[\phi(tY)] + \kappa(1 - t^2)\xi^\top \Gamma \xi$. All the preceding curvature arguments apply. Lemma A.11 gives

$$N^{-1} \mathbb{E}_{\mu_t}[\text{tr}[X^{2j}]] \leq \text{df}(j) a^j \omega_j, \quad 0 \leq \omega_{j+1} \leq \omega_j \leq 1. \quad (116)$$

Whitening ν and applying scalar Poincaré to linear forms gives $\text{Cov}_{\mu_t}(\xi) \preceq C_\Gamma$.

We first bound operator energies, without using a trace-dependent covariance improvement. Set

$$W_\pm^{(1)} := Y_\pm^{-1}, \quad W_\pm^{(2)} := Y_\pm^{-1} \otimes Y_\pm^{-1}, \quad W_e^{(s)} := (W_+^{(s)} + W_-^{(s)})/2.$$

Let L, J bound the expectations of the one-slot and two-slot even weights, respectively, and let k_1, k_2 bound their arithmetic-harmonic gaps. The even weights dominate I .

We combine the mean-gap and truncated-series bounds in a finite recurrence that improves the operator energy coefficients.

Definition A.23 (Finite operator-energy refinement). For a valid tuple (p, J_*, A_*, k_*) , initialize $V_0 = Z_0 = J_*$. For $1 \leq r \leq 9$, define

$$\begin{aligned} D_r &:= \frac{p}{2}(V_{r-1} + Z_{r-1}), \quad m_r := \frac{c_{r/2}^2}{(r!)^2} \quad (r \text{ even}), \\ (V_r, Z_r) &:= (A_* D_r, (A_* - 1)D_r) \quad (r \text{ odd}), \\ Z_r &:= (\sqrt{J_* m_r} + \sqrt{k_* D_r})^2, \quad V_r := D_r + Z_r \quad (r \text{ even}), \\ C_r(p, J_*, A_*, k_*) &:= (r!)^2 V_r. \end{aligned} \quad (117)$$

Here the c_s are from Lemma A.19. Use (p_1, L, L, k_1) and $(p_2, J, J(1+a), k_2)$ for slots one and two. Initialize L, J at the smaller roots of Definition A.6, with $L = L_1$, and set $k_1 = L - 1$, $k_2 = J - 1$.

At each pass compute $C_r^{(1)}, C_r^{(2)}$ from the current tuple for $1 \leq r \leq 4$, then set

$$\begin{aligned} L' &:= \min\{L, 1 + \sum_{j=1}^{r-1} c_j a^j + a^r C_r^{(1)} : 1 \leq r \leq 4\}, \\ J' &:= \min\{J, \mathcal{J}(a, p_2, L'), 1 + \sum_{j=1}^{r-1} (2j+1)c_j a^j + (2r+1)a^r C_r^{(2)} : 1 \leq r \leq 4\}, \\ k'_1 &:= \min\{k_1, L' - 1, a^2 C_2^{(1)}\}, \\ k'_2 &:= \min\{k_2, J' - 1, 27a^2 + (13/4)a^3 C_3^{(2)}\}. \end{aligned}$$

The function $\mathcal{J}(a, p_2, L')$ is the smaller-root expression for J in Definition A.6, with $c_1 = L' - 1$. All right-hand energies use the preceding tuple. Perform exactly six passes and denote the resulting coefficients by $\widehat{C}_r^{(1)}(a)$ and $\widehat{C}_r^{(2)}(a)$.

We verify that the finite refinement preserves the required energy and resolvent bounds at every pass.

Lemma A.24 (Validity of the finite refinement). *At each pass the bounds L, J, k_1, k_2 remain valid. In particular,*

$$\begin{aligned}\mathbb{E}_{\mu_t}[X^{2r}(I - X^2)^{-1}] &\preceq a^r \widehat{C}_r^{(1)}(a)I, \\ \mathbb{E}_{\mu_t}[(X^{2r} \otimes I)W_e^{(2)}] &\preceq a^r \widehat{C}_r^{(2)}(a)I, \\ \mathbb{E}_{\mu_t}[(I \otimes X^{2r})W_e^{(2)}] &\preceq a^r \widehat{C}_r^{(2)}(a)I, \\ N^{-1}\mathbb{E}_{\mu_t}[\text{tr}[X^{2r}Y_+^{-2}]] &\leq a^r \widehat{C}_r^{(2)}(a).\end{aligned}$$

These quantities are bounded for $1 \leq r \leq 9$. All computed scalar bounds are nondecreasing in a throughout the strict-root range.

Proof. For an ordered derivative array of X^ℓ , residual even degree $r = \ell - j$ has squared operator mean norm at most $(\ell!/r!)^2 c_{r/2}^2 a^\ell$, by Lemma A.19. This also holds for $X^\ell \otimes I$ and $I \otimes X^\ell$ acting on fixed vectors. The top constant derivative has squared norm at most $(\ell!)^2 a^\ell$; odd derivative means vanish. Apply Lemma A.18 to successive derivatives and normalize by $(\ell!)^2 a^\ell$, with an additional factor N for trace energies. For an even field use Lemma A.20. For odd fields, the identity $W_o W_e^{-1} W_o = W_e - I$ in one slot and the proof of Lemma A.8 in two slots give $\mathcal{R} \leq A_* \mathcal{D}$ and $\mathcal{P} \leq (A_* - 1)\mathcal{D}$, with $A_* = L$ or $J(1 + a)$. These are precisely the two branches of Eq. (117). Vectorization gives the trace assertion.

Expand $(I - X^2)^{-1}$ through degree $2r - 2$ and use the one-slot energy on its remainder to obtain L' . For J' use Eq. (113) in the joint eigenbasis of $X \otimes I$ and $I \otimes X$. Each of the $2j + 1$ terms in H_j is a word of degree $2j$, whose expected norm is at most $c_j a^j$. The two-slot operator energies bound its remainder. The improved smaller-root bound is also valid: reducing L increases the larger root, which was already excluded by the preceding bound on J , and increases the discriminant.

For the one-slot gap, $(W_e^{(1)} - I)^2 (W_e^{(1)})^{-1} = X^4 / (I - X^2)$. For the two-slot gap, Eq. (112) contributes at most $27a^2 I$ from the fourth moments, since $c_2 = 3$, and $(13/4)a^3 C_3^{(2)} I$ from the weighted sixth moments. Lemma A.20 proves the stated gap updates without an extra factor L' or J' . Thus every pass preserves validity. The initial roots are nondecreasing in a , by their Catalan-series representation in Lemma A.10. The recurrence and updates use nonnegative sums, products, square roots, and minima of nondecreasing functions. Induction proves monotonicity. Only six finite passes are used; no limiting fixed point is assumed. $\square$

The next estimate retains the covariance powers in derivative means. We use its conservative constants for the trace recurrence here and its general form in Section B.

Lemma A.25 (Covariance-sensitive derivative means). *Let $X := \sum_{i=1}^m \xi_i D_i$ under an even convex Gaussian tilt, and let $S_D := \sum_{i=1}^m D_i^2 \preceq aI_N$. Raw derivative directions N_i may differ from D_i , but satisfy $\sum_{i=1}^m N_i^2 \preceq aI_N$. Along the whitened radial path, let $\omega_q := \text{tr}[\Gamma C_\Gamma^q] / (N\eta)$ be as in Lemma A.11, with $a := \eta t^2$ and $C_\Gamma := (I + 2\kappa\Gamma)^{-1}$, where $\kappa > 0$ is the quadratic barrier coefficient. Denote by $\nabla_N^j(X^\ell)$ the ordered derivatives at $h = 0$ of $(X + \sum_{i=1}^m h_i N_i)^\ell$. For $r := \ell - j > 0$ even, define*

$$k(r) := \begin{cases} 2, & r = 2, \\ r/2, & r \geq 4, \end{cases} \quad m_j := (\ell!/r!)^2 \min\{\text{df}(r), b_{r/2} \text{df}(r/2)\}.$$

Then

$$\|\mathbb{E}_\mu[\nabla_N^j(X^\ell)]\|_{\text{HS}}^2 \leq N a^\ell m_j \omega_{k(r)}.$$

Odd ordinary means vanish. At $j = \ell$ the bound is $Na^\ell(\ell!)^2$, with no covariance factor.

For $r \in \{2, 4, 6, 8\}$, this gives the bounds used in the existence construction:

$$\frac{\|\mathbb{E}_\mu[\nabla_N^j(X^\ell)]\|_{\text{HS}}^2}{Na^\ell} \leq (\ell!/r!)^2 M_r \omega_2, \quad (M_2, M_4, M_6, M_8) := (1, 9, 270, 16065). \quad (118)$$

Proof. We first prove an auxiliary bound. If Ψ is completely positive, $S := \Psi(I)$, and a random matrix B satisfies $\mathbb{E}[B^\top B], \mathbb{E}[BB^\top] \preceq bI$, then

$$\mathbb{E}[\|\Psi(B)\|_F^2] \leq b \operatorname{tr}[S^2].$$

For $A := S + \epsilon I$, the map $\Psi_A(B) := A^{-1/2} \Psi(B) A^{-1/2}$ is subunital and completely positive. Its Schwarz bounds follow by applying it to the positive block matrix with blocks $B^\top B, B^\top, B, I$, enlarging the lower-right block to I , and taking a Schur complement. Thus $Y := \Psi_A(B)$ satisfies $\mathbb{E}[Y^\top Y], \mathbb{E}[YY^\top] \preceq bI$. We have

$$\mathbb{E}[\|\Psi(B)\|_F^2] = \mathbb{E}[\operatorname{tr}[AY^\top AY]] \leq \frac{1}{2} \mathbb{E}[\operatorname{tr}[A^2(Y^\top Y + YY^\top)]] \leq b \operatorname{tr}[A^2],$$

where the first step uses $\Psi(B) = A^{1/2} Y A^{1/2}$, the second follows from $\|AY - YA\|_F^2 \geq 0$, and the third uses the two second-moment bounds. Let $\epsilon \downarrow 0$.

Write $\Sigma := \mathbb{E}_\mu[\xi \xi^\top] \preceq I$ and $\Psi(B) := \sum_{i=1}^m \sum_{j=1}^m \Sigma_{i,j} D_i B D_j$. This map has symmetric Kraus matrices and $\Psi(I) \preceq S_D$. In a word $AXBXC$ with two common factors X , let every other factor be a distinct independent raw Gaussian direction. Integrating the outer factors and applying the preceding bound to the middle product gives

$$\mathbb{E}[\|\mathbb{E}_\mu[AXBXC]\|_F^2] \leq a^{\ell-2} \operatorname{tr}[S_D^2] \leq Na^\ell \omega_2.$$

The first step also uses $\operatorname{tr}[\Psi(I)^2] \leq \operatorname{tr}[S_D^2]$, which follows by taking the trace of $(S_D - \Psi(I))(S_D + \Psi(I))$. The second is the covariance-power trace bound of Lemma A.11.

For $r = 2s \geq 4$, encode the raw derivative labels in the orthonormal word space of grades $0, \dots, j$, with $T := \sum_{i=1}^m N_i \otimes L_i$ and $\|T\| \leq \sqrt{a}$. Let V embed the original space into the vacuum grade. Split a coefficient word as LR with s common factors on each side, and $j_L + j_R = j$ raw factors. The expected block Gram matrix gives

$$\|\mathbb{E}_\mu[LRV]\|_{\text{HS}}^2 \leq \|\mathbb{E}_\mu[LL^\top]\| \operatorname{tr}[V^\top \mathbb{E}_\mu[R^\top R] V] \leq Nb_s \operatorname{df}(s) a^\ell \omega_s.$$

The first step is its Schur-complement bound. For the second, Lemma A.19 bounds the first factor by $b_s a^{s+j_L}$. Vacuum compression of the second is exactly contraction against distinct independent raw directions. The completely positive word argument in Lemma A.10 bounds it by $a^{j_R} \mathbb{E}_\mu[\operatorname{tr}[X^{2s}]]$, and Lemma A.11 bounds this trace by $N \operatorname{df}(s) a^s \omega_s$.

There are $\binom{\ell}{j}$ placements. The full ordered derivative array is $j!$ times their orthogonal symmetrization, including repeated derivative indices. The triangle inequality therefore contributes $(j! \binom{\ell}{j})^2 = (\ell!/r!)^2$. The ordinary second-moment bound also gives coefficient $(\ell!/r!)^2 \operatorname{df}(r) \omega_r$, and $\omega_r \leq \omega_{r/2}$. Taking the smaller bound proves the assertion. At the terminal derivative no common random factor remains, so no ω_q is inserted. For $r = 2, 4, 6, 8$, the coefficients before the factorial factor are at most 1, 9, 225, 13545, respectively. Since $\omega_{k(r)} \leq \omega_2$, these imply Eq. (118) with the stated conservative constants. $\square$

We insert the covariance-sensitive derivative means into the trace recurrence while retaining the established operator bounds.

Definition A.26 (Covariance-sensitive trace recurrence). Fix the final operator tuple of Definition A.23 and $\rho \geq 0$. Use Eq. (117) with

$$(p, J_*, A_*, k_*) := (p_2, J, J(1+a), k_2), \quad m_r := M_r \rho / (r!)^2 \quad (r \text{ even}).$$

Define $F_\ell(a, \rho) := (\ell!)^2 V_\ell$ for $1 \leq \ell \leq 9$.

Concavity in the covariance parameter provides affine upper bounds that can be integrated and certified with nonnegative coefficients.

Lemma A.27 (Concave trace energies and affine bounds). For $1 \leq \ell \leq 9$,

$$N^{-1} \mathbb{E}_{\mu_t} [\text{tr}[X^{2\ell} Y_+^{-2}]] \leq a^\ell F_\ell(a, \omega_2). \quad (119)$$

For fixed a , $F_\ell(a, \rho)$ is nonnegative, increasing, and concave in $\rho \geq 0$. For fixed ρ it is nondecreasing in a . At any $\rho_0 > 0$, put

$$\mathcal{A}_\ell := F_\ell(a, \rho_0) - \rho_0 \partial_\rho F_\ell(a, \rho_0), \quad \mathcal{B}_\ell := \partial_\rho F_\ell(a, \rho_0). \quad (120)$$

Both coefficients are nonnegative and $F_\ell(a, \rho) \leq \mathcal{A}_\ell + \mathcal{B}_\ell \rho$ for all $\rho \geq 0$.

Proof. Apply Lemma A.18 to ordered derivative arrays. The top array is constant, and the odd projected-mean estimate is unchanged. For even arrays use Lemmas A.25 and A.20. Normalization by $N(\ell!)^2 a^\ell$ gives Definition A.26, proving Eq. (119).

For concavity, expand the even branch as $Z_r = Jm_r + k_2 D_r + 2\sqrt{Jk_2 m_r D_r}$. The geometric mean is concave and nondecreasing in its nonnegative arguments. Induction proves nonnegativity, monotonicity, and concavity in ρ . The same recurrence is nondecreasing in the valid operator tuple, proving monotonicity in a . A differentiable concave function lies below its tangent; its increasing property gives $\mathcal{B}_\ell \geq 0$, and evaluation at zero gives $\mathcal{A}_\ell \geq F_\ell(a, 0) \geq 0$. At a degenerate zero variance use continuous limits. $\square$

On a radial cell we first increase a to its right endpoint and then take the tangent there. Separate monotonicity of $\mathcal{A}_\ell$ and $\mathcal{B}_\ell$ is not required. Replacing either coefficient by an outward upper bound preserves the affine estimate.

Combining the two spectral edges gives a sharper bound for the untruncated radial remainder.

Lemma A.28 (Reflected radial remainder). Let $d := 2K - \ell \geq 2$ and $L := d - 1$. If $D_\ell(a)$ is either $\widehat{C}_\ell^{(2)}(a)$ or $F_\ell(a, \omega_2)$, then

$$R_K(t) := N^{-1} \mathbb{E}_{\mu_t} [\text{tr}[X^{2K} (I - X^2)^{-1}]] \leq a^K \sqrt{\frac{\text{df}(d) D_\ell(a)}{1 + (2d - 1)a}} \sqrt{\omega_L}. \quad (121)$$

Proof. Let ϱ be the expected normalized absolute-eigenvalue distribution of X , and put $m_j := \int r^{2j} d\varrho(r)$. Evenness and the trace energy bound give

$$\int \frac{r^{2\ell}(1+r^2)}{(1-r^2)^2} d\varrho(r) \leq a^\ell D_\ell(a).$$

Cauchy–Schwarz bounds $R_K(t)^2$ by this quantity times $\int r^{2d}/(1+r^2) d\varrho$. Concavity of $z/(1+z)$ under the probability measure $r^{2d-2} d\varrho/m_{d-1}$ gives

$$\int \frac{r^{2d}}{1+r^2} d\varrho \leq \frac{m_{d-1} m_d}{m_{d-1} + m_d} \leq \frac{\text{df}(d) a^d}{1 + (2d - 1)a} \omega_{d-1}.$$

The second step uses monotonicity of $xy/(x+y)$, Eq. (116), and $\omega_d \leq \omega_{d-1}$. The zero-moment case is immediate. Multiplying the bounds and using $d + \ell = 2K$ proves Eq. (121). $\square$

A.10 Positive radial majorants and a rank-sensitive Gaussian transfer

Write the barrier as $\phi(u) = \sum_{j=1}^{K-1} b_j u^{2j}/j + b_T \sum_{j=K}^{\infty} u^{2j}/j$, with nonnegative coefficients, $\kappa = b_1 > 0$, and $b_T > 2$. With $v = t^2$, radial differentiation gives

$$-N^{-1} \log Z_1 = \int_0^1 N^{-1} \mathbb{E}_{\mu_{\sqrt{v}}}[\text{tr}[g(X^2)]] \frac{dv}{v}, \quad g(x) := \sum_{j=2}^{K-1} b_j x^j + \frac{b_T x^K}{1-x}.$$

Boundary vanishing and the integrable bounds below justify differentiation and integration, as in Lemma A.12.

Positive scalar majorants decompose the radial integrand into terms controlled by ordinary moments and resolvent energies.

Lemma A.29 (Positive scalar majorants). *Suppose nonnegative rational coefficients $\alpha_j, \beta_\ell, \gamma_\ell$ satisfy*

$$g(x) \leq \sum_j \alpha_j x^j + \sum_\ell \beta_\ell \frac{x^\ell(1+x)}{(1-x)^2} + \sum_\ell \gamma_\ell \frac{x^\ell}{1-x} \quad (0 \leq x < 1), \quad (122)$$

where ordinary degrees satisfy $j \geq 2$ and weighted degrees satisfy $1 \leq \ell \leq 9$. On a radial interval $[v_-, v_+]$, evaluate the tangents of Lemma A.27 at $a_+ := \eta v_+$ and any fixed positive tangent point. The contributions to an upper bound on the radial integral over this interval are

$$\begin{aligned} & \alpha_j \text{df}(j) \eta^j (v_+^j - v_-^j) \omega_j / j, \\ & \beta_\ell \eta^\ell (v_+^\ell - v_-^\ell) (\mathcal{A}_\ell + \mathcal{B}_\ell \omega_2) / \ell, \\ & \gamma_\ell \eta^\ell (v_+^\ell - v_-^\ell) \widehat{C}_\ell^{(1)}(a_+) / \ell. \end{aligned}$$

Proof. Apply Eq. (116) to the ordinary terms. Evenness gives

$$N^{-1} \mathbb{E}[\text{tr}[X^{2\ell}(I + X^2)(I - X^2)^{-2}]] = N^{-1} \mathbb{E}[\text{tr}[X^{2\ell} Y_+^{-2}]].$$

For the two-slot terms use Lemma A.27, first increasing the variance to a_+ and then taking its tangent. For one-slot terms use Lemma A.24. Integrating the remaining powers exactly against dv/v gives the three expressions. All coefficients are nonnegative.

To verify Eq. (122), it suffices to check the following polynomial on the entire closed interval $[0, 1]$:

$$\begin{aligned} & (1-x)^2 \sum_j \alpha_j x^j + (1+x) \sum_\ell \beta_\ell x^\ell + (1-x) \sum_\ell \gamma_\ell x^\ell \\ & - (1-x)^2 \sum_{j=2}^{K-1} b_j x^j - b_T x^K (1-x) \geq 0. \end{aligned}$$

This is multiplication by the positive denominator $(1-x)^2$ on $[0, 1)$. Removing a common non-negative power of x and applying the Bernstein criterion from Lemma A.16 proves the inequality without omitting the spectral edge. $\square$

For the first three stages, apply Lemma A.29 on $[1/4, 1]$. On $[0, 1/4]$ retain the finite moments and use Lemma A.28 with $\widehat{C}_\ell^{(2)}$. For $v_i := (i/256)^2/4$, the early remainder is at most $T_{\text{early}} \sqrt{\omega_L}$, where $d := 2K - \ell$, $L := d - 1$, and

$$T_{\text{early}} := \frac{b_T \eta^K \sqrt{\text{df}(d)}}{K} \sum_{i=1}^{256} (v_i^K - v_{i-1}^K) \sqrt{\frac{\widehat{C}_\ell^{(2)}(\eta v_i)}{1 + (2d-1)\eta v_{i-1}}}.$$

Its finite moment coefficients are $b_j \text{df}(j) \eta^j 4^{-j}/j$. The numerator is bounded at the right endpoint and the reciprocal denominator at the left; the power integral is exact. Young's inequality gives $T_{\text{early}} \sqrt{\omega_L} \leq T_{\text{early}} \omega_L / (2z) + T_{\text{early}} z/2$ for $z > 0$.

For the last four stages, use 2048 quadratic cells $v_i := (i/2048)^2$ on $[0, 1]$. Take tangents at $\rho_0 = 0.65$. Let A_i, B_i be nonnegative upper bounds for the degree- ℓ tangent coefficients at ηv_i , and put

$$h_i := \frac{b_T \eta^K \sqrt{\text{df}(d)}}{K} \frac{v_i^K - v_{i-1}^K}{\sqrt{1 + (2d-1)\eta v_{i-1}}}, \quad T_i := h_i \sqrt{A_i + B_i},$$

$$T := \sum_i T_i, \quad T_A := \sum_i \frac{T_i A_i}{A_i + B_i}, \quad T_B := \sum_i \frac{T_i B_i}{A_i + B_i}.$$

A zero denominator contributes zero and is omitted. Lemma A.28 bounds the cell remainder by $T_i \sqrt{\omega_L (A_i + B_i \omega_2) / (A_i + B_i)}$. Young's inequality on each cell and summation therefore give

$$\text{remainder} \leq \frac{T}{2z} \omega_L + \frac{z}{2} (T_A + T_B \omega_2). \quad (123)$$

The finite terms have coefficients $b_j \text{df}(j) \eta^j / j$. Combining either construction gives

$$-N^{-1} \log \nu[\|Y\| < 1] \leq C + \sum_{j \geq 2} P_j \omega_j + \frac{T}{2z} \omega_L, \quad C, P_j, T \geq 0. \quad (124)$$

For a majorant stage, C includes $T_{\text{early}} z/2$ and $T = T_{\text{early}}$. For a reflected stage, $C = zT_A/2$ and $zT_B/2$ is added to P_2 ; there is no further $Tz/2$ term. The passage from Z_1 to the probability uses $Z_1 \leq \nu[\|Y\| < 1]$, since the residual potential is nonnegative inside and infinite outside the domain.

For either radial construction, apply Lemma A.13 with $Q_j = P_j + \mathbb{1}[j = L]T/(2z)$ and $\text{rank } \Gamma \leq m \leq wN$. Define

$$\mathcal{P}(x) := L_H(x) + \frac{H}{(H-1)\kappa\eta} \left(\sum_{j \geq 2} \frac{P_j}{(1+x)^j} + \frac{T}{2z(1+x)^L} \right),$$

the scalar condition is

$$\mathcal{P}(x) - b/x \leq B_{\text{cap}} \quad (x > 0). \quad (125)$$

The constant C remains outside the spectral average; the exponent is Eq. (69).

A.11 Exact data for the sub-10 construction

The seven stages use fixed spectral parameters and radial budgets. We specify the first three majorant families by finite rational linear systems after the stage parameters.

Definition A.30 (Existence-stage parameters). In the remainder of this appendix, the stage parameters refer to the following separate family. For $k = 0, 1, 2$, define

$$\phi^{[k]}(u) := \sum_{j=1}^{K_k-1} b_{k,j} u^{2j}/j + b_{k,T} \sum_{j=K_k}^{\infty} u^{2j}/j.$$

The nonzero finite coefficients and tail coefficients are

$$k = 0 : \quad b_{0,1} = 0.07085499, \quad b_{0,3} = 0.58222347,$$

$$\begin{aligned}
b_{0,6} &= 22.10545334, & b_{0,7} &= 14.50511282, & b_{0,T} &= 22.83787582; \\
k = 1 : & \quad b_{1,1} = 0.001322, & b_{1,3} &= 0.02099, & b_{1,14} &= 61900, & b_{1,T} &= 1057; \\
k = 2 : & \quad b_{2,1} = 0.00001267, & b_{2,2} &= 0.00003561, & b_{2,3} &= 0.00004397, \\
& \quad b_{2,22} = 58162000, & b_{2,23} &= 54687000, & b_{2,T} &= 91750.
\end{aligned}$$

For stage zero use $\zeta = 0.47982971$, $\gamma = 1.02221683$, and

$$d(u) := \min\{\gamma, \sqrt{(1-u)/\zeta}\}, \quad a_0 := 1 - \zeta\gamma^2 = 0.498612861844063305535181.$$

For stages one and two use Definition A.4 with

$$(\zeta_1, \theta_1) := (0.32288046, 0.0009507852), \quad (\zeta_2, \theta_2) := (0.27359432, 0.00001093325).$$

For $k = 3, 4, 5, 6$, use exactly the two-component spectral barrier parameters $(\tau_k, \zeta_k, \kappa_{0,k}, \kappa_{\text{tail},k}, \theta_k)$ of Definition A.15. Only these spectral parameters are inherited. At such a stage, set $b_{k,j} := \kappa_{0,k} + \kappa_{\text{tail},k} \mathbb{1}[j \geq \tau_k]$ and $b_{k,T} := \kappa_{0,k} + \kappa_{\text{tail},k}$. At every stage $\kappa = b_{k,1}$.

The radial degrees are

$$(K_k, \ell_k)_{k=0}^6 := ((8, 4), (17, 7), (25, 9), (34, 9), (44, 9), (56, 9), (63, 5)).$$

The variance, retention, and cost parameters are in Table 5; the transfer parameters are in Table 4. Define the rank-correction parameters

$$\begin{aligned}
b_0^{\text{rk}} &:= 8.257020779696646 \cdot 10^{-6}, \\
b_1^{\text{rk}} &:= 1.260355005129572 \cdot 10^{-6}, \\
b_2^{\text{rk}} &:= 1.598055954762951 \cdot 10^{-7},
\end{aligned}$$

and $b_3^{\text{rk}} = b_4^{\text{rk}} = b_5^{\text{rk}} = b_6^{\text{rk}} := 0$.

For stages zero, one, and two, use respectively 192, 64, 64 uniform majorant intervals on $[1/4, 1]$, subdividing each into 8, 16, 16 equal radial subintervals for the energy upper sums. Definition A.31 specifies their exact coefficients and tangent points. Stages three through six use Eq. (123) with 2048 quadratic cells and tangent point 0.65. Every operator refinement uses exactly six passes.

k	H_k	z_k	$B_{\text{cap},k}$
0	5.45444597	0.73257320	0.461415
1	5.53920133	0.73298851	0.446524
2	5.54909934	0.69131064	0.464189
3	5.33109076	0.72583701	0.439229
4	2.61968753	0.56550013	0.338323
5	1.15232869	0.18305585	0.238652
6	1.02752192	0.03589337	0.222120

Table 4: Exact transfer parameters for the existence construction.

The radial coefficients are specified by three rational linear systems. Their constraints combine polynomial domination with the scalar budgets needed by the Gaussian transfer.

k	η_k	q_k	c_k
0	0.04419261	0.0578850350	7.052711159929
1	0.0217136	0.0223838957	2.420740060029
2	0.01367132	0.0119572705	0.456432632054
3	0.00869674	0.0067284670	0.062577714042
4	0.00646514	0.0046741066	0.005953427019
5	0.00493509	0.0040355636	0.000465862518
6	0.0005	0.0016523300	0.000092976331
Entire tail	—	—	0.000591907040
Total	—	—	9.999565738962

Table 5: Exact stage parameters and upward discrepancy costs divided by $\sqrt{n}$.

Definition A.31 (A finite construction of the radial majorants). Set $(m_0, m_1, m_2) := (192, 64, 64)$, $(d_0, d_1, d_2) := (24, 40, 56)$, and $(s_0, s_1, s_2) := (8, 16, 16)$. The tangent points are

$$(\rho_0, \rho_1, \rho_2) := (0.6709927841045108, 0.625, 0.625).$$

Here m_k is the number of radial intervals, d_k bounds the ordinary degree, and s_k is the number of energy subintervals. At a fixed stage suppress k and put $v_i := 1/4 + 3i/(4m)$ for $0 \leq i \leq m$. The unknowns on interval i are

$$0 \leq \alpha_{i,j}, \beta_{i,\ell}, \gamma_{i,\ell} \leq 10^{20} \quad (2 \leq j \leq d, \quad 1 \leq \ell \leq 9, \quad 0 \leq i < m).$$

Every unlisted coefficient is zero. Define the cleared polynomial

$$\begin{aligned} p_i(x) := & (1-x)^2 \sum_{j=2}^d \alpha_{i,j} x^{j-1} + (1+x) \sum_{\ell=1}^9 \beta_{i,\ell} x^{\ell-1} + (1-x) \sum_{\ell=1}^9 \gamma_{i,\ell} x^{\ell-1} \\ & - (1-x)^2 \sum_{j=2}^{K-1} b_j x^{j-1} - b_T x^{K-1} (1-x). \end{aligned} \tag{126}$$

Require every degree- $(d+1)$ Bernstein coefficient of p_i on every interval $[a2^{-48}, (a+1)2^{-48}]$, $0 \leq a < 2^{48}$, to be nonnegative. These coefficients are the rational linear expressions obtained from Lemma A.16 by replacing the cell count 32 by 2^{48} and the degree by $d+1$.

We next specify the rational coefficients of the radial constraints. For a real number y , put $\mathcal{U}(y) := 10^{-60} \lceil 10^{60} y \rceil$. The enclosure rules of Lemma A.16, refined as needed, determine these upward endpoints. Let $\mathcal{A}_\ell(a; \rho)$ and $\mathcal{B}_\ell(a; \rho)$ be the exact tangent coefficients of Lemma A.27, and put

$$t_{i,r} := v_i + (v_{i+1} - v_i)r/s \quad (0 \leq r \leq s).$$

Define the following nonnegative rational weights:

$$o_{i,j} := \mathcal{U}(\text{df}(j)\eta^j(v_{i+1}^j - v_i^j)/j), \tag{127}$$

$$a_{i,\ell} := \mathcal{U}\left(\frac{\eta^\ell}{\ell} \sum_{r=1}^s (t_{i,r}^\ell - t_{i,r-1}^\ell) \mathcal{A}_\ell(\eta t_{i,r}; \rho)\right), \tag{128}$$

$$b_{i,\ell}^{\text{lin}} := \mathcal{U}\left(\frac{\eta^\ell}{\ell} \sum_{r=1}^s (t_{i,r}^\ell - t_{i,r-1}^\ell) \mathcal{B}_\ell(\eta t_{i,r}; \rho)\right), \tag{129}$$

$$g_{i,\ell} := \mathcal{U}\left(\frac{\eta^\ell}{\ell} \sum_{r=1}^s (t_{i,r}^\ell - t_{i,r-1}^\ell) \widehat{C}_\ell^{(1)}(\eta t_{i,r})\right). \quad (130)$$

With the early remainder from Section A.10, set $T^+ := \mathcal{U}(T_{\text{early}})$, $L := 2K - \ell_k - 1$, and

$$P_j := \mathbb{1}[j < K] b_j \text{df}(j) \eta^j 4^{-j} / j + \sum_{i=0}^{m-1} o_{i,j} \alpha_{i,j} + \mathbb{1}[j = 2] \sum_{i=0}^{m-1} \sum_{\ell=1}^9 b_{i,\ell}^{\text{lin}} \beta_{i,\ell}, \quad (131)$$

$$C := T^+ z / 2 + \sum_{i=0}^{m-1} \sum_{\ell=1}^9 (a_{i,\ell} \beta_{i,\ell} + g_{i,\ell} \gamma_{i,\ell}). \quad (132)$$

Here $2 \leq j \leq d$, and $b_j := 0$ for $K \leq j \leq d$ in the first term. Define

$$Q(x) := \frac{H}{(H-1)\kappa\eta} \left(\sum_{j=2}^d \frac{P_j}{(1+x)^j} + \frac{T^+}{2z(1+x)^L} \right), \quad \varepsilon := 10^{-8}, \quad \delta := 10^{-12}.$$

For $x_a := \varepsilon + (100 - \varepsilon)a2^{-28}$, $0 \leq a \leq 2^{28}$, impose

$$H\varepsilon/2 + Q(0) - b_k^{\text{rk}}/\varepsilon \leq B_{\text{cap}} - \delta, \quad (133)$$

$$\mathcal{U}(L_H((x_a + x_{a+1})/2)) + H(x_{a+1} - x_a)/4 + Q(x_a) - b_k^{\text{rk}}/x_{a+1} \leq B_{\text{cap}} - \delta \quad (0 \leq a < 2^{28}), \quad (134)$$

$$\mathcal{U}(\log(101)/100) + Q(100) \leq B_{\text{cap}} - \delta. \quad (135)$$

The final linear constraint is

$$\kappa\eta B_{\text{cap}} + b_k^{\text{rk}} w_k / 2 + HC / (H - 1) \leq \bar{e}_k, \quad (136)$$

where $w_0 := 1$, $w_{k+1} := q_k w_k$, and

$$(\bar{e}_0, \bar{e}_1, \bar{e}_2) := (0.00167627421655443, 0.0000145025346222408, 9.26288824926953 \cdot 10^{-8}).$$

These bounds define a rational polytope $\mathfrak{P}_k$. Order its coordinates by increasing i , then by α , β , γ , and then by increasing degree. The majorant coefficients are the lexicographically least point of $\mathfrak{P}_k$, obtained by minimizing one coordinate at a time. Lemma A.32 establishes nonemptiness. Boundedness gives a minimum at every step, and rational linear optimization gives a unique rational point. Equivalently, enumerate the full-rank sets of active constraints, solve their rational linear systems, retain the feasible vertices, and select the least vertex in this order.

For example, stage zero begins with $[v_0, v_1] = [1/4, 65/256]$ and eight equal energy subintervals. Its first majorant is

$$\sum_{j=2}^{24} \alpha_{0,j} x^j + \sum_{\ell=1}^9 \beta_{0,\ell} \frac{x^\ell (1+x)}{(1-x)^2} + \sum_{\ell=1}^9 \gamma_{0,\ell} \frac{x^\ell}{1-x}.$$

Its coefficients are the first block of the point specified in Definition A.31. The same rule gives the next 191 blocks and the 64 blocks at each of stages one and two.

The certificate checks curvature on complete spectral domains and verifies the rational systems that define the radial bounds. The resulting exponents fit the fixed coding budgets.

Lemma A.32 (All-domain certificates for the existence stages). *Put $w_0 := 1$, $w_{k+1} := q_k w_k$, and $E_k := q_k^2/2 + q_k^4/12$, with the parameters of Definition A.30. The polytopes of Definition A.31 are nonempty. Every stage satisfies the curvature and strict-root hypotheses, the majorant or reflected radial bounds, and Eq. (125) on its entire domain. Its exponent in Eq. (69), using $b = b_k^{\text{rk}}$ and $w = w_k$, satisfies $e_k < w_k E_k$. The stage costs satisfy $c_k > \sqrt{w_k/\eta_k}/s_\star$.*

Proof. We give bounded recursions for the curvature checks and rational feasibility conditions for the radial construction. All arithmetic comparisons use the enclosure rules of Lemma A.16.

Step 1: curvature on all spectral triples. For stage zero use Lemma A.22. The multiplier obeys $d(u)^2/(1-u) \leq \zeta^{-1}$ even though $\gamma > 1$. If all coordinates are regular, meaning at most a_0 , then $l_u = l_v = l_w = 1 - \gamma^4$. Set $t_0 := \gamma^4 - 1$. The bound $G(u, v) \geq t_0$ on regular pairs implies

$$M = t_0 \mathbf{1}\mathbf{1}^\top + \text{diag}(G(u, v) - t_0, G(u, w) - t_0, G(v, w) - t_0) \succeq 0.$$

As in Lemma A.14, G is a weighted average of $(1-u)^2\phi''(u)$. For $u \leq 0$ this quantity is at least $2b_{0,1} > t_0$. For $0 \leq u \leq a_0$, clearing $(1+u)^2$ gives the polynomial

$$\begin{aligned} & \sum_{j < K} 2b_j(2j-1)u^{2j-2}(1-u^2)^2 \\ & + 2b_T u^{2K-2}((2K-1) - (2K-3)u^2) - t_0(1+u)^2. \end{aligned}$$

After $u = a_0 s$, its Bernstein coefficients on the two half-intervals are positive.

For triples with an active coordinate, use the parameterizations

$$\text{N : } u = -s, \quad \text{P : } u = a_0 s, \quad \text{A : } u = 1 - \zeta\gamma^2 s^2, \quad 0 \leq s \leq 1.$$

The corresponding multipliers are $\gamma, \gamma, \gamma s$. Permutations reduce the domain to the six cubes NNA, NPA, PPA, NAA, PAA, AAA. Here are explicit diagonal lower bounds on each box. Write $p := 2K - 2$ and $Q_r(x, y) := \sum_{i=0}^r x^i y^{r-i}$. If the nonnegative moduli lie in $[x_0, x_1]$ and $[y_0, y_1]$, set

$$q_* := \min\{x_0 y_0(1 - x_0 y_0), x_1 y_1(1 - x_1 y_1)\}, \quad B_* := x_0^p + y_0^p + q_* Q_{p-2}(x_0, y_0).$$

For two nonnegative eigenvalues a lower bound for G is

$$2(1-x_1)(1-y_1) \sum_{j < K} b_j Q_{2j-2}(x_0, y_0) + \frac{2b_T B_*}{(1+x_1)(1+y_1)}.$$

For two nonpositive eigenvalues a lower bound is

$$2(1+x_0)(1+y_0) \sum_{j < K} b_j Q_{2j-2}(x_0, y_0) + \frac{2b_T B_*}{(1-x_0)(1-y_0)}.$$

These follow from $Q_p - x^2 y^2 Q_{p-2} = x^p + y^p + xy(1-xy)Q_{p-2}$, positivity of the coefficients, and concavity of $z(1-z)$ on the product interval. For a nonnegative eigenvalue x and a nonpositive eigenvalue $-y$, use

$$\begin{aligned} & 2(1-x_1)(1+y_0)(b_1 + \sum_{2 \leq j < K} b_j \frac{x_0^{2j-1} + y_0^{2j-1}}{x_1 + y_1}) \\ & + 2b_T (\frac{x_0^{p+1}(1+y_1)}{(x_0+y_1)(1+x_0)} + \frac{y_0^{p+1}(1-x_1)}{(x_1+y_0)(1-y_0)}). \end{aligned}$$

The two tail terms are respectively increasing in x , decreasing in y , and increasing in y , decreasing in x . A zero endpoint numerator contributes the lower bound zero. All denominators evaluated on accepted nondegenerate cells are positive.

Let $D_1, D_2, D_3 > 0$ be the resulting lower bounds for $G(u, v), G(u, w), G(v, w)$. Enclose l_u, l_v, l_w in intervals I_1, I_2, I_3 , put $s_i := \max_{x \in I_i} x^2$, and let P_* be the largest product of three interval endpoints, one from each I_i . A leaf is accepted only if

$$\begin{aligned} D_1 D_2 &> s_1, & D_1 D_3 &> s_2, & D_2 D_3 &> s_3, \\ D_1 D_2 D_3 &> D_1 s_3 + D_2 s_2 + D_3 s_1 + 2P_*. \end{aligned} \quad (137)$$

These comparisons prove positivity of all principal minors with diagonal D_i and the actual off-diagonal entries. Raising the diagonal to the actual G preserves positive definiteness. The product P_* retains its sign.

These rational box comparisons determine the subdivision directly. Let $\chi(B) := 1$ when all denominators are positive, all $D_i > 0$, and Eq. (137) holds on B ; otherwise put $\chi(B) := 0$. If $B_{j,-}$ and $B_{j,+}$ are its two halves in coordinate j , define

$$\begin{aligned} v(B, 0) &:= \chi(B), \\ v(B, d) &:= \max\{\chi(B), \max_{j \in \{1, 2, 3\}} v(B_{j,-}, d-1) v(B_{j,+}, d-1)\} \quad (d \geq 1). \end{aligned} \quad (138)$$

Induction on d shows that $v(B, d) = 1$ exactly when B has a verified cover of depth at most d : either B passes or both halves pass at depth $d-1$. Retain the region parameterization and accept passing boxes immediately; otherwise choose the first successful coordinate in the order 1, 2, 3.

Exact rational box comparisons give

$$v([0, 1]^3, d_{\mathcal{R}}) = 1, \quad (d_{\text{NNA}}, d_{\text{NPA}}, d_{\text{PPA}}, d_{\text{NAA}}, d_{\text{PAA}}, d_{\text{AAA}}) := (15, 20, 28, 34, 23, 17). \quad (139)$$

The recursion starts from each entire cube. Unresolved comparisons and zero denominators trigger subdivision, or return zero at depth zero.

Step 2: the remaining spectral barriers. At stages one through six it suffices to verify Eq. (62). For the custom barriers, multiplying its left side by $(1+u)(1+v)$ gives

$$2(1-u^2)(1-v^2) \sum_{j < K} b_j Q_{2j-2}(u, v) + 2b_T(Q_{2K-2}(u, v) - u^2 v^2 Q_{2K-4}(u, v)).$$

For a two-component barrier it gives $2\kappa_0(1+uv) + 2\kappa_{\text{tail}}(Q_{2\tau-2} - u^2 v^2 Q_{2\tau-4})$. Subtract each of the two cleared branches of F_θ on the following three squares:

	u	v	$d_\zeta(u)d_\zeta(v)$	
regular-regular	$-1 + (2-\zeta)s$	$-1 + (2-\zeta)t$	1	
active-regular	$1 - \zeta s^2$	$-1 + (2-\zeta)t$	s	
active-active	$1 - \zeta s^2$	$1 - \zeta t^2$	st	$(0 \leq s, t \leq 1).$

The resulting 36 rational bivariate polynomials are nonnegative by tensor Bernstein subdivision. Accept a box when all coefficients are nonnegative; otherwise bisect the coordinate with fewer previous bisections, choosing the first coordinate in a tie. For a univariate polynomial always bisect its sole coordinate. All these checks terminate within total depth 40. Boundary zeros are permitted because the original spectral domain is open. Symmetry and reflection cover the other orderings and edge.

Step 3: root and radial certificates. The initial root quantities at the largest variance of every stage satisfy

$$1 - 2\eta/\zeta^2 > 0.4823, \quad 1 - 4p_1\eta > 0.7812, \quad A_J > 0.9962, \quad \Delta_J > 0.6701.$$

Thus all roots and six refinement passes are valid at every smaller radial variance. For the first three stages, the systems in Definition A.31 have respectively 7872, 3648, and 4672 variables. Exact rational feasibility gives

$$\mathfrak{P}_0 \neq \emptyset, \quad \mathfrak{P}_1 \neq \emptyset, \quad \mathfrak{P}_2 \neq \emptyset.$$

Apply the finite vertex-selection rule of Definition A.31 to construct the coefficients.

Test the Bernstein constraints by midpoint de Casteljau averaging. A box with nonnegative coefficients certifies all its descendants at depth 48; otherwise subdivide to that depth. This evaluates exactly the prescribed constraints. Multiplication by $x \geq 0$ then gives the polynomial criterion of Lemma A.29.

For the radial weights, differentiate the recurrence in ρ with the operator tuple fixed. Product and square-root interval rules determine the upward endpoints $\mathcal{U}$; Lemma A.27 proves their non-negativity. Lemma A.29 gives C, P_j on $[1/4, 1]$, and the early reflected sums cover $[0, 1/4]$. The last four stages use the 2048 cells of Section A.10.

Step 4: the complete Gaussian-transfer half-line. Write $Q(x) := \mathcal{P}(x) - L_H(x)$; it is nonnegative and decreasing. The integral representation from Lemma A.16 gives $L_H(x) \leq Hx/2$ and $|L'_H(x)| \leq H/2$. For $b > 0$ and $\epsilon := 10^{-8}$, an upper bound on $(0, \epsilon]$ is $H\epsilon/2 + Q(0) - b/\epsilon$. On a cell $[a, c] \subset [\epsilon, 100]$ use

$$L_H((a+c)/2) + H(c-a)/4 + Q(a) - b/c.$$

For $b = 0$, start at zero and include $Q(0)$. For $x \geq 100$, drop the nonpositive correction and use $L_H(x) \leq \log(1+x)/x$; this follows from $\log(1+Hx) \geq \log(1+x)$. Both this envelope and Q decrease, so the endpoint suffices. For the first three stages, these are the constraints of Definition A.31, with reserve 10^{-12} . Evaluate the 2^{28} cell constraints by bisection from $[\epsilon, 100]$. The Lipschitz bound and monotonicity of Q and $-b/x$ let a passing cell certify its descendants, allowing 10^{-60} for their upward-rounded logarithmic terms.

For the last four stages, start from $[0, 100]$ and bisect every unresolved cell at its midpoint. Outward evaluation accepts every cell by depth 28. The comparisons at zero and beyond 100 are checked separately by the stated endpoint bounds. All scalar comparisons use rational enclosures for square roots and logarithms; refining the enclosures determines the strict inequalities and the prescribed upward rounding.

Step 5: entropy and costs. The following are outward rational upper bounds on the computed exponents:

k	$\bar{e}_k$
0	0.00167627421655443
1	0.0000145025346222408
2	$9.26288824926953 \cdot 10^{-8}$
3	$3.50703055350984 \cdot 10^{-10}$
4	$1.13872520877497 \cdot 10^{-12}$
5	$3.96760383059514 \cdot 10^{-15}$
6	$2.68421117846414 \cdot 10^{-18}$

For stages zero through two, Eq. (136) gives $e_k \leq \bar{e}_k$ for the constructed coefficients. For stages three through six, outward evaluation gives the displayed upper bounds. Exact rational comparison then gives $\bar{e}_k < w_k(q_k^2/2 + q_k^4/12)$ for every stage. Each phase-cost rounding is checked by the exact rational comparison $c_k^2 s_\star^2 \eta_k > w_k$. The coding entropy series in Lemma A.3 gives

$$\log 2 - h((1-q)/2) > q^2/2 + q^4/12 \quad (0 < q < 1).$$

This proves the required strict coding reserve and completes the certificate. $\square$

A.12 Hereditary scaling and the complete sub-10 signing

The certified stage bounds must remain valid for every smaller active set encountered during partial signing.

Lemma A.33 (Hereditary existence-stage interface). *Let $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ be symmetric contractions. For the parameters of Definition A.30 and every nonempty active set S of size $m \leq w_k n$, the closed symmetric convex set*

$$\mathcal{K}_{S,k} := \{x \in \mathbb{R}^S : \|\sum_{i \in S} x_i A_i\| \leq \sqrt{w_k n / \eta_k}\}$$

has Gaussian measure at least e^{-mE_k} , where $E_k = q_k^2/2 + q_k^4/12$.

Proof. Normalize by $B_i := A_i / \sqrt{w_k n / \eta_k}$ and put $\alpha := m / (w_k n) \leq 1$. The variance is at most $\alpha \eta_k I$. In the scalar-majorant construction retain the same polynomials and freeze the tangent bounds at their original right endpoints. Monotonicity of the energy functions makes them valid at the smaller variance. Ordinary radial contributions scale as α^j for $j \geq 2$, and weighted contributions as α^ℓ for $\ell \geq 1$. Dividing their coefficients by the new variance $\alpha \eta_k$ therefore does not increase the scalar profile, including when a weighted contribution is assigned to ω_2 rather than ω_ℓ .

For a reflected remainder the inequality $1 + c\alpha\eta v \geq \alpha(1 + c\eta v)$ gives scaling at least $\alpha^{K-1/2}$. With the original affine coefficients frozen, this applies separately to T, T_A, T_B . Since $K \geq 2$, the nonnegative radial constants and profile contributions all decrease by at least a factor α . The determinant part also scales by α . Finally the rank contribution in Lemma A.13 is at most $b_k^{\text{rk}} m / (2n) = \alpha b_k^{\text{rk}} w_k / 2$. Thus the complete exponent at the smaller variance is at most αe_k , and

$$\gamma_m[\|Y\| < 1] \geq e^{-n\alpha e_k} \geq e^{-mE_k}.$$

The first step is the rank-sensitive transfer with the scaled bounds, and the second uses $e_k < w_k E_k$ and $n\alpha w_k = m$. Passing to the closed ball can only increase measure. This covers every actual integer active count below the nominal schedule. $\square$

We now combine the seven partial signings with the uniform tail estimate and sum their discrepancy costs.

Lemma A.34 (Complete signing and exact sum). *The seven existence stages and the tail produce a complete signing with discrepancy at most $9.999565738962\sqrt{n}$.*

Proof. Start with all n coordinates active. At stage k , apply Lemma A.33 and then Lemma A.3, using the quartic reserve of Lemma A.32. Fix the nonzero partial signs. The new active count is less than $q_k m \leq w_{k+1} n$, and the increment norm is at most $\sqrt{w_k n / \eta_k} / s_\star \leq c_k \sqrt{n}$. Stop when the active set is empty; the supports are disjoint.

After seven stages the active count is less than $w_7 n$, where the exact product of the rational retentions in Table 5 is

$$w_7 = 3.2490003013687142768816922077757548252229058651306 \cdot 10^{-15}.$$

Set $U := -\log w_7$, $q_T := 0.011$, $\rho_T := \sqrt{q_T}$, and $\lambda := -\log q_T$. The enclosure rules of Lemma A.32 verify $U > 2$ and $U > 731/256$. Since $0.000051714 < q_T^2/2 = 0.0000605$, Lemmas A.17 and A.3 apply at every subsequent nonempty active set. They give increment at most $(185/s_\star)u\sqrt{m}$ and reduce the active count by a factor less than q_T .

The function $ue^{-u/2}$ decreases for $u > 2$. At the j th tail phase the actual aspect ratio is at least $U + j\lambda$, including when earlier phases made extra progress. Consequently the entire remaining discrepancy, divided by $\sqrt{n}$, is at most

$$\begin{aligned} \frac{185}{s_\star} \sqrt{w_7} \sum_{j=0}^{\infty} \rho_T^j (U + j\lambda) &= \frac{185}{s_\star} \sqrt{w_7} \left(\frac{U}{1 - \rho_T} + \frac{\lambda \rho_T}{(1 - \rho_T)^2} \right) \\ &\leq 0.000591907040. \end{aligned}$$

The equality sums the geometric series and its derivative. The inequality uses outward enclosures of the square roots and logarithms, giving an unrounded upper bound less than 0.000591907039679. Every nonempty phase fixes a coordinate, so the construction terminates.

By the triangle inequality, the seven upward-rounded costs in Table 5 and the tail bound give total discrepancy, divided by $\sqrt{n}$, at most

$$\sum_{k=0}^6 c_k + 0.000591907040 = \frac{4999782869481}{500000000000} = 9.999565738962 < 9.9996 < 10.$$

The first equality adds the rational costs; the second expands the decimal. Both comparisons are exact. $\square$

B A smaller constant for the cubic-time algorithm

We combine a sharp Gaussian-width estimate with coupled weighted energies and unbiased binary repair at four fixed levels. The rounding calls a hereditary signing subroutine obtained from Gaussian small-ball estimates and projection toward a point sampled from a Euclidean ball. All projections are implemented by the certified multiplier wrapper and recursive solver of Section 9.

Theorem B.1 (A cubic-time signing with constant 12.40). *There is a randomized algorithm, procedure MAINCUBICSMALLCONSTANT in Algorithm 13, that, for symmetric contractions $A_1, \dots, A_n \in \mathbb{R}^{n \times n}$ and $p \in (0, 1/2)$, returns signs $\varepsilon_1, \dots, \varepsilon_n \in \{-1, 1\}$ satisfying*

$$\left\| \sum_{i=1}^n \varepsilon_i A_i \right\| < 12.40\sqrt{n}$$

with failure probability at most p , using $n^{3+o(1)} \text{polylog}(1/p)$ arithmetic operations in the real-arithmetic model. The same statement holds for $N \times N$ input matrices with $N \leq n$.

Proof. Lemma B.30 constructs a feasible point with small Gaussian-weighted fractional mass, using the Gaussian-width bound of Lemma B.14. Lemma B.31 gives the exact unbiased dyadic identity used in the four-level repair, with the suffix signing subroutine of Lemma B.12 at depth zero. Lemma B.35 proves the discrepancy, failure probability, and running time; Lemma B.37 certifies its scalar bounds. Zero padding handles $N < n$. $\square$

Remark B.2. The small-ball estimates use the spectral allocation and analytic inequalities of Section A, together with the covariance-sensitive refinements proved below. The algorithm computes Euclidean projections; polar bodies and coordinate sections appear only in its analysis. The arithmetic statement does not assert new fixed-point word bounds for Section 10.

B.1 Hereditary radii and a geometric schedule

Every terminating decimal in this appendix denotes an exact rational number. We use the seven small-ball estimates already certified in Appendix A to define the radii for the terminating signing procedure.

Definition B.3 (Inherited small-ball parameters). For $0 \leq j \leq 6$, let q_j and η_j be the coefficients specified in Definition A.15, and put

$$\bar{e}_j := \frac{q_j^2}{2} \prod_{l=0}^{j-1} q_l, \quad \varrho_j := \eta_j^{-1/2}.$$

The empty product equals one. These parameters satisfy $\bar{e}_0 > \dots > \bar{e}_6 > 0$ and $\varrho_0 < \dots < \varrho_6$.

The inherited estimates remain valid when the active set is smaller than its prescribed capacity.

Lemma B.4 (Inherited hereditary Gaussian small balls). *For each $0 \leq j \leq 6$, symmetric matrices $B_i \in \mathbb{R}^{n \times n}$ satisfying $\sum_i B_i^2 \preceq v\eta_j I$ with $0 < v \leq 1$ obey*

$$\Pr\left[\left\| \sum_i \xi_i B_i \right\| < 1\right] \geq \exp(-nv\bar{e}_j), \tag{140}$$

where the ξ_i are independent standard Gaussians.

Proof. Lemma A.16 verifies all seven specializations of Lemma A.13 and gives exponent $e_j < \bar{e}_j$. At variance $v\eta_j$, each finite moment coefficient of degree r decreases by v^r . The radial remainder decreases by at least v^{K_j} , because the energy coefficient is nondecreasing. The determinant term scales linearly, so the same transfer bound gives exponent at most $ve_j < v\bar{e}_j$. $\square$

Linear interpolation between consecutive small-ball estimates defines the finite-radius regimes. A single projection tuple is used in every regime and in the tail.

Definition B.5 (A geometric projection schedule). Fix

$$N_0 := 3 \cdot 10^8, \quad N_* := 10^{24}, \quad \kappa_{\text{pr}} := 10^{-4}, \quad s_{\text{sm}} := \chi_{\text{sm}} := 5 \cdot 10^{-7}. \quad (141)$$

Set $e_{\text{pr}} := 1/32$, $b_j := \bar{e}_j/e_{\text{pr}}$ for $0 \leq j \leq 6$, and

$$(\delta, \alpha, a, v, t) := (1/4, 29/100, 9/20, 481/2000, 3/10). \quad (142)$$

For $0 < w \leq 1$, define the radius coefficient

$$\varrho(w) := \begin{cases} \varrho_0, & b_0 < w \leq 1, \\ \frac{(w - b_{j+1})\varrho_j + (b_j - w)\varrho_{j+1}}{b_j - b_{j+1}}, & b_{j+1} < w \leq b_j, \quad 0 \leq j < 6, \\ 185 \log(1/w), & 0 < w \leq b_6. \end{cases} \quad (143)$$

A call with initial capacity w uses $w_k := w(3/4)^k$ and $R_k := \varrho(w_k)\sqrt{w_k n}$ at stage $k \geq 0$. Its smoothed Gaussian exponent is bounded by

$$E_* := e_{\text{pr}} - \log(1 - 10^{-6}). \quad (144)$$

The radius regimes are selected by rational comparisons with b_j . The parameters a, v in the projection tuple are distinct from the capacities and the transfer parameters of the small-ball estimates.

The radius function supplies the same mass guarantee at every capacity, including after a projection fixes extra coordinates.

Lemma B.6 (Hereditary schedule). *For $0 < w \leq 1$ and any nonempty active set S with $m := |S| \leq wn$, the body*

$$K_S(w) := \{x \in \mathbb{R}^S : \|\sum_{i \in S} x_i A_i\| \leq \varrho(w)\sqrt{wn}\}$$

has Gaussian measure at least $e^{-e_{\text{pr}} m}$. Its radius $R := \varrho(w)\sqrt{wn}$ satisfies $R^2 > 22m$ and $R < 22\sqrt{n}$. The function ϱ is nonincreasing, and

$$34.4693 < \log(1/b_6) < 34.4694. \quad (145)$$

Proof. For an inherited row, put $B_i := \sqrt{\eta_j/(wn)}A_i$ and $v := m/(wn)$. Then $\sum_i B_i^2 \preceq v\eta_j I$, so Lemma B.4 gives Gaussian mass at least $\exp(-m\bar{e}_j/w)$ at radius $\varrho_j\sqrt{wn}$. If $w > b_0$, this exponent is at most e_{pr} .

For $b_{j+1} < w \leq b_j$, put $\lambda := (w - b_{j+1})/(b_j - b_{j+1})$. Convexity gives

$$\lambda K(\varrho_j\sqrt{wn}) + (1 - \lambda)K(\varrho_{j+1}\sqrt{wn}) \subseteq K(\varrho(w)\sqrt{wn}),$$

where $K(R) := \{x : \|\sum_{i \in S} x_i A_i\| \leq R\}$. Gaussian log-concavity, by Prékopa's theorem, gives the mass bound with exponent

$$\frac{\lambda \bar{e}_j + (1 - \lambda) \bar{e}_{j+1}}{w} = e_{\text{pr}}.$$

If $w \leq b_6$, then $u := \log(n/m) \geq \log(1/b_6) > 43 \cdot 17/256$. Lemma A.17 supplies radius $185u\sqrt{m}$ with Gaussian exponent $0.000051714 < e_{\text{pr}}$. The function $s \mapsto \sqrt{s} \log(1/s)$ is increasing on $(0, e^{-2})$. Since $m/n \leq w \leq b_6 < e^{-2}$, the prescribed radius is at least $185u\sqrt{m}$.

The seven variances give $\varrho_j^2 > 22$ and $\varrho_j < 17$. Rational outward bounds give Eq. (145), $185 \log(1/b_6) > \varrho_6$, and $185\sqrt{b_6} \log(1/b_6) < 0.001$. Thus the interpolation pieces and the tail join into a nonincreasing function. The finite pieces give $R^2 > 22wn \geq 22m$ and $R < 17\sqrt{n}$; monotonicity of the tail radius gives its upper bound, and $185 \log(1/w) > \varrho_6$ gives its lower bound. $\square$

B.2 Projection toward a Euclidean-ball sample

For this subsection, let $m \geq N_0$ and let $D \subseteq \mathbb{R}^m$ be symmetric, closed and convex with $\gamma_m(D) \geq e^{-Em}$. Define

$$h(q) := -q \log q - (1 - q) \log(1 - q), \quad J(s) := (s - 1 - \log s)/2,$$

where $h(0) := h(1) := 0$, and put

$$D_{\text{KL}}(p\|z) := p \log(p/z) + (1 - p) \log((1 - p)/(1 - z)).$$

Choose $0 < \delta < \alpha - 1/N_0$, $\alpha < 1/2$, $a \in (0, 1 - \alpha)$, and $v, t > 0$. Write

$$\begin{aligned} \ell &:= \lfloor \alpha m \rfloor, & d &:= m - \ell, & p_* &:= 1 - \alpha, \\ \alpha_- &:= \alpha - 1/N_0, & p_0 &:= p_* N_0 / (N_0 + 2), & z &:= v^2/a, \\ \Gamma &:= \tfrac{1}{2} D_{\text{KL}}(p_0\|z) - h(\delta) + \alpha_- h(\delta/\alpha_-). \end{aligned} \tag{146}$$

For every coordinate set $I \subseteq [m]$, set

$$H_I := \{x : x_i = 0 \ (i \in I)\}, \quad L_I := D \cap H_I \cap \sqrt{2m} B_2^m.$$

All these sections use the same ambient cutoff.

Draw independent standard Gaussians $\xi_1, \dots, \xi_{m+2}$ and put

$$W := \sum_{i=1}^{m+2} \xi_i^2, \quad g := \sqrt{m} \frac{(\xi_1, \dots, \xi_m)}{\sqrt{W}}, \quad b_g := g/\kappa_{\text{pr}}. \tag{147}$$

Before rejection, g is uniform in $\sqrt{m} B_2^m$: its squared radial fraction has distribution $\text{Beta}(m/2, 1)$ and is independent of its uniform direction. Reject if $W \notin [m/2, 2(m+2)]$; the estimates below include rejection. For $y \in (-1, 1)^m$, use the shifted box $Q_y(t) := t([-1, 1]^m - y)$.

A Gaussian mass lower bound gives volume in each coordinate section, which in turn controls support in a random ball direction.

Lemma B.7 (Gaussian volume and polar support). *Suppose $0 < z < p_0$ and*

$$N_0(J(2) - E) > \log 2, \quad N_0\left(\frac{a - p_* - p_* \log(a/p_*)}{2} - E\right) > \log 2. \tag{148}$$

For every $J_0 \subseteq [m]$ of size ℓ , the set L_{J_0} is a symmetric convex body in H_{J_0} , and

$$\text{vol}_d(L_{J_0}) > \text{vol}_d(B_2^d)(am)^{d/2}.$$

Writing $h_L(w) := \sup_{x \in L} \langle w, x \rangle$, we also have

$$\Pr[h_{L_{J_0}}(g_{H_{J_0}}) \leq vm] \leq \exp(-mD_{\text{KL}}(p_0\|z)/2).$$

Proof. Gaussian mass and volume. The central-section fact used in Lemma 5.4 gives $\gamma_{H_{J_0}}(D \cap H_{J_0}) \geq e^{-Em}$. A standard Gaussian in H_{J_0} has squared norm χ_d^2 , dominated by χ_m^2 . Thus

$$\gamma_{H_{J_0}}(L_{J_0}) \geq e^{-Em} - e^{-mJ(2)} > \frac{1}{2}e^{-Em}.$$

The first step uses the chi-square upper-tail bound, and the second uses Eq. (148) and $m \geq N_0$. The cutoff gives compactness. Positive intrinsic Gaussian measure and symmetry put the origin in the relative interior.

Among sets of a fixed volume, a centered ball maximizes mass under any nonincreasing radial density. Indeed, for a set A and an equal-volume ball B_R , the density on $A \setminus B_R$ is at most its boundary value, and on $B_R \setminus A$ it is at least that value. The two differences have equal volume, so comparison of their integrals proves the assertion. The same conclusion holds when $|A| \leq |B_R|$.

Put $p_d := d/m \geq p_*$. The chi-square lower-tail bound gives

$$\gamma_{H_{J_0}}(\sqrt{am}B_2^d) \leq \exp(-m \frac{a - p_d - p_d \log(a/p_d)}{2}) < \frac{1}{2}e^{-Em}.$$

The first step uses $a < p_d$. For the second, the derivative of the rate in p_d is $-\log(a/p_d)/2 > 0$, and Eq. (148) applies at p_* . Radial rearrangement under the Gaussian density now proves the volume lower bound.

Polar probability. The classical symmetric Blaschke–Santaló inequality is

$$\text{vol}_d(L) \text{vol}_d(L^\circ) \leq \text{vol}_d(B_2^d)^2;$$

see the introduction of [MY25]. It applies to every symmetric convex body; no unconditionality assumption on L is needed. Apply it to $L = L_{J_0}$. The volume bound just proved implies

$$\text{vol}_d(vmL_{J_0}^\circ) < \text{vol}_d(B_2^d)(v\sqrt{m}/\sqrt{a})^d.$$

The marginal law of $g_{H_{J_0}}$ has density proportional to

$$(1 - \|x\|_2^2/m)^{\ell/2} \mathbb{1}[\|x\|_2 < \sqrt{m}],$$

which is radial and nonincreasing. Apply radial rearrangement to this density and use $\{w : h_{L_{J_0}}(w) \leq vm\} = vmL_{J_0}^\circ$. The support probability is at most $\Pr[X/(X+Y) \leq z]$, where $X \sim \chi_d^2$ and $Y \sim \chi_{\ell+2}^2$ are independent.

For $p_J := d/(m+2)$, exponential Markov applied to $-\theta((1-z)X - zY)$ with $\theta = (p_J - z)/(2z(1-z))$ gives

$$\Pr[X/(X+Y) \leq z] \leq \exp(-(m+2)D_{\text{KL}}(p_J\|z)/2) \leq \exp(-mD_{\text{KL}}(p_0\|z)/2).$$

The first step uses the moment-generating functions of the two independent chi-square variables. The second uses $p_J \geq p_0 > z$ and the fact that $D_{\text{KL}}(p\|z)$ increases in $p > z$. This includes the two extra Gaussian sampling coordinates. $\square$

Counting supersets converts the bound for a fixed coordinate section into a simultaneous bound over all small coordinate sets.

Lemma B.8 (Simultaneous support by supersets). *Under the preceding hypotheses,*

$$\Pr[\exists I \subseteq [m], |I| \leq \delta m : h_{L_I}(g_{H_I}) \leq vm] \leq (m+1)e^{-m\Gamma}.$$

Proof. Put $k_0 := \lfloor \delta m \rfloor$. For fixed g , let $Z(g)$ be the fraction of ℓ -subsets J_0 with $h_{L_{J_0}}(g) \leq vm$. If an I with $|I| \leq k_0$ has this property, every $J_0 \supseteq I$ does too, since $L_{J_0} \subseteq L_I$. The support argument is the same ambient vector g ; projecting it onto the section does not change the inner product. The fraction of such supersets is at least $\binom{\ell}{k_0} / \binom{m}{k_0}$, because this ratio decreases with the size of I . Therefore Markov's bound and Lemma B.7 give

$$\Pr[\text{some such } I] \leq \frac{\binom{m}{k_0}}{\binom{\ell}{k_0}} \exp(-mD_{\text{KL}}(p_0 \| z)/2).$$

The binomial entropy bounds are

$$\binom{m}{k_0} \leq e^{mh(k_0/m)}, \quad \binom{\ell}{k_0} \geq \frac{e^{\ell h(k_0/\ell)}}{\ell+1}.$$

For the lower bound, the k_0 th term is a mode of the binomial law with parameter k_0/ℓ , so its probability is at least $1/(\ell+1)$. The cases $k_0 = 0$ follow directly.

The function $C(q, b) := h(q) - bh(q/b)$ increases in q and decreases in b for $0 < q < b < 1$, since

$$\partial_q C = \log((1-q)/(b-q)) > 0, \quad \partial_b C = \log(1-q/b) < 0.$$

Now $k_0/m \leq \delta$ and $\ell/m \geq \alpha - 1/N_0 = \alpha_-$. Hence the binomial ratio is at most $(m+1)e^{mC(\delta, \alpha_-)}$, which proves the assertion. No independence among the section events is used. $\square$

The simultaneous support estimate forces a projection to meet many faces of the shifted box.

Lemma B.9 (Ball-target projection). *Assume $0 < \delta < \alpha - 1/N_0$, $\alpha < 1/2$, $0 < a < 1 - \alpha$, $v, t > 0$, $v^2/a < p_0$, Eq. (148), and*

$$v - \kappa_{\text{pr}} - 4t/5 > 0, \quad \Gamma > 1/(N_0 + 1), \quad N_0\Gamma - \log(N_0 + 1) > \log 10000. \quad (149)$$

For every $m \geq N_0$ and $y \in (-1, 1)^m$, the Euclidean projection x_ of b_g onto $D \cap Q_y(t)$ has more than δm tight coordinate constraints with probability greater than 0.999, including rejection. Thus $y + x_*/t$ has more than δm sign coordinates.*

Proof. The shifted box. The function $A_y(\xi) := \sum_{i=1}^m (|\xi_i| - y_i \xi_i)$ has mean $m\sqrt{2/\pi} < 0.798m$ and Lipschitz constant at most $2\sqrt{m}$. Gaussian concentration and the chi-square lower-tail bound give

$$\Pr[A_y(\xi) > 0.799m] \leq e^{-m/(8 \cdot 10^6)}, \quad \Pr[W < 0.999m] \leq e^{-mJ(0.999)}.$$

For the second bound, retain the first m squared coordinates. Off these events,

$$\|g\|_1 - \langle g, y \rangle \leq \frac{0.799m}{\sqrt{0.999}} < 4m/5.$$

The first step substitutes the numerator and denominator bounds into Eq. (147); the second follows by squaring, since $0.799^2 < 0.8^2 \cdot 0.999$.

Projection toward b_g is equivalent to maximizing $F_g(x) := \langle g, x \rangle - \kappa_{\text{pr}} \|x\|_2^2/2$. Every $x \in Q_y(t)$ has $F_g(x) < 4tm/5$. On the simultaneous support event of Lemma B.8, every I of size at most δm has a witness $x_I \in L_I$ with

$$F_g(x_I) > vm - \kappa_{\text{pr}} m > 4tm/5.$$

The first step uses the support bound and $\|x_I\|_2^2 \leq 2m$; the second uses Eq. (149).

If x_* had at most δm tight coordinates, let I be their set and discard all nontight box halfspaces. The same point remains optimal on the relaxed set: a sufficiently short segment toward any better relaxed point preserves every discarded strict inequality. The relaxed set contains $D \cap H_I$, because zero satisfies each retained halfspace. It contains x_I , contradicting the objective comparison.

Failure and rejection. The function $(m+1)e^{-m\Gamma}$ decreases for $m \geq N_0$ by Eq. (149), and its value at N_0 is less than 10^{-4} . Lower rejection at $m/2$ is covered by $W < 0.999m$. Upper rejection adds at most $e^{-mJ(2)}$. The remaining three terms satisfy

$$e^{-m/(8 \cdot 10^6)} + e^{-mJ(0.999)} + e^{-mJ(2)} < 10^{-4}.$$

Indeed, at $m = N_0$ their exponents exceed or equal 37.5, 75, and 45000000, respectively, since $J(0.999) > 10^{-6}/4$ and $J(2) > 0.15$. Thus their sum is less than $3e^{-37.5} < 3/2^{37} < 10^{-4}$, and it decreases thereafter. The total exceptional probability is below $2 \cdot 10^{-4} < 0.001$. Tight coordinates become signs under the affine update. $\square$

B.3 Certified suffix signing

Algorithm 11 Cubic-time signing from a prescribed capacity

```

1: procedure SUFFIXSIGNING( $A_1, \dots, A_{m_0}, n, w, p$ ) ▷ Lemma B.12
2:   Use Definition B.5 and the caps of Lemma B.11
3:    $y \leftarrow 0 \in \mathbb{R}^{m_0}$ ,  $S \leftarrow [m_0]$ ,  $z \leftarrow w$ 
4:   while  $|S| \geq N_0$  do
5:      $m \leftarrow |S|$ ;  $R \leftarrow \varrho(z)\sqrt{zn}$ ;  $(\delta, t) \leftarrow (1/4, 3/10)$ 
6:     Compute  $\hat{R} \in [R, (1 + 10^{-8})R]$ ; form  $D$  and  $Q_y(t)$ 
7:     repeat
8:       Draw Eq. (147); reject if  $W$  is outside its allowed range
9:       Project toward  $b_g$  via RECURSIVEMULTIPLIER ▷ Algorithm 9
10:      Apply radial correction and snap within  $2\rho$ ; call the increment  $\bar{x}$ 
11:    until more than  $m/4$  coordinates are snapped, or the trial cap is reached
12:    if the cap is reached without success then
13:      return FAIL
14:    end if
15:     $y_i \leftarrow y_i + \bar{x}_i/t$  for  $i \in S$ ;  $S \leftarrow \{i : |y_i| < 1\}$ ;  $z \leftarrow 3z/4$ 
16:  end while
17:  Round each remaining  $y_i$  to its nearest sign, with a fixed tie rule at zero
18:  return the resulting signing
19: end procedure

```

The two projection procedures use the same fixed-target solver. We record its radius and accuracy requirements before verifying their different geometric guarantees.

Lemma B.10 (Certified fixed-target projection). *Let $n > N_*$, $N_0 \leq m \leq n$, and let $B(x) := \sum_{j=1}^m x_j M_j$ for symmetric contractions $M_j \in \mathbb{R}^{n \times n}$. Put $a_{\text{sm}} := 2 \cdot 10^6 \log(2n)$, $s_{\text{sm}} := 5 \cdot 10^{-7}$, and*

$$h_{\hat{R}}(x) := a_{\text{sm}}^{-1} \log(\text{tr}[e^{a_{\text{sm}} B(x)/\hat{R}}] + \text{tr}[e^{-a_{\text{sm}} B(x)/\hat{R}}]), \quad D := \{x : h_{\hat{R}}(x) \leq 1 - s_{\text{sm}}\}.$$

Let $Q \subset [-1, 1]^m$ be a box containing zero whose side lengths are bounded below by a positive absolute constant. Suppose $\hat{R}^2 > 15m$, $\hat{R} \leq n^{O(1)}$, and the fixed target satisfies $\|b\|_2^2 \leq c_b m$ for an absolute constant c_b . For $0 < \epsilon < 1/2$ and $0 < \rho \leq 1$ with $\rho^{-1} \leq n^{O(1)}$, the certified wrapper and recursive solver return an exactly feasible point within distance ρ of $\text{proj}_{D \cap Q} b$, with failure probability at most ϵ . Their cost is

$$(mn^{2+o(1)} + mn^{\omega-1/2+o(1)} + n^{\omega+o(1)}) \text{polylog}(1/\epsilon).$$

Proof. The origin has $h_{\hat{R}}(0) = 5 \cdot 10^{-7}$, so the Slater gap is absolute. On the multiplier branch, the spectral-feasibility calculation of Section 7.5, with $\tau_w \leq 1/1024$, gives

$$\|b\|_2 \geq \|b_Q\|_2 \geq (1 - s_{\text{sm}} - 5 \cdot 10^{-7} - \tau_w/(2a_{\text{sm}}))\hat{R}/\sqrt{m} > 3.$$

The first step uses $0 \in Q$, the second is the wrapper's level test, and the last uses $\hat{R}^2 > 15m$. The target cap gives initial objective, multiplier bracket and guarded level $H = O(m)$. Since $H/\hat{R}^2 = O(1)$, coordinate curvature is $O(\log^2 n)$ and all requested tolerances have polynomial scales. Lemmas 9.3 and 9.9 give the stated cost, including multiplier search and the radial feasibility correction. The target is fixed before their fresh internal probes are drawn. $\square$

The suffix procedure uses the projection accuracy to snap tight coordinates and reduce the active set.

Lemma B.11 (Finite implementation and failure budget). *Let $n > N_*$, $0 < w \leq 1$, and $0 \leq m_0 \leq wn$. For symmetric contractions $A_1, \dots, A_{m_0} \in \mathbb{R}^{n \times n}$, Algorithm 11 has total failure probability at most p . Every accepted projection increment has matrix norm at most $\hat{R}/t$, its active size decreases to less than $(1 - \delta)m$, and its terminal rounding costs less than $0.0003\sqrt{n}$. At stage k , the active size satisfies $m \leq w(3/4)^k n$. Its arithmetic cost is $n^{3+o(1)} \text{polylog}(1/p)$. For $n \leq N_*$, independent sign trials with the retained norm certificate instead give a $12\sqrt{n}$ signing within the same cost and failure budget when $m_0 = n$.*

Proof. Radius selection and smoothing. Fix the initial capacity w before drawing randomness and start with zero partial coloring on the m_0 supplied matrices. Keep their ambient dimension n throughout; in particular, do not substitute m_0 for n in the small-ball estimates, radii, smoothing, or matrix-function costs. If $m_0 = 0$, return the empty signing. Evaluate the radius formula in Definition B.5 outward to obtain $\hat{R}$; the interpolation weights are nonnegative rationals. In the tail, evaluate $\log(1/w_k)$ with a certified upper bound. These operations preserve the relative allowance 10^{-8} .

Use $h_{\hat{R}}$ and D from Lemma B.10. The smoothing inclusions are $(1 - 10^{-6})K(R) \subseteq D \subseteq (1 - s_{\text{sm}})K(\hat{R})$. For $0 < c \leq 1$, changing variables in the Gaussian integral gives $\gamma_m(cK) \geq c^m \gamma_m(K)$. Hence Lemma B.6 gives $\gamma_m(D) \geq e^{-E_* m}$ at every stage. Lemma B.36 verifies every hypothesis of Lemma B.9 for this exponent and the fixed projection tuple.

Target accuracy and snapping. The target obeys $\|b_g\|_2^2 \leq 10^8 m$. The box $Q_y(t)$ contains zero and has side lengths in $[t, 2t]$. Since $\hat{R}^2 > 22m$ and $\hat{R} < 23\sqrt{n}$, Lemma B.10 applies with $c_b = 10^8$.

Set

$$\rho := \min\{1/8, t/8, s_{\text{sm}}\hat{R}/(4(2m + \sqrt{m}))\}. \quad (150)$$

On an accepted draw, approximate $\sqrt{m/W}/\kappa_{\text{pr}}$ from below, preserving the target cap and making its Euclidean error at most $\rho/4$. Since $W \in [m/2, 2(m+2)]$, inverse-polynomial scalar accuracy suffices. Euclidean projection onto a closed convex set is nonexpansive in the target. Allocate the remaining $3\rho/4$ to the certified wrapper, inner solve and radial feasibility correction. The returned point lies in $D \cap Q_y(t)$ and is within ρ of the ideal projection. Every ideal tight coordinate is snapped within 2ρ , and $4\rho < 2t$ prevents opposite-face ambiguity. The snapped point satisfies

$$\|B(\bar{x})\| \leq (1 - s_{\text{sm}})\hat{R} + 2m\rho < \hat{R}.$$

The first step uses smooth feasibility and $\|A_i\| \leq 1$; the second uses Eq. (150). Thus all numerical perturbations are included in the increment bound $\hat{R}/t$. A successful stage gives $m' < 3m/4 \leq w_{k+1}n$, proving the capacity invariant even when extra coordinates are fixed. Lemma B.6 applies to every subsequent capacity, including the tail.

During a call, $w_k \geq m/n \geq N_0/n$, so radius evaluation and target normalization require only inverse-polynomial accuracy; the latter adds $O(m)$ arithmetic operations. Lemma B.10 supplies the feasible projection and its cost.

Confidence and total work. Every successful round fixes more than $m/4$ coordinates. Thus $P := 1 + \lceil \log n / (-\log(3/4)) \rceil$ bounds the number of rounds, and their active sizes sum to at most $4n$. Lemma B.9 gives ideal success probability greater than 0.999. Allocate total probability $p/4$ to numerical failures, with each trial's numerical budget at most 0.02. A trial then succeeds with conditional probability at least 0.9. Permit at most $\lceil \log(4P/p) / \log 10 \rceil$ fresh trials per round; exhaustion costs at most $p/4$. Allocate the internal budgets over this deterministic call cap. These bounds hold conditional on the past, without conditioning geometric events on future certificate success.

Summing the cost of Lemma B.10 over the active sizes and trial caps gives

$$(n^3 + n^{\omega+1/2} + n^\omega)n^{o(1)} \text{polylog}(1/p) = n^{3+o(1)} \text{polylog}(1/p),$$

using $\omega < 5/2$ as in Section 9. Matrix-image maintenance, input formation and final summation fit this cost.

All-dimension cleanup. For $n \leq N_* = 10^{24}$, use independent uniform signs from the outset. The matrix Rademacher-series bound [Tro12, Theorem 4.1] gives

$$\Pr\left[\left\|\sum_{i=1}^n \varepsilon_i A_i\right\| \geq 11\sqrt{n}\right] \leq 2ne^{-121/2} < 0.02.$$

The first step uses $\|\sum_i A_i^2\| \leq n$; the second substitutes $n \leq 10^{24}$. The norm certificate of Section 6.6 has excess at most $0.03\sqrt{n}$ on its successful event. Accept at $12\sqrt{n}$: every trial with true norm below $11\sqrt{n}$ is accepted on that event. Use $O(\log(1/p))$ trials with total certification and exhaustion probability at most p . Forming each signed sum costs $O(n^3)$, and the retained norm certificates fit the same exponent.

For $n > N_*$, stop projection when $m < N_0$ and round each remaining coordinate to its nearest sign. Each change has absolute value at most one, so the error is at most $m < N_0 < 0.0003\sqrt{n}$. This cleanup is deterministic and costs at most $O(n^3)$. The two branches cover every positive integer n . $\square$

Recursive repair uses the terminating signing procedure at several capacities. An integral bounds every geometric continuation at once.

Lemma B.12 (Suffix discrepancy bounds). *For $0 < w \leq 1$, define*

$$c_{\text{suf}}(w) := \frac{1 + 10^{-8}}{t(1 - \sqrt{3/4})} \mathcal{J}(w) + 0.0003, \quad \mathcal{J}(w) := \int_0^w \frac{\varrho(s)}{2\sqrt{s}} ds. \quad (151)$$

Under the hypotheses of Lemma B.11, a successful call with initial capacity w has discrepancy at most $c_{\text{suf}}(w)\sqrt{n}$. The function c_{suf} is increasing and $c_{\text{suf}}(1) < 141$. These bounds hold conditional on any prior history when the supplied matrices depend on that history and the call uses fresh randomness.

Proof. Put $q := 3/4$. Lemma B.6 gives $\varrho(s) \geq \varrho(wq^k)$ for $wq^{k+1} \leq s \leq wq^k$. Integrating on this interval yields

$$\mathcal{J}(wq^k) - \mathcal{J}(wq^{k+1}) \geq (1 - \sqrt{q})\varrho(wq^k)\sqrt{wq^k}. \quad (152)$$

The integral is finite because its tail is a constant multiple of $\sqrt{s} \log(1/s)$. Sum Eq. (152) over $k \geq 0$ and use $\mathcal{J}(s) \rightarrow 0$ as $s \downarrow 0$. Lemma B.11 bounds each accepted increment by $(1 + 10^{-8})\varrho(wq^k)\sqrt{wq^k n}/t$. The resulting sum, together with terminal rounding, proves Eq. (151). Early termination only omits nonnegative charges.

Here is a finite expression for the integral. For $0 \leq j < 6$ and $s > 0$, put

$$\Phi_j(s) := \frac{(b_j \varrho_{j+1} - b_{j+1} \varrho_j)\sqrt{s} + (\varrho_j - \varrho_{j+1})s^{3/2}/3}{b_j - b_{j+1}}.$$

On (b_{j+1}, b_j) , differentiation gives $\Phi'_j(s) = \varrho(s)/(2\sqrt{s})$. Set $z := \min\{w, b_6\}$. Integration of the tail and the six affine pieces gives

$$\begin{aligned} \mathcal{J}(w) &= 185\sqrt{z}(2 - \log z) + \varrho_0(\sqrt{w} - \sqrt{\min\{w, b_0\}}) \\ &\quad + \sum_{\substack{0 \leq j < 6 \\ w > b_{j+1}}} (\Phi_j(\min\{w, b_j\}) - \Phi_j(b_{j+1})). \end{aligned} \quad (153)$$

The first term integrates $185 \log(1/s)/(2\sqrt{s})$ from zero to z . The second integrates the constant-radius piece, and each summand integrates one affine piece. This formula uses only the seven inherited parameters, square roots, logarithms, and rational arithmetic. Lemma B.36 verifies $c_{\text{suf}}(1) < 141$ using rational outward bounds.

Positivity of the integrand proves monotonicity. The argument preserves the original ambient dimension n and starts from $m_0 \leq wn$. Every estimate holds for each fixed input family, so conditioning on earlier history and using fresh randomness proves the final assertion. $\square$

B.4 Entropy and Gaussian width

For a nonempty compact set $C \subseteq \mathbb{R}^n$, write

$$w(C) := \mathbb{E}_\xi[\sup_{x \in C} \langle \xi, x \rangle], \quad \xi \sim N(0, I_n).$$

Let φ_G and F_G be the standard Gaussian density and distribution function, respectively, and put $c_G := \sqrt{2/\pi}$. Retain the binary entropy h and $s_\star = 0.67448$ from Section A.1.

The size of a sign family controls how much Gaussian width it loses relative to the full cube.

Lemma B.13 (Entropy deficit controls width deficit). *For every nonempty $\mathcal{F} \subseteq \{-1, 1\}^n$,*

$$w(\mathcal{F}) \geq c_G n - (n \log 2 - \log |\mathcal{F}|)/c_G. \quad (154)$$

Proof. We first prove the scalar comparison

$$2\varphi_G(F_G^{-1}(q)) \geq c_G - \frac{\log 2 - h(q)}{c_G}, \quad 0 < q < 1. \quad (155)$$

For $x \geq 0$ put $U(x) := 2F_G(x) - 1$ and $T(x) := U(x)^2 + e^{-x^2/2} - 1$. Differentiation gives

$$T'(x) = e^{-x^2/2}b(x), \quad b(x) := 2c_GU(x) - x, \quad b'(x) = \frac{4}{\pi}e^{-x^2/2} - 1.$$

The function b' decreases strictly from $4/\pi - 1 > 0$ to -1 . Since $b(0) = 0$ and $b(x) \rightarrow -\infty$, b is positive until a unique positive zero and negative thereafter. Thus T first increases and then decreases. Both $T(0)$ and $\lim_{x \rightarrow \infty} T(x)$ equal zero, proving $T(x) \geq 0$. For $q = F_G(x)$ this implies

$$q(1-q) = \frac{1-U(x)^2}{4} \leq \frac{1}{4}e^{-x^2/2}, \quad \frac{d}{dx} \log \frac{F_G(x)}{1-F_G(x)} = \frac{\varphi_G(x)}{q(1-q)} \geq 2c_G.$$

The first step is the definition of U , the second uses $T \geq 0$, the third differentiates the logarithm, and the fourth substitutes the Gaussian density and the preceding bound. Integration from zero gives $\log(q/(1-q)) \geq 2c_G x$. The derivative with respect to $q \geq 1/2$ of the left side of Eq. (155) minus its right side is

$$-2F_G^{-1}(q) + \frac{1}{c_G} \log \frac{q}{1-q} \geq 0.$$

The difference vanishes at $q = 1/2$. Symmetry proves the other half, and continuity covers the endpoints, interpreting the density term as zero.

Let X be uniform on $\mathcal{F}$. Reveal its coordinates successively, writing $q_i := \Pr[X_i = 1 \mid X_1, \dots, X_{i-1}]$. Couple this reveal to independent $\xi_i \sim N(0, 1)$ by choosing $X_i = 1$ exactly when $\xi_i \geq F_G^{-1}(1 - q_i)$. Deterministic choices cover $q_i = 0, 1$. At each prefix the next bit has the required conditional law, so the resulting X is uniform on $\mathcal{F}$. Moreover,

$$\mathbb{E}[\xi_i X_i \mid X_1, \dots, X_{i-1}] = 2\varphi_G(F_G^{-1}(q_i)).$$

Pointwise $\sup_{x \in \mathcal{F}} \langle \xi, x \rangle \geq \langle \xi, X \rangle$. Sum Eq. (155) and use the entropy chain rule $\sum_i \mathbb{E}[h(q_i)] = \log |\mathcal{F}|$ to obtain Eq. (154). This coupling is used only to prove a width bound; the algorithm does not enumerate $\mathcal{F}$ or compute the conditional probabilities q_i . $\square$

The translated sign family connects Gaussian mass to the entropy estimate for width.

Lemma B.14 (Small-ball mass gives nearly maximal width). *Let $K \subseteq \mathbb{R}^n$ be closed, symmetric, and convex, with $\gamma_n(K) \geq e^{-\epsilon n}$. If C is a compact set containing $s_\star^{-1}K \cap [-1, 1]^n$, then*

$$w(C) \geq (c_G - \epsilon/c_G)n. \quad (156)$$

Proof. The translated-density argument in Lemma A.2 gives a vector z for which

$$\mathcal{F}_z := \{\varepsilon \in \{-1, 1\}^n : z + s_\star \varepsilon \in K\}, \quad |\mathcal{F}_z| \geq 2^n \gamma_n(K).$$

Indeed, its proof averages this cardinality over the independent translated-density variables; a cardinality at least its average occurs. Symmetry and convexity give $(\mathcal{F}_z - \mathcal{F}_z)/2 \subseteq s_\star^{-1}K \cap [-1, 1]^n$, because the translation cancels. Gaussian reflection gives

$$w((\mathcal{F}_z - \mathcal{F}_z)/2) = \frac{1}{2} \mathbb{E}[\max_{\varepsilon \in \mathcal{F}_z} \langle \xi, \varepsilon \rangle + \max_{\varepsilon \in \mathcal{F}_z} \langle -\xi, \varepsilon \rangle] = w(\mathcal{F}_z).$$

The first step maximizes independently over the two members of the family, and the second uses that ξ and $-\xi$ have the same law. Monotonicity of width and Lemma B.13 prove Eq. (156). Neither the translate nor its sign family is computed. $\square$

B.5 Strengthened weighted energies

We use the projection-aware variance bound of Lemma A.18, with the energies $\mathcal{R}_A, \mathcal{P}_A, \mathcal{D}_A$ defined there and $\Pi := \Pi_A$. It applies componentwise to the derivative arrays below.

We track derivative energies and their projected means together, so each differentiation step can use the strengthened variance bound.

Definition B.15 (Coupled energy and projected-mean recurrence). Use a spectral family of Definition A.15, with its unchanged barrier order τ and parameters $\zeta, \kappa_0, \kappa_{\text{tail}}, \theta$. At variance a , let $p_2, J, c := 1 - J^{-1}$ be as in Definition A.6. For an integer $\ell \geq 1$, initialize $V_\ell = P_\ell := (\ell!)^2 J$. For $j = \ell - 1, \dots, 0$, set

$$r := \ell - j, \quad b_j := \frac{p_2}{2}(V_{j+1} + P_{j+1}), \quad M_j := \left(\frac{\ell!}{r!}\right)^2 \begin{cases} 1, & r = 2, \\ \text{df}(r), & r \neq 2. \end{cases}$$

Define

$$(V_j, P_j) := \begin{cases} (J(1+a)b_j, (J(1+a)-1)b_j), & r \text{ odd}, \\ (\mathcal{F}_c(M_j, b_j), \mathcal{F}_c(M_j, b_j) - b_j), & r \text{ even}, \end{cases} \quad (157)$$

where $\mathcal{F}_c$ is defined in Eq. (64). Write $C_\ell^+(a) := V_0$.

Applying the coupled recurrence to the derivative arrays of a matrix power yields the resolvent moment estimate needed for radial integration.

Lemma B.16 (Coupled polynomial energies). *Let $X = \sum_{i=1}^m \xi_i N_i$ with $\sum_{i=1}^m N_i^2 \preceq aI_N$, under the even tilted measure of Lemma A.10. If the strict root conditions of Definition A.6 hold, then*

$$\frac{1}{N} \mathbb{E}_\mu[\text{tr}[X^{2\ell} Y_\sigma^{-2}]] \leq a^\ell C_\ell^+(a), \quad \sigma \in \{-1, 1\}.$$

The coefficient $C_\ell^+(a)$ is nondecreasing throughout its admissible variance interval.

Proof. Apply Lemma A.18 to the ordered arrays $\nabla^j(X^\ell)$, normalized by Na^ℓ . Their variance budget is $b_j = p_2(V_{j+1} + P_{j+1})/2$. Lemma A.10 supplies the ordinary mean bounds M_j and the terminal bound $(\ell!)^2 J$; odd means vanish. The two componentwise bounds of Lemma A.8 give exactly Definition B.15. Backward induction and vectorization at $j = 0$ prove the trace estimate; evenness treats both resolvent signs. Monotonicity follows from the same induction, since the root quantities and both parity maps are nondecreasing. $\square$

Averaging the two resolvent signs lets us express the radial tail through a scalar quadratic inequality.

Lemma B.17 (Sign-averaged radial remainder). *In the preceding setting put $W := (I - X^2)^{-1}$, and define*

$$R_j := N^{-1} \mathbb{E}_\mu[\text{tr}[X^{2j} W]], \quad m_j := N^{-1} \mathbb{E}_\mu[\text{tr}[X^{2j}]], \quad D_\ell := N^{-1} \mathbb{E}_\mu[\text{tr}[X^{2\ell} Y_+^{-2}]].$$

For $K \geq \ell$ and $L := 2K - \ell$, let $G := m_L$ and $S := \sum_{j=\ell}^{K-1} m_j$. Then

$$R_K \leq \frac{G + \sqrt{G^2 + 8G(D_\ell + S)}}{4}. \quad (158)$$

Along the covariance-preconditioned radial path of Lemma A.12, this implies

$$R_K \leq \frac{\text{df}(L)a^L}{4}\omega_L + a^K \sqrt{\text{df}(L)\widehat{C}_{\ell,K}(a)}\sqrt{\omega_L}, \quad (159)$$

where

$$\widehat{C}_{\ell,K}(a) := \frac{1}{2}(C_\ell^+(a) + \sum_{j=\ell}^{K-1} \text{df}(j)a^{j-\ell} + \frac{\text{df}(L)}{8}a^{L-\ell}).$$

This coefficient is nonnegative and nondecreasing.

Proof. Evenness and scalar functional calculus give $D_\ell = N^{-1}\mathbb{E}_\mu[\text{tr}[X^{2\ell}(2W^2 - W)]]$. Cauchy–Schwarz for the normalized expected trace and the geometric-series identity give

$$R_K^2 \leq m_L N^{-1}\mathbb{E}_\mu[\text{tr}[X^{2\ell}W^2]] = \frac{G}{2}(D_\ell + R_\ell) = \frac{G}{2}(D_\ell + S + R_K).$$

The first step splits the powers as $L + \ell = 2K$. The second uses the sign-averaged identity, and the third uses $R_\ell = S + R_K$. Solving this scalar quadratic proves Eq. (158).

Let Γ be the coefficient Gram matrix before radial scaling, and $C_\Gamma := (I + 2\kappa_0\Gamma)^{-1}$. Lemma A.11 gives $m_j \leq \text{df}(j)a^j\omega_j$, where $\omega_j := \text{tr}[\Gamma C_\Gamma^j]/(N\eta) \in [0, 1]$ and $a = \eta t^2$. Use Lemma B.16 for D_ℓ , $\omega_j \leq 1$ in S , and $\omega_L^2 \leq \omega_L$ in the radicand. Factoring out $a^{2K}\text{df}(L)\omega_L$ gives Eq. (159). Every power in $\widehat{C}_{\ell,K}$ is nonnegative because $K \geq \ell$ and $L \geq \ell$; its monotonicity follows from Lemma B.16. $\square$

We specify the common construction for the coupled small-ball rows used by the repair algorithm.

Definition B.18 (Coupled small-ball row construction). Choose $f \in \{0, \dots, 6\}$, integers $K \geq \max\{\ell, \tau_f\}$ with $\ell \geq 2$, and $r, v_{\text{tr}}, \bar{e} > 0$, $H > 1$. The family f selects the unchanged spectral parameters in Definition A.15. Set $\eta := r^{-2}$, $L := 2K - \ell$ and $u_i := (i/512)^2$. Define

$$T^+ := \frac{(\kappa_0 + \kappa_{\text{tail}})\eta^K \sqrt{\text{df}(L)}}{K} \sum_{i=1}^{512} (u_i^K - u_{i-1}^K) \sqrt{\widehat{C}_{\ell,K}(\eta u_i)}.$$

Initialize $p_j^+ = 0$ for $2 \leq j \leq L$, set

$$p_j^+ := \frac{\text{df}(j)\eta^j}{j}(\kappa_0 + \kappa_{\text{tail}}\mathbb{1}[j \geq \tau]) \quad (2 \leq j < K),$$

and add $(\kappa_0 + \kappa_{\text{tail}})\text{df}(L)\eta^L/(4L)$ to p_L^+ . The added degree- L coefficient accounts for the linear term in Eq. (159).

The refined coefficients enter the determinant transfer argument through a scalar profile whose bound persists under variance reduction.

Lemma B.19 (Refined hereditary Gaussian small balls). Suppose a parameter choice in Definition B.18 satisfies the strict root conditions of Definition A.6 at η , and for every $x \geq 0$,

$$\begin{aligned} & \kappa_0\eta \frac{H \log(1+x) - \log(1+Hx)}{(H-1)x} \\ & + \frac{H}{H-1} \left(\sum_{j=2}^L p_j^+ (1+x)^{-j} + \frac{T^+}{2v_{\text{tr}}} (1+x)^{-L} + \frac{T^+ v_{\text{tr}}}{2} \right) < 0.999\bar{e}. \end{aligned} \quad (160)$$

The first term is interpreted continuously at zero. Then, for symmetric $B_i \in \mathbb{R}^{n \times n}$ with $\sum_i B_i^2 \preceq v\eta I$ and $0 < v \leq 1$, independent standard Gaussians ξ_i satisfy

$$\Pr[\|\sum_i \xi_i B_i\| < 1] \geq \exp(-nv\bar{e}).$$

Proof. Integrate Lemma B.17 along the radial path of Lemma A.12. Its linear term adds p_L^+ , while exact monomial integration with right-endpoint bounds gives $T^+ \sqrt{\omega_L}$. The radial loss is therefore $\sum_{j=2}^L p_j^+ \omega_j + T^+ \sqrt{\omega_L}$. Apply Young's inequality with v_{tr} and Lemma A.13; the resulting exponent is bounded by the left side of Eq. (160). At variance $v\eta$, the finite coefficients decrease by v^j and T^+ by at least v^K . The scaling criterion in Lemma A.13 proves the hereditary assertion. $\square$

B.6 Covariance-sensitive polynomial barriers

The next estimates allow the finite coefficients of the barrier to vary independently. We use the same spectral allocation condition as in Appendix A.

Definition B.20 (Polynomial and logarithmic barriers). Let $K \geq 2$, $c_1 > 0$, $c_K > 2$, and $c_j \geq 0$ for $1 \leq j \leq K$. Define

$$\phi(x) := \sum_{j=1}^{K-1} \frac{c_j x^{2j}}{j} + c_K \sum_{j=K}^{\infty} \frac{x^{2j}}{j} \quad (|x| < 1),$$

and extend ϕ by $+\infty$ outside $(-1, 1)$. Put $\Phi(X) := \text{tr}[\phi(X)]$. For $0 < \zeta, \theta < 1$, let d_ζ and F_θ be as in Definition A.4. We require

$$(1-u)(1-v)\phi'[u, v] \geq F_\theta(d_\zeta(u)d_\zeta(v)) \quad (-1 < u, v < 1). \quad (161)$$

Here $\phi'[u, v]$ is the divided difference of ϕ' , with its continuous value on the diagonal.

The following finite reduction checks this condition without a monotonicity assumption on the coefficients c_j .

Lemma B.21 (Curvature for a polynomial barrier). Put $a_0 := 1 - \zeta$ and $G(u) := (1-u)^2 \phi''(u)$. Suppose $c_1 > \theta/(1+\theta)$ and, for $0 \leq t \leq 1$,

$$G(a_0 t) > \frac{2\theta}{1+\theta}, \quad G(a_0 + \zeta t) > \frac{2(\theta+t)}{1+\theta}, \quad G(a_0 + \zeta t) > 3t + (\theta^{-1} - 1)t^2.$$

Choose $w_0 \in (0, 1)$ such that

$$\frac{w_0^{2K-1}}{1-w_0} > 2K-1, \quad c_K > 2 + \theta^{-1}.$$

For $u := 1 - \zeta t^2$ and $v := -w_0 + (a_0 + w_0)s$, define

$$\begin{aligned} F_1(t) &:= 2 - 2t/(1+\theta), \\ F_2(t) &:= \theta^{-1} + 2 - (2\theta^{-1} + 1)t + (\theta^{-1} - 1)t^2, \\ P_i(t, s) &:= 2c_K(1+uv) + 2(1-u^2)(1-v^2) \sum_{j=1}^{K-1} (c_j - c_K) \sum_{q=0}^{2j-2} u^{2j-2-q} v^q \\ &\quad - (1+u)(1+v)F_i(t) \quad (i = 1, 2). \end{aligned}$$

If $P_1, P_2 > 0$ on $[0, 1]^2$, then Eq. (161) holds. All resolvent-root and weighted Poincaré conclusions of Lemmas A.5 and A.7 consequently hold for this barrier.

Proof. For two spectral gaps $h := 1 - u$ and $k := 1 - v$, the left side of Eq. (161) is the average of $G(1 - s)$ with density proportional to s^{-2} between h and k . Under this probability measure, $\mathbb{E}[s] \leq \sqrt{hk}$ and $\mathbb{E}[s^2] = hk$. The first assertion follows from $\sinh z \geq z$ after setting $k/h = e^{2z}$; the second follows by direct integration. Both extend continuously to $h = k$.

When both endpoints are regular, the required lower bound is $F_\theta(1) = 2\theta/(1 + \theta)$. The first displayed condition covers nonnegative endpoints, while $G(u) \geq 2c_1$ for $u \leq 0$. When both endpoints are active, average each of the other two bounds with $t = 1 - s/\zeta$. Each right side is affine or quadratic in s , with nonpositive linear coefficient and nonnegative quadratic coefficient. The preceding two moments give its value at $s = \sqrt{hk}$, proving both branches of the required bound.

For a mixed pair $u \geq a_0$ and $v = -w \leq -w_0$, write $d := 2K - 1$. The unit logarithmic tail has divided-difference contribution

$$\frac{2u^d(1+w)}{(u+w)(1+u)} + \frac{2w^d(1-u)}{(u+w)(1-w)} \geq u^d + \frac{w^d}{1-w}(1-u) \geq 1.$$

The first step uses $2(1+w) \geq (u+w)(1+u)$ and $u+w \leq 2$. The second uses $u^d \geq 1 - d(1-u)$ and the condition on w_0 . The finite monomials have nonnegative divided differences, and $F_\theta \leq 2 + \theta^{-1} < c_K$.

The remaining mixed pairs form the rectangle parametrized by $(t, s) \in [0, 1]^2$. Multiplying the difference between the two sides of Eq. (161) by $(1+u)(1+v)$ gives P_i for the corresponding branch. Reflection treats the other spectral edge. For the three diagonal checks, the exact polynomial numerator is

$$(1+u)^2 G(u) = 2(1-u^2)^2 \sum_{j=1}^{K-1} (2j-1)c_j u^{2j-2} + 2c_K u^{2K-2}((2K-1) - (2K-3)u^2).$$

Thus all the stated checks are polynomial checks on compact intervals or rectangles. The allocation proof uses the barrier only through its spectral Hessian and boundary integrability. Here $\phi - c_1 x^2$ is even and convex, and the boundary order is $c_K > 2$, so the same exhaustion argument applies. $\square$

The ordinary means in the following refinements use Lemmas A.19 and A.25. In the latter, take $\kappa = c_1$; its separate common-field and raw-direction coefficients account for whitening.

B.7 Finite resolvent refinements

For the weighted means, a finite expansion of the even resolvent improves the initial root bounds. The expansion is used only for commuting monomials, not for arbitrary derivative fields.

Definition B.22 (Finite refinement of the resolvent bounds). Fix a barrier satisfying Eq. (161) and an admissible variance $a > 0$. Let p_2, J be the initial quantities of Definition A.6, put $\delta := 9a^2 C_2^+(a)$, and require $\delta < 1$. The recurrence defining C_2^+ is unchanged; Lemma B.21 permits this barrier in that recurrence.

For any already established J, δ with $J > 1$ and $0 < \delta < 1$, set $c := 1 - J^{-1}$ and define

$$\begin{aligned} R_0(M, b) &:= \mathcal{F}_c(M, b), & P_0(M, b) &:= \mathcal{F}_c(M, b) - b, \\ R_1(M, b) &:= \frac{(\sqrt{\delta M} + \sqrt{M + (1-\delta)b})^2}{(1-\delta)^2}, & P_1(M, b) &:= (\sqrt{M} + \sqrt{\delta R_1(M, b)})^2. \end{aligned}$$

For degree ℓ , initialize $V_\ell = P_\ell := (\ell!)^2 J$. Descend with $r := \ell - j$ and $b := p_2(V_{j+1} + P_{j+1})/2$. At odd r , use the odd branch of Eq. (157). At even r , set $M := (\ell!/r!)^2 b_{r/2}^2$ and use

$$V_j := \min\{R_0(M, b), R_1(M, b)\}, \quad P_j := \min\{P_0(M, b), P_1(M, b)\}.$$

Write $c_\ell^{\text{raw}} := V_0$.

Perform the following refinement five times. First put $c_0^{\text{op}} := J$ and $c_\ell^{\text{op}} := c_\ell^{\text{raw}}$ for $1 \leq \ell \leq 8$. In descending order $\ell = 7, 6, \dots, 0$, replace c_ℓ^{op} by the minimum of its current value and

$$\sum_{r=0}^{k-1} (2r+1)b_{\ell+r}a^r + (2k+1)a^k c_{\ell+k}^{\text{op}}, \quad 1 \leq k \leq 8 - \ell. \quad (162)$$

Finally replace J by $\min\{J, c_0^{\text{op}}\}$ and δ by the minimum of its current value, $9a^2 c_2^{\text{op}}$, and the five numbers

$$9b_2a^2 + 3b_3a^3 + 4b_4a^4 + \sum_{r=4}^{k-1} (2r+1)b_r a^r + (2k+1)a^k c_k^{\text{op}}, \quad 4 \leq k \leq 8. \quad (163)$$

The sum is zero when $k = 4$. Denote the final bounds by $J_*(a), \delta_*(a)$.

The next lemma verifies that every refinement uses only bounds already proved at the preceding step.

Lemma B.23 (Resolvent means and the even defect). *Put $Z_1 := X \otimes I$, $Z_2 := I \otimes X$, $Y_\sigma := I + \sigma X$, and*

$$\mathcal{W}_\sigma := Y_\sigma^{-1} \otimes Y_\sigma^{-1}, \quad \mathcal{W}_e := (\mathcal{W}_+ + \mathcal{W}_-)/2, \quad Q := I - \mathcal{W}_e^{-1}.$$

Under the even tilted measure of Lemma B.16, the final bounds of Definition B.22 satisfy

$$I \preceq \mathbb{E}_\mu[\mathcal{W}_e] \preceq J_* I, \quad \mathbb{E}_\mu[Q^2 \mathcal{W}_e] \preceq \delta_* I.$$

Each intermediate c_ℓ^{op} bounds $\mathbb{E}_\mu[P(Z_1, Z_2)^2 \mathcal{W}_e] \preceq a^\ell c_\ell^{\text{op}} I$ for every monomial P of degree ℓ .

Proof. On a simultaneous spectral pair (u, v) , the scalar weight is

$$w(u, v) := \frac{1 + uv}{(1 - u^2)(1 - v^2)} \geq 1.$$

The initial bound on its expectation is Lemma A.7. The pointwise bound $(1 - w^{-1})^2 \leq \frac{9}{2}(u^4 + v^4)$ follows from $0 \leq 1 - w^{-1} \leq u^2 + v^2 + |uv|$ and Cauchy–Schwarz. The operator degree-two coupled recurrence gives the initial defect $9a^2 C_2^+(a)$. For this recurrence its quadratic derivative mean is bounded in operator norm by covariance domination, and its terminal derivative by the coefficient variance, exactly as in the proof of Lemma B.16; a trace-only estimate is not used here.

Write $S := \mathbb{E}_\mu[\mathcal{W}_e]$. For an even field f , its projected-mean energy obeys

$$\mathcal{P}(f) \leq (\|\mathbb{E}_\mu[f]\| + \sqrt{\delta \mathcal{R}(f)})^2.$$

Indeed $S \succeq I$, and weighted Cauchy–Schwarz bounds $\mathbb{E}_\mu[(\mathcal{W}_e - I)f]$ by the defect and $\mathcal{R}(f)$. Combining this with $\mathcal{R}(f) \leq \mathcal{P}(f) + b$ gives R_1 and P_1 . The bounds R_0, P_0 and the odd branch follow from Lemma B.16 with the improved upper bound J .

Apply the recurrence to all derivatives of a fixed ordered word in Z_1, Z_2 , acting on a fixed vector. The word-space encoding in Lemma A.19 bounds its even ordinary derivative mean by $(\ell!/r!)^2 b_{r/2}^2$ after division by a^ℓ . Each differentiated position uses its own coefficient creation operator of norm at most $\sqrt{a}$. The terminal array has squared norm at most $(\ell!)^2 a^\ell$ on a unit input. This proves the raw operator bounds, including mixed monomials.

Define $h_{2r} := H_r$ as in Lemma A.21. Its truncated-series inequality applies on every spectral pair.

Multiply Eq. (113) by P^2 . Lemma A.19 bounds each finite monomial; the two remainder terms use the already proved degree- $\ell + k$ bound. This gives Eq. (162).

For the stronger defect estimate, put $t := uv$. The identities

$$Q = h_2 - \frac{t(u+v)^2}{1+t} \geq h_2 - t(u+v)^2, \quad w^{-1} = (1-u^2)(1-v^2)(1-t) + t^2 w^{-1}$$

imply

$$Q^2 w \leq P_4 + P_6 + P_8 + w - h_0 - h_2 - h_4 - h_6,$$

where

$$P_4 := u^4 + 2u^3v + 3u^2v^2 + 2uv^3 + v^4, \\ P_6 := u^6 + u^5v - u^3v^3 + uv^5 + v^6, \quad P_8 := (uv)^3(u+v)^2.$$

The difference in the first step of the preceding identity for Q is $t^2(u+v)^2/(1+t) \geq 0$. Substituting the resulting upper bound for $w^{-1} = 1 - Q$ only in the last term of the second identity gives the displayed polynomial bound. Weighted arithmetic-geometric mean gives $P_4 \leq \frac{9}{2}(u^4 + v^4)$ and $P_8 \leq 2(u^8 + v^8)$; for $uv < 0$ the latter is immediate. Also

$$3(u^6 + v^6) - 2P_6 = (u-v)^2(u^4 - u^2v^2 + v^4) \geq 0.$$

Apply these bounds, Lemma A.19, and Eq. (113) to obtain Eq. (163). All moment coefficients substituted at this step are nonnegative. Descending degree and then the five finite repetitions prove every update; no limiting or fixed-point assertion is required. $\square$

We also use a scalar spectral function to obtain a covariance-independent alternative. This estimate is used only for trace energies.

Lemma B.24 (An odd spectral-field bound). *In the same setting, for every integer $\ell \geq 0$,*

$$N^{-1} \mathbb{E}_\mu[\text{tr}[X^{2\ell} Y_+^{-2}]] \leq a^\ell J_*(\ell!)^2 [p_2 J_*(1+a)]^\ell.$$

Proof. Let $w_2(x) := (1+x^2)/(1-x^2)^2$ and $f_\ell(x) := x|x|^{\ell-1}$ for $\ell \geq 1$. Set $b(x) := \ell^2|x|^{2\ell-2}$. Cauchy-Schwarz and convexity give $f_\ell[u, v]^2 \leq (b(u) + b(v))/2$. The sign-averaged weight kernel is at most $(w_2(u) + w_2(v))/2$. Since b and w_2 increase with absolute argument, their product bound improves to

$$\frac{1}{2}((1-u)^{-1}(1-v)^{-1} + (1+u)^{-1}(1+v)^{-1}) f_\ell[u, v]^2 \leq \frac{1}{2}(w_2(u)b(u) + w_2(v)b(v)).$$

In an eigenbasis of X , the symmetric array $\sum_{i=1}^m |(N_i)_{j,k}|^2$ has row sums at most a . Thus the sum of the sign-averaged derivative energies is at most $a\ell^2 N^{-1} \mathbb{E}_\mu[\text{tr}[X^{2\ell-2} w_2(X)]]$. Apply the odd-field weighted variance bound, with coefficient $p_2 J_*(1+a)$, to $f_\ell(X)$. Its energy equals that of X^ℓ after sign averaging. Induction starts from $N^{-1} \mathbb{E}_\mu[\text{tr}[w_2(X)]] \leq J_*$. For even ℓ , smooth odd approximation of f_ℓ and its first derivative is justified by the integrable inverse-weight envelope. Only first derivatives are used. $\square$

The improved ordinary means and resolvent bounds give an affine expression in several covariance powers. A fixed reference value selects among bounds, but is not an assumption on the covariance.

Definition B.25 (Covariance-power energy coefficients). Fix $\ell \geq 1$ and $\omega_0 \in (0, 1)$, and put $d_\ell := \ell + 11$. For a formal expression $F := F_0 + \sum_{q=1}^{d_\ell} F_q \omega_q$ with nonnegative coefficients, define $\langle F \rangle_0 := F_0 + \omega_0 \sum_{q=1}^{d_\ell} F_q$. Use $J := J_*(a)$, $\delta := \delta_*(a)$, $c := 1 - J^{-1}$, $\iota := (1 - c)^{-1}$, and $\alpha := c\iota^2$. Initialize $R_\ell = P_\ell := (\ell!)^2 J$, a constant expression. At $j = \ell - 1, \dots, 0$ put $r := \ell - j$ and $B_j := p_2(R_{j+1} + P_{j+1})/2$.

For odd r , use $R_j := J(1 + a)B_j$ and $P_j := (J(1 + a) - 1)B_j$. For even r , let $M_j := m_j \omega_{k(r)}$ be as in Lemma A.25. Put $M_{\text{ref}} := m_j \omega_0$, $B_{\text{ref}} := \langle B_j \rangle_0$, and

$$h := \sqrt{\frac{\alpha M_{\text{ref}} + \iota B_{\text{ref}}}{\alpha M_{\text{ref}}}}, \quad h_1 := \sqrt{\frac{M_{\text{ref}} + (1 - \delta) B_{\text{ref}}}{\delta M_{\text{ref}}}}.$$

There are two candidate pairs:

$$\begin{aligned} R_j^{(0)} &:= (1 + 2\alpha + \alpha h + \alpha/h)M_j + (1 + c\iota + \iota/h)B_j, \\ P_j^{(0)} &:= (1 + 2\alpha + \alpha h + \alpha/h)M_j + (c\iota + \iota/h)B_j, \\ R_j^{(1)} &:= \frac{1 + \delta + \delta h_1 + 1/h_1}{(1 - \delta)^2} M_j + \frac{1 + 1/h_1}{1 - \delta} B_j, \\ P_j^{(1)} &:= \frac{1 + \delta + \delta h_1 + 1/h_1}{(1 - \delta)^2} M_j + \frac{\delta + 1/h_1}{1 - \delta} B_j. \end{aligned}$$

Either pair is valid. The certificate selects a pair by comparing outward upper bounds for $\langle R_j^{(i)} \rangle_0$. At $j = 0$, it may instead select the constant bound of Lemma B.24.

Let R be the resulting expression. Define $A(a) := A_0(a) + \sum_{q=1}^{d_\ell} A_q(a) \omega_q$ by selecting among R and the following twelve expressions, again by their outward reference bounds:

$$\frac{s+1}{2s+1} R + \sum_{j=0}^{s-1} \frac{s-j}{2s+1} \text{df}(\ell+j) a^j \omega_{\ell+j}, \quad 1 \leq s \leq 12.$$

Here A_0 denotes a genuine constant coefficient; it is not multiplied by the reference value. Certified outward enclosures may be used for all scalar bounds and positive Young parameters; each selected expression remains a valid upper bound. On a radial cell, every selected alternative and every Young parameter is frozen at its upper endpoint.

The affine energy coefficients yield a radial remainder bound that stays valid throughout each integration cell.

Lemma B.26 (Affine radial remainder). *All coefficients in Definition B.25 are nonnegative, and*

$$N^{-1} \mathbb{E}_\mu[\text{tr}[X^{2\ell}(I - X^2)^{-2}]] \leq a^\ell (A_0 + \sum_{q=1}^{d_\ell} A_q \omega_q).$$

For $L := 2K - \ell \geq 1$, put $z_0 := \sqrt{A_0 + \omega_0 \sum_{q=1}^{d_\ell} A_q}$. For every $v > 0$, the radial remainder satisfies

$$R_K \leq a^K \sqrt{\text{df}(L)} \left(\frac{z_0}{2v} \omega_L + \frac{v}{2z_0} (A_0 + \sum_{q=1}^{d_\ell} A_q \omega_q) \right).$$

These upper bounds hold throughout a radial cell when the endpoint choices and bounds are retained.

Proof. Apply Lemma A.18 to the complete derivative arrays. Lemma A.25 supplies their ordinary means. Young's inequality applied to R_0, P_0, R_1, P_1 of Definition B.22 gives exactly the two affine pairs. Each pair is valid for every covariance spectrum, regardless of its reference value. The terminal derivative remains a constant expression. Lemma B.24 proves the additional constant alternative.

For $0 \leq z < 1$, coefficient comparison gives

$$\frac{1}{(1-z)^2} = \frac{s+1}{2s+1} \frac{1+z}{(1-z)^2} + \sum_{j=0}^{s-1} \frac{s-j}{2s+1} z^j - \frac{z^{s+1}}{(2s+1)(1-z)^2}.$$

Use $z = X^2$, multiply by $X^{2\ell}$, take the trace expectation, and discard the last nonpositive term. The covariance-power moments give the twelve additional expressions. This proves the inverse-square bound.

Cauchy–Schwarz followed by Young's inequality gives

$$R_K \leq a^K \sqrt{\text{df}(L)} \sqrt{\omega_L(A_0 + \sum_{q=1}^{d_\ell} A_q \omega_q)} \leq a^K \sqrt{\text{df}(L)} \left(\frac{z_0 \omega_L}{2v} + \frac{v}{2z_0} (A_0 + \sum_{q=1}^{d_\ell} A_q \omega_q) \right),$$

where the first step also uses the ordinary moment of order $2L$, and the second uses the positive parameter v/z_0 . In particular the term containing A_0 stays outside the covariance-spectrum average.

For uniformity on a cell, retain its endpoint root bounds, its five finite refinement steps, and its Young parameters. Every variance power is nonnegative. The two mean bounds before linearization are monotone in the derivative budget and in the upper bounds J, δ ; their displayed nonnegative linear coefficients remain upper bounds when these endpoint values are kept. Repeating the finite inductions at smaller variance proves the claimed uniformity. This also justifies hereditary variance reduction; no numerical monotonicity test is being substituted for an argument. $\square$

B.8 Small-ball parameters for the repair algorithm

We now specify all small-ball rows used by the repair algorithm. The seven coupled rows use the existing spectral families; the remaining rows use Definition B.20.

Definition B.27 (Small-ball parameters). The coupled rows below use the construction of Definition B.18 with the following parameters. In these seven rows $\ell := 8$, and f selects the unchanged family of Definition A.15.

j	f	K	r	H	v_{tr}	$\bar{e}$
10	4	43	12.4253328	5	0.63814040595167	$1.64426167798038 \cdot 10^{-12}$
11	4	45	12.6016924	2	0.41463781721018	$9.63355424219867 \cdot 10^{-13}$
12	5	53	13.8058984	8	0.6924088219785	$1.23685377953941 \cdot 10^{-14}$
13	5	54	14.0018531	2	0.42802566212566	$5.96384567598748 \cdot 10^{-15}$
14	6	61	15.0310766	64	0.92955343498961	$1.46422223322188 \cdot 10^{-16}$
15	6	63	15.2444209	3	0.47398701149934	$4.49242947951826 \cdot 10^{-17}$
16	6	65	15.4577652	2	0.1201300838154	$3.10214657627498 \cdot 10^{-17}$

For the polynomial barriers, let $Q_f(x) := \sum_{j=1}^{K_f} c_{f,j} x^j$. Their coefficients and spectral parameters

are defined by

f	K_f	ζ_f	θ_f	$w_{0,f}$
0	16	0.32788819012121	0.00099235404674344	0.99
1	24	0.27164064793644	$3.9303380938806 \cdot 10^{-6}$	0.995
2	32	0.25223376708465	$8.2076618223836 \cdot 10^{-9}$	0.995
3	25	0.26648109410379	$1.1758093057131 \cdot 10^{-6}$	0.995
4	23	0.27771534880081	$1.0271147781981 \cdot 10^{-5}$	0.995
5	20	0.28941555026725	$4.1043137363215 \cdot 10^{-5}$	0.995
6	16	0.323318851003971	0.00114517152259975	0.995
8	15	0.327489586910613	0.00136622561436123	0.995

and

$$\begin{aligned}
Q_0(x) &:= 0.0012971998262828x + 0.00108053781547036x^2 + 0.0167961417087915x^3 \\
&\quad + 67384.2888392018x^{14} + 1017.78250320307x^{16}, \\
Q_1(x) &:= 4.45853065136505 \cdot 10^{-6}x + 1.54956242346197 \cdot 10^{-5}x^2 \\
&\quad + 3.90591602891705 \cdot 10^8x^{23} + 256977.368144358x^{24}, \\
Q_2(x) &:= 9.2313324416040203 \cdot 10^{-9}x + 3.6840159552683464 \cdot 10^{-8}x^2 \\
&\quad + 4.1078726253280973 \cdot 10^{12}x^{31} + 1.2428631184136885 \cdot 10^8x^{32}, \\
Q_3(x) &:= 1.350531304170612252 \cdot 10^{-6}x + 4.61336362877522568 \cdot 10^{-6}x^2 \\
&\quad + 1.765510371838126866 \cdot 10^9x^{24} + 867489.646233362826x^{25}, \\
Q_4(x) &:= 1.193189235308612298 \cdot 10^{-5}x + 3.80349289860350346 \cdot 10^{-5}x^2 \\
&\quad + 1.170901061230449042 \cdot 10^8x^{22} + 99309.344465954397x^{23}, \\
Q_5(x) &:= 4.89975704756345085 \cdot 10^{-5}x + 0.000124612924019477631x^2 \\
&\quad + 0.000232078050348111516x^3 + 9041303.3805839526x^{19} \\
&\quad + 24853.9413294090849x^{20}, \\
Q_6(x) &:= 0.0015817842109148x + 0.0266135748044715x^3 \\
&\quad + 1462.38917137446x^{13} + 49195.4507263369x^{14} + 878.630519626377x^{16}, \\
Q_8(x) &:= 0.00187892435927635x + 0.0321444735341866x^3 \\
&\quad + 26489.4033655906x^{13} + 735.301392353928x^{15}.
\end{aligned}$$

Unlisted coefficients are zero. The polynomial rows have the following parameters, with $K := K_f$,

$\eta := r^{-2}$, and $L := 2K - \ell$:

j	f	r	ℓ	H	$\bar{e}$
47	0	6.9341848886221	7	3	$1.06246838945167 \cdot 10^{-5}$
48	1	8.8969026358097	9	7	$3.32527918144844 \cdot 10^{-8}$
49	1	8.961658572716	9	5	$2.74812565752633 \cdot 10^{-8}$
50	2	10.848478627797	9	10	$6.4452182929912 \cdot 10^{-11}$
51	2	10.928017999311	9	7	$4.48713786809528 \cdot 10^{-11}$
52	2	11.007557370825	9	4	$3.50850362196492 \cdot 10^{-11}$
53	2	11.087096742339	9	3	$2.95683086718661 \cdot 10^{-11}$
54	3	9.3227647790481	9	7	$9.06881270363679 \cdot 10^{-9}$
55	3	9.3906203454778	9	4	$7.42610703648346 \cdot 10^{-9}$
56	3	9.4584759119076	9	3	$6.29054415051436 \cdot 10^{-9}$
57	4	8.5567996057163	9	7	$9.56174097968345 \cdot 10^{-8}$
58	4	8.6190801091756	9	5	$7.94415211555075 \cdot 10^{-8}$
59	5	8.0042077279365	8	10	$5.32989244296269 \cdot 10^{-7}$
60	5	8.0628933440482	8	7	$4.31710821397348 \cdot 10^{-7}$
61	5	8.1215789601599	8	5	$3.6661043387213 \cdot 10^{-7}$
62	6	6.7656242147914	7	4	$1.56154518925382 \cdot 10^{-5}$
64	0	6.85	7	4	$1.27040007532298 \cdot 10^{-5}$
74	8	6.6502	7	5	$2.09898502707834 \cdot 10^{-5}$

The remaining two parameters, indexed by the same row j , are

j	ω_0	v_{tr}	j	ω_0	v_{tr}
47	0.47444922858642	0.56480634333094	48	0.76005798813903	0.79287413588591
49	0.51500208083266	0.5538808561055	50	0.98498934766445	0.99709206607674
51	0.64824484757477	0.68083521295303	52	0.48572444770912	0.54813724530658
53	0.34041109721597	0.38323145564369	54	0.72579432975927	0.76107946950087
55	0.74017734896939	0.80051195164946	56	0.54323893666379	0.63179148789931
57	0.76478080218808	0.79837667702084	58	0.52649369770235	0.56882775629284
59	0.98498751273893	0.99759206660314	60	0.7205186743821	0.75565126945003
61	0.55201066327663	0.60073408210815	62	0.56948186288992	0.65123767285517
64	0.71348301308164	0.78081031514684	74	0.63166	0.693944505231636

For a polynomial row, take $N_{\text{grid}} := 128$, except $N_{\text{grid}} := 512$ for row 74. Put $u_i := 1 - (1 - i/N_{\text{grid}})^3$ and $q_i := (u_i^K - u_{i-1}^K)/K$. At $a_i := \eta u_i$, let $A_{i,q}$ be the coefficients of Definition B.25 and put $z_i := \sqrt{A_{i,0} + \omega_0 \sum_{q=1}^{d_\ell} A_{i,q}}$. With $D := \max\{K - 1, L, d_\ell\}$ and coefficients outside their ranges set to zero, define the polynomial

$$\begin{aligned}
T(s) := & \sum_{j=2}^{K-1} \frac{c_j \text{df}(j) \eta^j}{j} s^j \\
& + c_K \eta^K \sqrt{\text{df}(L)} \sum_{i=1}^{N_{\text{grid}}} q_i \left(\frac{z_i}{2v_{\text{tr}}} s^L + \frac{v_{\text{tr}}}{2z_i} \sum_{q=0}^{d_\ell} A_{i,q} s^q \right).
\end{aligned} \tag{164}$$

This polynomial has degree at most D and nonnegative coefficients. The menu for the recursive policy consists of the 25 rows with indices $\mathcal{J} := \{10, \dots, 16, 47, \dots, 62, 64, 74\}$.

The full-domain scalar bounds imply the same hereditary interface for both types of rows.

Lemma B.28 (Additional hereditary Gaussian small balls). *Every row in Definition B.27 has the strict curvature and root conditions. For a polynomial row, let g_H be the continuous function on $[0, 1]$ defined in the proof of Lemma B.37. Then*

$$\frac{c_1\eta}{H-1}g_H(s) + \frac{H}{H-1}T(s) < 0.999\bar{e} \quad (0 \leq s \leq 1). \quad (165)$$

For every menu row, symmetric matrices $B_1, \dots, B_m \in \mathbb{R}^{n \times n}$ with $\sum_{i=1}^m B_i^2 \preceq v\eta I$, $0 < v \leq 1$, and independent standard Gaussian variables $\xi_1, \dots, \xi_m$ satisfy

$$\Pr\left[\left\|\sum_{i=1}^m \xi_i B_i\right\| < 1\right] \geq \exp(-nv\bar{e}).$$

Proof. For coupled rows, use Lemma B.19 with the seven displayed choices. For polynomial rows, separate the quadratic term c_1x^2 , whiten by $(I + 2c_1\Gamma)^{-1}$, and apply Lemma B.26. The integral of each radial monomial is exactly q_i . Holding the other coefficients at the cell's upper endpoint gives Eq. (164) as an upper bound for the radial loss. The determinant Hölder calculation in Lemma A.13 then gives the profile in Eq. (165). Its constant term $HT(0)/(H-1)$ is retained outside the covariance-spectrum average. The full-domain checks are in Lemma B.37.

At variance $v\eta$, keep all endpoint choices. The determinant coefficient decreases by v , every finite moment coefficient by v^j , and every remainder coefficient by at least v^K . Thus the complete exponent decreases by at least v . The ambient matrix dimension remains n , independently of m , which proves the hereditary statement. $\square$

B.9 Nearly integral projections

For recursive repair, we organize calls by their maximum active size and specify the projection accuracy at each state.

Definition B.29 (Near-integral states and accuracy). Put $a_i := 10^{-40} \lceil 10^{40}(999/1000)^i \rceil$ for $0 \leq i \leq 33584$. A state (h, i) receives $m \leq a_i n$ input matrices $M_1, \dots, M_m$ of ambient size $n \times n$. The policy of Definition B.34 either selects a suffix or specifies a row $(r, e) := (r, \bar{e})$ and a positive rational dual scale v . For a projection state define

$$E := e/a_i, \quad \beta := 10^{-14}, \quad t_0 := 0.4, \quad \epsilon_w := 10^{-6}, \quad \sigma_{\text{sm}} := 10^{-6}.$$

Let $K := \{z : \|\sum_j z_j M_j\| \leq r\sqrt{a_i n}\}$, and set

$$\begin{aligned} R &:= \frac{t_0 r \sqrt{a_i n}}{s_\star(1 - \sigma_{\text{sm}})}, & R \leq \hat{R} &\leq (1 + 10^{-8})R, \\ D &:= \{x : h_{\hat{R}}(x) \leq 1 - \sigma_{\text{sm}}\}, & C &:= (D/t_0) \cap [-1, 1]^m, \\ \rho &:= \min\{t_0/8, t_0\epsilon_w E, s_{\text{sm}}\hat{R}/(4(2m + \sqrt{m}))\}. \end{aligned} \quad (166)$$

Here $h_{\hat{R}}$, s_{sm} and a_{sm} are as in Lemma B.10. Let

$$y_\star(\xi) := \arg \max_{y \in C} \{\langle \xi, y \rangle - \beta \|y\|_2^2/2\}, \quad \xi \sim N(0, I_m).$$

The Gaussian width estimate controls the distance to signs in a weighted average, and the projection tolerance preserves this control.

Lemma B.30 (Weighted deficit and certified projection). *At every projection state,*

$$\mathbb{E}[\sum_{j=1}^m |\xi_j| \frac{1 - |y_{*,j}|}{2}] \leq (\frac{E}{2c_G} + \frac{\beta}{4})m. \quad (167)$$

For $m \geq N_0$, a certified trial uses the prescribed error ρ and numerical failure budget ϵ . Its arithmetic cost is $n^{3+o(1)} \text{polylog}(1/\epsilon)$. It rejects $\|\xi\|_2^2 > 2m$. On a valid nonrejected solve its feasible point $y = x/t_0$ satisfies

$$\|\sum_j y_j M_j\| \leq I_{h,i} \sqrt{n}, \quad I_{h,i} := \frac{(1 + 10^{-8})r\sqrt{a_i}}{s_\star(1 - \sigma_{\text{sm}})},$$

and its weighted half-deficit exceeds the ideal one by at most $\epsilon_w Em$.

Proof. Normalize $B_j := M_j/(r\sqrt{a_i n})$ and use $v = m/(a_i n) \leq 1$ in Lemma B.28. This gives $\gamma_m(K) \geq e^{-Em}$. The smoothed norm inclusion gives $(t_0/s_\star)K \subset D$: its value there is at most $1 - 10^{-6} + 5 \cdot 10^{-7} = 1 - s_{\text{sm}}$. In particular smoothing is paid by radius enlargement, not by reducing the possibly small exponent e .

Lemma B.14 gives $w(C) \geq (c_G - E/c_G)m$. Comparison of the regularized optimizer with a support maximizer yields

$$\mathbb{E}[\|\xi\|_1 - \langle \xi, y_\star \rangle] \leq (E/c_G + \beta/2)m.$$

Since $|\xi_j|(1 - |y_{*,j}|) \leq |\xi_j| - \xi_j y_{*,j}$ pointwise, division by two proves Eq. (167).

The exact physical projection of $t_0 \xi / \beta$ onto $D \cap [-t_0, t_0]^m$ is $t_0 y_\star$. Request an exactly feasible output within Euclidean distance ρ , including numerical approximation and the downward-rounded radial correction. On a nonrejected target, Cauchy–Schwarz gives

$$\sum_j |\xi_j| \left| \frac{1 - |y_j|}{2} - \frac{1 - |y_{*,j}|}{2} \right| \leq \frac{\sqrt{2m\rho}}{2t_0} \leq \epsilon_w Em.$$

The first step uses the target norm cap and Euclidean accuracy. The second uses Eq. (166) and $m \geq 1$. Feasibility gives the spectral norm bound. Gaussian concentration bounds norm rejection by $e^{-m(1-\log 2)/2} < 10^{-6}$ for $m \geq N_0$.

Apply Lemma B.10 with $Q = [-t_0, t_0]^m$ and $c_b = 3.2 \cdot 10^{27}$. Indeed, the side length is 0.8, the target cap follows from $\|\xi\|_2^2 \leq 2m$, and every selected row has $R^2 > 15.55435876527199m > 15m$. The finite menu gives $R = O(\sqrt{n})$, and $E > 2.03 \cdot 10^{-12}$ makes ρ^{-1} polynomial in n . Thus the lemma supplies the required accuracy and cost. $\square$

Algorithm 12 A certified near-integral projection trial

- 1: **procedure** NEARINTEGRAL($M_1, \dots, M_m, n, h, i, \epsilon$) $\triangleright$ Lemma B.30
 - 2: Form the body, target scale and accuracy in Eq. (166)
 - 3: Draw a fresh $\xi \sim N(0, I_m)$
 - 4: **if** $\|\xi\|_2^2 > 2m$ **then**
 - 5: **return** FAIL
 - 6: **end if**
 - 7: Compute the certified projection of $t_0 \xi / \beta$, with error ρ and budget ϵ
 - 8: **if** the solve fails **then**
 - 9: **return** FAIL
 - 10: **end if**
 - 11: **return** x/t_0 for its exactly feasible output x
 - 12: **end procedure**
-

B.10 Unbiased binary repair and a four-level algorithm

We round the remaining fractional masses one binary digit at a time, using a fresh random orientation for each helper signing.

Lemma B.31 (Unbiased dyadic repair). *Let $y \in [-1, 1]^m$, put $s_j := 1$ if $y_j \geq 0$ and $s_j := -1$ otherwise, and let $q_j := (1 - |y_j|)/2$. Truncate each q_j downward to a multiple of 2^{-B} , where $B := \lceil \log_2 n \rceil + 32$. Process the bits $b = B, B-1, \dots, 1$. At bit b , let S_b consist of the coordinates whose current numerator on the 2^{-b} grid is odd. Choose any signing $\sigma^{(b)}$ of the matrices $(s_j M_j)_{j \in S_b}$, then draw a fresh independent fair sign ϵ_b and replace*

$$q_j \leftarrow q_j + 2^{-b} \epsilon_b \sigma_j^{(b)} \quad (j \in S_b).$$

The final vector $z_j := s_j(1 - 2q_j)$ is a signing. Conditional on the initial truncated masses, even if every helper signing depends on the preceding history,

$$\mathbb{E}[|S_b|] = \sum_{j=1}^m \tau_b(q_j^{\text{init}}), \quad \tau_b(q) := \text{dist}(2^b q, 2\mathbb{Z}). \quad (168)$$

If the helper at bit b has discrepancy at most $D_b \sqrt{n}$, the total change in the matrix sum is at most

$$\left(\sum_{b=1}^B 2^{1-b} D_b + 10^{-8} \right) \sqrt{n}.$$

Proof. An odd grid numerator is not a boundary numerator, so both adjacent even numerators lie in $[0, 2^b]$. Every update therefore stays in $[0, 1]$ and moves to the next coarser grid. The final masses lie in $\{0, 1\}$. The orientation is drawn after the helper output is fixed, so each coordinate update has conditional mean zero, regardless of dependencies among the helper signs.

Fix a coordinate and a bit b . Until the b th bit is processed, the coordinate cannot cross a boundary of the 2^{-b} grid: a coordinate on such a boundary is inactive at every finer step. Between consecutive grid points, the final value on that grid is one of the two endpoints, and its expectation is its initial value by the bounded martingale property. The probability of the odd endpoint is consequently the linear interpolation between endpoint parities. This interpolation is exactly τ_b . Summing over coordinates proves Eq. (168). No independence among coordinates is used.

At bit b the matrix increment is $-2^{1-b} \epsilon_b \sum_{j \in S_b} \sigma_j^{(b)} s_j M_j$. The triangle inequality bounds its norm by $2^{1-b} D_b \sqrt{n}$. Truncation changes the initial matrix sum by at most $2n2^{-B} \leq 2^{-31} < 10^{-8} \sqrt{n}$. Adding these bounds proves the assertion. $\square$

To estimate several binary repair levels together, we combine their tent functions into one piecewise-linear function and bound it by a concave polygon.

Definition B.32 (A fixed dyadic shape). For $1 \leq b \leq 15$ set $\theta_b := 2^{1-b}$ and take the following exact rational scales, in increasing order of b :

$$\begin{aligned} (s_1, \dots, s_8) &:= (1, 0.81, 0.71, 0.65, 0.59, 0.56, 0.54, 0.50), \\ (s_9, \dots, s_{15}) &:= (0.48, 0.47, 0.43, 0.40, 0.55, 0.48, 0.055). \end{aligned}$$

On $[0, 1/2]$ put $f_0(q) := \sum_{b=1}^{15} \theta_b s_b \tau_b(q)$. Form its least concave majorant from the 16385 points $q = k/32768$, and replace any decreasing terminal part by its maximum. Denote the resulting nondecreasing concave polygon by H_0 . Define

$$I_0 := \frac{1}{2} \int_0^{1/2} (H_0'(q))^2 dq = \frac{1721493679}{655360000} = 2.62679089202880859375. \quad (169)$$

The concave polygon converts the Gaussian weighted mass bound into a joint estimate for the coarse repair levels.

Lemma B.33 (A joint Gaussian conjugate bound). *Suppose the possibly dependent random masses $q_j \in [0, 1/2]$ satisfy*

$$\mathbb{E}\left[\sum_{j=1}^m |\xi_j| q_j\right] \leq b_* m, \quad \xi \sim N(0, I_m).$$

For every $v > 0$,

$$\frac{1}{n} \mathbb{E}\left[\sum_{j=1}^m v f_0(q_j)\right] \leq 2 \frac{m}{n} v \sqrt{c_G b_* I_0}. \quad (170)$$

In particular, if $m \leq wn$ and $b_* := e/(2c_G w) + \beta/4 + \epsilon_w e/w$, its right side is at most

$$v \sqrt{I_0((2 + 0.0000032)ew + 0.8\beta w^2)}.$$

Proof. Let $F_v(x) := \max_{0 \leq q \leq 1/2} (v f_0(q) - xq)$ for $x \geq 0$. Since $f_0 \leq H_0$ and $H_0(0) = 0$, integration of the conjugate polygon gives

$$\int_0^\infty F_v(x) dx \leq \frac{v^2}{2} \int_0^{1/2} (H'_0(q))^2 dq = v^2 I_0.$$

For completeness, on each linear segment the nonnegative slope of H_0 is constant. The maximizing abscissa in its conjugate is the measure of the set where $vH'_0 > x$. Integrating the area under the conjugate, or applying Fubini to these level sets, gives the displayed identity for H_0 . The first step uses pointwise domination by its conjugate; the second uses Eq. (169).

For every $\lambda > 0$, the pointwise inequality $v f_0(q_j) \leq \lambda |\xi_j| q_j + F_v(\lambda |\xi_j|)$ applies even when q_j depends on all of ξ . The density of $|\xi_j|$ is at most c_G , whence

$$\mathbb{E}\left[\sum_j v f_0(q_j)\right] \leq m(\lambda b_* + \frac{c_G v^2 I_0}{\lambda}).$$

Minimizing in λ proves Eq. (170); the zero-budget case follows by a limit. Finally use $m/n \leq w$, $c_G < 0.8$, $\epsilon_w = 10^{-6}$, and the stated value of b_* . This proof uses the Gaussian marginal density, not independence between the masses and the target. $\square$

We specify the choices at each recursion state and account for their discrepancy through the costs of the child calls.

Definition B.34 (Finite recursive policy and exact charges). Append $a_{33585} := 0$ to the capacities in Definition B.29. Let c_{suf} be the function in Lemma B.12. At depth zero, define

$$C_{0,i} := 10^{-12} \lceil 10^{12} c_{\text{suf}}(a_i) \rceil, \quad 0 \leq i \leq 33584.$$

Lemma B.36 specifies the rational enclosure test for these ceilings. The policy has all 33585 states at depths 0, 1, 2, 3, and only state (4, 0) at depth four.

The following finite rational rule specifies every remaining choice. For a nonnegative rational x , define the upward square root

$$\text{sqrt}_{128}^+(x) := 2^{-128} \min\{z \in \mathbb{Z}_{\geq 0} : z^2 \geq 2^{256} x\}. \quad (171)$$

It lies between $\sqrt{x}$ and $\sqrt{x} + 2^{-128}$. Use the fixed set of positive rational dual scales

$$\mathcal{V} := \{2^{j-16}(65536 + k) : -64 \leq j \leq 64, \quad 0 \leq k < 65536, \quad j, k \in \mathbb{Z}\}. \quad (172)$$

This set has 8454144 elements. At depth $h \geq 1$, assume the charges at depth $h - 1$ have been defined. Put

$$u_{h-1}(z) := \max_{0 \leq l \leq 33584} (C_{h-1,l} - za_{l+1}).$$

For state (h, i) set $w := a_i$. A choice (r, e) of Definition B.27 is admissible if $e/w > 2.03 \cdot 10^{-12}$. For each admissible row and $v \in \mathcal{V}$, define

$$\bar{I}_i(r) := \text{sqr}_{128}^+((\frac{(1 + 10^{-8})r}{s_\star(1 - \sigma_{\text{sm}})})^2 w), \quad (173)$$

$$\bar{Z}_i(e) := \text{sqr}_{128}^+(I_0((2 + 0.0000032)ew + 0.8\beta w^2)), \quad (174)$$

$$\bar{B}_{h,i}(e, v) := \sum_{b=1}^{15} \theta_b u_{h-1}(vs_b) + v\bar{Z}_i(e) + 2^{-14}C_{h-1,i}, \quad (175)$$

$$d_{h,i}(r, e, v) := \bar{I}_i(r) + 1.005\bar{B}_{h,i}(e, v) + 10^{-8}. \quad (176)$$

Select the suffix if $C_{0,i}$ is at most every candidate $d_{h,i}(r, e, v)$, including when there is no admissible row. Its charge is $C_{h,i} := C_{0,i}$. Otherwise select a candidate with least $d_{h,i}$, breaking ties by the smaller row index and then the smaller v .

For a selected projection row and scale, define

$$\begin{aligned} U_{h,i} &:= \sum_{b=1}^{15} \theta_b u_{h-1}(vs_b), \\ Z_{h,i} &:= \sqrt{I_0((2 + 0.0000032)ew + 0.8\beta w^2)}, \\ B_{h,i} &:= U_{h,i} + vZ_{h,i} + 2^{-14}C_{h-1,i}. \end{aligned} \quad (177)$$

Use $\bar{I}_{h,i} := \bar{I}_i(r)$ and $\bar{B}_{h,i} := \bar{B}_{h,i}(e, v)$ as the outward upper bounds in the algorithm, and set

$$C_{h,i} := 10^{-12} \lceil 10^{12} d_{h,i}(r, e, v) \rceil. \quad (178)$$

Integer square comparisons, rational arithmetic, finite maxima and finite minima determine these quantities. All choices are made before the input matrices, n , and p are supplied.

For a child support of relative size $z \in (0, 1]$, choose a minimum-charge state among all l with $z \leq a_l$, breaking ties by the smaller index, at the preceding depth. Denote its charge by $c_{h-1}(z)$ and set $c_{h-1}(0) := 0$. Thus, for every $z \in [0, 1]$ and $t > 0$,

$$c_{h-1}(z) \leq u_{h-1}(t) + tz, \quad c_{h-1}(z) \leq C_{h-1,i} \quad \text{if } z \leq a_i. \quad (179)$$

An induction on the recursion depth combines the projection bound, unbiased repair, and suffix guarantees with explicit failure budgets.

Lemma B.35 (Complete discrepancy, confidence, and running time). *For $n > N_*$, every state (h, i) signs any $m \leq a_i n$ symmetric contractions of ambient size $n \times n$, with discrepancy at most $C_{h,i}\sqrt{n}$, failure probability at most its requested budget p_c , and cost $n^{3+o(1)} \text{polylog}(1/p_c)$. At the root,*

$$C_{4,0} \leq 12.396142442412 < 12.40.$$

For $n \leq N_$, the certified independent-sign branch of Lemma B.11 suffices.*

Proof. The suffix case is Lemma B.12. For $m < N_0$, arbitrary signs suffice, since

$$\left\| \sum_j \varepsilon_j M_j \right\| \leq m < N_0 < 0.0003\sqrt{n}.$$

The first step uses the triangle inequality and the contraction hypothesis, the second is the short-case condition, and the third uses $n > N_*$. Every state charge exceeds 0.0003. Fix a projection state and assume the assertion for all preceding-depth states.

We analyze a reference trial without conditioning on numerical success. On target-norm rejection or an invalid projection output, set all reference masses to zero. Otherwise use the masses of its feasible point, truncated as in Lemma B.31. These masses satisfy the expected weighted bound in Lemma B.33, with $b_* = E/(2c_G) + \beta/4 + \epsilon_w E$. Indeed, on valid outcomes Lemma B.30 applies, on the others the weighted mass is zero, and the ideal weighted mass is always nonnegative. Truncation only decreases it.

During reference repair, replace a failed helper by arbitrary signs and still draw its fair orientation after its output is fixed. Eq. (168) therefore remains valid without conditioning on any helper's success. Charge every support its prescribed child-state bound, irrespective of that success. If $\mathcal{D}$ denotes the resulting sum of bit charges, Eqs. (168) and (179) give

$$\mathbb{E}[\mathcal{D}] \leq U_{h,i} + \frac{1}{n} \mathbb{E}\left[\sum_{j=1}^m v f_0(q_j)\right] + 2^{-14} C_{h-1,i} \leq B_{h,i}.$$

The first step sums the affine support bounds at bits 1 through 15 and uses $\sum_{b=16}^{\infty} 2^{1-b} = 2^{-14}$ for the remaining bits. The second is Lemma B.33. In particular Markov's inequality gives $\Pr[\mathcal{D} > 1.005\bar{B}_{h,i}] \leq 1/1.005$.

The actual trial rejects a failed projection, any failed child, or a repair charge exceeding $1.005\bar{B}_{h,i}$. On all other outcomes, the reference construction agrees with it, every helper has its certified discrepancy, and Lemma B.31 and Eq. (178) prove the claimed norm bound.

Here are explicit confidence caps, including adaptive calls. Put

$$M_c := \left\lceil \frac{\log(4/p_c)}{-\log(249/250)} \right\rceil, \quad \epsilon_c := \min\{10^{-5}, p_c/(8M_c)\}, \quad p'_c := \min\{10^{-5}/B, p_c/(4M_c B)\}.$$

Use at most M_c independent trials, projection budget ϵ_c , and budget p'_c for each of its at most B child calls. Uniformly conditional on all preceding history, a trial succeeds with probability at least

$$1 - 1/1.005 - 10^{-6} - 2 \cdot 10^{-5} > 1/250.$$

The terms pay respectively for excessive reference charge, target-norm rejection, and the projection and child failure budgets. Thus exhaustion has probability at most $(249/250)^{M_c} \leq p_c/4$. Over all attempts, projection failure has probability at most $p_c/8$, and child failure at most $p_c/4$. Their union, including possible invalid accepted numerical outputs, has probability at most $5p_c/8 < p_c$. Fresh targets, internal numerical probes, and orientations are used after each adaptive input is fixed. This closes the induction.

There are exactly four repair levels. Each call makes $O(\log n)$ children per trial and $O(\log(1/p_c))$ trials. Descendant budgets have logarithms bounded by a polynomial in $\log n$ and $\log(1/p_c)$, since the depth is fixed. Lemma B.30 and the suffix cost therefore give $n^{3+o(1)}$ polylog($1/p_c$). Computing signed sums, changing signs of input matrices, and norm certificates use at most $O(n^3)$ per call. Dyadic truncation can be implemented by $O(\log n)$ comparisons per coordinate; no unit-cost floor primitive is required. The finite scalar choices are determined before the input matrices are supplied. Lemma B.37 proves the root value. The small-input branch gives discrepancy $12\sqrt{n} < 12.40\sqrt{n}$ with the stated failure and cost. $\square$

Algorithm 13 Cubic-time signing with constant below 12.40

```
1: procedure MAINCUBICSMALLCONSTANT( $A_1, \dots, A_n, p$ ) ▷ Theorem B.1
2:   if  $n \leq N_*$  then
3:     Run capped independent sign trials with the  $12\sqrt{n}$  norm certificate ▷ Lemma B.11
4:     return the accepted signing, or FAIL if the cap is reached
5:   end if
6:   return RECURSIVEREPAIR( $A_1, \dots, A_n, n, 4, 0, p$ ) ▷ Algorithm 14
7: end procedure
```

Algorithm 14 Four-level signing by unbiased binary repair

```
1: procedure RECURSIVEREPAIR( $M_1, \dots, M_m, n, h, i, p_c$ ) ▷ Lemma B.35
2:   if  $m < N_0$  then
3:     return arbitrary signs
4:   end if
5:   if  $h = 0$  or the policy selects a suffix then
6:     return SUFFIXSIGNING( $M_1, \dots, M_m, n, a_i, p_c$ ) ▷ Algorithm 11
7:   end if
8:   Set  $B, M_c, \epsilon_c, p'_c$  as in Lemma B.35
9:   for  $a = 1, \dots, M_c$  do
10:     $y \leftarrow$  NEARINTEGRAL( $M_1, \dots, M_m, n, h, i, \epsilon_c$ ) ▷ Algorithm 12
11:    if  $y = \text{FAIL}$  then
12:      Continue to the next trial
13:    end if
14:    Set nearest signs  $s$ , truncated masses  $q$ , and charge  $D \leftarrow 0$ 
15:    for  $b = B, B - 1, \dots, 1$  do
16:      Form the odd-numerator support  $S_b$ 
17:      if  $S_b \neq \emptyset$  then
18:        Select its minimum-charge covering child index  $j$  at depth  $h - 1$ 
19:         $\sigma \leftarrow$  RECURSIVEREPAIR( $(s_l M_l)_{l \in S_b}, n, h - 1, j, p'_c$ )
20:        if  $\sigma = \text{FAIL}$  then
21:          Reject this trial and continue with the next trial
22:        end if
23:        Draw a fresh fair  $\epsilon_b \in \{-1, 1\}$  and update the masses
24:         $D \leftarrow D + 2^{1-b} C_{h-1, j}$ 
25:      end if
26:    end for
27:    if  $D \leq 1.005 \bar{B}_{h, i}$  then
28:      return  $(s_l (1 - 2q_l))_{l=1}^m$ 
29:    end if
30:  end for
31:  return FAIL
32: end procedure
```

B.11 Full-domain scalar certificates and exact data

The projection tuple is fixed, so its geometric hypotheses require one scalar verification. The seven inherited small-ball estimates are already certified in Appendix A.

Lemma B.36 (Certified projection and suffix inequalities). *The tuple in Definition B.5, with exponent E_* , satisfies every hypothesis of Lemma B.9. The suffix cost of Lemma B.12 satisfies*

$$140.9914310872 < c_{\text{suf}}(1) < 140.9914310873 < 141. \quad (180)$$

For every capacity a_i of Definition B.29, rational outward evaluation determines $\lceil c_{\text{suf}}(a_i) \rceil_{12}$, where $\lceil x \rceil_{12} := 10^{-12} \lceil 10^{12} x \rceil$.

Proof. Substitute $(\delta, \alpha, a, v, t) = (1/4, 29/100, 9/20, 481/2000, 3/10)$ in Eq. (146). The domain conditions follow from

$$\alpha - 1/N_0 - \delta > 0.03999, \quad p_* = 0.71, \quad a = 0.45, \quad 0.12853 < z < 0.12854 < p_0.$$

Let $V := (a - p_* - p_* \log(a/p_*))/2$. The logarithm bounds from Lemma A.16 give

$$J(2) - E_* > 0.12217, \quad (181)$$

$$V - E_* > 0.00063517, \quad (182)$$

$$\Gamma > 0.00118766. \quad (183)$$

The first two inequalities imply both volume conditions since $N_0 = 3 \cdot 10^8$ and $\log 2 < 1$. The third gives $\Gamma > 1/(N_0 + 1)$ and

$$N_0 \Gamma - \log(N_0 + 1) - \log 10000 > 356000.$$

Here $\log(N_0 + 1) < 20$ and $\log 10000 < 10$. Finally,

$$v - \kappa_{\text{pr}} - 4t/5 = 1/2500 > 0,$$

which proves the objective-separation condition. Thus Lemma B.9 applies uniformly to every projection call.

Evaluate the finite expression in the proof of Lemma B.12 using rational outward bounds for square roots and logarithms. At $w = 1$ this gives Eq. (180). At each a_i , refine the bounds until the two enclosing endpoints have the same upward ceiling at scale 10^{12} . All 33585 evaluations satisfy this test. In particular the smallest ceiling is $0.00580335075 > 0.0003$.

The Gaussian mean comparison follows from $\pi > 3.14159$ and $3.14159(0.798)^2 > 2$, so $\sqrt{2/\pi} < 0.798$. The positive Taylor sum $\sum_{j=0}^{128} (121/2)^j / j! > 18828 \cdot 10^{22}$ gives $2 \cdot 10^{24} e^{-121/2} < 0.010623 < 0.02$. The identity $N_0^2 = (0.0003)^2 N_*$ verifies the terminal-rounding allowance. These are the remaining scalar inputs to Lemma B.11. $\square$

The remaining certificate checks the refined rows and the recursive charges that determine the final discrepancy constant.

Lemma B.37 (Certified energies and four-level repair). *The rows of Definition B.27 satisfy their stated curvature, root, and full-domain profile bounds, including the hypotheses of Lemma B.19 for its seven coupled rows. Definition B.32 gives Eq. (169). All 134341 states in Definition B.34 are well-defined and meet the projection and cleanup hypotheses. At the root,*

$$C_{4,0} \leq \frac{12396142442412}{1000000000000} = 12.396142442412 < 12.40. \quad (184)$$

Proof. Lemma B.36 certifies the unchanged geometric suffix rule. Use the displayed rational parameters and the enclosure rules of Lemma A.16 for the remaining checks.

Curvature and resolvent conditions. The coupled rows retain the seven spectral families already certified in Appendix A. For each polynomial barrier, apply Lemma B.21. The three univariate polynomials are obtained by multiplying each diagonal difference by $(1 + u)^2$. The two mixed polynomials have bidegree at most $(4K - 4, 2K - 2)$. Apply the Bernstein criterion of Lemma A.16 in each coordinate. Start from $[0, 1]^2$; accept precisely when every coefficient is positive, otherwise bisect the first coordinate at even depth and the second at odd depth. Midpoint de Casteljau averaging gives exact rational child coefficients. For a univariate polynomial always bisect its interval. The far-negative strip uses the two explicit scalar conditions of that lemma.

All eight barriers pass. In the order $f = 0, 1, 2, 3, 4, 5, 6, 8$, the numbers of accepted univariate leaves are at most 5, 4, 4, 4, 4, 3, 5, 6, and the corresponding bivariate counts are at most 12, 19, 14, 11, 11, 11, 17, 12. The maximum bivariate depth is at most 11. The accepted cells cover the full spectral domain, including diagonal limits.

For each row, check the four positive root margins $1 - 2\eta/\zeta^2$, $1 - 4p_1\eta$, A_J , and Δ_J , together with $J > 1$. For the polynomial rows also check $9\eta^2 C_2^+(\eta) < 1$ and carry out the five finite passes of Definition B.22. Lemmas B.23 and B.26 justify every improved operator bound and every frozen radial-cell bound.

Full-domain scalar profiles. The seven coupled rows use 3584 radial cells in total. In their displayed order, the profile checks accept 12, 13, 12, 13, 11, 12, and 15 leaves. The 18 polynomial rows use $17 \cdot 128 + 512 = 2688$ radial cells; each monomial is integrated by Eq. (164).

For completeness, set

$$g_H(s) := \frac{s}{1-s}(-(H-1)\log s - \log(H-(H-1)s)), \quad g_H(0) := g_H(1) := 0.$$

This function is concave on $[0, 1]$, since

$$g_H(s) = (H-1) \int_0^1 \frac{s}{s+t(1-s)} dt - \int_1^H \frac{s}{s+t(1-s)} dt.$$

The first integrands are concave and the subtracted integrands are convex. For the polynomial profile, a complete upper bound on $[a, b]$ is

$$\frac{c_1\eta}{H-1}(g_H((a+b)/2) + \frac{b-a}{2}|g'_H((a+b)/2)|) + \frac{H}{H-1}T(b).$$

The tangent bound controls the first term, and the nonnegative coefficients of T control the second. Bisect every unresolved interval at its midpoint until the outward bound is strictly below $0.999\bar{e}$. The coupled profiles are checked in the same way after $s = (1+x)^{-1}$. Both compactified endpoints are included. For the polynomial rows, the accepted-leaf counts in the displayed row order are

$$36, 37, 36, 14, 36, 34, 35, 38, 38, 39, 36, 36, 18, 38, 36, 33, 36, 54,$$

with maximum depth 15. In particular the constant term is checked before averaging over the covariance spectrum. The accepted intervals certify the stated reserve on the full domain.

Exact shape and all support fractions. Construct the upper concave hull of the 16385 dyadic knots of f_0 by rational cross-products and flatten its decreasing terminal part. Check every original

knot against the hull and check nonnegative, nonincreasing slopes. For consecutive vertices (x_j, y_j) , with $0 \leq j \leq J_0$, exact summation gives

$$I_0 = \frac{1}{2} \sum_{j=0}^{J_0-1} \frac{(y_{j+1} - y_j)^2}{x_{j+1} - x_j} = \frac{1721493679}{655360000},$$

where the first step integrates the squared piecewise-constant slope and the second is rational arithmetic. Affineness between knots proves domination everywhere.

At a repair depth, form the upper concave hull of the rational points $(a_{j+1}, C_{h-1,j})$. A supporting line of slope vs_b evaluates $u_{h-1}(vs_b)$. Verify that every original point is below the hull, that its slopes decrease, and that the selected vertex dominates its two neighbors after subtracting $vs_b x$. Concavity proves the maximum over all vertices. Thus the affine charge bound holds for every real support fraction, including those between adjacent capacities.

Constructing all recursive charges. Construct the capacities and base charges by Definition B.29 and Lemma B.36. Then apply Definition B.34 in increasing depth: form the preceding layer's hull, test all 25 menu rows, minimize over admissible dual scales, compare with the suffix, and round upward. Depths one through three each have 33585 states; depth four needs only $(4, 0)$.

Convexity reduces each dual-grid minimum to a binary search. Put

$$W_{h-1}(v) := \sum_{b=1}^{15} \theta_b u_{h-1}(vs_b),$$

and list $\mathcal{V}$ increasingly as $v_0, \dots, v_{N-1}$, where $N := 8454144$. At a fixed row and state, only $W_{h-1}(v) + v\bar{Z}_i(e)$ depends on v . The function W_{h-1} is convex and piecewise linear because each u_{h-1} is a finite maximum of affine functions. Consequently the rational divided differences

$$D_l := \frac{W_{h-1}(v_{l+1}) - W_{h-1}(v_l)}{v_{l+1} - v_l} + \bar{Z}_i(e) \quad (0 \leq l < N-1) \quad (185)$$

are nondecreasing. Find the smallest l with $D_l \geq 0$ by binary search and choose v_l ; if all differences are negative, choose v_{N-1} . This gives the smallest minimizing scale, including ties. Each comparison uses rational hull values and the integer-square definition of $\bar{Z}_i(e)$.

For example, to construct $(1, 0)$, use $a_0 = 1$ and the entire depth-zero suffix layer in W_0 . Test the admissible menu rows and their minimizing scales, compare the resulting costs with $C_{0,0}$, and round upward at scale 10^{12} . To construct $(2, 0)$, repeat with the completed first layer in W_1 . The same rule gives the third layer and the root. Exact rational evaluation gives

$$C_{1,0} \leq 17.727678996319, \quad (186)$$

$$C_{2,0} \leq 13.157731621060, \quad (187)$$

$$C_{3,0} \leq 12.414301141168. \quad (188)$$

All 33585 charges, including these examples, enter the next layer's hull.

At the root it suffices to test row 74 with

$$r := 6.6502, \quad e := 2.09898502707834 \cdot 10^{-5}, \quad v := \frac{110955}{1024} = 2^{6-16}(65536 + 45419) \in \mathcal{V}.$$

Define $U^{\text{test}} := W_3(v)$ and $B^{\text{test}} := U^{\text{test}} + v\bar{Z}_0(e) + 2^{-14}C_{3,0}$. Integer square comparisons and the rational hull evaluation give

$$\bar{I}_0(r) < 9.859753760984339, \quad (189)$$

$$U^{\text{test}} < 1.385176068142620, \quad (190)$$

$$v\overline{Z}_0(e) < 1.137836045362629, \quad (191)$$

$$2^{-14}C_{3,0} < 0.000757708809886, \quad (192)$$

$$B^{\text{test}} < 2.523769822315134, \quad (193)$$

$$1.005B^{\text{test}} < 2.53638867142670967. \quad (194)$$

The bound on B^{test} uses its full rational value. The root rule selects this candidate or one of smaller cost. We have

$$9.859753760984339 + 1.005 \cdot 2.523769822315134 + 10^{-8} = 12.39614244241104867 < 12.396142442412.$$

The first step is exact rational addition. The second uses the upward ceiling at scale 10^{12} . Monotonicity of that ceiling proves Eq. (184). The child charges include all subsequent repair and cleanup.

Uniform numerical and solver conditions. The menu radii satisfy $r \geq 6.6502$, so every admissible projection state has

$$\left(\frac{t_0 r}{s_\star(1 - 10^{-6})}\right)^2 > 15.55435876527199 > 15.$$

The first step uses the smallest menu radius and the second is a rational comparison. Every selected projection row satisfies $e/a_i > 2.03 \cdot 10^{-12}$ by the admissibility test in Definition B.34. The smallest suffix charge is 0.00580335075. Every affine intercept is at least the smallest child charge because $a_{33585} = 0$. The coarse and fine coefficients sum to 2, so a projection candidate cannot decrease the minimum charge below the minimum suffix charge. Induction gives $C_{h,i} \geq 0.00580335075 > 0.0003$ at every state.

Finally,

$$e^{-N_0(1-\log 2)/2} < 10^{-6}, \quad 1 - 1/1.005 - 10^{-6} - 2 \cdot 10^{-5} > 0.00495412 > 1/250.$$

These inequalities verify the rejection and confidence budgets. These finite constructions, the full-domain scalar checks, and Definition B.34 establish all numerical inputs. $\square$